DESIGN AND DEVELOPMENT OF A WEBXR-ENABLED DIGITAL HERITAGE PLATFORM FOR THE NHCP MUSEO NG LIBINGAN SA ILALIM NG LUPA NG NAGCARLAN

A Capstone Project
Presented to the Faculty
of the College of Computer Studies
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Bachelor of Science in Information Technology

LAGUNA STATE POLYTECHNIC UNIVERSITY
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ABSTRACT

The study designed, developed, implemented, tested, and evaluated a WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The project responded to the need for a more integrated digital environment for heritage information, learning resources, immersive exploration, accessibility-oriented content, visitor services and administrative management. Requirements were discovered through interviews, onsite observation, document examination and conversation with museum staff. The platform was developed iteratively using the Agile Software Development Model and includes digital exhibits, a Heritage Trail, an Interactive Site Map, a 360-degree Virtual Tour, selected AR and WebXR experiences, bilingual and accessibility-oriented features, community and feedback functions, booking services and administrative tools. The Black Box Functional Testing consisted of 54 test cases in 13 module groups. All documented instances received a final result of Pass under the testing conditions reported. The user evaluation consisted of 190 completed responses and eight evaluation criteria. The platform achieved an overall weighted mean of 4.77 read as Strongly Agree. The results demonstrate positive user perceptions of usefulness, ease of use, immersive experience, and accessibility, while also identifying opportunities for continued refinement in digital-exhibit interaction, information-access pathways, and control usability.

Keywords: *WebXR, digital heritage, virtual museum, augmented reality, 360-degree virtual tour, accessibility, technology acceptance*

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DEFINITION OF TERMS

This section presents the important terms used in the study. These terms are defined operationally to provide a clear understanding of how they are used in the development and discussion of the WebXR-Enabled Digital Heritage Platform.

OPERATIONAL TERMS

Operational terms refer to words and concepts defined according to their specific application, implementation, or measurement within the context of the study. They establish a consistent interpretation of the terms as applied throughout the research.

3D Model	A digital three-dimensional representation of the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan, artifacts, structures, or selected heritage elements used in the WebXR-enabled platform for visualization, exploration, and interactive learning.
Agile Methodology	The iterative software development approach used in the study to plan, design, develop, test, and improve the WebXR-enabled digital heritage platform through repeated evaluation, feedback, and system refinement.
Artifact	Objects, materials, images, historical items, or cultural resources presented in the digital heritage platform to provide information about the history, culture, architecture, and burial practices related to the Nagcarlan Underground Cemetery.
Augmented Reality	A feature of the developed system that overlays digital heritage content such as 3D objects, information, or visual elements onto the real-world environment through supported mobile devices.

Booking Schedule	A feature of the developed WebXR-Enabled Digital Heritage Platform that allows visitors to request, schedule, and manage museum visit reservations through an online system.
Interactive	The capability of the platform to allow users to actively engage with digital content through clickable hotspots, navigation controls, guided heritage trails, maps, forms, multimedia, and WebXR-based features.
Museum	Refers to the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan, the heritage institution and historical site that serves as the primary subject and content source of the developed digital heritage platform.
Tourist	Individuals who visit or access the platform onsite or offsite for the purpose of exploring, learning, and experiencing the historical and cultural significance of the Nagcarlan Underground Cemetery.
System Testing	The process conducted by the researchers and IT experts to evaluate the functionality, performance, reliability, compatibility, and major features of the developed WebXR-enabled digital heritage platform before user acceptance evaluation.

TECHNICAL TERMS

Technical terms refer to specialized words and concepts related to the technologies, tools, frameworks, and processes utilized in the development of the system. These terms are defined according to their established meanings in the field of Information Technology to provide readers with a fundamental understanding of the technical components of the study.

3D Model	A digital three-dimensional representation of an object, artifact, structure, or heritage element that users can view and interact with through the developed platform.
Black Box Testing	A software testing method used in the study to evaluate the functions and features of the WebXR-enabled digital heritage platform without examining its internal source code. It focuses on checking whether the system produces the expected output based on the given input.
Content Management System (CMS)	The administrative component of the system that allows authorized museum staff or administrators to create, edit, publish, update, and manage digital content.
Flask	A lightweight Python web framework used in the study to develop the backend of the system, manage web routes, process user requests, and render dynamic web pages.
Online Gallery	A public-facing feature of the system that allows users to browse museum items, images, descriptions, and related digital content such as optional 3D models.
Supabase	A cloud-based backend platform used in the system for database management, data storage, authentication support, and access to system records.

Virtual Tour	A digital simulation of a real place that allows users to explore selected areas of the museum or heritage site online using panoramic or interactive visual content.
WebXR	A web technology standard used to support immersive 3D, virtual reality, and augmented reality experiences through compatible web browsers and devices.

CHAPTER I

INTRODUCTION

This chapter introduces the study and provides the foundation for the “Design and Development of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. It presents the project context, research objectives, conceptual framework, project purpose, and scope and limitations of the study that define the direction and boundaries of the study.

1.1 Project Context

Museums operate in the physical and digital venues that advocate for heritage preservation, education, research, cultural involvement and public access. (International Council of Museums, 2022). Museums have a key role in terms of accessibility and inclusivity. Digital interpretation offers the potential to extend these tasks beyond the physical constraints of a museum, allowing for new ways for visitors to access, interpret and relate with cultural material. But its worth is not just based on the amount of technology used, but also on the reliability of the information delivered, appropriate institutional administration, usable design, accessibility and relevant learning experiences.

International innovation in virtual museums and digital heritage demonstrates that panoramic environments, three dimensional visualisations and browser based immersive technologies can provide increased access and participatory ways of viewing cultural assets. However, these technologies have limitations, including

device and browser compatibility, Internet connectivity, accessibility, complexity of interface, and the possibility of immersive media privileged visual display over accurate historical interpretation. The most acceptable application of immersive technology is as a supplement to digital heritage display rather than an alternative for accessible standard web material.

The Philippines continues to develop internet technology for cultural and historical access, as seen by the growing number of digital collections, virtual museum projects, and institutional repositories. However, the level of services at the national and collection-wide level does not always respond to the individual cultural site's specific demands in interpretation, education, accessibility, visitor services and administration. Therefore, a site-specific digital heritage platform should consider the institution's authorised historical narratives, local context, language, target users, visitor experience, accessibility demands, available technology resources, and standards for storing digital material.

The NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan (Nagcarlan Underground Cemetery) is a National Historical Landmark associated with local history, architecture, religious ceremonies, burial practices, and collective memory. The value of history requires that knowledge be presented in ways accessible to all kinds of visitors. The museum personnel requirements-gathering operations also identified practical options to manage cultural information, visitor access and selected museum services within a more integrated digital environment.

The problem identified is definitely more than the lack of a certain feature on the website. Heritage information, spatial orientation, immersive exploration, accessible learning, visitor-service requests, community engagement and

administrative content management necessitate different user needs and operational processes. If these are addressed independently, through multiple channels or resources, users may find it difficult to go from initial discovery to major heritage learning, while museum personnel may have additional efforts in maintaining a unified digital experience.

To overcome these challenges, our project developed a site-specific WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The platform integrates five major functional areas: Digital Heritage and Educational Content; Interactive and Immersive Experiences; Inclusive Accessibility Support; Visitor Engagement and Museum Services; and Administrative Management and Monitoring. Features include digital museum and educational resources; a 360 degree Virtual Tour; selected AR and WebXR experiences; an Interactive Site Map; a guided Heritage Trail; bilingual and accessibility features; feedback and participation features; visit-request scheduling and administrative content-management features.

The developed platform is an integrated digital environment that presents heritage interpretation, learning, immersive exploration, inclusive access, visitor interaction, museum services and administrative management and informational website. The project aims to enrich the physical museum experience for onsite and offsite users, while maintaining the museum’s function in managing and presenting its heritage treasures.

1.2 Research Objectives

The research objectives define the intended outcomes of the study and outline the design, development, implementation, testing and evaluation of the proposed WebXR-Enabled Digital Heritage Platform. The objectives were designed to fulfill the recognised needs of the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan and to ensure the developed platform corresponds to the requirements of its intended users and museum stakeholders.

1.2.1 General Objectives

This study aimed to design, develop, test, and evaluate a WebXR-enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan that supports accessible, interactive, and immersive heritage learning for onsite and offsite users.

1.2.2 Specific Objectives

1. **Design and develop** an integrated WebXR-enabled platform that presents curated heritage information through digital museum functions, bilingual content, inclusive access, and appropriately scoped immersive experiences.
2. **To implement** the identified functional modules of the developed platform based on the requirements of the intended users and museum stakeholders.
    2.1. Digital Heritage and Educational Content
    2.2. Interactive and Immersive Experiences

2.3. Inclusive Accessibility Support

2.4. Visitor Engagement and Museum Services

2.5. Administrative Management and Monitoring

3. **To test** whether the major public and administrative features operate according to their specified functional requirements through documented black-box test cases, accessibility checks, and a device/browser compatibility record through covering:

3.1. Homepage

3.2. User Registration and Login

3.3. Digital Exhibits

3.4. Heritage Trail

3.5. Interactive Site Map

3.6. 360° Virtual Tour

3.7. AR Content

3.8. WebXR 3D Environment

3.9. Community Submissions Form

3.10. Online Booking Schedule

3.11. Visitor Account

3.12. Admin Login and Function

3.13. Navigation/Error Handling

4. **To evaluate** the user's acceptance of the WebXR-enabled Digital Heritage Platform in terms of:

4.1. Selected Software Quality Characteristics

4.2. Perceived Ease of Use

4.3. Perceived Usefulness

4.4. Attitude Toward Using

4.5. Behavioral Intention to Use

4.6. Actual System Interaction

4.7. Immersive Digital Experience

4.8. Accessibility

1.3 Conceptual Framework

This section presents the conceptual framework used in the study. The framework follows the Input-Process-Output (IPO) model to show how the needs, resources, and requirements of the project are transformed into a WebXR-enabled digital heritage platform. The framework also serves as a guide for the development and evaluation of the system.

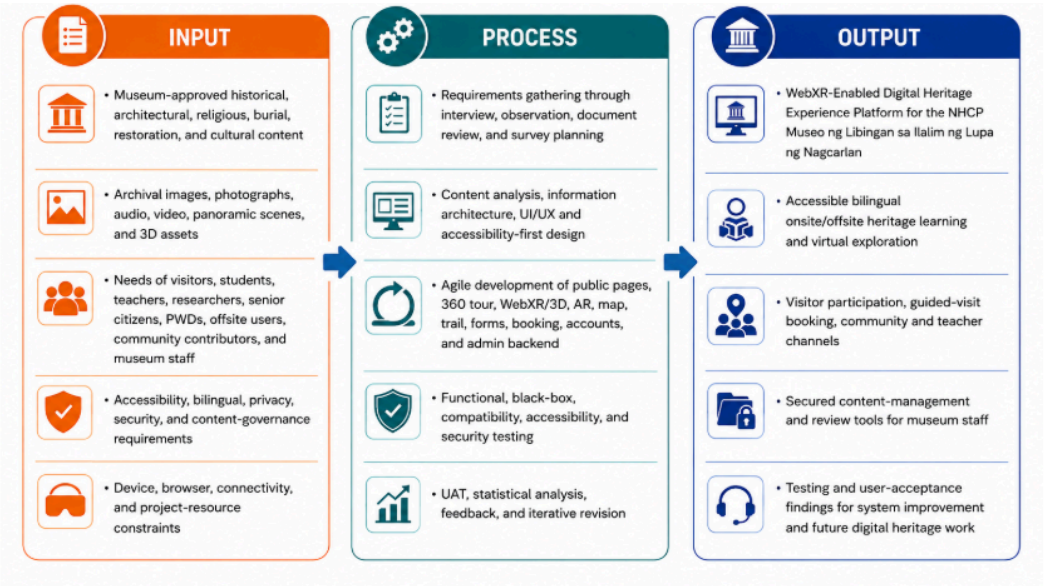

Figure 1.1. Conceptual Framework of the Study

The figure 1.1 above shows the input phase, the study looked at validated museum information and media, the needs of stakeholders and users, accessibility requirements, bilingual content, privacy and governance issues, immersive-media

assets, and the technical limitations of common devices and browsers. These inputs were necessary to ensure the platform was culturally responsible, technically feasible and usable by a diverse audience.

The process stage included requirements analysis, content planning, user interface and information architecture design, development of the public and administrative modules, integration of the 360-degree tour, WebXR/3D environment, selected AR content, site map, heritage trail, accounts, booking, participation forms and accessibility controls. The agile iterations enabled the researchers to evaluate, examine, modify and refine the platform, based on technical discoveries and stakeholder input.

The output included the implemented WebXR-Enabled Digital Heritage Platform, technical test results, user-acceptance findings, and documented limitations. The output supports systematic museum interpretation, engaging and inclusive learning, visitor involvement, guided-visit management and sustainable content management.

1.4 Project Purpose

The study aimed to develop a digital extension of the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan to promote cultural preservation, museum education, public participation, and inclusive access to heritage information. The platform was designed to enhance the physical museum experience with well-organised, interactive and accessible digital materials for onsite and offsite users, while maintaining the role of museum professionals, official tours and permitted institutional services.

1.4.1 Intended Beneficiaries

Visitors to museums and tourists. The platform offers another option to examine museum material before, during or after an onsite visit.

Students & teachers. It can be used as an additional resource for classroom discussion, self-study, and heritage related activities.

History enthusiasts and researchers. The structured digital content can enable an initial investigation of site history, architecture, cultural importance and museum interpretation.

Senior citizens, persons with disabilities, and offsite users. Accessibility and alternate content pathways features are meant to address the demands of users with varying access requirements and device capabilities.

People of the community. Feedback and moderated participation functions offer ways for the community to engage, but not replace museum assessment and content regulation.

Museum staff and management. Administrative tools are provided for the administration of selected content, submissions, bookings, comments and other platform records.

Future researchers and developers. The project may also serve as a reference for future work on WebXR, digital heritage, accessibility, immersive media and web-based museum systems.

1.4.2 Expected Benefits

The developed platform is expected to provide the following benefits:

Expanded heritage access. Supported computers, tablets and mobile devices will give people access to selected museum information and digital experiences.

Improved heritage interpretation. Its organized narratives, multimedia exhibitions, interactive maps, virtual tours and guided activities offer users several options to explore and appreciate the value of the heritage site.

More inclusive access. Bilingual content, text controls, captions, narration, responsive layouts and non-immersive alternatives accommodate users with diverse capabilities, devices and learning preferences.

Support for museum education. The platform also offers other resources for classroom discussion, independent study, visitor orientation and heritage-awareness activities.

Support for visitor engagement and services. The museum offers structured opportunities for user interaction: feedback forms, community participation channels, user accounts and guided-visit requests.

Improved content management. The administrative features provide authorized museum staff with the capabilities to handle specific heritage content, exhibitions, educational resources, submissions, bookings, and visitor feedback.

Contributing to the preservation of digital heritage. The initiative offers a structured digital environment for the presentation of museum-approved materials, while ensuring institutional content governance and realising that the platform does not substitute physical preservation or professional care.

1.5 Scope and Limitation of the Study

This study focused on the design and development, functional testing, and descriptive user evaluation of a WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The study covered the following:

1.5.1 Scope of the Study

- Within the functionalities developed in the project the platform is available for public visitors, registered users, museum personnel, students, teachers, researchers, senior citizens, individuals with disabilities, community members and offsite users.
- The project is site-specific at NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan located in Nagcarlan, Laguna.
- The platform that is implemented includes museum and heritage information, digital exhibits, educational content, a Heritage Trail, an Interactive Site Map, a 360-degree Virtual Tour, selected AR and 3D/WebXR experiences, accessibility-oriented controls, mechanisms for feedback and community participation, scheduling for visit-request, user-account functionalities, and administrative management tools.
- The study encompasses requirement gathering, system planning and design, iterative development, content integration, Black Box Functional Testing, descriptive user evaluation and refinement based on documented findings.

- The system stores some content authorised by the museum, and user account, booking, feedback, submission, and content administration information.

- The web-based platform is built on technologies and services that were adopted in the final system, HTML, CSS, JavaScript, Python/Flask, Supabase, Browser-based 3D/WebXR technologies, and supported multimedia resources.

1.5.2 Limitations of the Study

- The platform is restricted to heritage information and multimedia elements that were provided to or approved for use by the researchers during the development period.
- The user evaluation is confined to those respondents who are eligible and selected according to the approved sampling design during the data-collection period. The findings must be interpreted in the light of the given demographic, sample and conditions of the evaluation.
- Immersive features may not work the same in all user contexts and require support from the device, browser, operating system, graphics, camera, sensor and permissions.
- Hosting availability, database and storage services, and applicable third-party services, require Internet connectivity, and online content and services.
- The study does not claim to have conducted extensive penetration-security testing, formal performance benchmarking, full cross-browser/device certification, or extensive WCAG conformance

unless otherwise expressly supported by independent published processes.

- User-perceived accessibility is not formal accessibility-conformance assessment. Behavioural Intention to Use is not observed long-term adoption.
- Other advanced elements not included in the examined build such as full digital twin, real-time multiplayer interaction or other future developments are out of scope for this study.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter reviews the scholarship, standards, and related systems that establish the research basis for the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The discussion is organized around established knowledge, recurring limitations in previous work, the site-specific and integration-oriented gap that remains unresolved, and the contribution of the present study. The reviewed evidence informs the project requirements, design decisions, development boundaries, testing approach, and user-evaluation procedures used in the study.

2.1 Studies and Literature

This section presents scholarly literature, research studies, theories, concepts, technologies, standards, and other credible sources related to the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The reviewed materials are connected to the research problem, objectives, proposed solution, development requirements, major system components, and evaluation needs of the study.

The discussion uses a thematic approach to connect findings across digital heritage, immersive technologies, accessibility, learning, visitor services, institutional governance, and system evaluation. The findings, limitations, and contributions of the

reviewed sources are analyzed and synthesized according to their relevance to the present project.

2.1.1 Selection of Related Literature

Sources were selected for their relevance to the research problem and objectives of the study, particularly digital heritage preservation, virtual museums, WebXR and panoramic tours, accessibility, mobile and experiential learning, content governance, human-computer interaction, privacy, and evaluation. Academic studies were distinguished from product pages and system documentation. Official standards and institutional sources were used for terminology, accessibility, privacy, and platform specifications. The selection and organization of literature were guided by the study's research problem, scope, and evaluation requirements rather than by the presence of individual proposed features.

2.1.2 Review and Analysis of Studies

Analytical review of representative studies. Previous studies were examined according to their purpose, methodological approach, major findings, limitations, and relevance to the present project. Representative studies are discussed here to show how development-oriented research, systematic reviews, and user-evaluation studies contribute different kinds of evidence to the design and evaluation of the Nagcarlan platform.

Virtual-museum development. Besoain et al. (2021) examined the development of a virtual museum through a process that included content selection, 3D digitization, iterative design, and user testing. The study demonstrates that a virtual museum requires coordination between cultural content, technical implementation, and user experience rather than the simple conversion of museum materials into digital objects. Its context differs from the present study and does not by itself address the combined WebXR, accessibility, visitor-service, and local-governance requirements of the Nagcarlan Underground Cemetery. Nevertheless, its development process informs the present project by supporting iterative design, curated content, and user-oriented organization of the digital visitor journey.

Immersive-technology reviews. Ramtohul and Khedo (2024) conducted a systematic review of augmented-reality systems in cultural heritage, while Rodriguez-Garcia et al. (2024) systematically reviewed virtual 3D reconstructions of cultural heritage in immersive virtual reality. The AR review identified contextualization, reconstruction, and mobile learning as important uses while also reporting tracking stability and device compatibility as recurring constraints. The VR reconstruction review reinforces the value of immersive representation but also highlights the need for careful technical and interpretive design. Together, these studies support the selective use of AR, WebXR, and 3D content in the present project while justifying fallback access and the decision not to claim a complete digital twin.

Accessibility and mobile-learning reviews. Pirrone et al. (2023) used a systematic literature review to examine cultural accessibility for people with disability in digital-humanities contexts, while Pedraja-Rejas et al. (2024) systematically reviewed the effects of mobile learning on learning outcomes and critical thinking. Although their research purposes differ, both studies show that access depends on more than the availability of technology: inclusive participation requires attention to barriers, and mobile learning is most useful when instructional design is intentional. For the present study, these findings support accessibility as a cross-cutting requirement and support a mobile-first design with readable content and non-immersive alternatives. They do not justify a claim of full WCAG conformance or measured learning gain, because those outcomes require forms of evaluation beyond the present research design.

Learning and engagement evaluation. Xu et al. (2024) investigated gamified museum learning using a mixed research approach and reported effects on visitors' knowledge and awareness. The study is relevant because it shows that guided and game-based activities can contribute to museum learning when they are connected to meaningful content. However, its method differs from the present study, which does not use a pretest-posttest or experimental learning design. Accordingly, the present project uses guided and self-check activities as design elements for engagement and interpretation, while avoiding claims that the platform itself has experimentally produced learning gains.

Visitor-service system development. Yudhanto et al. (2024) designed and developed a museum ticketing system using the Design Thinking method, supported by use cases, database design, and UI/UX mockups. The study illustrates how a visitor-service process can be translated into a structured digital workflow through user-centered design and system modeling. Its operational scope differs from the present project because the Nagcarlan platform uses visit-request or guided-visit scheduling rather than full commercial ticketing. The study nevertheless informs the organization of visitor-service requests, administrative review, and interface design within the project's actual scope.

WebXR platform development. Fanini et al. (2021) presented ATON, an open-source framework for creating immersive and collaborative web applications for cultural heritage. The work demonstrates the feasibility of browser-based immersive heritage experiences and supports storytelling and collaborative access through web technologies. As a framework-focused contribution, however, it does not resolve the site-specific content governance, accessibility, visitor-service, and administrative requirements of one local heritage institution. For the present study, it provides a technical basis for treating WebXR as an enabling technology while retaining standard web content and alternative access paths.

Technology-acceptance evaluation. Puspasari et al. (2023) assessed an innovative virtual-museum application using the Technology Acceptance Model. Its purpose and evaluation approach

are relevant because TAM provides established constructs for examining users' perceptions of a digital museum system. The present study adopts this contribution only for the acceptance-related dimensions to which TAM applies and uses those constructs descriptively. Technical verification remains separate through black-box testing, and reported interaction during the evaluation session is not treated as evidence of long-term system use.

Comparison of research approaches. Across the representative studies, the methods range from iterative system development and design thinking to systematic reviews, mixed-method evaluation, and technology-acceptance assessment. Development studies explain how digital museum and visitor-service systems can be structured, systematic reviews identify recurring technical and accessibility constraints across many implementations, and evaluation studies show how user experience or acceptance can be assessed. This comparison informs the present study's separation of development, functional testing, and user evaluation.

Theoretical framework. The theoretical basis of the study distinguishes technology-acceptance constructs from learning principles so that the evaluation does not overstate what the selected theories can establish.

Technology Acceptance Model. The Technology Acceptance Model (TAM) provides the theoretical basis for the acceptance-related portion of this study. Recent museum studies have applied TAM to examine perceived usefulness, perceived ease of use, attitudes, and

behavioral intentions in virtual environments (Li & Lv, 2024; Puspasari et al., 2023). In the present study, four TAM-related constructs were applied: Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use. The questionnaire items were adapted to the context of the WebXR-Enabled Digital Heritage Platform while retaining the intended meaning of each construct. Perceived Ease of Use refers to how understandable, learnable, and easy the platform is perceived to be; Perceived Usefulness concerns its perceived value for accessing, exploring, and learning about heritage information; Attitude Toward Using reflects respondents' favorable or unfavorable evaluation of using the platform; and Behavioral Intention to Use concerns their expressed willingness to use the platform again for relevant museum and heritage-learning activities. TAM is used descriptively in the study; weighted means and standard deviations summarize the relevant dimensions, and no causal relationships among TAM constructs are claimed. Actual System Interaction is treated separately and refers only to respondents' reported interaction during the evaluation session rather than long-term system use.

Constructivist learning principles. Digital heritage learning can be supported when users actively explore, interpret, connect information with place, and receive feedback (Li & Zhang, 2025). Guided heritage trails, panoramic scenes, contextual markers, and short learning activities reflect these principles. The study does not claim to prove learning gains because it does not use a pretest-posttest learning

experiment. Instead, constructivist learning principles inform the design of the learning experience and help explain the relevance of perceived usefulness and immersive digital experience within the user evaluation.

Conceptual and technical framework. The conceptual and technical framework translates the research problem, scope, and accessibility requirements into design principles that guide implementation and evaluation.

User-centered design and human-computer interaction. User-centered design and HCI require the platform to be organized around users' goals, abilities, contexts, and constraints. These concepts inform navigation, task clarity, readable content, feedback, comfort, trust, and the design of both public and administrative interfaces.

Progressive enhancement and responsive web design. Progressive enhancement and responsive web design support access across common phones, tablets, and computers. Core information remains available through standard web content when advanced immersive capabilities are unsupported. This is consistent with the study's scope, which recognizes differences in device, browser, operating-system, camera, hardware, and connectivity conditions.

Accessibility and WCAG. WCAG 2.2 provides normative guidance for perceivable, operable, understandable, and robust web content. Relevant principles guide contrast, keyboard access, text resizing, captions, labels, language, and fallback paths. The study uses WCAG-informed accessibility practices and checks but does not claim

complete WCAG conformance without a formal accessibility audit (World Wide Web Consortium, 2024).

Evaluation framework. The evaluation framework separates technical verification from user assessment and connects the literature to the testing and user-evaluation procedures defined by the research objectives.

The evaluation framework deliberately separates technical verification from user assessment. Structured black-box testing is used to determine whether the specified public and administrative functions produce their expected results. Relevant accessibility checks and a device/browser compatibility record address access and technical-environment constraints identified in the research objectives and scope. These procedures provide the basis for verifying the functional and technical requirements of the developed platform.

User assessment is conducted separately through the evaluation dimensions defined in the research objectives: Selected Software Quality Characteristics, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, Actual System Interaction, Immersive Digital Experience, and Accessibility. TAM provides the theoretical basis only for the acceptance-related constructs to which it applies. “Actual System Interaction” refers to self-reported interaction during the evaluation session rather than long-term actual use. ISO/IEC 25010:2023 serves as the reference for the selected software-quality indicators used in the user evaluation. The study focuses on applicable indicators associated with functional suitability,

interaction capability, and performance efficiency because these qualities correspond to observable aspects of the developed platform that intended users can reasonably assess during the evaluation session. Functional suitability is reflected in whether the platform's major functions operate as intended; interaction capability is reflected in the clarity and understandability of navigation and user interaction; and performance efficiency is reflected in the respondents' perception of whether the platform responds within an acceptable amount of time. Only these selected and researcher-adapted indicators are used for user assessment. The study does not claim to evaluate the complete ISO/IEC 25010 quality model or all of its characteristics and subcharacteristics (International Organization for Standardization, 2023).

2.1.3 Thematic or Chronological Organization

Thematic approach. The study adopts a thematic approach and synthesizes the related studies and literature into twenty-four thematic areas. Each theme identifies what previous work establishes, the limitations or conditions that repeatedly appear, and the implications of those findings for the integrated and site-specific approach of the present study. Collectively, the themes support the research problem, objectives, development scope, technical testing, and user-evaluation procedures of the study.

Digital heritage preservation. Digital heritage preservation now extends beyond converting physical records into digital files.

Recent studies emphasize interpretation, public access, sustainability, and cultural responsibility as central goals of digital heritage work. Lian and Xie (2024) showed that digital cultural heritage research has expanded around digitization, virtual reconstruction, visualization, communication, and user participation, while Siliutina et al. (2024), Suaidi (2026), and Tan et al. (2026) connected preservation with community value and cultural continuity.

Across these studies, preservation is treated as both a technical and interpretive responsibility. The implication for the present study is that museum-approved narratives, images, and other heritage materials must remain understandable, contextualized, and manageable by the institution. This supports the project objective of presenting curated heritage information without reducing the site to a collection of disconnected digital objects.

Cultural heritage digitization and digital archives. Digitization becomes useful when cultural materials are arranged in ways that support retrieval, interpretation, and reuse. Daga et al. (2022) and Carvalho et al. (2024) examined how citizen experiences and institutional heritage archives may be connected, while Nappi et al. (2024) emphasized the organization of cultural knowledge through digital tools and adaptive structures. Ma et al. (2025) further showed that architectural heritage documentation depends on structured data and validated content.

The recurring concern is that digitized content can become fragmented or lose institutional context when records, public

contributions, and presentation tools are handled separately. For the Nagcarlan project, this literature supports separating museum-controlled content from moderated community submissions while maintaining administrative workflows for reviewing and updating heritage information.

Virtual restoration and digital reconstruction. Virtual restoration and reconstruction literature stresses accuracy and interpretive transparency. Pietroni and Ferdani (2021) explained that digital reconstruction should distinguish documented evidence from interpretive elements, and Cardozo and Papadopoulos (2021) noted that digital models preserve cultural value when placed in a meaningful context. Basu et al. (2023) and Rodriguez-Garcia et al. (2024) likewise emphasized that computational restoration and immersive reconstruction require expert validation and careful design.

These findings caution against presenting visually impressive reconstructions as verified history. In the present study, any 3D environment, AR overlay, object display, or visual enhancement is treated as interpretive support and should remain traceable to reviewed museum content. A complete digital twin is outside the project scope, which keeps the implementation consistent with available resources and documented evidence.

Architectural heritage documentation. Architectural heritage studies show that digital documentation can preserve spatial and structural information, but advanced methods such as photogrammetry, laser scanning, and HBIM may require resources that smaller

institutions do not always possess. Ma et al. (2025) identified these methods as valuable but technically demanding, while Buragohain et al. (2024) noted that metaverse-related heritage applications raise issues of infrastructure, sustainability, and cultural accuracy. Nappi et al. (2024) added that digital knowledge structures should make heritage information understandable to users rather than only technically complete.

The recurring limitation is therefore the balance between documentation quality and practical institutional capacity. This supports the study's manageable combination of 360-degree tours, site-map-linked information, hotspots, and selected WebXR or AR elements rather than a resource-intensive complete digital reconstruction.

Virtual museums and online museum experiences. Virtual museums extend public access to cultural content, but their value depends on planning, usability, and curatorial control. Kamariotou et al. (2021) emphasized strategic planning for virtual exhibitions, while Besoain et al. (2021) showed that virtual museum development involves content selection, 3D digitization, iterative design, and user testing. Recent reviews also show that virtual exhibitions increasingly combine immersive storytelling, curatorial workflows, and interactive access (Akabli & Essamoud, 2026; Cui & Wu, 2025; Gherheș et al., 2025).

The literature indicates that remote access is most meaningful when users are guided through reliable interpretation rather than left

with isolated media. For the present study, this supports an organized onsite/offsite visitor journey that connects digital exhibits, orientation information, virtual-tour content, and guided heritage learning under museum-approved interpretation.

Digital museum education and distance learning. Museum learning is strengthened when online platforms connect reliable content with educational tasks and accessible navigation. Zakaria (2024) demonstrated that digital museum resources can support distance learning, while Corrales-Serrano et al. (2024) linked VR heritage activities with usability, learning, and emotional response. Hu and Ng (2026) further showed that low tech-barrier VR content creation can support heritage education, especially when learning outcomes and pedagogy are considered.

These studies establish the educational potential of digital museum environments but do not imply that technology alone produces measurable learning gains. The present study therefore treats the platform as a learning resource before, during, and after a museum visit, while evaluating users' perceptions and interaction rather than claiming experimentally measured learning improvement.

WebXR in cultural heritage platforms. WebXR is relevant to digital heritage because it enables immersive experiences through a browser rather than through a dedicated application. The WebXR Device API provides access to compatible VR and AR capabilities through web applications (World Wide Web Consortium, 2026), while MDN Web Docs (2026) explains that support still depends on device,

browser, and operating-system capability. Fanini et al. (2021), Zöllner and Gemeinhardt (2024), and Shawash et al. (2025) demonstrated that web-based XR can support cultural heritage storytelling, collaborative viewing, and cross-platform evaluation.

The recurring limitation is uneven technical support across user environments. Accordingly, WebXR in the present project is an appropriately scoped immersive option rather than the only route to content. Standard pages, captions, transcripts, and non-immersive alternatives remain necessary so that the system can serve users whose devices cannot support immersive functions.

Augmented reality in museums and heritage sites. Augmented reality can enrich museum interpretation by placing digital content in relation to physical objects or spaces. Ramtohul and Khedo (2024) reported that AR in cultural heritage can support contextualization, reconstruction, and mobile learning, although tracking stability and device compatibility remain limitations. Chen et al. (2024) found that usefulness, trust, authenticity, and satisfaction influence AR adoption, while Nikolarakis and Koutsabasis (2024), Xu et al. (2022), and Cho (2025) emphasized interaction patterns, retention, and youth-oriented interpretation.

These findings suggest that AR should be used selectively where it adds interpretive value and where content authenticity can be maintained. In the Nagcarlan platform, AR is treated as an enhancement rather than the main access method, which allows users

without AR-capable devices to continue learning through ordinary pages, hotspots, maps, and guided trails.

Virtual reality and immersive heritage experiences. Virtual reality can increase presence and spatial understanding, but it must be evaluated in terms of learning value, comfort, and accessibility. Zhao et al. (2025) identified design considerations for VR heritage education, while Jangra et al. (2025) examined immersion and visitor experience. Tsita et al. (2023), Trunfio et al. (2022), and Xu et al. (2025) similarly showed that virtual experiences can extend access but cannot fully replace the meaning of onsite visits. Related mixed-reality studies also emphasize spatial exploration, contextual interaction, and adoption considerations in cultural settings (Chin et al., 2024; Nunes et al., 2025; Yangting & Yahaya, 2024).

Taken together, this literature supports a hybrid model rather than substitution of the physical site. The present study uses immersive experiences selectively to strengthen spatial interpretation while preserving the cultural importance of onsite visitation and maintaining accessible, non-immersive content for users whose devices or preferences do not support immersive functions.

360-degree virtual tours in museums and heritage sites. A 360-degree virtual tour is practical for heritage institutions because it provides spatial orientation without requiring a full VR headset. Wiryawan and Nuraisyah (2023) developed a 360-degree museum tour and showed that panoramic navigation can support remote exploration. Borucka and Picchio (2026), Rodríguez et al. (2025), Liu et al. (2024),

Torre et al. (2025), and Torre (2026) also showed that virtual tours can support storytelling, education, accessibility, and cultural interpretation.

These studies agree on the value of panoramic access but also imply that a tour becomes more meaningful when navigation is connected with interpretation. For the Nagcarlan Underground Cemetery, 360-degree scenes therefore function as both spatial orientation and offsite learning support rather than as an isolated visual walkthrough.

Interactive hotspots and clickable exhibit markers. Panoramic tours become more educational when users can open contextual information through hotspots, markers, captions, maps, and guided prompts. Karakuş Yılmaz et al. (2026) found that guided and gamified support can influence learning in museum virtual tours, while Liu et al. (2024), Rodríguez et al. (2025), and Torre (2026) connected virtual-tour navigation with storytelling and learning pathways.

The implication for the present study is that interactive markers should connect locations with credible interpretation rather than function as decorative controls. Hotspots and clickable markers are therefore treated as navigation-to-content mechanisms that can link locations with text, archival images, audio, captions, transcripts, and related heritage information.

Interactive site maps and location-based digital access. Location-based digital access allows users to move from spatial orientation to heritage information through maps, markers, trail stops,

and standard web links. Virtual-tour and mobile-learning studies show that digital content becomes more useful when access points are structured, readable, and connected to learning tasks (Karakuş Yılmaz et al., 2026; Lin et al., 2024; Pedraja-Rejas et al., 2024; Rodríguez et al., 2025).

These findings support the broader principle of connecting place with interpretation. In the present study, site-map-linked content, location markers, tour hotspots, guided trail stops, and standard web links provide multiple paths to the same heritage information, supporting both onsite navigation and offsite learning.

Accessibility and inclusive design in digital heritage. Digital heritage platforms should address diverse users from the beginning of the design process. International Council of Museums (2022) identifies museums as accessible and inclusive institutions, while UNESCO (2025) highlights the role of digital technologies in improving access to culture and knowledge and supporting inclusive and sustainable development. Academic studies further show that cultural accessibility remains uneven for persons with disabilities and other users with access barriers (Anastasovitis et al., 2024; Kruczek et al., 2024; Pirrone et al., 2023; Tamim et al., 2025).

The recurring problem is that digital access can reproduce exclusion when immersive media, interface controls, or content formats assume a single type of user. The present study therefore treats accessibility as a cross-cutting design and evaluation concern. Implemented accessibility functions and content alternatives are

distinguished from extensible provisions, such as sign-language or tactile-oriented resources that depend on the availability of appropriate materials.

WCAG, web accessibility, and assistive features. Accessibility standards translate inclusive design into practical development requirements. WCAG 2.2 describes criteria related to perceivability, operability, understandability, and robustness (World Wide Web Consortium, 2024), while the W3C Web Accessibility Initiative (2026) explains how accessibility benefits users with visual, auditory, physical, cognitive, language, learning, and neurological differences. Uribe-Quevedo (2024), Montusiewicz et al. (2022), and Fiorenza et al. (2025) also show that immersive and 3D content should include accessible alternatives.

These sources support the use of headings, contrast, text alternatives, captions, transcripts, keyboard-friendly interaction, readable language, and fallback paths. In the present study, these principles guide both development and accessibility checking; however, the research does not present the platform as fully WCAG-conformant without a formal conformance audit.

Mobile-first and web-based learning platforms. Because the developed platform is intended for common smartphones and web browsers, mobile-first and web-based access can reduce barriers to use. Pedraja-Rejas et al. (2024) found that mobile learning can improve outcomes when instructional design is intentional, while Chin et al. (2024), Hu and Ng (2026), Zakaria (2024), and Lin et al. (2024)

showed that mobile and low-barrier digital tools can support heritage learning and engagement.

The recurring condition is that mobile access must still account for readability, loading behavior, connectivity, and different device capabilities. Accordingly, the present study prioritizes responsive pages, concise readable content, accessible media, and ordinary web links while recognizing stable internet access and device/browser compatibility as project limitations.

Interactive learning and self-check activities. Interactive learning helps users move beyond passive viewing. Xu et al. (2024) showed that gamified museum learning can influence knowledge and awareness, while Li and Zhang (2025), Zakaria (2024), Lin et al. (2024), and Karakuş Yılmaz et al. (2026) connected digital museum learning with guided activities, reflection, and learner engagement.

These findings support brief self-check and trail-based activities when they remain connected to heritage interpretation. The present study uses such interactions to encourage comprehension and cultural appreciation but does not treat them as evidence of measured learning gain, consistent with the descriptive and evaluative design of the research.

Gamification and progress indicators in museum learning. Gamification can motivate users when it supports meaning rather than competition. Sangamuang et al. (2025) found that VR museum gamification can affect both enjoyment and meaningful experience, while Yolthasart et al. (2024), García-Milon et al. (2026), Alatrash et

al. (2025), and Li and Zhang (2025) showed that game-based elements can improve engagement when aligned with cultural learning goals.

For a solemn heritage site, the design requirement is restraint. In the present study, gamified elements are limited to progress and completion indicators across the platform's guided and immersive experiences. These indicators are intended to reinforce participation and heritage awareness rather than emphasize competition or formal certification.

Community participation and digital storytelling. Heritage is shaped not only by institutional records but also by community memory. Daga et al. (2022) identified requirements for integrating citizen experiences into heritage archives, and Carvalho et al. (2024) examined how such experiences can be linked to cultural records. Kasemsarn and Nickpour (2025), Tan et al. (2026), and Roque et al. (2024) also emphasized storytelling, community engagement, and institutional transformation in digital heritage practice.

The common limitation is that public participation can introduce accuracy, sensitivity, and governance concerns when contributions are published without review. The present study therefore supports moderated submissions and community stories subject to administrative approval, preserving participation while maintaining institutional responsibility for heritage content.

Feedback systems and visitor analytics dashboards. Feedback and analytics help museums improve services using evidence rather than assumption. Wen and Ma (2024) showed that multimedia and

data-driven technologies can enhance museum experience, while Tian et al. (2025), Ivanov (2024), and Avlonitou et al. (2025) discussed visitor prediction, profiling, and smart-museum technologies.

The reviewed work demonstrates the value of visitor data but also shows that sophisticated analytics can introduce additional technical, privacy, and operational requirements. Within the scope of the present study, analytics remain descriptive and practical, focusing on available feedback and service records. Richer predictive analytics and automated sentiment classification are treated as future enhancements rather than as required contributions of the developed platform.

Online booking, visit scheduling, and digital reservation systems. Online booking and digital reservation systems help cultural and tourism institutions improve visitor convenience, staff preparation, and service monitoring. Yudhanto et al. (2024) designed a museum ticketing system using design thinking, use cases, database design, and UI/UX mockups. Attanasio et al. (2022) presented visitor-flow management at the Uffizi Gallery through predictive and prescriptive models for assigning entry times.

These studies indicate that scheduling is useful when it supports institutional planning and user convenience, but recurring concerns include privacy, connectivity, reliability, accessibility, and additional staff workload. The present project therefore frames booking as visit-request or guided-visit scheduling rather than full

commercial ticketing, keeping the service consistent with the museum's operational context and project scope.

Administrative backend and content management systems. Digital heritage platforms require administrative control so that content remains accurate, current, and institutionally appropriate. Cesário et al. (2025) emphasized the value of authoring tools that allow heritage professionals to design and manage interactive exhibitions. Roque et al. (2024), Gherheș et al. (2025), Avlonitou et al. (2025), and Cui and Wu (2025) likewise connected museum digitalization with institutional workflow and curatorial responsibility.

The literature suggests that sustainability depends not only on public-facing content but also on the institution's ability to review, update, and govern it. This supports an administrative backend for content management, contribution review, visit scheduling, feedback monitoring, and practical reporting, helping keep public heritage content and visitor-service records under authorized museum control.

Human-computer interaction, usability, and user experience. Human-computer interaction is central to digital heritage because historically accurate content may still fail to serve users if navigation, readability, feedback, comfort, or task flow are poor. Bonel et al. (2023) showed that user expectations shape digital museum experiences, while Lyu (2024), Quiñones et al. (2026), Bonino et al. (2025), and Trunfio et al. (2022) emphasized usability, accessibility, cognitive load, immersion, comfort, and evaluation in digital and immersive environments.

These findings make HCI a cross-cutting requirement rather than a justification for one feature. In the present study, HCI principles inform the clarity of public and administrative tasks, navigation, readable content, interactive controls, immersive comfort, and the users' overall experience of the integrated platform.

Software quality evaluation and system validation. System validation in the present study requires both technical checking and user evaluation. Recent evaluation studies connect usability, engagement, accessibility, comfort, and learning relevance with the success of virtual and immersive heritage systems (Bonino et al., 2025; Chen et al., 2024; Fiorenza et al., 2025; Quiñones et al., 2026; Rodriguez-Garcia et al., 2024; Tsita et al., 2023; Xu et al., 2022). ISO/IEC 25010:2023 is also useful as a product-quality reference because it supports the specification and evaluation of software quality characteristics across the system life cycle (International Organization for Standardization, 2023).

The key methodological implication is that technical verification should not be confused with users' perceptions. The present study therefore uses documented black-box test cases for specified public and administrative functions and separately uses selected, researcher-adapted software-quality indicators for user assessment. It does not claim measurement of the complete ISO/IEC 25010 model.

Security, privacy, and content governance. Security, privacy, and content governance are necessary because the developed platform

includes user-related records, feedback, community or educator submissions, visit requests, and administrative functions. NPC Circular No. 2023-06 identifies security of personal data as a requirement for processing systems (National Privacy Commission, 2023), while National Privacy Commission guidance emphasizes privacy engineering across planning, development, testing, deployment, and maintenance (National Privacy Commission, 2025).

These requirements intersect with heritage governance because submitted stories, digital exhibits, and service records require both data protection and institutional review. For the present study, this supports secure administrative access, appropriate authorization, moderation of submissions, privacy protection for visitor records, and content-approval workflows while avoiding unnecessary collection of personal information.

2.1.4 Relevance to the Present Project

Across the reviewed literature, previous studies consistently support four broad conclusions: digital heritage can widen access when content is meaningfully interpreted; immersive and panoramic technologies can strengthen engagement and spatial understanding when they remain purposeful and usable; accessibility is necessary for inclusive cultural participation; and active digital learning can support engagement when it is connected to credible content and appropriate pedagogy. The literature also emphasizes institutional governance,

usability, and sustainable administration as conditions for meaningful digital heritage work.

At the same time, recurring limitations include device and browser compatibility, connectivity requirements, uneven accessibility, infrastructure and sustainability constraints, cultural-accuracy concerns, fragmented systems, limited local governance, and weak integration among heritage interpretation, learning, visitor services, participation, and administration. These findings indicate that the research problem is not the absence of isolated technologies. Rather, the unresolved issue is how these capabilities can be responsibly integrated for one local heritage institution while remaining accessible, manageable, and aligned with its approved content and operational needs. These findings provide the basis for the system-development, testing, and user-evaluation procedures used in the present study.

2.1.5 Citation and Referencing

The study uses APA 7th Edition conventions for in-text citations and the Literature Cited section. All sources cited in the review correspond to entries in the Literature Cited section, with author or organization names, publication years, titles, and source information presented consistently. Academic, professional, government, institutional, standards, and official platform sources are distinguished according to their role in the review.

2.2 Related Systems and Technology Gap Analysis

This section examines existing software applications, online platforms, virtual-tour tools, digital archives, and institutional heritage systems that are closely related to the present project. The purpose of the comparison is to identify documented capabilities, contextual limitations, and opportunities that remain for the Nagcarlan Underground Cemetery. Ten related systems are considered, and all comparisons are limited to the platform versions and documentation reviewed for this study so that differences are presented as evidence-based observations rather than permanent claims.

2.2.1 Identification of Related Systems

Related systems. Ten existing systems, websites, applications, and platforms were examined in relation to the developed WebXR-Enabled Digital Heritage Platform. Official institutional and platform documentation, together with relevant scholarly studies, is used here to describe the reviewed systems and support the technology-gap comparison.

The systems are used as benchmarks for online collections, 3D visualization, panoramic tours, virtual exhibition building, institutional heritage presentation, and digital archiving. They are compared with the site-specific requirements of the present study, including local interpretation and governance, immersive or spatial access, inclusive design, learning support, community participation, visitor services, feedback, and administrative management.

Google Arts & Culture / Museum Views. Google Arts & Culture supports remote cultural exploration through digital content that includes

photographs, artifacts, and 360-degree Street View imagery from museums and cultural sites (Niccolai, 2020). The reviewed documentation emphasizes broad cultural access across multiple institutions rather than a locally administered workflow for one NHCP-managed site. It did not identify a Nagcarlan-specific workflow combining local interpretation, community review, visitor-service management, and local administrative monitoring. The platform nevertheless provides a useful benchmark for digital cultural access and panoramic viewing.

Matterport Museum and Gallery Tours. Matterport has been applied to indoor spatial digitization and virtual-museum environments for the management and presentation of cultural and academic heritage (Bocconcino et al., 2024). Its relevance to the present study lies primarily in spatial digitization and virtual presentation. Accessibility, local-language interpretation, museum-specific learning design, visitor services, and local governance would still depend on the broader implementation. Matterport therefore serves as a benchmark for spatial presentation rather than as evidence of an integrated Nagcarlan heritage-management workflow.

Sketchfab Cultural Heritage Collections. Sketchfab hosts extensive cultural-heritage 3D collections and supports virtual visits, VR and AR viewing, annotations, embedded models, and reusable digital heritage content (Flynn, 2020). The reviewed documentation is primarily object-centered and did not identify a complete site-specific visitor journey integrating spatial orientation, local heritage interpretation, visitor services, moderated participation, accessibility management, and local administrative monitoring. Its strength for the present study lies in interactive 3D object presentation.

Smithsonian 3D Digitization. The Smithsonian's 3D program uses institutional workflows for creating, managing, preserving, and publicly delivering 3D digital assets, including delivery through its Voyager 3D viewer (Zorich, 2023). The reviewed documentation demonstrates institutional-scale 3D digitization and public access. Because it serves a different institution, collection, and governance context, it is not a site-specific system for the history, architecture, religious practices, and community memory of the Nagcarlan Underground Cemetery. It therefore provides a benchmark for 3D digitization and delivery rather than a direct model for the project's local visitor and administrative workflow.

Europeana. Europeana is a European digital cultural-heritage portal and API ecosystem that supports searchable collections, metadata aggregation, digitized cultural materials, and educational reuse (Wuyts, 2024). Based on the reviewed documentation, its emphasis is broad discovery and aggregation across many collections. The reviewed version does not represent a locally administered, site-specific visitor journey for Nagcarlan. Europeana is therefore useful as a benchmark for digital-heritage organization and reuse, while the present study addresses a narrower local interpretation and governance context.

Artsteps. Artsteps has been used as a virtual-museum authoring application in educational settings, allowing users to design and organize virtual museum environments (Kayaalp et al., 2024). This supports its relevance as a benchmark for virtual exhibition authoring. The reviewed study did not establish an integrated Nagcarlan-oriented workflow combining local site mapping, visitor-service requests, moderated participation, accessibility

management, and museum administrative monitoring. It is therefore most relevant as a benchmark for virtual exhibition construction.

3DVista Virtual Tour Pro. 3DVista Virtual Tour Pro supports immersive virtual-tour authoring with VR interfaces, multimedia elements, navigation systems, guided flows, and hotspot-related controls (3DVista, 2026). The reviewed documentation shows strong support for tour production and interactive panoramic presentation. It does not by itself establish a culturally grounded learning, participation, visitor-service, and local governance model; those elements depend on the implementation built around the tour. The product is therefore a useful technical benchmark for panoramic authoring rather than a complete reference model for the present study.

National Museum of Fine Arts 360 Virtual Tour. The National Museum of the Philippines launched the 360-degree virtual tour of the National Museum of Fine Arts in 2020, featuring nine galleries with interactive hotspots containing information about exhibited collections (National Museum of the Philippines, 2020). This initiative demonstrates the feasibility of Philippine museum virtual viewing. The reviewed documentation did not identify the full Nagcarlan-specific workflow required by the present project, including local cemetery interpretation, moderated community participation, visitor-service management, and local administrative monitoring. The comparison is therefore contextual rather than a claim that such capabilities could never be added.

UPLB Museum of Natural History in VR. The UPLB Museum of Natural History launched a virtual reality tour that allows visitors to enter the museum gallery through their screens and view its dioramas and collections

(Padayon UP, 2021). It demonstrates a Philippine institutional use of virtual access, but it addresses a different institution, subject matter, audience context, and governance environment. Its relevance lies in showing the feasibility of remote museum exploration in the Philippine setting, while the Nagcarlan project requires a site-specific historical and operational approach.

NHCP National Memory Project. The National Memory Project is an official NHCP digital initiative that aims to democratize access to Philippine historical materials and was launched on August 1, 2023 (National Historical Commission of the Philippines, 2023). Its documented emphasis is on historical materials and digital access. The reviewed documentation did not identify the complete visitor-journey, spatial interpretation, visitor-service, participation, and localized administrative workflow required for the Nagcarlan Underground Cemetery. The present project is therefore positioned as complementary to broader NHCP digital initiatives rather than as a replacement for them.

2.2.2 System Feature Comparison

Table 2.1 compares the reviewed systems through their primary capability and an evidence-qualified observation relative to the present study. The comparison focuses on functions, technologies, processes, and contextual requirements that are directly related to the research objectives. A descriptive comparison is used instead of binary claims about undocumented features because commercial and institutional platforms may change or support different configurations over time.

Table 2.1 Concise comparison of related systems

Reviewed System	Primary Capability	Observed Limitation Relative to the Project
Google Arts & Culture / Museum Views	Digital cultural content and panoramic viewing	Broad cultural access; reviewed documentation emphasizes multi-institution access rather than a locally administered Nagcarlan workflow.
Matterport	Indoor spatial digitization and virtual-museum presentation	Strong spatial capture; accessibility and local interpretation depend on implementation.
Sketchfab	3D model viewing and annotations	Object-centered; reviewed documentation did not identify a complete local visitor-service and governance workflow.
Smithsonian 3D	Institutional 3D archive	High-quality object records in a different institutional and site context.
Europeana	Aggregated cultural collections and metadata	Discovery-centered rather than a site-specific local museum journey.
Artsteps	Virtual museum and exhibition authoring	Exhibition creation; reviewed documentation did not identify the project's integrated local services and governance workflow.
3DVista	360-degree tour authoring	Strong tour production; broader learning, participation, services, and governance depend on implementation.
NMFA 360 Virtual Tour	Official Philippine panoramic museum tour	Virtual viewing; reviewed documentation did not identify the full Nagcarlan-specific local workflow.
UPLB Museum of Natural History in VR	Philippine institutional virtual access	Different institution, subject matter, and governance context.
NHCP National Memory Project	Official digital archive and collections	Archive-centered; reviewed documentation did not identify a complete Nagcarlan visitor-journey and local administration workflow.

2.2.3 Analysis of Existing Systems

Across the ten reviewed systems, several capabilities are already well established. Broad cultural platforms and institutional archives demonstrate online collection access and digital archiving; 3D repositories and digitization initiatives demonstrate object-centered visualization; and virtual-tour tools and

Philippine museum platforms demonstrate panoramic or spatial viewing. These capabilities provide useful technical and presentation benchmarks for the present study.

The main differences appear in integration and context rather than in the existence of individual technologies. Based on the reviewed documentation, accessibility, local-language interpretation, learning design, visitor-service workflows, moderated community participation, and local administrative governance are not consistently presented as one integrated workflow across the reviewed systems. In several platforms, such capabilities may depend on how an implementation is configured; therefore, the analysis does not treat their absence as absolute.

Target users and operational environments also differ. Some reviewed systems serve many institutions or large collections, while others focus on 3D objects, panoramic-tour authoring, or virtual exhibition creation. The Nagcarlan project instead addresses one NHCP heritage site whose requirements combine verified local interpretation, inclusive access, spatial and immersive viewing, learning support, visitor services, community participation, and staff administration. The opportunity identified by the comparison is therefore the responsible local integration of these requirements rather than a claim of absolute technological novelty.

2.2.4 Technology Gap

Existing systems provide individual capabilities, but the reviewed literature and platform documentation indicate an unresolved need for an integrated, site-specific, locally governed digital heritage platform addressing

the particular heritage interpretation, accessibility, learning, visitor-service, community-participation, and administrative requirements of the Nagcarlan Underground Cemetery. The technology gap is therefore not that no other system contains any of the project's individual functions. Rather, it concerns the integration and local governance of these requirements within one web-based environment designed for this specific NHCP heritage site.

2.2.5 Proposed Response to the Gap

The proposed platform responds to the identified limitations through a set of connected, scope-controlled solutions. First, where broad collection and archive platforms emphasize discovery across many institutions, the present project organizes museum-approved information around one heritage site and links interpretation with the site map, heritage trail, 360-degree scenes, and selected immersive content. Second, where immersive technologies are affected by device, browser, and accessibility constraints, the platform retains responsive standard web pages, readable content, captions, transcripts, and non-immersive fallback paths.

Third, where related platforms primarily emphasize viewing, authoring, or archiving, the project integrates visit-request scheduling, feedback, and moderated community participation within the same browser-based environment. Fourth, the need for local governance is addressed through staff content management, approval workflows, visitor-service monitoring, and practical reporting. These responses correspond to the actual research objectives and documented system scope;

capabilities that remain future enhancements are not presented as completed contributions.

2.3 Synthesis

The synthesis integrates the major findings from the reviewed literature, previous studies, and related systems. Rather than repeating the preceding discussions, the synthesis identifies common findings, important differences, useful approaches, recurring limitations, and the specific gap that provides the basis for the present project.

2.3.1 Integration of Findings

The reviewed studies consistently agree that digital heritage can widen access to cultural information, that immersive and panoramic technologies can strengthen engagement and spatial understanding, that accessibility is necessary for inclusive cultural participation, and that virtual museum environments can support remote and self-paced learning. The literature also emphasizes that institutional governance, credible interpretation, usability, and sustainable administration are essential if digital technologies are to contribute meaningfully to heritage work.

At the same time, recurring limitations appear across the literature: device and browser compatibility, connectivity requirements, uneven accessibility, infrastructure and sustainability constraints, cognitive or interaction barriers, cultural-accuracy concerns, fragmented digital services, limited local governance, and weak integration between interpretation, learning, visitor services, participation, and administrative processes. These

limitations show that the research problem cannot be addressed simply by adding more immersive functions. A useful heritage platform must remain informative without immersion, manageable by the institution, and grounded in verified local content.

Important differences also appear in the purposes and methods of the reviewed work. Development-focused studies such as Besoain et al. (2021), Fanini et al. (2021), and Yudhanto et al. (2024) demonstrate how virtual-museum, WebXR, and visitor-service solutions can be designed and implemented, while systematic reviews such as Pirrone et al. (2023), Ramtohul and Khedo (2024), Pedraja-Rejas et al. (2024), and Rodriguez-Garcia et al. (2024) reveal recurring accessibility, device, design, and implementation constraints across broader bodies of work. User-oriented studies such as Xu et al. (2024) and Puspasari et al. (2023) further show the value of evaluating learning-related engagement and technology acceptance. Taken together, these approaches support a browser-based, mobile-first, accessibility-aware, and locally governed platform, while also supporting the study's separation of technical verification from descriptive user evaluation.

2.3.2 Identified Gap

The evidence supports primarily a contextual and integration-oriented technology gap. Based on the reviewed literature and systems, the central unresolved need is for a site-specific, locally governed Philippine digital heritage platform that integrates immersive interpretation, accessible learning, visitor services, community participation, and administrative content management in one web-based environment for the NHCP Museo ng Libingan

sa Ilalim ng Lupa ng Nagcarlan. The gap is defined by the specific combination of local heritage interpretation, inclusive access, institutional governance, visitor support, and integrated administration required by this site.

2.3.3 Contribution of the Proposed Project

The present study addresses the identified gap through an implemented and documented WebXR-Enabled Digital Heritage Platform whose public and administrative requirements can be technically verified and whose intended users can evaluate the platform using the dimensions defined in the research objectives. The study connects the literature and technology gap to documented black-box functional testing, relevant accessibility and device/browser compatibility checks, and descriptive user evaluation of Selected Software Quality Characteristics, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, Actual System Interaction, Immersive Digital Experience, and Accessibility.

The contribution of the study is therefore both contextual and evaluative. It provides a locally governed digital environment for the Nagcarlan Underground Cemetery while explicitly recognizing device, accessibility, connectivity, content, and implementation limitations. Immersive functions are treated as appropriately scoped enhancements to interpretation rather than the sole means of access, and future capabilities are distinguished from functions included within the developed system. This alignment keeps the literature, technology gap, system design, testing, user evaluation, results, and conclusions grounded in the same research problem and objectives.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the methods and procedures that were used in the design, development, testing, and evaluation of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan.

3.1 Materials and Methods

The study has carried out research, system development, testing and evaluation activities in an effort to achieve the objectives of the study. Relevant fact-finding strategies were used to acquire information on the current status of the museum, the needs of its visitors, the heritage material and the operational needs. The collected information was the basis for determining the functional requirements, the structure of the content, the design of the interface and the main modules of the proposed platform.

The creation of the WebXR-Enabled Digital Heritage Platform followed an iterative process, where requirements, designs, system components and content were progressively developed and refined. The development procedure used the suitable hardware, software, web technologies, development tools and digital heritage materials. The constructed platform was tested for functionality and evaluated by users to determine if the primary aspects of the system performed according to their intended requirements and to assess the users' acceptability and experience of the system after the implementation.

3.1.1 Research and Fact-Finding Activities

The researchers conducted interviews to understand the current process of the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan and to identify the requirements for the proposed platform. Museum personnel were consulted regarding the museum's current accessibility, historical and interpretative resources, museum regulations and policies, content priorities, common questions from visitors, accessibility issues, and relevant administrative procedures.

The information gathered from these activities was analysed and translated into functional, content, accessibility and administrative requirements for the proposed system. The historical and museum-related information included in the platform was based on reviewed and authorised sources to help preserve the accuracy and appropriateness of the presented heritage content.

3.1.2 System Development Activities

The researchers carried out the major system development activities using an iterative Agile-based development process. The activities included requirements analysis, system and interface design, database and workflow design, construction of system modules, integration of heritage and multimedia content, internal testing, refinement and preparation for system evaluation.

The requirements analysis stage was the step in which the information gathered from the fact-finding activities was compiled into the functional requirements and the primary components of the platform. Design activities encompassed the development of information architecture, user interface, navigation structure, content classification, administrative procedures, and immersive interaction mechanisms.

Development was carried out in phases so that the single modules and functionalities could be built and reviewed before being implemented into the fully developed platform. These featured digital heritage and educational content, interactive and immersive experiences, inclusive accessibility features, visitor engagement and museum services, and administrative management and monitoring.

Internal reviews were conducted during development to catch flaws, inconsistencies, and areas for improvement. The major public and administrative features that were implemented were then subjected to Black Box Testing to determine if they worked according to their established functional criteria. After functional testing, the platform was ready for user evaluation for acceptability and overall user experience.

3.1.3 Resources and Technologies

The development of the proposed platform utilized web design, immersive-media, database and content creation technologies that are suitable for a responsive and WebXR-enabled digital heritage system. The main development environment was built with the most common web technologies,

HTML, CSS and JavaScript, which were utilized to create the platform's structure, display, interaction and client-side functionality.

Python and the Flask web framework were utilized for the server-side application development and implementation of the backend processes. Supported web browsers displayed and interacted with three-dimensional and immersive content using JavaScript-based technologies such as Three.js and libraries related to WebXR. 3D assets were converted into web-friendly formats such as GLB, enabling interactive viewing and immersive experiences.

Database management system was used to store and manage the relevant system information such as user related records, museum content, submissions, bookings and administrative data. Development and testing were performed on desktops, mobile devices, modern web browsers and internet connection to verify the behaviour of the platform in its intended access contexts.

The technologies and resources were selected according to the functional requirements of the study, particularly the need to provide a browser-based platform capable of delivering conventional web content together with interactive, three-dimensional, and immersive heritage experiences without requiring the installation of a dedicated application.

3.2 Research Design

The study carried out a descriptive and evaluative developmental research design. The descriptive part was used to detail the museum context, user and

stakeholder requirements, current practices, available heritage resources, and accessibility and administrative issues. The developmental component included the design, development, deployment and iterative refinement of the WebXR-Enabled Digital Heritage Platform. The evaluation component was composed of Black Box Functional Testing and a descriptive system user evaluation.

The research design was adequate because the investigation did not need manipulation of experimental variables or testing of causal links. Instead, it set specifications, developed a workable technological solution, verified the described functions worked as intended, and summarised how intended consumers perceived and interacted with the resulting platform.

3.2.1 Rationale for the Selected Research Design

The developmental research approach was appropriate for the study since it aimed to develop and evaluate a workable technical solution to an identified need in digital heritage and accessibility at the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The study aimed to identify relevant requirements, design a system to be built from those requirements and to evaluate whether the resulting platform fulfilled its intended functions and was acceptable to its intended users.

Descriptive research is concerned with systematically describing existing characteristics, conditions, experiences, or phenomena without manipulating the variables being examined (McCombes, 2023). In this study, the descriptive component provided the contextual information needed to

identify the requirements and priorities that guided the development of the proposed platform.

The developmental component addressed the design and creation of the WebXR-Enabled Digital Heritage Platform. It involved translating the identified requirements into the system's information architecture, user interface, database structure, administrative workflows, heritage-content organization, accessibility mechanisms, and interactive and immersive components. Development was conducted iteratively, allowing the researchers to implement, review, refine, and integrate the major functional modules of the platform. This component is consistent with developmental approaches that involve systematic needs analysis, design, development, implementation, and evaluation in producing a technological product or solution (Govindasamy et al., 2023).

The evaluative component was applied once the major system functionalities had been implemented. The technical evaluation was carried out using Black Box Testing to establish whether the major public and administrative features worked as per their specified functional criteria. Then a user evaluation was conducted to evaluate the respondents' acceptability and experience of the developed platform according to the evaluation criteria defined in the study. This distinction enabled the researchers to differentiate between the system's performance as defined and the system's perception and experience by the target users. The selected research design was appropriate to the research objectives of the study. The developmental part concerned the design, development and implementation of the platform and the evaluating

part with the functional testing and user evaluation of the platform. The descriptive component supported these operations with the contextual and requirements-related information needed to design a platform that was responsive to the museum, its stakeholders, and the needs of the intended users.

3.3 Population of the Study

The population of the study consisted of the intended users and relevant stakeholders identified for the system user evaluation. Based on the consolidated population record provided by the museum for the approved reference period, the study used a total population of 2,633, composed of 2 administrative personnel, 2,259 museum visitors, 371 senior citizens, and 1 person with disability (PWD).

3.4 System Testers

The system testers who participated in the study were qualified people with relevant backgrounds in information technology, computing, software development, system testing, web-based systems or similar technical domains. They were chosen separately from the general respondents, as their duty was to evaluate the functional behaviour of the developed platform and not its perceived utility, usability, acceptance, or overall user experience.

Black Box Testing was done by the system testers on the primary public and administrative features of the research. The testing was done to find out if each function was providing the intended outcome as per its functional requirements specification. Testers tested by watching the inputs and system responses and outputs

and not looking at the internal source code of the platform. Each test case was documented as per its expected outcome, actual result and related Pass, Fail, N/A or Retest status with appropriate technical remarks or recommendations.

System testers were selected where members were picked based on qualifying criteria relevant to the technical evaluation required by the study. Preference was given to those having at least one (1) year of relevant professional or technical expertise in software development, software testing, web systems, system analysis or comparable computing fields.

Table 3.1 Qualification Criteria for System Testers

Qualification	Requirement
IT, Computer Science, Information Systems, Computer Engineering, Electronics Engineering, or related computing/technology background	Required
Knowledge or experience in software development, system development, or software testing	Required
Relevant professional or technical experience	At least one (1) year
Experience with web-based applications or information systems	Preferred
Experience in system or software testing	Preferred
Experience with XR, WebXR, AR, VR, 3D applications, or other immersive technologies	Advantage

Table 3.1 presents the qualification criteria that were established to ensure that the individuals conducting the Black Box Testing possessed sufficient technical knowledge and experience to evaluate whether the developed system operated according to its intended functional requirements. Experience with immersive technologies was considered an advantage rather than a mandatory requirement because the Black Box Testing primarily focused on functional behavior rather than the internal implementation of WebXR technologies.

3.4.1 Profile of System Testers

The professional and technical backgrounds of the selected system testers are presented in Table 3.2. Their educational background, credentials, current position or organizational affiliation, and years of relevant experience were documented to establish the basis of their qualification to participate in the technical evaluation of the developed platform.

Table 3.2 Profile of System Testers

Tester Name	Educational Background/ Credentials	Current Position/ Affiliation	Years of Relevant Experience
Jerem Micah Rivera	Bachelor of Science in Information Technology with MSIT Units	Information Analyst II / Municipality of Nagcarlan	10 Years
Gerica Urriza	Bachelor of Science in Information Technology	IT Staff - ICT Unit / Municipality of Nagcarlan	2 Years
Noreen Matienzo	Bachelor of Science in Computer Science	Clerk / Municipality of Nagcarlan	1 Year
Bernard Arciaga	Bachelor of Science in Information Technology	MIS Staff / FUFRB Inc. - Santa Cruz	1 Year
Janssen Espino	Bachelor of Science in Information Technology	IT Support / Municipality of Santa Cruz	1 Year
Jonas Santos	Bachelor of Science in Information Technology	IT Staff / Municipality of Santa Cruz	3 Years
Christian Dale Padilla	Bachelor of Science in Information Technology	Head IT / FUFRB Inc. Santa Rosa City	4 Years
Glady May America	Bachelor of Science in Information Technology	IT Personnel / FUFRB Inc. - San Pablo City	3 Years
April Mendoza	Bachelor of Science in Electronics and Communication Engineering	Data and Systems Manager / Tree to Tub / United States	5 Years
Cris Orencia	Bachelor of Science in Information Technology	Application Engineer / Rockaway, New Jersey	5 Years

3.5 Respondents for User Evaluation

The system user evaluation involved intended users and relevant stakeholders who directly interacted with the completed platform before answering the questionnaire. Using the confirmed population of 2,633 and a 7% margin of error, Slovin's formula produced a minimum sample requirement of approximately 189.40, which was rounded upward to 190 respondents.

The final respondent distribution consisted of 2 administrative personnel, 161 museum visitors, 26 senior citizens, and 1 PWD, for a total of 190 respondents. Because the administrative and PWD populations were very small, the resulting subgroup observations are descriptive and should not be generalized beyond the individuals represented.

Slovin's Formula

Slovin's Formula was used to determine the required sample size from the total identified population of the study. The formula was applied using a 7% margin of error to obtain a sample that was feasible for the available data-gathering period while maintaining representation of the identified respondent groups.

The formula is expressed as:

$$n = \frac{N}{1 + Ne^{2}}$$

Equation 3.1. Slovin's Formula

Where:

- (n) = sample size;
- (N) = total population;
- (e) = margin of error; and
- (1) = constant.

The 7% margin of error was arrived at based on the practical conditions of the study such as accessibility and availability of qualified respondents, the time allocated for data gathering and system evaluation, as well as respondent groups such as senior citizens and persons with disabilities. It provided a manageable sample size and allowed the researchers to acquire representation of the identified population groupings.

The users for user evaluation were separated from the qualified system testers. The system testers performed Black Box Testing to verify the functionality of the primary public and administrative features by the functional requirements and the respondents for user assessment for acceptance and experience of the built platform.

3.5.1 Distribution of Respondents

Table 3.3 Distribution of Respondents for User Evaluation

Respondent Group	Total Population	Sample Size	Sampling Design
Administrative Personnel	2	2	Total Enumeration
Museum Visitors	2,259	161	Stratified Random Sampling
Senior Citizens	371	26	Stratified Random Sampling
Person with disabilities	1	1	Total Enumeration
Total	**2,633**	**190**	

Table 3.3 presents the distribution of respondents for the user evaluation. The study identified a total population of 2,633, consisting of 2 administrative personnel, 2,259 museum visitors, 371 senior citizens, and 1 person with disability (PWD). From this population, a total sample of 190 respondents was determined, composed of 2 administrative personnel, 161 museum visitors, 26 senior citizens, and 1 PWD. The corresponding sampling design for each respondent group is also presented in the table.

3.6 Sampling Design

The study used a stratified sampling design because the target population consisted of distinct respondent groups with different roles and characteristics. The population was divided into four strata: administrative personnel, museum visitors, senior citizens, and persons with disabilities (PWDs). Because the administrative-personnel and PWD strata contained only two and one identified members, respectively, these strata were included through total enumeration. The museum-visitor and senior-citizen strata were sampled using stratified random sampling, with the required sample allocated proportionally between these strata.

Where n is the required sample size, N is the population, and e is the selected margin of error.

$$n = \frac{2{,}633}{1 + 2{,}633(0.07^2)}$$

$$n = \frac{2{,}633}{13.9017}$$

$$n = 189.40$$

Equation 3.2. Implemented Slovin's Formula

Minimum sample = 190 respondents

After the 2 administrative personnel and 1 PWD were included through total enumeration, 187 slots remained for proportional allocation between museum visitors and senior citizens. The allocation was computed as follows:

$$Museum\ Visitors\ =\ (\frac{2{,}259}{2{,}630})(187)\ =\ 160.62\ \approx 161$$

$$Senior\ Citizens\ =\ (\frac{371}{2{,}630})(187)\ =\ 26.38 \approx 26$$

Equation 3.3 Allocated Slovin's Formula

After the proportional allocation was determined, respondents within the museum-visitor and senior-citizen strata were selected randomly from the applicable sampling frame. The same stratification and random-selection procedure was used to preserve the planned representation of the two sampled strata. The researchers should state the exact randomization procedure actually used (for example, random-number selection from an eligible list) and any replacement procedure for selected individuals who did not participate.

Eligibility Criteria for User Evaluation

- Participants belonged to one of the respondent groups defined in the approved sampling plan.
- Participants were available during the approved evaluation period and voluntarily agreed to participate.
- Participants directly interacted with the relevant sections or functions of the working system before completing the questionnaire.

- Age classification was recorded through the respondent profile. Senior citizens were identified using the study's approved age rule, and PWD status was recorded only according to the actual identification procedure used during data gathering.
- Responses that were incomplete, duplicated, or ineligible under the approved procedure were excluded from analysis.

3.7 Data Collection Instrument

The researchers used various data-collection tools to gather the information essential for the design, development, testing, and assessment of the WebXR-Enabled Digital Heritage Platform. These included an interview guide, a Black Box Testing device, and a user-evaluation questionnaire. Each instrument was designed to serve a specific purpose, and was linked with the research goals and the different phases of the study. The interview guide was used during consultations with museum staff to gather information about the current museum environment, visitor issues, heritage and interpretive content, administrative processes, accessibility considerations, and requirements relevant to the proposed platform.

The system testers with the required qualification tested the produced platform's principal public and administrative features using an instrument for Black Box Testing to see if they work according to the functional requirements they claim. The instrument had formal test cases that contained the feature or function to be tested, the test technique or input, the expected result, the actual result and the Pass, Fail, N/A, or Retest status. Space for observations and recommendations was also offered where appropriate.

The assessment of the respondents took place after they used the relevant functions of the built platform, through a separate user-evaluation questionnaire. To measure the evaluation criteria defined in the research objectives, the questionnaire was used. These criteria included the chosen software quality characteristics, perceived ease of use, perceived usefulness, attitude toward using, behavioural intention to use, actual system interaction, immersive digital experience, and accessibility. Responses were collected on a five-point Likert scale and processed for statistical analysis.

3.7.1 Instrument Development

The interview guide and observation activities were used to identify museum processes, content requirements, visitor needs, accessibility considerations, and administrative requirements. The Black Box Functional Testing instrument was developed from the documented functions of the completed platform and contained 54 test cases distributed across 13 module groups.

The system user-evaluation questionnaire contained 26 statements under eight criteria: Selected Software Quality Characteristics, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, Actual System Interaction, Immersive Digital Experience, and Accessibility. The TAM-related items were adapted to the digital-heritage context, while the selected software-quality items were based on applicable user-observable indicators rather than the complete ISO/IEC 25010 quality model. The immersive-experience, actual-interaction, and accessibility items were researcher-developed or adapted for the specific platform context.

Table 3.4 Specific Platform Context

Questionnaire Section	Basis	Items
A	Selected Software Quality Characteristics — selected user-observable indicators informed by ISO/IEC 25010	A1–A3
B	Perceived Ease of Use — Technology Acceptance Model	B1–B3
C	Perceived Usefulness — Technology Acceptance Model	C1–C3
D	Attitude Toward Using — Technology Acceptance Model	D1–D3
E	Behavioral Intention to Use — Technology Acceptance Model	E1–E3
F	Actual System Interaction — researcher-developed for evaluation-session interaction	F1–F3
G	Immersive Digital Experience — researcher-developed/adapted to implemented immersive functions	G1–G4
H	Accessibility — researcher-developed/adapted accessibility perception items	H1–H4

Table 3.4 presents the 26-item user-evaluation questionnaire organized into eight sections (A–H). The instrument incorporates selected ISO/IEC 25010 quality characteristics, Technology Acceptance Model (TAM) constructs, and researcher-developed or adapted items to assess the platform's software quality, user acceptance, actual interaction, immersive digital experience, and accessibility.

3.7.2 Instrument Validation

The user-evaluation questionnaire was reviewed by the research adviser for clarity, relevance, alignment with the research objectives, and

appropriateness for the intended respondents. Revisions were made before administration based on the review. Unless separate accomplished expert-validation forms and validity results are available, the manuscript should describe this process as adviser review and should not present it as statistical or formal expert content validation.

The Black Box Testing instrument was likewise reviewed before use to ensure that its test cases represented observable functional requirements and that expected results could be compared with actual system behavior. Any formal validation procedure beyond adviser or technical review should be reported only when supported by completed documentation.

3.7.3 Administration of the Instrument

The interview and observation activities were conducted during the requirements-gathering phase. Black Box Functional Testing was conducted after the platform had reached a sufficiently operational build. The 10 qualified system testers executed the documented cases and recorded the actual behavior and final status of each test.

The user-evaluation questionnaire was administered after eligible participants had interacted directly with the working platform. Respondents were briefed on the purpose of the evaluation and given the instructions necessary to access and use the relevant functions. Assistance was limited to operational guidance and was not intended to influence the respondents' ratings. Completed responses were checked for completeness and eligibility before encoding and analysis.

The manuscript must state the actual consent, assent, privacy, and institutional-permission procedures that were followed. Because the dataset includes respondents below 18 years old, the researchers should specifically document the procedure required by the institution for minor participants rather than assuming that ordinary adult consent was sufficient.

3.8 Project Design

The project design defines the structure and technological setup of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. It was developed from the requirements identified in interviews, consultations with museum personnel, and the research and fact-finding activities undertaken as part of the study.

The design took into account the needs of general users and authorized museum personnel. The platform for the public users was designed to provide accessible heritage information, educational resources, interactive and immersive experiences, and visitor-based museum services. Administrative services were built in allowing authorised people to manage and maintain selected digital material, submissions, bookings and other platform data.

The system was designed as a responsive web-based platform capable of supporting conventional web content together with multimedia, 360-degree, three-dimensional, and WebXR-enabled experiences. The project design also considered bilingual presentation, accessibility, security, compatibility with commonly used devices and browsers, maintainability of museum content, and centralized management of system data.

3.8.1 System Requirements

The system requirements were derived from the information gathered during the research and fact-finding activities and were classified into functional and non-functional requirements.

Functional Requirements

The functional requirements describe the services and operations that the developed platform is expected to perform. The major functional requirements were organized into the following groups:

1. Digital Heritage and Educational Content

The system shall provide users with access to curated museum and heritage information, digital exhibits, gallery content, educational resources, learning materials, museum information, and related interpretive content in English and Filipino.

2. Interactive and Immersive Experiences

The system shall provide interactive and immersive components, including the heritage trail, interactive site map, 360-degree virtual tour, three-dimensional content, and supported augmented reality or WebXR experiences. These components shall allow users to explore selected heritage areas and multimedia content through appropriate navigation and interaction mechanisms.

3. Inclusive Accessibility Support

The system shall provide accessibility-oriented functions such as adjustable text size, high-contrast presentation, bilingual English and Filipino content, narration, and captions or subtitles where applicable. These functions are intended to improve access to heritage information for users with varying preferences and accessibility requirements.

4. Visitor Engagement and Museum Services

The system shall provide functions that allow users to register and log in, access museum-related information, submit feedback or community contributions, request or access learning materials, view relevant news and events, and submit guided-visit or booking requests through available museum services.

5. Administrative Management and Monitoring

The system shall provide a secured administrative interface through which authorized personnel can manage selected platform content and records. Administrative functions include management of digital exhibits, gallery materials, events, educational resources, virtual-tour content, community submissions, visitor feedback, bookings, and other supported museum information. Access to administrative functions shall be restricted according to authorized user roles.

Non-Functional Requirements

The non-functional requirements describe the conditions and quality considerations that guide the operation of the developed platform.

Usability. The platform should provide clear navigation, understandable controls, readable content, and consistent interface elements to support users with varying levels of technical experience.

Accessibility. The interface should support accessibility-oriented presentation through features such as text-size adjustment, high-contrast display, bilingual content, narration, and captions or subtitles where applicable.

Performance. The platform should load and respond within a reasonable period under normal operating conditions while considering the size and complexity of multimedia, 3D, and 360-degree resources.

Compatibility. The platform should operate through modern web browsers and adapt to commonly used desktop computers, tablets, and mobile devices. WebXR and other immersive functions may depend on the capabilities and permissions supported by the user's device and browser.

Security. User and administrative functions should incorporate appropriate authentication, authorization, access control, and protection of stored information. Administrative functions should only be accessible to authorized personnel.

Reliability. The system should provide consistent operation of its implemented functions during normal usage and should appropriately handle invalid input or unsuccessful requests where applicable.

Maintainability. The platform should allow authorized personnel to update supported museum content and records without requiring direct modification of the application's source code for routine content-management activities.

The identified functional and non-functional requirements served as the basis for the design, implementation, testing, and evaluation of the developed platform. Functional requirements were specifically used as the basis for the Black Box Testing of the system's major public and administrative functions.

3.8.2 System Design

The system design took the defined requirements and transformed them into the processes, interfaces, data structures, user interactions and technological components necessary to create the platform. The design of the system evaluated two key access environments: the public user interface for visitors and other intended users and the administrative interface for authorized museum workers.

The public user interface has been created to structure heritage information and digital experiences in accessible and clear portions. Its key elements are museum and visitor information, digital exhibitions, educational materials, heritage-learning activities, interactive and immersive experiences,

accessible features, community engagement and visitor services. Navigation and responsive interface features were built to provide access to these resources using popular desktop and mobile devices.

The administrative interface was developed separately from the public user environment, which allows for proper administration functions for authorised workers. It provides creation, update, review and management of selected platform content and user generated data based on the functions provided in the system.

The data design structured the records needed by the platform: user information, heritage and museum content, multimedia references, events, learning resources, contributions, feedback, bookings and administrative information. These relationships between the records were designed to support retrieval, and update and maintenance through the public and administrative interfaces.

The main design models used to describe the structure and behaviour of the system include relevant diagrams and models of the users and their interactions, system operations, information flow, database relationships and overall system architecture. These representations were used as reference points during development and helped to ensure that the components deployed were consistent with the identified system requirements.

3.8.3 System Architecture

The system architecture presents the overall technical organization of the WebXR-Enabled Digital Heritage Platform and illustrates how users, application components, data services, multimedia resources, and external services interact with one another.

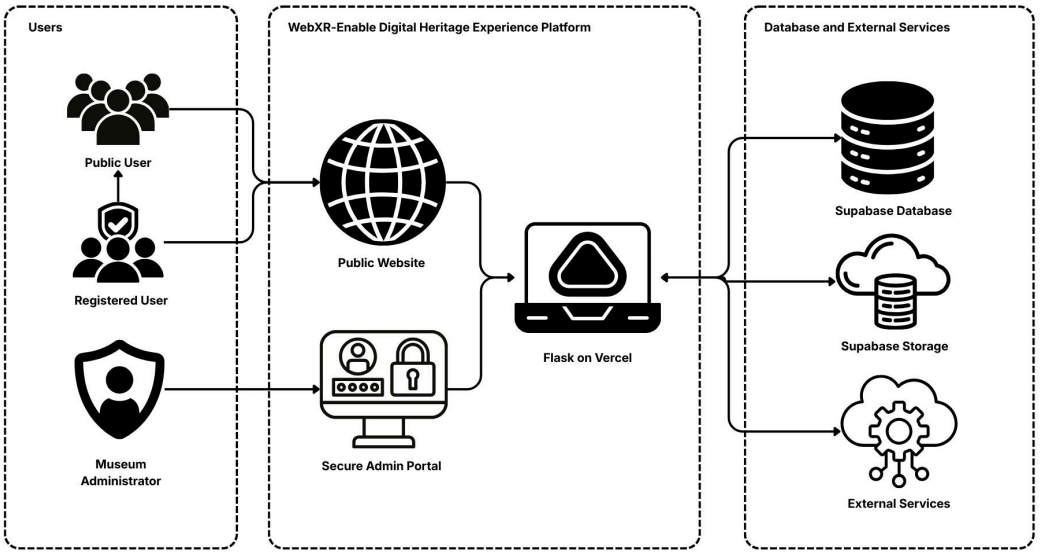

Figure 3.1 System Architecture

Figure 3.1 presents that public and registered users access the platform through supported web browsers using desktop computers, tablets, or mobile devices. Through the public user interface, users can access museum information, digital exhibits, educational resources, interactive and immersive experiences, accessibility functions, and supported visitor services. Museum administrators access a secured administrative interface for the management of selected system content, submissions, bookings, and administrative records.

Both the public user and administrative interfaces communicate with the Flask-based web application hosted on Vercel. The application serves as the main processing layer of the system by receiving user requests, processing application logic, enforcing supported authentication and authorization rules, retrieving or updating information, and coordinating communication with the platform's connected services.

The application communicates with Supabase for the management of system data and digital resources. The Supabase Database stores structured information required by the platform, including user-related records, museum content, submissions, feedback, bookings, and other supported data. Supabase Storage is used for applicable multimedia resources such as photographs, audio files, videos, 360-degree panoramic images, and three-dimensional assets.

Information moves between the user interface, Flask application, Supabase services, and applicable external services according to the action requested by the user. Public users primarily retrieve heritage information and submit permitted records, while authorized administrators can perform supported content and record-management operations. This architecture separates the presentation, application-processing, data-management, and external-service components of the platform, allowing the developed system to support both public digital heritage experiences and administrative museum functions.

3.9 Software Development Model

The researchers employed the Agile Software Development Model in the development of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. Agile was selected because the project required the progressive development, review, testing, and refinement of multiple interconnected system components, including conventional web functions, administrative features, multimedia content, accessibility mechanisms, and interactive and immersive experiences.

The development process followed six major phases: Planning, Design, Development, Testing, Deployment, and Feedback and Refinement. Although presented sequentially, these phases were carried out iteratively. Findings obtained during development, testing, and evaluation could require the researchers to return to an earlier phase to revise requirements, interface designs, workflows, system functions, or content before proceeding.

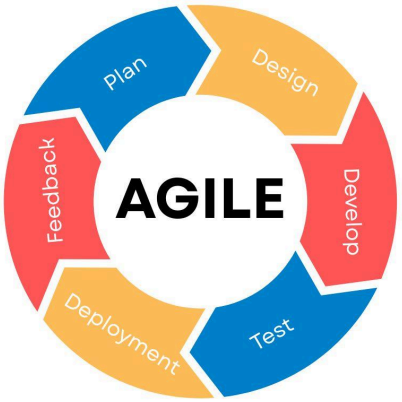

Figure 3.2 Agile Software Development Model

The figure 3.2 above illustrates the process of the agile software development methodology. Below is the process of how each step in the cycle works.

Planning

During the Planning phase, the researchers identified the existing museum situation, user and stakeholder needs, project objectives, system scope, and major functional requirements. Information gathered through interviews, onsite observations, document and content review, and consultation with museum personnel was analyzed and translated into system requirements and development priorities.

The activities conducted during this phase included requirements gathering, identification of intended users, definition of system modules, review of available heritage content and digital resources, and determination of technical requirements and project constraints.

The major outputs of the Planning phase included the system requirements, identified user groups, initial feature requirements, content requirements, project scope, and development priorities.

Design

During the Design phase, the identified requirements were translated into the structural and visual design of the proposed platform. The researchers planned the system's information architecture, user navigation, interface layout, database structure, public and administrative workflows, content organization, accessibility mechanisms, and interactive and immersive components.

The design process also considered responsive presentation across mobile devices, tablets, and desktop computers, as well as the organization of multimedia, 360-degree, three-dimensional, and WebXR-enabled resources.

The major outputs of this phase included the system architecture, interface designs, navigation structure, database design, workflow representations, system models, and planned organization of the platform's functional modules.

Development

During the Development phase, the researchers implemented the approved designs and requirements using the selected programming languages, frameworks, database services, development tools, and multimedia technologies.

Development was conducted incrementally so that individual functions and modules could be implemented, reviewed, and refined before full integration. The researchers developed the public-facing pages, user registration and authentication functions, digital heritage and educational content, digital exhibits, interactive site map, heritage trail, 360-degree virtual tour, AR and 3D components, accessibility controls, visitor-engagement functions, booking services, and administrative management features.

The major outputs of the Development phase included working system modules, integrated frontend and backend components, database functionality, multimedia and immersive content, and an operational version of the platform prepared for technical testing.

Testing

During the Testing phase, the developed platform underwent internal checking and formal Black Box Testing. The major public and administrative functions were tested to determine whether they operated according to their specified functional requirements.

Qualified system testers used structured test cases that identified the required actions or inputs, expected results, actual results, and corresponding Pass or Fail outcomes. Errors, inconsistencies, and technical observations identified during testing were documented. Necessary corrections were implemented, and affected functions were retested when required.

The major outputs of this phase included completed Black Box Testing results, documented functional issues, corrected system components, and a refined version of the platform.

Deployment

During the Deployment phase, a working and sufficiently operational version of the developed platform was made available for the user-evaluation activities of the study. The selected respondents were provided access to the relevant system functions and were given an opportunity to interact with the platform before completing the user-evaluation questionnaire.

Deployment in the context of the study referred to making the functional system available for evaluation and intended use within the scope of the capstone project. It did not necessarily represent full institutional or organization-wide adoption by the NHCP unless formally approved by the concerned institution.

The major outputs of this phase included the accessible working platform and the system version used for respondent evaluation.

Feedback and Refinement

During the Feedback and Refinement phase, the researchers reviewed the comments, observations, Black Box Testing results, and user-evaluation findings obtained from the system testers and respondents.

Technical issues and feasible recommendations were evaluated and used as the basis for additional corrections and improvements to the platform. When necessary, the researchers returned to the appropriate Planning, Design, Development, or Testing activities to revise affected system components.

The major outputs of this phase included documented feedback, identified areas for improvement, implemented revisions, and the final refined version of the developed platform within the scope of the study.

Rationale for the Use of the Agile Software Development Model

The Agile Software Development Model was appropriate for the study because the platform consisted of multiple interconnected modules that required continuous development, integration, testing, and refinement. The project also involved heritage content, interactive interfaces, accessibility functions, multimedia resources, administrative workflows, and immersive technologies that could require adjustments as development progressed.

The iterative nature of Agile allowed the researchers to incorporate findings from museum consultations, internal reviews, technical testing, and user evaluation

into subsequent revisions of the platform. This approach was particularly suitable because issues identified in one component could be addressed without requiring the researchers to restart the entire development process.

Through the use of Agile, the researchers were able to progressively transform the identified requirements into a functional system, evaluate the implemented features, correct identified problems, and refine the platform based on technical and user-related findings. The model therefore supported the achievement of the study's objectives involving the design, development, implementation, functional testing, and evaluation of the WebXR-Enabled Digital Heritage Platform.

3.10 System Architecture

The system architecture defined the planned organization of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. It served as the technical blueprint for integrating the public website, registered-user services, immersive experiences, administrative functions, database, multimedia storage, and external services into a unified web-based platform.

3.10.1 Hardware and Software Requirements

The tables 3.5 and 3.6 consist of the hardware and software requirements that were identified according to the resources needed to develop, operate, administer, maintain, and deploy the proposed platform. Because the system was designed as a web-based application, most public

functions could be accessed through an existing browser without installing a separate desktop or mobile application.

Table 3.5 Hardware Resources for System Development and Operation

Hardware resource	Intended purpose
Development computer	Source-code development, local execution, database configuration, multimedia preparation, testing, and maintenance
Administrative computer	Content management, file uploads, visitor-record management, virtual-tour editing, and site-map configuration
Visitor desktop or laptop	Access to public content, educational resources, virtual tours, maps, videos, and three-dimensional experiences
Smartphone or tablet	Mobile access, touchscreen interaction, camera-based augmented reality, and device-motion viewing where supported
Optional immersive device	Access to compatible WebXR or virtual-reality experiences
Conventional and 360-degree cameras	Capture of photographs, object images, site views, and equirectangular panoramas
Microphone and audio equipment	Recording, editing, reviewing, and playing narration or other audio content
Stable network connection	Communication with the hosted application, database, storage, and external services

Table 3.5 presents the hardware resources required for the development, administration, and operation of the platform. These include computers for development and content management, user devices for accessing digital heritage content, and multimedia equipment for capturing images, panoramas, and audio. A stable network connection supports communication with the hosted application, while optional immersive devices provide additional WebXR and virtual reality experiences.

Table 3.6 Software Resources for System Development and Deployment

Category	Selected software or technology	Intended purpose
Server-side language	Python 3	Application logic, validation, database operations, and service integration
Web framework	Flask	Routing, sessions, authentication, request processing, and template rendering
Template engine	Jinja	Generation of dynamic HTML interfaces
Front-end languages	HTML5, CSS3, and JavaScript	Page structure, visual presentation, responsiveness, and interaction
Interface tools	Tailwind CSS, Lucide, GSAP, and custom styles	Responsive design, icons, animation, and interface consistency
Panorama viewer	Pannellum	Presentation and navigation of 360-degree scenes
Mapping library	Leaflet	Interactive aerial-image maps and positioned markers
Three - dimensional rendering	Three.js and WebGL	Browser-based rendering of three-dimensional environments
WebXR support	WebXR Device API, SparkJS, and three-mesh-bvh	Immersive three-dimensional environments and optimized interaction
Augmented reality	AR.js and Google <model-viewer>	Marker-based AR, model viewing, and compatible mobile AR
Database	Supabase PostgreSQL	Storage of structured system records and relationships
Multimedia storage	Supabase Storage and packaged static assets	Storage and delivery of images, audio, panoramas, documents, and models
Deployment	Vercel	Hosting and execution of the Flask web application
Version control	Git and GitHub	Source-code management and project revision tracking
Development tools	Source-code editor, browser developer tools, SQL editor, Python package manager, and Node.js utilities	Development, debugging, database configuration, and dependency management
External services	Google OAuth, SMTP-compatible email, and text-to-speech services	Optional login, email delivery, and narration generation

Table 3.6 presents the software technologies used to develop, manage, and deploy the platform. These include Python and Flask for server-side operations, HTML, CSS, and JavaScript for the user interface, and specialized libraries for interactive maps, 3D rendering, WebXR, and augmented reality. Supabase provides database and multimedia storage services, while Vercel supports deployment. Additional development tools and external services support version control, maintenance, authentication, and accessibility features.

3.10.2 System Components and Modules

The proposed system components and modules were organized according to the five functional areas identified in the second specific objective of the study. Each functional area contained related modules designed to address the requirements of the intended users and museum stakeholders. Common technical services, including authentication, database access, security, analytics, and multimedia management, were designed to support the five functional areas rather than function as separate objective-level categories.

Digital Heritage and Educational Content

The Digital Heritage and Educational Content component was designed to organize and present curated museum information, historical narratives, digital exhibits, gallery materials, educational resources, community highlights, and related interpretive content. The Online Gallery module was designed to present categorized heritage objects through bilingual

descriptions, images, and applicable three-dimensional content. The Educational Resources module was designed to provide learning materials, documents, worksheets, and instructional resources for students, teachers, researchers, and other visitors.

Interactive and Immersive Experiences

The Interactive and Immersive Experiences component was designed to provide alternative methods of exploring the museum and its cultural resources. It included the interactive site map, 360-degree virtual tour, heritage trail, three-dimensional models, augmented-reality content, and WebXR environments. The Site Map module was designed to present aerial and 360-degree views containing selectable location hotspots, while the Virtual Tour module was designed to connect panoramic scenes through informational and navigational hotspots.

Inclusive Accessibility Support

The Inclusive Accessibility Support component was designed to improve access to digital heritage information for users with different preferences, abilities, devices, and levels of technical experience. Its planned functions included adjustable text presentation, high-contrast display, bilingual English and Filipino content, narration, captions or subtitles, readable labels, responsive layouts, and keyboard-operable conventional controls. Applicable alternative viewing methods were also considered when a device or browser could not support an advanced immersive function.

Visitor Engagement and Museum Services.

The Visitor Engagement and Museum Services component was designed to support interaction between visitors and the museum. It included

public-user registration, authentication, account management, bookmarks, guided-tour booking, visitor feedback, community participation, and Heritage Keeper progress. The account functions were designed to provide secure access to personalized services, while the booking, feedback, and participation modules were designed to allow visitors to communicate and interact with the museum through the platform.

Administrative Management and Monitoring

The Administrative Management and Monitoring component was designed to provide authorized museum personnel with secured tools for maintaining the platform. It included the administrative dashboard, user management, booking review, community-submission review, feedback moderation, content management, virtual-tour editing, site-map editing, notifications, analytics, and account-security settings. Administrative functions were designed to require authentication and applicable authorization, request validation, and confirmation controls.

3.10.3 Database Structure

The database was designed using a relational model to organize structured system information and define the relationships among accounts, content, visitor activities, administrative records, and immersive configurations. PostgreSQL through Supabase was selected because the proposed system required relationships, validation constraints, transactions, indexing, and controlled access to structured data.

Account and Security Data

The account and security group was designed to contain administrator accounts, public-user accounts, pending registrations, email-verification information, password-recovery records, administrator login challenges, and notification records.

Administrator authentication was separated from public-user authentication because administrators required access to protected management functions. Passwords were designed to be stored as secure hashes instead of plain text. Temporary verification, recovery, and login codes were also designed to use limited validity periods and controlled consumption.

Digital Content Data

Digital content entities were designed to store community highlights, gallery categories, gallery items, and educational resources. Each content record could contain descriptive metadata, language-specific information, publication status, multimedia references, timestamps, and other values required by its corresponding module.

Gallery items were designed to reference a gallery category, allowing related materials to be grouped. Publication-status fields were included to distinguish active, draft, archived, or otherwise controlled content where applicable.

Visitor Engagement Data

Visitor-engagement entities were designed to contain bookmarks, guided-tour bookings, community submissions, visitor feedback, consent records, and moderation history. These records could reference a registered user when appropriate while retaining the information necessary for the corresponding museum service.

Feedback moderation was separated from the original feedback record to preserve a history of administrative review. Booking records were designed to contain visitor information, requested schedule details, status, and administrative decisions.

Progress and Analytics Data

Progress-related entities were designed to contain Heritage Keeper activities, earned awards, WebXR-collected items, and viewed exhibit records. Analytics records were designed to store selected event information required for administrative summaries while avoiding unnecessary storage of raw network identifiers.

Virtual-Tour Data

The virtual-tour database design included the virtual-tour environment, revision, scene, and hotspot entities. Each tour could contain draft and published revisions. A revision could contain multiple panoramic scenes, and each scene could contain informational or navigational hotspots.

Informational hotspots were designed to store a title, position, bilingual description, images, audio, three-dimensional objects, or nested information where applicable. Navigation hotspots were designed to reference another scene and could include a target pitch, yaw, and field of view.

The draft-and-publish design was intended to prevent partially edited configurations from being immediately displayed to visitors. Administrators could prepare and validate a working revision before it became the public revision.

Site-Map Data

The site-map database design used site-map and revision entities. A revision was designed to contain the aerial-map and 360-degree-map configuration, starting view, visibility settings, hotspot coordinates, hotspot relationships, bilingual information, audio, photographs, and optional destination links.

3.10.4 User Interface Design

The user interface was designed to provide a consistent and responsive experience across the platform's public, registered-user, immersive, and administrative areas. The design considered the varying technical experience, device capability, and accessibility preferences of visitors.

The public interface was planned around a common navigation structure that connected the landing page, museum information, digital exhibits, online gallery, educational resources, interactive experiences,

community participation, visitor services, and account functions. Cards, headings, images, search controls, filters, buttons, tabs, modals, and detail pages were used to organize information into understandable sections.

The registered-user interface was designed to extend the public interface rather than function as a separate application. Account pages were planned to contain profile information, security settings, bookmarks, bookings, preferences, and Heritage Keeper progress.

The immersive interfaces required controls that differed from conventional content pages. The 360-degree virtual-tour interface was designed to support panorama dragging, zooming, navigational hotspots, informational hotspots, scene transitions, language controls, narration, device-motion viewing, and a location minimap. The site-map interface was designed to provide aerial and 360-degree views, selectable markers, location descriptions, photographs, narration, and links to related experiences.

The augmented-reality and WebXR interfaces were designed to request device or camera access only when required by the selected function. The interface was also designed to communicate compatibility limitations and provide an alternative viewing method where applicable.

The administrative interface was designed around a dashboard, side navigation, searchable record lists, editing forms, preview areas, status controls, validation messages, confirmation dialogs, and publication controls. Specialized virtual-tour and site-map editors were planned to allow administrators to position hotspots visually rather than estimate coordinates without a preview.

3.10.5 System Communication and Interaction

The platform was designed to use HTTPS as the principal communication mechanism between the user's browser and the hosted Flask application. Communication included page requests, form submissions, asynchronous JavaScript requests, database operations, multimedia transfers, and external-service requests.

When a visitor requested a page, the browser would send an HTTPS request to the Flask application. Flask would identify the requested route, retrieve the required information, render the appropriate Jinja template, and return an HTML response. JavaScript in the browser would then initialize the applicable interactive behavior.

For registration, login, booking, feedback, submissions, profile updates, and administrative actions, the browser would send submitted data to a protected application endpoint. The application would validate the session, authorization level, CSRF token, rate limit, and input values before performing the requested operation.

Database communication was designed to occur through the server-side application. The application would retrieve, create, update, or archive records according to the authorized request. Protected database credentials would remain in the server environment and would not be delivered to the visitor's browser.

Multimedia retrieval could occur through packaged static-resource routes or controlled Supabase Storage URLs. For supported administrative

uploads, the application would first validate the administrator, intended media type, requested size, and destination. The browser could then upload the authorized file to storage, after which the server would verify the uploaded object before associating it with a database record.

Virtual tours, interactive maps, augmented reality, and WebXR experiences were designed to perform real-time rendering on the client device. The application would provide a sanitized content configuration containing the scenes, hotspots, coordinates, text, and media references required by the public interface. The browser would use this information to initialize the panorama, map, model, or WebXR environment.

Google authentication was designed to use an OAuth redirection and callback sequence. Email verification, password recovery, booking updates, and applicable notifications were designed to use an SMTP-compatible email service. Narration generation could use an authorized text-to-speech request, after which the generated audio would be validated and associated with the corresponding content.

The platform was designed to coordinate interactions between the public interface and the administrative backend. Visitor activities, such as booking requests, feedback submissions, and community contributions, would be transmitted to the Flask application for processing and storage. Authorized museum personnel could then access and manage the corresponding records through the administrative interface. Any approved updates to museum information, exhibits, or multimedia content would be retrieved and displayed

through the public interface, ensuring that visitors could access the latest available content without requiring changes to the application's source code.

Table 3.7 System Communication and Interaction

Interaction	Planned communication	Intended purpose
Page retrieval	Browser-to-Flask HTTPS request	Retrieve and render public, account, or administrative pages
Form submission	HTTPS form or asynchronous request	Submit account, booking, feedback, participation, or administrative data
Database operation	Server-side Flask-to-Supabase request	Retrieve or modify authorized structured records
Multimedia retrieval	Static-resource or storage request	Load images, audio, video, panoramas, documents, and 3D models
Multimedia upload	Authorized upload request and subsequent verification	Store administrative media without exposing protected credentials
Virtual-tour navigation	Client-side panorama rendering using server-provided configuration	Display scenes, hotspots, narration, and scene transitions
Site-map interaction	Client-side map or panorama rendering	Display location markers, information, and related destinations
Augmented reality	Browser camera and model-rendering interaction	Place or display applicable digital content through supported devices
WebXR interaction	Browser and compatible immersive-device communication	Render and interact with supported immersive environments
Google authentication	OAuth redirection, authorization, and callback	Provide optional account authentication
Email delivery	Server-to-email-service communication	Deliver verification, recovery, booking, and notification messages
Narration generation	Server-to-text-to-speech request	Generate applicable bilingual narration audio
Analytics recording	Browser-to-application event request	Record selected activity for administrative summaries

Table 3.7 shows the communication mechanisms supporting performance by separating page rendering, database operations, multimedia delivery, and client-side immersive rendering. They supported security through HTTPS, server-side authorization, session controls, CSRF validation, protected credentials, input and upload validation, restricted administrative routes, and database-access policies. Reliability and maintainability were supported through modular services, database constraints, draft-and-publish workflows, and separation between source-controlled original content and database-managed updates.

3.11 System Testing and Evaluation Procedure

The system testing and evaluation procedure was conducted to determine whether the developed WebXR-Enabled Digital Heritage Platform satisfied its documented functional requirements and whether the intended users favorably evaluated the platform according to the criteria established in the research objectives.

The procedure consisted of two major activities: Black-Box Functional Testing and system user evaluation. Black-Box Functional Testing was conducted first by qualified system testers to determine whether the major public user and administrative functions of the developed platform produced the expected results based on the documented functional requirements. The testing instrument consisted of structured test cases specifying the module or function to be tested, test action or scenario, expected result, actual observed result, and corresponding test status.

The system testers performed the tests using appropriate computers or mobile devices, supported web browsers, and internet connectivity. Device or

browser-dependent capabilities were considered when testing immersive functions such as the 360-degree virtual tour, Augmented Reality (AR), and WebXR 3D environment. Relevant testing-environment information, including the device, operating system, browser version, network condition, testing date, and tester identification, was documented where applicable.

After the documented functional test cases had been completed and identified issues had been addressed, the sufficiently operational version of the platform proceeded to system user evaluation. The selected respondents were given an opportunity to interact with the relevant functions of the developed platform before completing the validated user-evaluation questionnaire. Their responses were used to evaluate the platform according to the criteria specified in the research objectives.

This procedure allowed the researchers to distinguish between technical functional performance, which was assessed by qualified system testers, and user acceptance and experience, which were assessed by the intended users.

3.11.1 Testing Plan

The testing and evaluation activities were conducted in the following sequence:

1. The researchers identified the major public and administrative functions to be tested based on the documented system requirements.
2. Structured Black Box test cases were prepared for each selected system function.
3. The testing instrument was reviewed by the research adviser to ensure that the test procedures, expected results, testing coverage, and

decision criteria were clearly stated and aligned with the system requirements.

4. Qualified system testers were provided access to the developed platform and corresponding test cases.
5. Each tester performed the specified actions and compared the actual system behavior with the documented expected results.
6. Test cases that produced the expected result were marked Pass, while those that produced results different from the expected behavior were marked Fail.
7. Identified issues and tester observations were documented and reviewed by the researchers for correction and refinement.
8. A sufficiently operational version of the system was subsequently presented to the selected respondents.
9. Respondents were oriented to and allowed to interact with the relevant functions of the platform before answering the user-evaluation questionnaire.
10. Completed responses were collected, checked, organized, encoded, and prepared for statistical analysis.

The technical testing activity addressed the functional behavior of the developed platform, while the user-evaluation activity addressed the users' acceptance and experience of the system.

3.11.2 Test Cases and Test Scenarios

The test cases were developed based on the documented functional requirements and major implemented features of the developed platform. Each test case identified the function or module being tested, the required action or condition, expected result, actual observed result, and corresponding Pass or Fail status.

The Black Box Testing covered the following major public and administrative functions:

- Homepage
- User Registration and Login
- Digital Exhibits
- Heritage Trail
- Interactive Site Map
- 360° Virtual Tour
- AR Content
- WebXR 3D Environment
- Community Submissions Form
- Online Booking Schedule
- Visitor Account
- Admin Login and Function
- Navigation/Error Handling

The testing scenarios represented typical actions that users or administrators were expected to perform while using the platform. These

included opening pages, navigating between modules, entering valid information, signing in, accessing multimedia content, interacting with 360-degree and three-dimensional environments, using accessibility controls, submitting forms, requesting bookings, and accessing authorized administrative functions.

Test cases were limited to requirements that could be directly and observably evaluated through the implemented platform. Non-functional characteristics were included only when an observable test condition had been specifically defined, such as responsive display on supported devices or compatibility of selected immersive functions.

3.11.3 Testing Data

The testing data consisted of representative information and inputs required to perform the documented test scenarios. These included sample user credentials, registration information, verification codes, search terms, multimedia content, registered AR markers, form entries, supported file uploads, booking information, and administrator credentials.

The test data were selected to represent realistic conditions that users could encounter while interacting with the developed platform. Where applicable, valid data were used to determine whether the system successfully performed the intended function, while invalid or incomplete inputs could be used to determine whether the system appropriately prevented or handled unsupported actions.

For multimedia and immersive modules, the researchers used the actual digital resources integrated into the platform, including photographs, audio narration, 360-degree panoramas, three-dimensional models, AR targets, and heritage content. Supported mobile devices and modern web browsers were also used when a test required device-specific functionality such as camera access, gyroscope interaction, AR, VR, or responsive presentation.

3.11.4 Testing Procedure

Black Box Testing was conducted by the qualified system testers identified in the study. The testers accessed the developed platform using appropriate computers or mobile devices and supported web browsers. Where necessary, internet connectivity, cameras, gyroscope-enabled devices, and other device capabilities were used to perform the required immersive-system test scenarios.

Each tester was provided with the structured Black Box Testing instrument. The tester followed the stated procedures, performed the required actions, observed the actual system response, and compared the observed behavior with the expected result.

A test case was marked Pass when the actual system behavior corresponded with the documented expected result. A test case was marked Fail when the observed system behavior differed from the expected result. Relevant observations, technical issues, and recommendations were recorded when applicable.

The completed testing instruments were collected and reviewed by the researchers. Identified functional issues were documented and used as the basis for system correction and refinement. Affected functions were retested when necessary before the platform proceeded to user evaluation.

3.12 System Testing

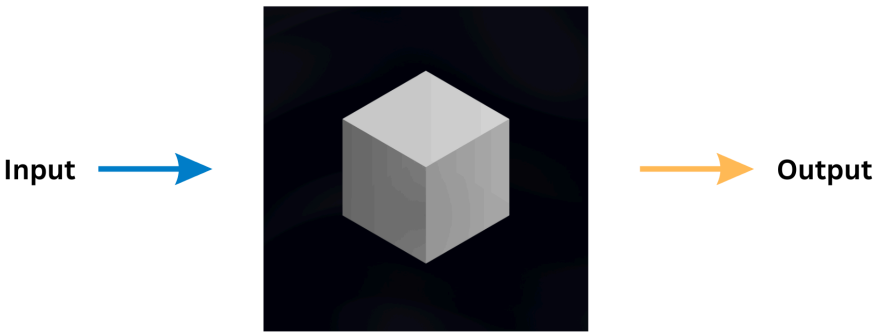

Figure 3.3 Blackbox Testing Process Used in the Study

Figure 3.3 presents the developed WebXR-Enabled Digital Heritage Platform underwent Black-Box Functional Testing to determine whether its major public and administrative functions operated according to their documented functional requirements.

Black-Box Functional Testing was selected because the technical evaluation focused on the observable behavior of the developed system and whether each tested function produced its expected result. The testing was conducted through the system's user interfaces without examining its internal source code or programming structure.

The testing primarily addressed the functional requirements of the platform. Selected observable conditions related to responsive presentation, device and browser compatibility, and immersive-feature operation were also included when these conditions formed part of a documented test case. This was particularly relevant to functions such as the 360-degree Virtual Tour, Augmented Reality (AR), and WebXR 3D environment, which may depend on the capabilities of the device, browser, camera, sensors, and operating system used during testing.

The Black-Box Functional Testing was not intended to independently measure all software-quality characteristics. Specialized performance benchmarking, penetration testing, long-term reliability testing, and formal accessibility-conformance testing were outside the scope of the technical testing unless a directly observable test condition was specifically included in the testing instrument.

3.12.1 Test Cases

The system test cases were developed from the documented functional requirements and major implemented features of the platform. Each test case provided a structured basis for determining whether the tested function produced its intended result.

Each test case identified the following information:

- Test Case ID
- System module or feature
- Test action or scenario
- Expected result
- Actual observed result

- Test status
- Remarks or technical observations

The test cases covered the major public and administrative functions identified in the research objectives, including the Homepage, User Registration and Login, Digital Exhibits, Heritage Trail, Interactive Site Map, 360-degree Virtual Tour, AR Content, WebXR 3D Environment, Community Submission Forms, Online Booking Schedule, Visitor Account, Administrative Login and Functions, and Navigation/Error Handling.

The test scenarios represented expected system-use conditions such as opening pages, navigating between system modules, submitting valid and invalid information, accessing digital heritage content, interacting with multimedia and immersive functions, using accessibility-oriented controls, submitting forms, requesting visit schedules, and performing authorized administrative operations.

The documented expected result for each test case served as the basis for comparing the actual observed system behavior and determining its corresponding test status.

3.12.2 Testing Instrument

The researchers used a researcher-made structured Black Box Testing checklist as the technical testing instrument.

The instrument provided a standardized format for documenting the testing procedure and comparing expected and actual system behavior. It

contained fields for the test case identification, scenario, description, preconditions, test steps, expected results, actual results, and Pass or Fail outcome.

The instrument also recorded relevant information about the system tester, including the tester's name, professional background or current position, relevant experience, and date of testing.

The Black Box Testing checklist provided the researchers with a systematic means of determining whether each tested function of the developed platform operated according to its documented functional requirement.

3.12.3 Validation of Testing Instrument

Before its actual administration, the Black Box Testing instrument was subjected to review and validation to determine whether the test cases were clear, relevant, complete, and aligned with the documented functional requirements of the developed platform.

The validation considered the appropriateness of the test scenarios, clarity of the testing steps, observability of the expected results, consistency of the Pass or Fail criteria, and coverage of the major functions identified in the research objectives.

Comments and recommendations obtained during validation were considered in revising the testing instrument. Ambiguous instructions, incomplete procedures, duplicated test steps, or expected results that could not

be directly observed were revised before the instrument was administered to the qualified system testers.

3.12.4 Interpretation of Test Results

The results of Black Box Testing were interpreted using a Pass or Fail classification.

A test case was interpreted as Pass when the observed actual result corresponded with the documented expected result and the tested function operated according to its specified requirement.

A test case was interpreted as Fail when the observed result differed from the expected result, when the intended function could not be completed, or when an error prevented the system from producing the specified output.

Failed test cases and technical observations were documented and reviewed by the researchers. Identified issues were used as the basis for correcting and refining the affected system functions. Where appropriate, corrected functions were subjected to retesting to determine whether the identified issue had been resolved.

The Black Box Testing results were used specifically to determine the functional behavior of the developed platform and were interpreted separately from the respondents' user-evaluation ratings.

3.13 System User Evaluation

The system user evaluation was conducted to determine how the intended users assessed and accepted the WebXR-Enabled Digital Heritage Platform after interacting with its relevant functions.

The evaluation was conducted after the major system functions had undergone technical testing and the platform was considered sufficiently operational for respondent use. The respondents for user evaluation were separate from the qualified system testers because their role was to assess the platform from the perspective of intended users rather than perform technical Pass or Fail testing.

The user evaluation addressed the criteria identified in the research objectives, including selected software quality characteristics, technology-acceptance constructs, actual system interaction during the evaluation session, immersive digital experience, and accessibility.

3.13.1 User Evaluation Instrument

The researchers used a structured user-evaluation questionnaire to collect the respondents' assessment of the developed platform. The questionnaire consisted of statements corresponding to the evaluation criteria established in the study.

The questionnaire also contained introductory information regarding the purpose of the study, voluntary participation, and data-privacy considerations. Relevant respondent information was collected only according to the requirements of the study and the approved data-gathering procedure.

English and Filipino versions or translations of the evaluation statements were provided where applicable to improve the respondents' understanding of the questionnaire.

3.13.2 Instrument Source and Validation

The user-evaluation questionnaire was adapted and researcher-developed using evaluation concepts appropriate to the objectives and characteristics of the developed platform.

Selected software-quality indicators were based on applicable characteristics of ISO/IEC 25010, while the indicators for Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use were adapted from the Technology Acceptance Model (TAM).

Additional indicators were incorporated to evaluate Actual System Interaction, Immersive Digital Experience, and Accessibility according to the characteristics and intended use of the developed WebXR-enabled heritage platform.

For this study, Actual System Interaction refers to the respondents' reported interaction with the platform during the evaluation session. It does not represent long-term or repeated system usage following full institutional deployment.

Before administration, the questionnaire was submitted for validation to qualified validators. The instrument was reviewed by the research adviser for alignment with the purpose of the study, intended respondents, evaluation

constructs, item clarity, response format, and appropriateness for use after interaction with the developed platform. The adviser review resulted in refinements to the wording and organization of selected items, particularly the software-quality, Actual System Interaction, immersive-experience, and accessibility-related indicators.

Comments and recommendations from the validators were incorporated into the final questionnaire. Statements that were unclear, repetitive, overly technical, or insufficiently aligned with the identified evaluation criteria were revised, simplified, reorganized, or removed before actual administration.

3.13.3 User Evaluation Procedure

The user evaluation was administered to the selected respondents identified in the study. Before completing the questionnaire, respondents were informed about the purpose of the evaluation and the voluntary nature of their participation. They were provided an opportunity to access and interact with the relevant functions of the developed platform so that their responses would be based on direct experience with the system.

Respondents explored the platform using the available device and supported web browser. Relevant functions were presented or made available according to the user-evaluation procedure. Assistance was limited to orientation and clarification of instructions when necessary so that the researchers would not influence the respondents' ratings.

After interacting with the platform, the respondents completed the user-evaluation questionnaire. Completed questionnaires were checked for completeness, organized according to the required respondent classifications, encoded, tabulated, and prepared for statistical analysis.

3.13.4 Evaluation Criteria

The developed platform was evaluated using the following criteria:

1. **Selected Software Quality Characteristics** – assessed selected observable qualities of the platform relevant to the user experience, including applicable indicators such as functional suitability, interaction capability, and performance efficiency.
2. **Perceived Ease of Use** – assessed the extent to which respondents perceived the platform as understandable, learnable, and easy to use.
3. **Perceived Usefulness** – assessed the extent to which respondents perceived the platform as beneficial for accessing, exploring, and learning about the heritage information of the museum.
4. **Attitude Toward Using** – assessed the respondents' favorable or unfavorable perception toward using the developed platform.
5. **Behavioral Intention to Use** – assessed the respondents' willingness or intention to use the platform again or use it for relevant museum and heritage-learning activities.
6. **Actual System Interaction** – assessed the respondents' reported experience while directly interacting with the system during the evaluation session rather than long-term system usage.

7. **Immersive Digital Experience** – assessed the respondents' experience with the platform's interactive, 360-degree, three-dimensional, AR, WebXR, and related immersive components.
8. **Accessibility** – assessed the respondents' perception of the platform's accessibility-oriented mechanisms, such as bilingual content, readable presentation, text-size controls, high-contrast presentation, narration, captions or subtitles, and other applicable access features.

The responses were analyzed using the statistical procedures specified in the study. The weighted mean was used to determine the average level of agreement for each criterion, while the standard deviation was used to describe the variation or consistency of the respondents' ratings.

The resulting statistical values and their corresponding verbal interpretations served as the basis for determining the intended users' assessment and acceptance of the developed platform.

3.14 Statistical Treatment

The quantitative data gathered from the system testing and user evaluation were organized, summarized, and analyzed using appropriate descriptive statistical techniques. The statistical procedures were selected according to the research objectives, type of data collected, and evaluation instruments used in the study.

For the Black Box Testing results, frequency and percentage were used to summarize the number of test cases classified as Pass or Fail. For the user-evaluation

questionnaire, frequency, percentage, weighted mean, and standard deviation were used to summarize and interpret the respondents' ratings.

The statistical treatment focused on descriptive analysis because the study aimed to determine the functional performance of the developed platform and describe the respondents' assessment and acceptance of the system. Inferential statistical procedures such as correlation, regression, and analysis of variance were not applied because the research objectives did not seek to determine relationships among variables, predict outcomes, or test significant differences among respondent groups.

3.14.1 Descriptive Statistics

Descriptive statistics were used to organize, summarize, and interpret the quantitative data gathered from the study. These statistical measures were appropriate because the study aimed to describe the results of the Black Box Functional Testing and the responses obtained from the system user evaluation rather than to test causal relationships or statistically compare respondent groups.

For the Black Box Functional Testing, frequency and percentage were used to summarize the number of test cases that obtained Pass or Fail results. For the system user evaluation, frequency and percentage were used to present the respondents’ profile, while weighted mean was used to determine the average rating for each questionnaire item and evaluation criterion. Standard deviation was also computed to describe the degree of variation or consistency among the respondents’ ratings. The use of descriptive statistics allowed the researchers to present the findings in a clear and understandable manner and to

determine how the respondents assessed the developed WebXR-Enabled Digital Heritage Platform based on the established evaluation criteria. The results were interpreted according to the five-point Likert scale used in the questionnaire.

Frequency and Percentage

Frequency was used to determine the number of responses falling under a particular category. Percentage was used to express the proportion of responses relative to the total number of observations or respondents.

For Black Box Testing, these measures were used to summarize the number and proportion of test cases that received Pass or Fail results. They may also be used to summarize appropriate respondent-profile information when applicable.

Percentage was computed using the formula:

$$PP = \frac{ff}{NN} \times 100$$

Equation 3.4. Percentage

Where:

- PP = percentage
- ff = frequency
- NN = total number of observations or respondents

Weighted Mean

The weighted mean was used to determine the average rating of the respondents for each questionnaire statement and evaluation criterion. It served as the primary measure for describing the respondents' level of agreement with the statements regarding the developed platform.

The user evaluation covered the following criteria:

1. Selected Software Quality Characteristics
2. Perceived Ease of Use;
3. Perceived Usefulness;
4. Attitude Toward Using
5. Behavioral Intention to Use
6. Actual System Interaction
7. Immersive Digital Experience
8. Accessibility.

The weighted mean was computed using the formula:

$$\overline{x}_w = \frac{\Sigma f x}{N}$$

Equation 3.5. Weighted Mean

Where:

- $\overline{x}_w$ = weighted mean
- f = frequency of responses

- x= assigned Likert-scale weight
- N = total number of responses

The resulting weighted mean was interpreted using the verbal interpretation corresponding to the five-point Likert scale adopted in the study.

Standard Deviation

Standard deviation was used to determine the degree of variation or consistency among the respondents' ratings. A lower standard deviation indicates that the respondents' ratings were more closely grouped around the mean, while a higher standard deviation indicates greater variation in their responses.

The sample standard deviation was computed using the formula:

$$s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{N - 1}}$$

Equation 3.6. Standard Deviation

Where:

- s = sample standard deviation
- x = individual response
- $\bar{x}$ = mean of the responses
- N = number of respondents

Standard deviation was interpreted together with the weighted mean to provide additional information regarding the consistency of the respondents' assessments.

Five-Point Likert Scale

A five-point Likert scale was used in the user-evaluation questionnaire to measure the respondents' level of agreement with each statement. The computed weighted means were interpreted according to the following ranges:

Table 3.8 Five-Point Likert Scale and Verbal Interpretation

Scale	Mean Range	Verbal Interpretation
5	4.21-5.00	Strongly Agree
4	3.41-4.20	Agree
3	2.61-3.40	Neutral
2	1.81-2.60	Disagree
1	1.00-1.80	Strongly Disagree

The table 3.8 above shows verbal interpretations were used to describe the respondents' assessment of each questionnaire statement and the overall evaluation criteria. Higher mean scores indicated more favorable assessments of the developed platform, while lower mean scores indicated less favorable assessments.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents and discusses the results of the design, development, system testing, and user-acceptance evaluation of the WebXR-Enabled Digital Heritage Platform. The discussion follows the research objectives and explains how the implemented features, technical findings, and respondent evaluations addressed the purpose of the study.

4.1 General Presentation of Results

This chapter presents the results obtained from the design, development, implementation, testing, and user evaluation of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The findings are organized according to the sequence of the specific objectives stated in Chapter I so that each objective can be traced to the corresponding activity, evidence, interpretation, and conclusion.

The first objective is represented by the completed system architecture, database structure, interface design, technical configuration, and operational platform. The second objective is represented by the implemented functional modules under the five major functional areas. The third objective is addressed through the Black Box Functional Testing results, while the fourth objective is addressed through the descriptive system user-evaluation findings.

4.1.1 Presentation of Objective-Based Results

The results of the study were presented in accordance with the established research objectives to ensure a systematic and coherent discussion of the development, testing, and evaluation of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. This objective-based approach established a direct relationship between the intended outcomes of the study, the activities undertaken by the researchers, the corresponding results, and the evidence supporting the accomplishment of each objective.

Each research objective was discussed individually, beginning with the design and development of the platform and its major functionalities, followed by the integration of interactive and immersive technologies, accessibility features, and visitor and administrative services. The presentation included relevant system screenshots, figures, tables, and technical descriptions to demonstrate the implementation of the identified requirements and illustrate the operational capabilities of the developed platform. These supporting materials provided a documented basis for examining the relationship between the proposed system design and its actual implementation.

The results of system testing and user evaluation were subsequently presented to provide empirical evidence regarding the functionality, software quality, and user acceptance of the platform. Black Box Testing results were used to examine whether the tested functionalities produced their expected outputs, while the findings from the user evaluation were analyzed using the

established assessment criteria and statistical procedures. The corresponding tables and interpretations were presented to facilitate a clear and objective discussion of the findings.

Table 4.1 Objective-Based Presentation of Results

Specific Objective	Method/Activity	Result/Evidence Obtained	Evidence of Achievement
1. To design and develop an integrated WebXR-enabled digital heritage platform	Requirements analysis, system planning, interface design, database design, architecture design, and system development	Completed WebXR-enabled digital heritage platform with public and administrative components	System architecture, database structure, user-interface design, technical configuration, and operational platform
2. To implement the identified functional modules of the developed platform	Incremental development, integration, content preparation, and implementation of system functions	Five major functional areas implemented in the completed system	Digital Heritage and Educational Content; Interactive and Immersive Experiences; Inclusive Accessibility Support; Visitor Engagement and Museum Services; and Administrative Management and Monitoring
3. To test whether the major public and administrative features operate according to their specified functional requirements	Black Box Functional Testing conducted using structured test cases	54 test cases across 13 system modules were tested and obtained final Pass results	54 Passed, 0 Failed, 100.00% final passing rate
4. To evaluate the users' acceptance and experience of the developed platform	User evaluation using a five-point Likert-scale questionnaire	190 completed evaluations covering eight evaluation criteria	Weighted mean, standard deviation, percentage, verbal interpretation, and overall user-evaluation results

Table 4.1 links each objective with its supporting evidence. These are complementary development and evaluation outputs, so a highest-to-lowest ranking does not apply. The separation of evidence is supported by ISO/IEC 25010:2023, which provides a product-quality evaluation framework; the International Software Testing Qualifications Board (ISTQB, 2024), which explains requirements-based testing; and Li and Lv (2024), who evaluate acceptance in a virtual museum. In this study, completing the platform, passing a functional case, and receiving a favorable user rating establish different aspects of the result.

4.2 System Architecture and Development Results

This section presents the development results corresponding to the first and second specific objectives of the study. These objectives focused on designing and developing an integrated WebXR-enabled digital heritage platform and implementing its identified functional modules. The results are presented through the completed system, final system architecture, operational requirements, database structure, user-interface design, system modules, and requirements-to-results assessment.

The completed platform integrated conventional web content, digital museum resources, visitor services, administrative content management, and interactive and immersive technologies into a single web-based environment. Its development resulted in a functional public interface, registered-user services, secured administrative interface, database-backed content-management system, interactive site map, 360-degree virtual tour, augmented-reality content, three-dimensional exhibits, WebXR environments, and visitor-engagement features.

4.2.1 System Overview

The completed WebXR-Enabled Digital Heritage Platform was developed to support the presentation, interpretation, management, and promotion of the cultural and historical resources of the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The system provided onsite and remote visitors with a centralized digital environment for accessing museum information, curated heritage content, educational materials, virtual-tour experiences, interactive maps, augmented-reality content, and other museum-related services.

The target users of the platform included general visitors, students, teachers, researchers, cultural-heritage enthusiasts, registered users, and authorized museum personnel. Public visitors could access the major informational and immersive features without creating an account. Registered users could use additional functions such as bookmarks, account settings, participation records, guided-tour bookings, and Heritage Keeper progress. Authorized administrators accessed a separate secured interface for managing users, content, submissions, feedback, bookings, virtual tours, site-map information, notifications, and analytical records.

The platform's public functions were organized into five major functional areas: Digital Heritage and Educational Content, Interactive and Immersive Experiences, Inclusive Accessibility Support, Visitor Engagement and Museum Services, and Administrative Management and Monitoring.

These functional areas were implemented as interconnected modules rather than as separate applications.

During normal operation, the user accessed the platform through a web browser. The browser requested the selected page or service from the Flask application. The application processed the request, retrieved the required information from the database or multimedia storage, and returned the corresponding interface. Interactive and immersive experiences were rendered in the browser using JavaScript, WebGL, panoramic-image technology, augmented-reality tools, and WebXR-compatible components.

The completed system used English and Filipino content where applicable. Accessibility-oriented controls included adjustable text presentation, high-contrast display, narration, captions or subtitles, and responsive interfaces. When an advanced immersive function was unavailable because of device or browser limitations, the platform provided conventional navigation or three-dimensional viewing alternatives where applicable.

Figure 4.1 Completed WebXR-Enabled Digital Heritage Platform.

Figure 4.1 presents the main public interface of the completed platform. The homepage serves as the primary entry point to the digital heritage experience and provides access to the major public modules of the system. Its implementation demonstrates that the identified design requirements were translated into an operational web-based platform.

4.2.2 System Architecture

The final system used a layered web-based client–server architecture. The architecture consisted of the client and presentation layer, application and service layer, data and content layer, and external-service layer. This separation allowed the interfaces, application rules, stored information, multimedia resources, and external integrations to perform distinct functions while remaining connected through controlled communication mechanisms.

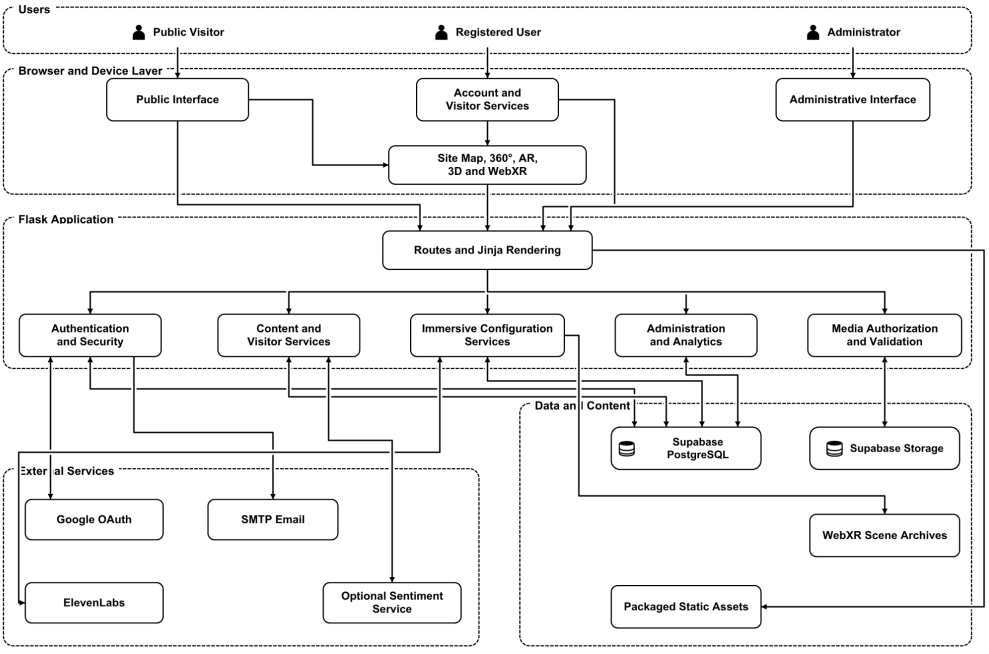

Figure 4.2 Final System Architecture

This figure 4.2 presents the implemented relationship among the platform's users, browser-based interfaces, Flask application services,

Supabase database and storage resources, immersive components, and external services.

4.2.3 Hardware and Software Requirements

The developed WebXR-Enabled Digital Heritage Platform was designed to operate through a cloud-based infrastructure, eliminating the need for a dedicated on-premises application or database server at the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. Its server-side application, database, and multimedia storage components utilized managed cloud services, allowing the platform to be accessed through compatible web browsers. The hardware requirements primarily concerned the devices used by visitors and administrators, including smartphones, tablets, laptops, and desktop computers.

The software requirements included compatible web browsers, a stable internet connection, and the technologies supporting the platform's server-side operations. The Flask application framework, together with database and storage services, facilitated request processing, information management, and multimedia delivery. The minimum and recommended specifications presented in the succeeding tables serve as practical operational baselines for accessing and administering the platform, considering the varying resource demands of its features.

Table 4.2 Minimum and Recommended Hardware Requirements

Resource	Minimum requirement	Recommended requirement
Visitor desktop or laptop	Modern dual-core 64-bit processor, 4 GB RAM, WebGL-capable integrated graphics, 1366 × 768 display, speakers or headphones	Modern quad-core processor, 8 GB RAM or higher, current integrated or dedicated graphics, and 1920 × 1080 display
Visitor smartphone or tablet	Current mobile device with at least 4 GB RAM, touchscreen, modern browser, and stable internet connection	Current device with at least 6 GB RAM, high-resolution display, updated browser, and hardware-accelerated graphics
AR-capable device	Rear-facing camera, camera permission, and supported mobile browser	Rear-facing camera, gyroscope, accelerometer, updated operating system, and browser-supported AR capability
WebXR device	Browser and hardware that support the selected WebXR mode	Compatible immersive headset, motion controllers, and WebXR-enabled browser
Administrator workstation	Quad-core processor, 8 GB RAM, modern browser, keyboard, mouse, and stable internet connection	Quad-core or higher processor, 16 GB RAM, 1920 × 1080 or higher display, and broadband connection
Audio equipment	Built-in speakers or headphones	Headphones or external speakers for clearer narration playback
Content capture equipment	Conventional digital camera and microphone where new content is required	360-degree camera, high-resolution camera, tripod, external microphone, and audio-editing equipment
Network connection	Stable broadband connection capable of loading images, panoramas, audio, and ordinary 3D content	High-speed broadband connection for faster loading of videos, large 3D models, and WebXR environments
Server infrastructure	No local server required; access to configured cloud services	Maintained Vercel and Supabase projects with monitoring, backups, and production environment variables

Table 4.2 distinguishes basic access from configurations intended for more demanding media and immersive functionalities. The minimum requirements establish a practical baseline for accessing the platform's

essential functions, including viewing museum information, navigating digital exhibits, submitting forms, and performing administrative tasks. In contrast, the recommended specifications identify hardware configurations intended to provide a more responsive experience when accessing panoramic images, multimedia content, interactive three-dimensional models, augmented reality, and WebXR environments. The table also considers the different operational requirements of visitors and administrators by identifying suitable specifications for desktop computers, smartphones, tablets, immersive devices, audio equipment, content capture equipment, and network connectivity.

Fanini et al. (2021) discuss browser-based immersive heritage applications, MDN Web Docs (2026) documents device-dependent WebXR support, and Ramtohul and Khedo (2024) review AR implementation constraints. Together, these sources contextualize the need to retain ordinary browser access alongside optional immersive modes; they do not validate the specific RAM or processor thresholds listed here. Consequently, the minimum and recommended specifications serve as practical operational guidelines rather than experimentally established hardware requirements. No comparative hardware benchmarking was conducted to determine performance differences among the proposed configurations. Actual system performance may vary depending on device capabilities, browser compatibility, network stability, and the complexity of the digital resources being accessed. These considerations provide a reference for selecting suitable equipment, identifying potential operational limitations, and planning future system deployment and maintenance.

Table 4.3 Operational Software Requirements

Category	Minimum requirement	Recommended requirement
Visitor browser	Current browser with JavaScript, cookies, HTML5, CSS, and WebGL enabled	Latest stable Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari
Immersive browser support	Browser support for the required camera, orientation, WebGL, or WebXR function	Updated WebXR-compatible browser and device
Server application	Python 3 and Flask 2.3 or later	Maintained Python 3 environment with current compatible package versions
Python packages	Flask, PyYAML, python-dotenv, Supabase client, Pillow, and their dependencies	Versions verified against the deployed application
Front end technologies	HTML5, CSS3, JavaScript, Jinja templates, and WebGL	Updated browser with hardware acceleration enabled
Interface libraries	Tailwind CSS, Lucide, GSAP, Leaflet, and Pannellum	Deployed and version-controlled compatible builds
Immersive libraries	Three.js, AR.js, Google <model-viewer>, SparkJS, and three-mesh-bvh	Versions packaged or referenced by the deployed application
Database	Supabase-hosted PostgreSQL	Production Supabase project with database policies and monitored capacity
Multimedia storage	Supabase Storage and packaged application assets	Configured storage buckets, access rules, and sufficient storage allocation
Deployment platform	Vercel-compatible Flask deployment	Production Vercel project with protected environment variables
External services	SMTP-compatible email service; other integrations where enabled	Configured Google OAuth, email delivery, ElevenLabs, and applicable monitoring
Administration tools	Modern browser and authorized administrator account	Updated browser, secure email access for login OTP, and password manager

Table 4.3 records the implemented software environment rather than comparative software scores. Fanini et al. (2021) provide a related example of web-based heritage delivery; MDN Web Docs (2026) explains the compatibility conditions affecting WebXR; and ISO/IEC 25010:2023 provides a framework for considering product quality. Their relevance is to deployment compatibility and evaluation scope, not proof that any listed package version is sufficient in every environment. The working deployment and its dependencies require continuing maintenance.

4.2.4 Database Structure

The resulting database was implemented using PostgreSQL through Supabase. The database contained 28 application tables organized into account security, visitor services, content management, engagement, analytics, virtual-tour, and site-map groups. This organization provided a structured approach to storing and managing the different types of information required by the developed study. The database supported the integration of public-facing functionalities and administrative operations by maintaining the information necessary for user accounts, museum content, visitor transactions, interactive experiences, and system activity records.

Each table used a primary key to identify its records. Foreign keys established relationships among associated records, while constraints validated required values, statuses, coordinate ranges, content formats, and supported relationships. Indexes were provided for frequently searched values and related records.

Table 4.4 Implemented Database Structure

Database group	Implemented tables	Supported functions	Database group
Administrator security	admin_users, admin_password_reset_requests, admin_password_reset_tokens, admin_login_otp_challenges	Administrator login, OTP, account status, password recovery, and secure password updating	Administrator security
Administrator notifications	admin_notifications	Persistent administrator notifications for bookings, submissions, and feedback	Administrator notifications
Public-user accounts	public_users, pending_public_registrations	Registration, email verification, login, profile information, preferences, and account status	Public-user accounts
User activity and consent	user_bookmarks, public_terms_acceptances, analytics_events	Saved gallery items, acceptance records, and privacy-conscious usage analytics	User activity and consent
Visitor bookings	tour_bookings	Guided-tour requests, administrative review, status tracking, and visitor updates	Visitor bookings
Community content	community_highlights, community_participation_submissions	Published community highlights and visitor or teacher submissions	Community content
Visitor feedback	visitor_feedback, visitor_feedback_moderation_log	Feedback collection, sentiment information, review, publication, and moderation history	Visitor feedback
Online gallery	online_gallery_categories, online_gallery_items	Categorized materials, bilingual descriptions, multimedia, and publication status	Online gallery
Educational resources	edu_resources	Learning materials, documents, media, information	Educational resources
Heritage Keeper	heritage_keeper_events, heritage_keeper_awards	Progress events, completed activities, collected fragments, and awarded achievements	Heritage Keeper
WebXR progress	webxr_player_items, webxr_player_museum_views	Collected WebXR objects and viewed museum content	WebXR progress
Virtual tours	virtual_tours, virtual_tour_revisions, virtual_tour_scenes, virtual_tour_hotspots	Virtual-tour environments, drafts, scenes, navigation, hotspots, coordinates, media, and narration	Virtual tours
Interactive site map	site_maps, site_map_revisions	Aerial and 360-degree map settings, hotspots, coordinates, bilingual content, audio, images, and destination links	Interactive site map

Table 4.4 describes storage responsibilities and does not provide a numerical ranking. Daga et al. (2022) examine the integration of citizen contributions into heritage archives, Nappi et al. (2024) address the organization of accessible heritage knowledge, and Kamariotou et al. (2021) discuss planning for virtual exhibitions. These works help explain the value of connecting content, participation, and administrative workflows. The schema supports that organization, but its existence alone does not establish security, data accuracy, or long-term preservation.

4.2.5 User-Interface Design and System Modules

The completed user interface was divided into public, registered-user, immersive, visitor-service, and administrative interfaces. A responsive design was used so that the interface could adjust to desktop, tablet, and mobile displays. Consistent colors, typography, cards, forms, buttons, dialogs, navigation controls, and feedback messages were used throughout the system.

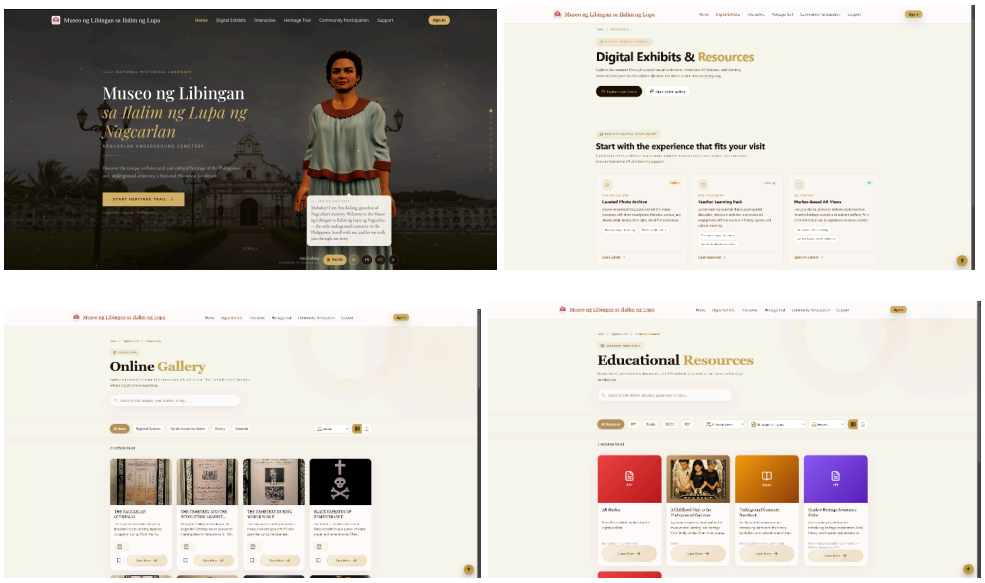

Figure 4.3. Digital Heritage and Educational Content Modules.

Figure 4.3 presents curated historical narratives, artifact information, archival photographs, digital exhibits, online gallery collections, educational resources, and teacher-oriented learning materials. English and Filipino presentations, captions, images, audio, and selected 3D media allow users to access heritage information through several formats and support learning before, during, or after a physical visit.

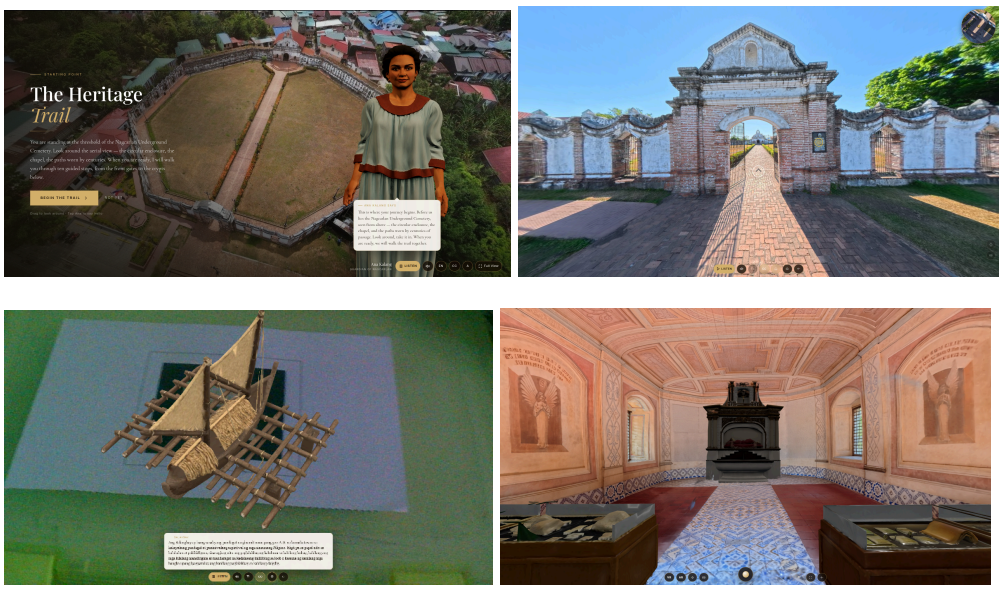

Figure 4.4 Interactive and Immersive Experiences

Figure 4.4 presents the guided ten-stop Heritage Trail, 360-degree Virtual Tour, Interactive Site Map, selected AR content, and WebXR 3D environments, which provide spatial and interactive ways to explore the site. Navigation hotspots, artifact viewers, gyroscope and supported VR modes, short learning activities, and bilingual narration encourage active exploration rather than passive reading alone.

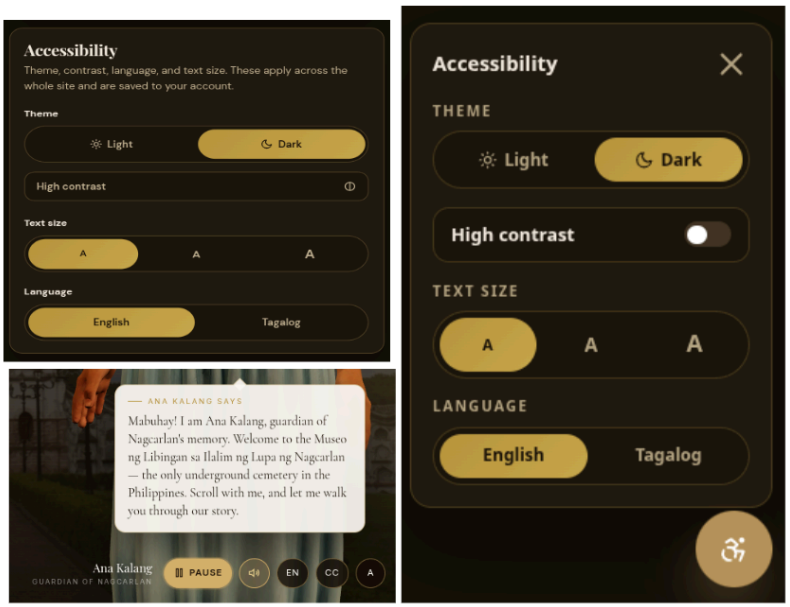

Figure 4.5 Inclusive Accessibility Support

Figure 4.5 presents the accessibility support features incorporated into the developed WebXR-Enabled Digital Heritage Platform. These include English and Filipino content, adjustable text-size controls, high-contrast viewing, narration, subtitles, audio controls, responsive layouts, and reduced-motion options. The features were designed to accommodate users with varying abilities, age groups, language preferences, and learning needs. The bilingual content allows users to access heritage information in their preferred language, while text-size adjustments and high-contrast viewing support readability. Narration, subtitles, and audio controls provide alternative ways of accessing and understanding museum content, allowing users to choose the presentation format that suits their preferences and circumstances.

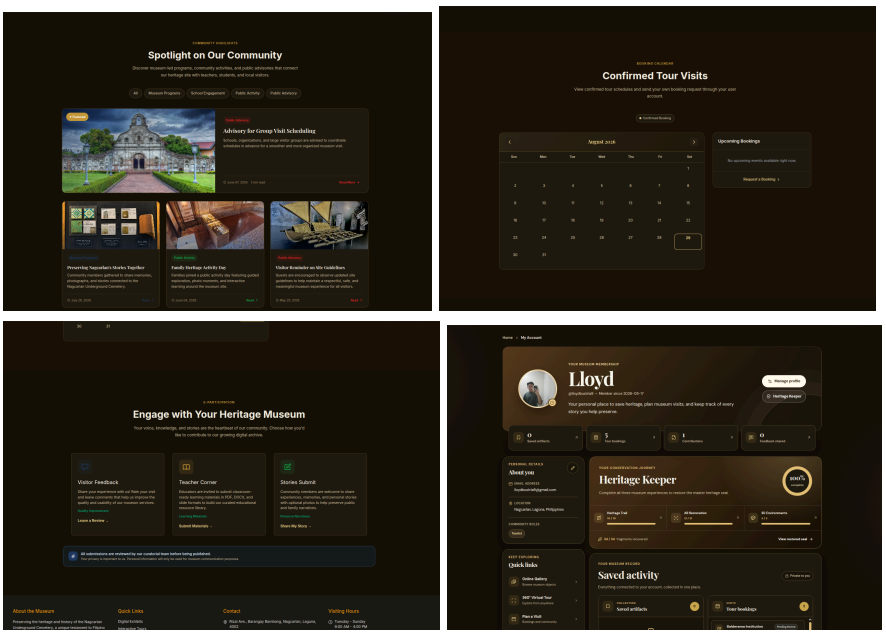

Figure 4.6. Visitor Engagement and Museum Services

Figure 4.6 presents visitor registration, verified accounts, profiles, bookmarks, community stories, learning-material submissions, feedback forms, notifications, and a booking calendar extending the platform beyond information delivery. These services support continued participation and provide structured channels for communication between visitors and museum personnel.

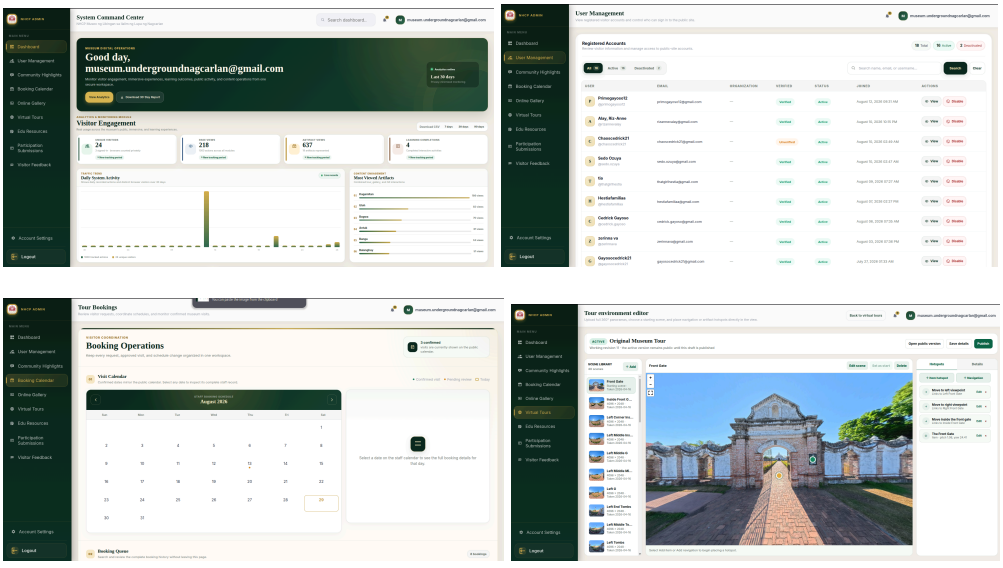

Figure 4.7. Administrative Management and Monitoring

Figure 4.7 presents authorized administrators who can manage users, exhibits, gallery items, educational resources, community highlights, virtual tours, submissions, bookings, feedback, and selected monitoring records. The administrative platform supports ongoing content maintenance and service coordination. Richer visual dashboards and automated sentiment categories remain under development and were therefore not treated as completed system results.

4.2.6 Functional and Non-Functional Requirements

The completed platform was compared with the functional and non-functional requirements. Functional implementation was assessed through Black Box Testing. The testing instrument contained 54 test cases across 13 system modules or feature groups, and all 54 obtained a final Pass result.

Functional Requirements

The functional requirements specify the essential operations and services that the developed WebXR-Enabled Digital Heritage Platform was expected to provide. These requirements served as the basis for determining whether the implemented system addressed the identified needs of museum visitors, administrators, and other intended users. They were organized into five major categories: digital heritage and educational content, interactive and immersive experiences, inclusive accessibility support, visitor engagement and museum services, and administrative management and monitoring.

Table 4.5 Functional Requirements-to-Results Matrix

Requirement	Implemented result	Evidence
Digital Heritage and Educational Content	The completed platform provided historical information, digital exhibits, online-gallery items, educational resources, museum information, interpretive content, and bilingual descriptions.	Completed content interfaces and relevant cases included in the 54-case Black Box Test
Interactive and Immersive Experiences	The platform implemented the heritage trail, interactive aerial and 360-degree site map, 360-degree virtual tour, three-dimensional models, augmented reality, and WebXR environments.	Operational immersive modules, representative screenshots, and relevant functional test cases
Inclusive Accessibility Support	The interface provided English and Filipino content, adjustable text presentation, high-contrast controls, narration, and captions or subtitles where applicable.	Accessibility controls visible in the completed public and immersive interfaces
Visitor Engagement and Museum Services	Users could register, verify an account, log in, manage a profile, save bookmarks, access materials, submit guided-tour requests, provide feedback, contribute community content, and review relevant museum information.	Completed user and visitor-service modules and corresponding functional test cases
Administrative Management and Monitoring	Authorized administrators could manage supported content, users, bookings, submissions, feedback, notifications, virtual tours, site maps, and analytical records.	Secured administrative dashboard, management interfaces, and corresponding functional test cases

The table 4.5 above shows that all five major functional groups identified during requirements analysis were represented in the completed system. The final passing result of the 54 functional test cases provided evidence that the tested public and administrative features produced their expected observable outputs.

Non-Functional Requirements

The non-functional requirements define the quality attributes and operational conditions of the developed platform, including performance efficiency, compatibility, accessibility, security, and reliability.

Table 4.6 Non-Functional Requirements-to-Results Matrix

Requirement	Implemented result	Evidence and limitation	Result
Usability	Consistent navigation, labeled controls, responsive layouts, readable forms, feedback messages, and confirmation dialogs were implemented.	Demonstrated by completed interfaces and assessed further through the user-evaluation criteria	Implemented ; user-evaluation findings should be cross-referenced
Accessibility	Text-size controls, high-contrast presentation, bilingual content, narration, and applicable captions or subtitles were provided.	Demonstrated through interface inspection and relevant observable test cases	Implemented
Performance	Interactive experiences used client-side rendering, browser hardware acceleration, managed hosting, separated multimedia delivery, and controlled media loading.	Selected observable response conditions were included where applicable; no independent load or stress benchmark was claimed	Implemented under normal tested conditions
Compatibility	The interface adapted to desktop, tablet, and mobile displays, while immersive features checked device and browser capabilities and provided applicable fallbacks.	Demonstrated on the devices and browsers used during testing; universal compatibility with every device was not claimed	Implemented for supported environments
Security	Separate public and administrator authentication, password hashing, email verification, login OTP, protected sessions, role checks, CSRF protection, rate limiting, upload validation, security headers, and database policies were implemented.	Functional security behavior was inspected and tested where represented by a test case; no independent penetration test was claimed	Security controls implemented
Reliability	Required-field validation, status controls, error messages, database constraints, revision-controlled publication, and protected content operations were implemented.	Functional tests produced 54 Pass and 0 Fail results; long-term availability testing was outside the reported Black Box Testing scope	Operational in tested scenarios
Maintainability	Database-backed administrative modules allowed routine updates to content, records, virtual tours, and the site map without direct source-code.	Demonstrated by the completed administrative editors and modular application structure	Implemented

Table 4.6 presents the results of the non-functional requirements, covering usability, accessibility, performance, compatibility, security, reliability, and maintainability. The findings indicate that these requirements were implemented or demonstrated under the tested conditions, with usability findings requiring cross-reference to the user-evaluation results.

4.3 System Testing Results

The system testing results present the findings obtained from the Black Box Functional Testing conducted on the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. This section corresponds to the third specific objective of the study, which aimed to determine whether the major public and administrative features of the developed platform operated according to their specified functional requirements.

The Black Box Testing instrument consisted of 54 functional test cases distributed across 13 major system modules or feature groups. The testing instrument required the system testers to directly interact with the working system, perform the specified test procedures, compare the actual observed result with the expected result, and classify each test case as Pass, Fail, N/A, or Retest. A Pass status indicated that the complete expected result was observed.

4.3.1 Test Results by Requirement

The 54 Black Box test cases were organized according to the major public, immersive, visitor-account, service-oriented, and administrative functions of the developed platform. The testing instrument allocated six test

cases to Homepage, Navigation, and Responsive Access; six to User Registration, Login, and Logout; four to Digital Exhibits; two to Heritage Trail; two to Interactive Site Map; three to the 360-Degree Virtual Tour; three to AR Content; three to the 3D/WebXR Environment; three to Community Submission Forms; three to the Online Booking Schedule; three to Visitor Account and Heritage Keeper; fifteen to Administrator Login and Functions; and one to Navigation/Error Handling.

Table 4.7 Test Case Results

System Module / Feature Group	Test Cases	Total Cases	Passed	Failed	Pass Rate
Homepage / Navigation / Responsive Access	TC-01 - TC-06	6	6	0	100.00%
User Registration, Login, and Logout	TC-07 - TC-12	6	6	0	100.00%
Digital Exhibits	TC-13 - TC-16	4	4	0	100.00%
Heritage Trail	TC-17 - TC-18	2	2	0	100.00%
Interactive Site Map	TC-19 - TC-20	2	2	0	100.00%
360-Degree Virtual Tour	TC-21 - TC-23	3	3	0	100.00%
AR Content	TC-24 - TC-26	3	3	0	100.00%
3D/WebXR Environment	TC-27 - TC-29	3	3	0	100.00%
Community Submission Forms	TC-30 - TC-32	3	3	0	100.00%
Online Booking Schedule	TC-33 - TC-35	3	3	0	100.00%
Visitor Account and Heritage Keeper	TC-36 - TC-38	3	3	0	100.00%
Administrator Login and Functions	TC-39 - TC-53	15	15	0	100.00%
Navigation / Error Handling	TC-54	1	1	0	100.00%
Overall	TC-01 - TC-54	54	54	0	100.00%

Table 4.7 shows that all 54 Black Box test cases distributed across the 13 system module or feature groups obtained a final Pass result, resulting in a 100.00% final passing rate. This indicates that, under the specified testing conditions, the tested public, immersive, visitor-service, account, and administrative functions produced the expected outputs documented in the Black Box Testing instrument. The 100.00% result may be attributed to the systematic development and refinement of the platform before the final testing results were recorded. The system was developed incrementally, allowing individual functions and modules to be implemented, reviewed, refined, and integrated before full evaluation. Govindasamy et al. (2023) support a developmental approach involving systematic analysis, design, development, implementation, and evaluation, while Besoain et al. (2021) demonstrate the value of iterative design and user testing in virtual-museum development. These studies support the interpretation that continuous review and refinement during development can help ensure that implemented functions are sufficiently operational before final functional testing.

Another possible reason for the 100.00% final passing rate is the direct alignment of the Black Box test cases with the documented functional requirements of the platform. Each test case specified the required action, expected result, and actual observed result, and a Pass was recorded only when the complete expected behavior was observed. The International Software Testing Qualifications Board (ISTQB, 2024) supports requirements-based testing in which test conditions and expected results are derived from specified requirements. In addition, ISO/IEC 25010:2023 identifies functional

suitability as an important software-quality consideration concerned with whether system functions meet specified needs and intended purposes (International Organization for Standardization, 2023). These sources support the use of structured, requirement-aligned functional testing as a basis for determining whether the implemented system functions operated as intended.

The final passing results also covered functions beyond basic page loading. The 360-Degree Virtual Tour test cases included panorama loading, navigation through supported pointer or touch input, and hotspot interaction, while the AR test cases included access to the AR feature, camera behavior, marker recognition, and display of the corresponding three-dimensional artifact. Administrative functions such as authentication, dashboard access, user management, content and submission management, visitor-feedback handling, account settings, logout, and access control were likewise included. The testing procedure also provided for identified functional issues to be corrected and affected functions to be retested when necessary before proceeding to user evaluation. Therefore, the 100.00% result should be interpreted as a final passing rate for the 54 documented test cases under the specified testing conditions, rather than as evidence that the entire system is completely free from defects. This interpretation is consistent with ISTQB (2024), which cautions that successful testing does not demonstrate the absence of all defects.

4.3.2 Testing Summary

The testing summary presents the consolidated results of Black Box Testing conducted to verify the functionality of the developed

WebXR-Enabled Digital Heritage Platform. The testing instrument consisted of 54 test cases distributed across 13 system modules or feature groups, with all 54 test cases obtaining a final Pass result. These findings indicate that the tested functionalities produced their expected outputs under the specified testing conditions, providing evidence of the system's functional compliance within the scope of the conducted tests.

Table 4.8 Presents the consolidated result of the Black Box Functional Testing

Test Status	Frequency	Percentage
Pass	54	100.00%
Fail	0	0.00%
N/A	0	0.00%
Retest	0	0.00%
Total	**54**	**100.00%**

Table 4.8 shows that Pass had the highest frequency at 54 cases (100.00%). Fail, N/A, and Retest were tied at the lowest frequency of zero (0.00%). These are final recorded statuses and do not establish that no problem occurred during earlier development.

ISTQB (2024) cautions that successful tests cannot demonstrate the absence of all defects. ISO/IEC 25010:2023 provides a wider quality framework, while the World Wide Web Consortium (2024) supplies accessibility criteria that require specific checks. Consequently, the result supports the expected behavior of the documented cases; it is not a substitute for independent performance, security, or accessibility assessment.

4.3.3 Discussion of Testing Results

The results provide functional evidence that the tested public, immersive, visitor-service, account, and administrative features operated according to the expected behavior documented in the final Black Box instrument. The result therefore supports the third specific objective within the coverage of the 54 test cases.

The 100.00% final passing rate should not be interpreted as proof that the system is defect-free under all conditions. The Black Box procedure did not, by itself, establish complete performance benchmarking, penetration-security assurance, full compatibility certification, formal accessibility conformance, or fault-free behavior across every possible browser, device, network, and future deployment condition.

4.4 System User Evaluation Results

The system user evaluation results present the findings obtained from the intended users and relevant stakeholders who evaluated the completed WebXR-Enabled Digital Heritage Platform. This section corresponds to the fourth specific objective of the study, which aimed to evaluate the users' acceptance and experience of the developed platform.

The user-evaluation questionnaire consisted of 26 statements organized under eight evaluation criteria: Selected Software Quality Characteristics, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, Actual System Interaction, Immersive Digital Experience, and Accessibility.

A total of 190 completed evaluations were included in the final analysis. The respondent records extended from RC-001 through RC-190.

4.4.1 Respondent Profile

For purposes of presenting the intended respondent groups, participants aged 60 years and above were classified as senior citizens. The PWD respondent was separately identified through the respondent-group information recorded during data gathering.

Table 4.9 Distribution of Respondents for User Evaluation

Respondent Classification	Frequency	Percentage
Administrative Personnel	2	1.05%
Museum Visitors	161	84.74%
Senior Citizens	26	13.68%
Person with Disability (PWD)	1	0.53%
Total	190	100.00%

Table 4.9 shows that Museum Visitors formed the largest group, with 161 respondents (84.74%), while the PWD classification had the smallest representation, with one respondent (0.53%). The findings therefore primarily reflect visitor evaluations and provide very limited evidence about the range of disability-related access needs.

The International Council of Museums (2022) places inclusion within the museum's purpose. Kruczek et al. (2024) examine access barriers for museum visitors with disabilities, while Pirrone et al. (2023) review cultural accessibility in digital humanities. These sources support interpreting the small PWD representation as a limitation, rather than treating a favorable aggregate

score as evidence of accessibility for all users. Consequently, the overall evaluation results should not be interpreted as a comprehensive assessment of the platform's accessibility across different disability groups. Future evaluations should involve a more diverse representation of persons with disabilities and targeted accessibility testing to provide stronger evidence regarding the platform's inclusivity and usability.

Table 4.10 Distribution of Respondents According to Age Group

Age Group	Frequency	Percentage
Below 18	120	63.16%
18–30	28	14.74%
31–59	15	7.89%
60 and above	27	14.21%
Total	**190**	**100.00%**

Table 4.10 shows that respondents below 18 formed the largest age group, with 120 participants (63.16%), whereas those aged 31–59 formed the smallest, with 15 (7.89%). The evaluation is consequently weighted toward younger participants; the aggregate findings do not establish equivalent experiences across ages.

Puspasari et al. (2023) evaluated a virtual museum with student participants, Pedraja-Rejas et al. (2024) reviewed mobile learning, and Kruczek et al. (2024) examined barriers to museum access. These different audience and access contexts support attention to the composition of the sample.

Table 4.11 Device Used During System Evaluation

Device Used	Frequency	Percentage
Smartphone	146	76.32%
Tablet	0	0.00%
Laptop/Desktop	44	23.16%
Others	1	0.53%
Total	**190**	**100.00%**

The reported device counts in Table 4.11 identify Smartphone as the largest category (146) and Tablet as the smallest (0). However, the category counts sum to 191 while the stated sample is 190, and 146/190 corresponds to 76.84%, not the reported 76.32%. The device records must be reconciled before a final percentage-based interpretation is made.

Fanini et al. (2021) discuss cross-device web heritage delivery, MDN Web Docs (2026) explains device and browser dependencies, and Nikolarakis and Koutsabasis (2024) review mobile AR interaction patterns. These sources make device context relevant to interpreting usability, but cannot resolve the discrepancy in the recorded counts. The absence of tablet evaluations also prevents conclusions about the tablet user experience.

Table 4.12 Internet Access During Evaluation

Internet Condition	Frequency	Percentage
Stable	183	96.32%
Intermittent	6	3.16%
Limited	1	0.53%
Total	**190**	**100.00%**

Table 4.12 shows that Stable access had the highest frequency, with 183 respondents (96.32%), while Limited access had the lowest, with one (0.53%). The evaluation therefore provides much stronger evidence about use with stable connectivity than about constrained access. The small percentage total difference is attributable to rounding.

Fanini et al. (2021) describe web delivery of immersive resources, Ramtohul and Khedo (2024) review technical constraints in heritage AR, and Li and Lv (2024) examine online virtual exhibitions. These sources establish the relevance of the access environment to digital heritage experiences. The present distribution does not measure loading performance or demonstrate dependable operation on slow connections.

4.4.2 Evaluation Results by Criterion

This section presents the system user evaluation results according to the eight criteria established in the study: Selected Software Quality Characteristics, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, Actual System Interaction, Immersive Digital Experience, and Accessibility. The weighted mean and standard deviation were used to summarize the respondents' ratings for each criterion and its corresponding indicators.

The results were interpreted using the five-point Likert scale defined in Chapter III. Rather than treating the numerical values as objective measures of system quality, the findings are presented as the respondents' perceptions and reported experiences during the evaluation of the developed WebXR-Enabled

Digital Heritage Platform. Each criterion is discussed separately to show the areas in which the platform was rated more favorably and the aspects that may still require further refinement.

Selected Software Quality Characteristics

Table 4.13 Selected Software Quality Characteristics

Indicator	Weighted Mean	SD	Interpretation
A1. The platform's major features work as intended.	4.93	0.25	Strongly Agree
A2. The platform's navigation is easy to understand.	4.76	0.47	Strongly Agree
A3. The platform responds within an acceptable amount of time.	4.76	0.46	Strongly Agree
Overall	**4.82**	**0.25**	**Strongly Agree**

Table 4.13 shows that A1, which concerns whether the platform's major features work as intended, obtained the highest weighted mean of 4.93 (SD = 0.25), interpreted as Strongly Agree. One possible reason for this high rating is that respondents were able to directly observe and use the platform's major functions during the evaluation, including its heritage-content, visitor-service, accessibility, and interactive features. The International Organization for Standardization (2023) identifies functional suitability as an

important aspect of software product quality, and the evaluation framework of the present study relates this characteristic to whether the platform's major functions operate as intended. This supports the interpretation that the high A1 rating reflects respondents' favorable perception of the functionality of the implemented features. However, this result represents perceived functional quality and should be interpreted separately from the technical Black Box Testing results.

In contrast, A2, which concerns the understandability of the platform's navigation, and A3, which concerns whether the platform responds within an acceptable amount of time, obtained the tied lowest weighted mean of 4.76, with standard deviations of 0.47 and 0.46, respectively. Although both indicators remained within the Strongly Agree range, their comparatively lower ratings may indicate that navigation clarity and responsiveness were more dependent on the respondents' interaction with different sections, functions, devices, and content than the basic observation that the platform's major features were working. Li and Lv (2024) emphasize navigation as an important consideration in online virtual-exhibition environments, while the International Organization for Standardization (2023) identifies interaction capability and performance efficiency as relevant aspects of software quality. These sources support the interpretation that clear navigation and acceptable response behavior are important to users' overall experience of a digital platform. Therefore, the lower A2 and A3 ratings should not be interpreted as negative evaluations, but rather as areas where navigation clarity and perceived responsiveness may be further refined. The ratings represent

respondents' perceptions and do not replace objective measurements of response time or reliability.

Perceived Ease of Use

Table 4.14 Perceived Ease of Use

Indicator	Weighted Mean	SD	Interpretation
B1. I find the platform easy to use.	4.76	0.44	Strongly Agree
B2. Learning to use the platform is easy for me.	4.72	0.49	Strongly Agree
B3. Using the platform's major features requires little effort from me.	4.70	0.49	Strongly Agree
Overall	**4.73**	**0.28**	**Strongly Agree**

Table 4.14 shows that B1, which concerns the respondents' general perception that the platform is easy to use, obtained the highest weighted mean of 4.76 (SD = 0.44), interpreted as Strongly Agree. One possible reason for this high rating is that the platform provides familiar web-based navigation, organized content, labeled interface elements, and conventional controls that may have made the overall system understandable to respondents during the evaluation. Puspasari et al. (2023) included perceived ease of use as an important component in the evaluation of virtual-museum acceptance, while Li and Lv (2024) emphasized navigation in online virtual-exhibition environments. Nikolarakis and Koutsabasis (2024) also examined interaction design patterns in heritage AR applications. These studies support the interpretation that clear navigation and understandable interaction mechanisms are relevant to users' perceptions of ease of use. The high rating for B1

therefore suggests that respondents generally found the platform manageable and understandable during their interaction with the system.

In contrast, B3, which concerns whether using the platform's major features requires little effort, obtained the lowest weighted mean of 4.70 (SD = 0.49), although it remained within the Strongly Agree range. The comparatively lower rating may indicate that some of the platform's major features required more interaction or familiarity than basic navigation, particularly because the system includes interactive maps, a 360-Degree Virtual Tour, AR/WebXR components, digital exhibits, and other interactive functions. Li and Lv (2024) emphasize the importance of navigation in online virtual exhibitions, while Nikolarakis and Koutsabasis (2024) examine interaction design patterns in heritage AR applications. These studies provide context for considering how the complexity of navigation and interaction may affect the effort required to use digital heritage features. Therefore, the lower B3 rating should not be interpreted as a negative evaluation, but rather as an indication that some major features may benefit from clearer instructions, interaction cues, or further usability refinement. The present results reflect respondents' perceived effort and do not identify which specific feature caused the comparatively lower rating.

Perceived Usefulness

Table 4.15 Perceived Usefulness

Indicator	Weighted Mean	SD	Interpretation
C1. The platform is useful for accessing	4.85	0.36	Strongly Agree

heritage information about the site.			
C2. The platform helps me understand the history and heritage of the site.	4.81	0.39	Strongly Agree
C3. The platform is a useful resource for learning about the heritage site.	4.87	0.34	Strongly Agree
Overall	**4.84**	**0.20**	**Strongly Agree**

Table 4.15 shows that C3, which concerns the platform's usefulness as a resource for learning about the heritage site, obtained the highest weighted mean of 4.87 (SD = 0.34), interpreted as Strongly Agree. One possible reason for this high rating is that the platform brings together heritage information, educational resources, digital exhibits, multimedia content, and interactive experiences within a single digital environment, providing respondents with several ways to access materials related to the heritage site. Li and Lv (2024) examined perceived usefulness as an important aspect of user acceptance in online virtual exhibitions, while Puspasari et al. (2023) applied the Technology Acceptance Model in evaluating a virtual-museum application. These studies support the interpretation that users' recognition of the practical and educational value of a digital museum system is an important component of their evaluation of its usefulness. The high rating for C3 therefore suggests that respondents particularly recognized the developed platform as a useful resource for heritage-related learning.

In contrast, C2, which concerns whether the platform helps respondents understand the history and heritage of the site, obtained the lowest

weighted mean of 4.81 (SD = 0.39), although it remained within the Strongly Agree range. The comparatively lower rating may indicate that providing access to heritage information does not automatically result in the same level of perceived understanding, since historical understanding may also depend on how clearly information is organized, contextualized, and connected across the platform. Nappi et al. (2024) emphasized the organization of cultural knowledge through digital tools and adaptive structures, while the constructivist learning principles discussed in the study indicate that heritage learning can be supported when users actively explore, interpret, and connect information with place. These sources support the continued refinement of interpretive explanations and learning pathways to help users develop a clearer understanding of the site's history and heritage. Therefore, the lower C2 rating should not be interpreted as a negative evaluation, particularly because it remained within the Strongly Agree range, but rather as an area where the platform's interpretive and educational presentation may be further strengthened. The questionnaire results reflect respondents' perceived usefulness and do not establish a measured improvement in historical knowledge.

Attitude Toward Using

Table 4.16 Attitude Toward Using

Indicator	Weighted Mean	SD	Interpretation
D1. I have a positive attitude toward using the platform.	4.82	0.39	Strongly Agree
D2. I think using the platform is a good idea.	4.84	0.37	Strongly Agree

D3. I like the idea of using the platform as a digital heritage resource.	4.83	0.38	Strongly Agree
Overall	**4.83**	**0.23**	**Strongly Agree**

Table 4.16 shows that D2, which concerns whether respondents considered using the platform a good idea, obtained the highest weighted mean of 4.84 (SD = 0.37), interpreted as Strongly Agree. One possible reason for this high rating is that respondents may have recognized the appropriateness and value of using a digital platform to access and engage with heritage-related information and experiences. The statement presents the use of the platform as a specific and practical idea, which may have been easier for respondents to evaluate positively based on their experience during the system evaluation. Puspasari et al. (2023) reported a relationship between attitude toward use and behavioral intention in a virtual-museum application, while Li and Lv (2024) examined user acceptance of online virtual exhibitions. Chen et al. (2024) also examined satisfaction and usage intention in museum AR experiences. These studies provide related evidence that users' favorable evaluations and acceptance are important considerations in the adoption of digital museum technologies. The high rating for D2 therefore indicates that respondents generally regarded the use of the developed platform as a favorable idea for a digital heritage environment.

In contrast, D1, which concerns respondents' overall positive attitude toward using the platform, obtained the lowest weighted mean of 4.82 (SD = 0.39), although it remained within the Strongly Agree range. The comparatively lower rating may reflect the broader nature of this statement,

since expressing an overall positive attitude involves a more general evaluation of the platform than simply agreeing that its use is a good idea. Puspasari et al. (2023) examined attitude toward use as part of virtual-museum acceptance, while Li and Lv (2024) investigated acceptance in online virtual-exhibition environments. Chen et al. (2024) likewise examined users' satisfaction and usage intention in heritage-museum AR experiences. These studies provide context for understanding attitude as one aspect of users' overall response to digital heritage technologies. Therefore, the lower D1 rating should not be interpreted as a negative attitude toward the platform, particularly because it remained within the Strongly Agree range and differed from the highest indicator by only 0.02. The results describe respondents' evaluations and do not establish that usefulness, ease of use, satisfaction, or other factors caused their attitudes.

Behavioral Intention to Use

Table 4.17 Behavioral Intention to Use

Indicator	Weighted Mean	SD	Interpretation
E1. I intend to use the platform again in the future.	4.79	0.40	Strongly Agree
E2. I intend to use the platform when I need information about the heritage site.	4.82	0.38	Strongly Agree
E3. I plan to use the platform for future museum or heritage-learning activities.	4.86	0.35	Strongly Agree
Overall	**4.82**	**0.24**	**Strongly Agree**

Table 4.17 shows that E3, which concerns the respondents' intention to use the platform for future museum or heritage-learning activities, obtained the highest weighted mean of 4.86 (SD = 0.35), interpreted as Strongly Agree. One possible reason for this high rating is that respondents may have perceived the platform as particularly relevant for a specific educational and heritage-related purpose rather than simply as a system to revisit without a defined reason. The platform provides heritage information, educational resources, digital exhibits, and interactive experiences that can support museum-related learning activities. Puspasari et al. (2023) examined behavioral intention in the context of a virtual-museum application, while Li and Lv (2024) investigated users' willingness to use online virtual exhibitions. Chen et al. (2024) also examined usage intention in heritage-museum AR experiences. These studies support the interpretation that users' intention to engage with digital museum technologies can be associated with the perceived relevance and purpose of the experience. The high rating for E3 therefore suggests that respondents particularly recognized the platform's potential for future heritage-learning and museum-related activities.

In contrast, E1, which concerns the general intention to use the platform again in the future, obtained the lowest weighted mean of 4.79 (SD = 0.40), although it remained within the Strongly Agree range. The comparatively lower rating may indicate that respondents were more certain about using the platform when they had a specific heritage-learning or museum-related purpose than about returning to it in general without a defined need. Li and Lv (2024) examined willingness to use online virtual exhibitions, while Puspasari et al. (2023) treated behavioral intention and actual use as

separate constructs in virtual-museum evaluation. These studies support distinguishing expressed willingness to use a digital heritage platform from actual future or repeated use. Therefore, the lower E1 rating should not be interpreted as a negative evaluation, but rather as an indication that respondents' future-use intention was somewhat stronger when connected to a specific museum or heritage-learning purpose. Follow-up usage records would still be required to determine whether respondents actually return to and use the platform over time.

Actual System Interaction

Table 4.18 Actual System Interaction

Indicator	Weighted Mean	SD	Interpretation
F1. I used more than one section of the platform during the evaluation.	4.68	0.53	Strongly Agree
F2. I interacted with the platform's digital heritage content during the evaluation.	4.70	0.50	Strongly Agree
F3. I used the platform to access heritage information during the evaluation.	4.63	0.52	Strongly Agree
Overall	**4.67**	**0.31**	**Strongly Agree**

Table 4.18 shows that F2, which concerns interaction with the platform's digital heritage content during the evaluation, obtained the highest weighted mean of 4.70 (SD = 0.50), interpreted as Strongly Agree. One possible reason for this high rating is that digital heritage content is a central and readily accessible part of the platform, allowing respondents to interact directly with exhibits, historical information, multimedia materials, and other

heritage-related resources during the evaluation session. Nappi et al. (2024) emphasized the importance of organizing cultural knowledge through digital tools and adaptive structures, while Nikolarakis and Koutsabasis (2024) examined interaction patterns in digital heritage applications. These studies support the interpretation that well-organized content and appropriate interaction mechanisms can encourage users to engage with digital heritage resources. The favorable rating therefore suggests that respondents generally interacted with the platform's heritage content during the evaluation.

In contrast, F3, which concerns using the platform specifically to access heritage information during the evaluation, obtained the lowest weighted mean of 4.63 (SD = 0.52), although it remained within the Strongly Agree range. The comparatively lower rating may indicate that purposeful information access required respondents to locate and navigate toward specific heritage content rather than simply interact with the available digital materials. Nappi et al. (2024) noted that digitized cultural materials become more useful when they are organized in ways that support retrieval and interpretation, while Nikolarakis and Koutsabasis (2024) emphasized the role of interaction design in guiding user engagement. These studies suggest that clearer information pathways, navigation cues, and connections between interactive elements and relevant heritage content may further support purposeful information access. Puspasari et al. (2023) also distinguished actual use from behavioral intention in virtual-museum evaluation, supporting the treatment of reported interaction as a separate aspect of system use. Therefore, the lower F3 rating should not be interpreted as a negative result, but rather as an area where access to specific heritage information may be further improved. The

findings are based on respondents' reported interaction during the evaluation session and do not establish session duration, completed tasks, or sustained system use.

Immersive Digital Experience

Table 4.19 Immersive Digital Experience

Indicator	Weighted Mean	SD	Interpretation
G1. The digital exhibits help me explore the heritage site.	4.69	0.49	Strongly Agree
G2. The 360° virtual tour helps me visualize the heritage site.	4.75	0.46	Strongly Agree
G3. The AR/WebXR features are engaging to use.	4.78	0.44	Strongly Agree
G4. The interactive features make the heritage experience engaging.	4.69	0.48	Strongly Agree
Overall	**4.73**	**0.27**	**Strongly Agree**

Table 4.19 shows that G3, which concerns engagement with the platform's AR/WebXR features, obtained the highest weighted mean of 4.78 (SD = 0.44), interpreted as Strongly Agree. One possible reason for this high rating is that the AR/WebXR features provide users with a more immersive and interactive way of experiencing heritage content by allowing them to engage with digital elements beyond conventional text and image presentation. Fanini et al. (2021) demonstrated that browser-based XR can support cultural heritage storytelling and interactive access, while Nikolarakis and Koutsabasis

(2024) emphasized the importance of interaction patterns in heritage AR applications. Li and Lv (2024) also examined immersion and user acceptance in online virtual-exhibition environments. These studies support the interpretation that immersive and interactive technologies can contribute to favorable user engagement when they are meaningfully integrated into digital heritage experiences.

In contrast, G1, which concerns the use of digital exhibits for exploring the heritage site, and G4, which concerns engagement through interactive features, obtained the tied lowest weighted mean of 4.69, with standard deviations of 0.49 and 0.48, respectively. Although both indicators remained within the Strongly Agree range, their comparatively lower ratings may suggest that the broader digital exhibits and interactive pathways could be further refined to make exploration and engagement more direct and meaningful. The effectiveness of these features may depend on how clearly content, navigation, hotspots, and other interactive elements guide users toward relevant heritage information. Nikolarakis and Koutsabasis (2024) emphasize the importance of interaction design in heritage applications, while related virtual-tour literature in the study indicates that interactive markers and navigation become more meaningful when they are connected to contextual interpretation and guided content. Therefore, the lower ratings of G1 and G4 should not be interpreted as negative evaluations, but rather as areas where the presentation and interaction pathways may be further improved. The results represent respondents' perceived engagement and do not establish experimentally measured presence or learning gains.

Accessibility

Table 4.20 Accessibility

Indicator	Weighted Mean	SD	Interpretation
H1. The platform's text is easy for me to read.	4.73	0.48	Strongly Agree
H2. The platform's controls are easy for me to use.	4.63	0.52	Strongly Agree
H3. The accessibility features are useful to me.	4.78	0.46	Strongly Agree
H4. The accessibility options help me navigate the platform.	4.73	0.48	Strongly Agree
Overall	**4.72**	**0.26**	**Strongly Agree**

Table 4.20 shows that H3, which concerns the usefulness of the platform's accessibility features, obtained the highest weighted mean of 4.78 (SD = 0.46), interpreted as Strongly Agree. One possible reason for this high rating is that the platform provides several accessibility-oriented features, including adjustable text size, high-contrast presentation, bilingual English and Filipino content, narration, captions or subtitles, responsive layouts, and alternative ways of accessing digital heritage content. These features provide users with different options for viewing, reading, listening to, and navigating heritage information according to their preferences and access needs. The World Wide Web Consortium (2024) emphasizes perceivability, operability, understandability, and robustness as important principles of web accessibility, while Pirrone et al. (2023) highlight the importance of addressing access barriers to support inclusive participation in digital cultural environments. These sources support the interpretation that providing multiple accessibility

options may have contributed to the respondents' favorable assessment of the usefulness of the platform's accessibility features.

In contrast, H2, which concerns the ease of using the platform's controls, obtained the lowest weighted mean of 4.63 (SD = 0.52), although it remained within the Strongly Agree range. The comparatively lower rating may indicate that while respondents generally valued the available accessibility features, some controls may have required greater familiarity or effort during interaction. This may be particularly relevant because the platform combines conventional web controls with interactive, multimedia, and immersive components that can differ in their methods of operation. The World Wide Web Consortium (2024) identifies operability as an important accessibility requirement, while Kruczek et al. (2024) and Pirrone et al. (2023) discuss barriers that may affect the accessibility of museum and cultural experiences for users with different needs. These studies support continued refinement and direct user testing of interface controls to improve ease of operation. However, because only one respondent in the study was classified as a person with disability, the favorable aggregate results should not be interpreted as evidence of accessibility for all disability groups or as confirmation of formal WCAG conformance.

4.4.3 Statistical Results

The statistical results present the analysis of the respondents' evaluations of the developed system using frequency, percentage, weighted mean, and standard deviation.

Table 4.21 Summary of System User Evaluation Results

Evaluation Criterion	Weighted Mean	Standard Deviation	Interpretation
Selected Software Quality Characteristics	4.82	0.25	Strongly Agree
Perceived Ease of Use	4.73	0.28	Strongly Agree
Perceived Usefulness	4.84	0.20	Strongly Agree
Attitude Toward Using	4.83	0.23	Strongly Agree
Behavioral Intention to Use	4.82	0.24	Strongly Agree
Actual System Interaction	4.67	0.31	Strongly Agree
Immersive Digital Experience	4.73	0.27	Strongly Agree
Accessibility	4.72	0.26	Strongly Agree
Overall	**4.77**	**0.12**	**Strongly Agree**

Table 4.21 shows that Perceived Usefulness had the highest criterion mean (4.84, SD = 0.20), while Actual System Interaction had the lowest (4.67, SD = 0.31). Respondents therefore rated the platform's value more favorably than their reported use during the session. Both remain within the Strongly Agree range. The overall mean of 4.77 and SD of 0.12 summarize a favorable evaluation; the descriptive ordering should not be interpreted as evidence of a statistically significant difference.

4.4.4 Discussion of User Evaluation

The System User Evaluation results demonstrate a consistently favorable assessment of the WebXR-Enabled Digital Heritage Platform. All eight evaluation criteria obtained weighted means within the Strongly Agree range, while the overall evaluation resulted in a weighted mean of 4.77.

Among the eight evaluation criteria, Perceived Usefulness obtained the highest weighted mean of 4.84 (SD = 0.20), interpreted as Strongly Agree. This suggests that respondents most strongly recognized the value of the platform as a resource for accessing and learning about the heritage of the site. One possible reason for this high rating is that the platform integrates heritage information, educational resources, digital exhibits, and interactive content within a single accessible environment, allowing users to obtain heritage-related information through different forms of digital content. This interpretation is further supported by the Perceived Usefulness indicators, particularly C3, which described the platform as a useful resource for learning about the heritage site and obtained the highest item mean of 4.87. Puspasari et al. (2023) assessed a virtual museum using the Technology Acceptance Model and considered perceived usefulness in evaluating user acceptance, while Li and Lv (2024) likewise examined perceived usefulness in the acceptance of online virtual exhibitions. These studies support the interpretation that the perceived educational and informational value of a digital museum platform can contribute to favorable user assessments. However, the present results represent respondents' perceptions and do not establish measured improvements in heritage knowledge.

In contrast, Actual System Interaction obtained the lowest weighted mean of 4.67 (SD = 0.31), although it remained within the Strongly Agree range. The comparatively lower rating may be explained by the nature of this criterion, which measured respondents' reported interaction with the platform during the evaluation session rather than their long-term or repeated use of the system. Among its indicators, F3, which concerned using the platform to access heritage information during the evaluation, obtained the lowest mean of 4.63. This suggests that although respondents strongly recognized the platform's usefulness, some may not have interacted with all available sections, content, or functions to the same extent during the evaluation period. Puspasari et al. (2023) distinguished actual use from other technology-acceptance constructs in evaluating a virtual-museum application, while Chen et al. (2024) examined factors associated with the adoption and use of augmented reality in heritage museums, including usefulness and satisfaction. These studies support treating perceived value and actual interaction as related but distinct aspects of digital museum evaluation. Therefore, the comparatively lower Actual System Interaction score should not be interpreted as a negative assessment, but rather as an indication that respondents' recognition of the platform's value was somewhat stronger than the extent of their reported interaction during the evaluation session.

The Selected Software Quality Characteristics criterion also received a favorable evaluation, with an overall weighted mean of 4.82, interpreted as Strongly Agree. Respondents gave the highest rating to the statement that the platform's major features worked as intended, which obtained a weighted mean of 4.93. This result indicates that respondents generally perceived the

major functions of the platform as operational during their interaction with the system. The International Organization for Standardization (2023) provides a framework for examining software product quality, while Puspasari et al. (2023) examined users' acceptance of a virtual-museum application. These studies provide relevant context for evaluating both the observable qualities of the developed platform and users' perceptions of their interaction with it. However, the ratings in the present study represent perceived software quality and should not be interpreted as substitutes for specialized technical performance, security, or reliability testing.

The favorable Perceived Ease of Use result indicates that respondents generally found the platform understandable and manageable during the evaluation. Similarly, the positive result for Attitude Toward Using indicates that respondents generally viewed the use of the platform as a digital heritage resource favorably. These findings suggest that the platform was not only perceived as useful but was also generally considered manageable and acceptable to use during the evaluation.

The results for Behavioral Intention to Use indicate that respondents expressed willingness to use the platform again and to access it when seeking heritage-related information. This suggests that respondents viewed the platform as potentially useful for future museum and heritage-learning activities. Nevertheless, these results represent reported future intention rather than evidence of actual long-term adoption or repeated system use.

The Immersive Digital Experience criterion also received a favorable evaluation. Engagement with the AR/WebXR features obtained the highest

rating within this criterion, suggesting that respondents particularly appreciated the platform's immersive and interactive features. In contrast, exploration through digital exhibits and engagement through the broader interactive features received the tied lowest ratings, although both remained within the Strongly Agree range. This pattern suggests that AR/WebXR was a relative strength of the platform, while the presentation of digital exhibits and the broader interaction pathways may still be refined to support more effective exploration and engagement.

The Accessibility findings also indicate a favorable assessment of the platform's accessibility-oriented mechanisms. The usefulness of the accessibility features received the highest rating within this criterion, while ease of using the platform's controls received the comparatively lowest rating. This pattern suggests that respondents generally valued the accessibility options provided by the system, although the operability and ease of use of some controls remain areas for further refinement. These findings should be interpreted as respondents' perceptions of accessibility rather than as evidence of formal accessibility conformance.

Taken collectively, the results provide evidence for the achievement of the fourth specific objective, which aimed to evaluate the users' acceptance and experience of the developed platform in terms of the established evaluation criteria.

4.5 Discussion and Interpretation of Findings

The findings show that the study produced an operational, site-specific digital heritage platform and generated separate technical and user-evaluation evidence for the objectives. The design and implementation results address the integration gap identified in Chapter II by combining heritage content, interactive and immersive experiences, accessibility-oriented presentation, visitor services, participation mechanisms, and administrative management within one platform.

The platform's development approach is consistent with the literature emphasizing that virtual-museum projects require coordination among curated content, technical implementation, and user experience rather than simple digitization of materials. The use of 360-degree scenes, selected AR/WebXR experiences, maps, and guided content also reflects literature that treats immersive media as a complement to heritage interpretation rather than a replacement for conventional access paths.

The Black Box result supports the functional objective because the 54 documented test cases produced the expected results within the recorded conditions. This technical evidence should be interpreted separately from the user evaluation: a functional Pass result demonstrates expected behavior for a case, while a favorable user rating describes perception or reported interaction.

The user evaluation showed favorable perceived usefulness, ease of use, attitude, immersive experience, and accessibility. The acceptance-related findings are consistent with the role of Perceived Ease of Use and Perceived Usefulness in the Technology Acceptance Model, but the present study uses those constructs descriptively and does not claim causal relationships. Likewise, the favorable

immersive ratings are consistent with literature describing the potential of virtual tours, AR, and WebXR for heritage exploration. The comparatively high rating for AR/WebXR engagement suggests that these features were a relative strength of the platform, while the lower ratings for digital-exhibit exploration and broader interactive features indicate areas where content presentation and interaction pathways may be further refined.

Taken together, the results indicate that the developed platform responded to the identified contextual and integration gap within the scope of the implemented system, documented Black Box cases, and descriptive user evaluation. The findings do not establish universal accessibility, long-term adoption, experimentally measured learning improvement, or complete technical reliability beyond the procedures actually performed.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of the major findings, conclusions derived from the results, and recommendations based on the design, development, implementation, testing, and user evaluation of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan.

5.1 Summary

The major purpose of this study was to design, develop, implement, test and evaluate the WebXR-Enabled Digital Heritage Platform to facilitate accessible, interactive and immersive heritage learning for onsite and offsite users of the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The results of the investigation are given following the stated objectives.

For the first specific purpose, the researchers planned and created an integrated web based digital heritage platform based on the requirements discovered throughout the research and fact seeking efforts. The final system consisted of a front-end interface, user registration capabilities, a secured administrative environment, database and storage services, multimedia resources and interactive and immersive web technologies supporting digital heritage information, educational resources, visitor services, accessibility features and administrative functions.

For the second specific objective, the identified functional modules were developed under five major areas: Digital Heritage and Educational Content, Interactive and Immersive Experiences, Inclusive Accessibility Support, Visitor Engagement and Museum Services, and Administrative Management and Monitoring. Components implemented: - Digital exhibits, - Educational resources, - Heritage Trail, - Interactive Site Map, - 360-degree virtual tour, - AR content, - 3D/WebXR experiences, - Accessibility controls, - Community participation functions, - Booking services, - Visitor-account functions, - Administrative management features.

Regarding the third specific target, Black Box Functional Testing was undertaken with 54 test cases for 13 system modules or feature groups. All 54 test cases passed, resulting in a 100.00% overall pass rate. This means that the functions under test behaved as documented against the documented expected results under the defined testing conditions.

In an effort to achieve the fourth specific objective, the developed platform was evaluated by 190 respondents through a five-point Likert scale questionnaire, which examined eight evaluation criteria. The overall weighted mean of the platform was obtained to be 4.77 (Strongly Agree). All eight criteria were evaluated as Strongly Agree. The highest weighted mean was achieved by Perceived Usefulness at 4.84, followed by Attitude Toward Using at 4.83. Selected Software Quality Characteristics and Behavioral Intention to Use both obtained weighted means of 4.82. Perceived Ease of Use and Immersive Digital Experience both obtained 4.73, while Accessibility obtained 4.72. Actual System Interaction obtained a weighted mean of 4.67.

Overall the respondents thought the site was useful, easy to use, engaging, accessible and acceptable for accessing and studying digital heritage content.

5.2 Conclusions

The conclusions were formed as follows on the results of the study:

1. The project successfully designed and created a WebXR-Enabled Digital Heritage Platform which integrates heritage information, educational resources, accessibility-oriented mechanisms, interactive and immersive experiences, visitor services, and administrative management functions.
2. The finished system indicates that the museum and user needs identified have been turned into a working web-based platform. The identified functional modules were successfully implemented in the five key system areas defined in this study. The finished modules make it possible for the platform to be a source of digital heritage information, a learning environment, an environment for immersive exploration, visitor interaction, museum services and administrative content management.
3. The results from the Black Box Functional Testing indicated that the tested public and administrative functions worked according to the functional requirements provided for them. The cumulative result of all 54 written test cases for the 13 system module groups was Pass. Thus, the third specific objective was achieved under the conditions of the described test cases and testing conditions.

4. The user evaluation results of the system revealed a high degree of user acceptance and experience. The aggregate weighted mean of 4.77 which is read as Strongly Agree implies that the respondents usually rated the produced platform positively in all the eight established criteria. The high score for Perceived Usefulness suggests that the respondents perceived the usefulness of the platform for accessing heritage material, as well as for enhancing their knowledge of the history and heritage of the site. This is in favour of the educational and informative direction of the established system.
5. Positive results for Perceived Ease of Use and Attitude Toward Using indicate that respondents generally considered the platform manageable to use and viewed its use as a digital heritage resource favorably. The favorable rating for Immersive Digital Experience also indicates that respondents positively assessed the digital exhibits, 360-Degree Virtual Tour, AR/WebXR features, and other interactive components. Within this criterion, engagement with the AR/WebXR features obtained the highest rating, while exploration through digital exhibits and engagement through the broader interactive features obtained the comparatively lowest ratings. This suggests that AR/WebXR was a relative strength of the immersive experience, while the broader digital exhibits and interaction pathways remain areas for further refinement.
6. The favorable result for Accessibility indicates that respondents generally rated the platform's readability, controls, and accessibility-oriented features positively. The usefulness of the

accessibility features received the highest rating within this criterion, while ease of using the platform's controls received the comparatively lowest rating. This suggests that although respondents valued the accessibility mechanisms provided, further refinement of control operability and ease of interaction may strengthen the platform's support for inclusive use.

7. To conclude, the results show that the constructed platform served the purpose of the study by providing a more accessible, interactive, immersive and manageable digital environment for heritage data, learning, visitor engagement and museum-related services within the project's boundaries.

5.3 Recommendations

Based on the findings and conclusions of the study, the following recommendations are proposed:

1. The museum and future system administrators can continue to manage and update the platform's heritage content, educational resources, exhibitions, events and visitor information to ensure that the digital resources are relevant, accurate and valuable.
2. The digital exhibits and broader interactive features may be further refined to improve exploration, navigation, and user engagement. Particular attention may be given to strengthening the connection between interactive elements and relevant heritage information so that users can move more clearly and meaningfully through the available content. AR/WebXR features may

continue to be optimized for compatibility and performance, but their user-engagement rating should be recognized as a relative strength of the current platform.

3. Accessibility controls and navigation techniques might be improved in terms of visibility, consistency, location, and usability to improve the platform's accessibility-driven navigation.
4. Future assessments may include a larger proportion of persons with disabilities and other user groups connected to accessibility to generate more targeted evidence on the platform performance for users with diverse access needs.
5. We may test further with a larger variety of devices, browsers, screen sizes and network circumstances, especially for 360-degree, AR, 3D and WebXR features that are device and browser dependent.
6. Future development may incorporate more specialized technical review, such as performance, compatibility, security, reliability and accessibility-compliance testing, when relevant tools, measurable criteria and technical resources are available. These should be handled as separate testing operations, not inferred from the existing Black Box Functional Testing findings.
7. To promote consistent usage of the administration system, the museum may provide relevant orientation or administrative advice to authorized personnel responsible for content maintenance, submission review, booking management and other backend operations.
8. Future researchers can evaluate the platform for a longer period after lengthier deployment to study recurrent usage, maintained behavioural intention, actual long-term use, and changes in user experience over time. This would enhance

the present study, which mainly examined system interaction throughout the evaluation session.

9. In future editions of the platform, the integration of digital heritage, learning, accessibility and immersive technologies could be further improved and new features should be closely linked to the defined museum and user requirements.
10. The established platform may be a reference for other comparable digital heritage projects, provided that future implementations are customized to the historical content, operational needs, technology assets and target user of the individual cultural institution in question.

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APPENDIX A

TECHNICAL BACKGROUND

Interview Transcript

The researchers conducted an face to face interview with the Officer in charge, or acting museum curator at the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan, Mr. Albert Vincent Barretto for the data collection in order to develop the study. In February 2026.

Date of Interview: February 2026

Researchers: Alay, Riz-Anne V., Bustria, John Lloyd J., Gayoso, Cedrick C.

Client: Mr. Albert Vincent Barretto

(Start of Interview)

Researcher: So good day po we're from BSIT students, Laguna State Polytechnic University, and nandito ako po ngayon para po mag-interview sa inyo, and para po magawa ng. Sa aming capstone po.

Client: Capstone, okay.

Researcher: So**.**

Client: Morning.

Researcher: Start na lang po tayo dito sa—ay, bago pa po ako lang mag-umpisa, ask ko lang po, is meron na po bang nagpunta dito ng naggaganto din po, like.. nag iinterview?

Client: Kayo palang

Researcher: Kami pa lang po. Sige, thank you po. So ask ko lang po dito, um, pwede po bang malaman yung historical significance ng underground cemetery?

Client: Historical significance? Ito yung, uh, nag-iisa at natatanging, uh, underground cemetery sa buong Pilipinas, architecturally, the earliest known na, uh, underground cemetery sa Philippines. At, uh, naging historical siya sapagkat dito, uh, ginanap yung mga, uh, secret meetings ng mga katipuneros, uh, during the, uh, Philippine Revolution. Tapos, during the World War II, ito rin ay naging, um, hiding place or meeting place din ng mga guerillas during the World War II. Oy, gerillas, sila yung

mga resistance against, uh, Japanese during the World War II. Tapos, uh, during the 1970s, uh, ito ay naging, uh, film location ng iba't ibang mga classic films sa Philippines na, uh, ilan-ilan sa mga aktor dito ay naging national artists and naging, uh, Pangulo ng Pilipinas. At yung mga director nito ay naging mga national artists for film. So yun yung ilan sa mga na-significance niya. Tapos, during the 1980s, ni-restore ang underground cemetery at ginawang historical landmark. So hanggang ngayon, uh, ang nagma-manage ngayon ng, uh, cemetery ay ang, uh, National Government through National Historical Commission of the Philippines. At yung property ay pagmamay-ari pa rin ng Roman Catholic Diocese of San Pablo.

Researcher: Sa San Pablo po?

Client: Oo, San Pablo pa rin. Oo. Tapos, um, tinayo siya ang cemetery around 1845.

Researcher: 1845?

Client: 1845 siya. Around 1845 siya, uh, sinimulang itayo. Oo, sinimulang itayo.

Researcher: So...

Client: In, uh, in, uh, anong tawag dito, to follow the order from the Spanish government na lahat ng mga cemeteries, lahat ng mga bagong cemeteries ay dapat nakatayo o itatayo outside the town proper or outside the residential area ng, uh, bayan during those times. Hindi lang Nagcarlan ang sumunod, marami pang bayan. Halos lahat ng mga bayan sa Philippines sumunod sa utos na yun na lahat ng cemeteries ay dapat itayo outside the, uh, city proper or town proper rather.

Researcher: So during that time po, is cemetery na po pala talaga siya noong mga kasunod?

Client: Oo. During the—talagang ang intensyon ng lugar na ito ay gawing sementeryo dahil ang cemetery noong unang panahon, uh, dati matatagpuan sa paligid ng simbahan. Either nasa loob ng simbahan, kaya may mga simbahan sa Pilipinas, may mga, uh, nitso o lapida yung mga pader ng simbahan, doon sila naglilibing dati. Tapos bukod doon, naglilibing din sila sa paligid. Oo. Kaya pag minsan may nagsasabi pag may mga Catholic schools o mga liceo na dating parte, dating kinatatayuan ng mga sementeryo. Kasi mga liceo o Catholic school, katabi lang ng simbahan, di ba? So yung mga usually, yung mga lugar kung saan nakatayo ang mga

liceo, yun yung mga dating sementeryo. Nasa bakuran lang ng simbahan ang sementeryo noon. Tapos nagkaroon ng epidemya, um, nakakamatay na sakit, nakakahawa, nakamamatay na sakit. Maraming namatay. So dahil maraming namatay, ilan sa kanila ay, um, wala namang ibang libingan kundi simbahan lang din. So doon sila inilibing. Tapos may ibang mga bangkay na nare-recover, naagnas na, namamaho na. Napuno ang mga sementeryo sa loob ng simbahan noon. Kaya simula noon, pinagbabawal na. Kasi imagine na sa dami ng mga namatay, ang baho ng simbahan, unhygienic, unhealthy, and unsanitary. So kaya nagbaba ang utos ng Espanya na dapat ang mga cemeteries ay mga bagong cemetery, uh, should be located outside the town proper, ideally 1 km or 2 km away. Kaya itong underground cemetery ng Nagcarlan ay, uh, about 2 km away from the church. Oo, ganoon siya kalayo. At ito ay isang Catholic cemetery kasi simbahan ang patayo nito. Mga to be specific ay Franciscan friar, sila yung nagpatayo nito.

Researcher: So ito pong simbahan na ito, unique po talaga. So ask ko lang po, bakit po itong simbahan na ito is underground po na compare po doon sa iba na nasa ground lang po?

Client: Simbahan o...

Researcher: Ay, itong sementeryo.

Client: Sementeryo. Ang paliwanag, walang ano, documentary, uh, explanation diyan. Ultimo mo, yung nagpagawa nito, walang naiwang available record bakit siya nagpatayo ng sementeryo na may crypt or underground cemetery. So ang pwede nating isipin ay, uh, dahil ang mga prayle ay mga Kastila, mga taga san ba sila, Europe, Spain, di ba? So sa Europe, ay doon uso ang mga catacombs.

Researcher: Sa Paris?

Client: Mga crypts. O, halimbawa, nasa Paris, sa France, may mga catacombs sa ilalim noon. Sa Italia, may mga catacombs, may crypts din. Espanya, meron din. So posible na yung paring iyon, nakakita siya ng mga catacombs ba sa ibang bansa, especially sa home continent niya, Europa, doon siya sinubukan niya na kopyahin kung anong meron sa Europa. Kaya niya naipatayo na nagawa niya yang may crypt. Oo. So ang nagpoproject ay Kastila pero ang gumawa nito ay mga Pilipino sa pamamagitan ng polo y servicio o community work.

Researcher: So next naman po is, ano po ang role ng NHCP nung pagma-manage po ng sementeryo?

Client: Kami yung nagme-maintain, kami yung nagre-restore. So sa structure, kami yung nagme-maintain ng kalinisan, kaayusan, restoration. Kami nagre-report kung anong mga pinsalang makikita. Yan yung ginagawa namin. Pagdating sa museum, kami rin nagma-manage ng museum. We educate the people, the Filipinos, and even foreigners through museum education. Tour guiding, yun din ginagawa namin. And public programs, halimbawa ay mga lectures, workshops. Yan yung ginagawa. Bibigay kami ng interviews para ma-educate natin yung mga Pilipino about sa underground cemetery.

Researcher: So move naman po tayo sa operations ng museum. Ask ko lang po, ilan pong average person ang pumunta po dito per week po?

Client: Per?

Researcher: Week po. Per day or per week po?

Client: So per day, ang pinakamababa per day, 50.

Researcher: 50?

Client: Oo. Tapos pinakamataas per day, 200 yun. Usually, yung 250 or 200 plus weekends. Oo, weekends siya.

Researcher: Saan po kayo mga lugar pong galing? Mostly po ba is mga karatig bayan?

Client: May local, may mga galing sa iba't ibang bahagi ng Pilipinas. Karamihan ay mga taga-Calabarzon din, NCR.

Researcher: May mga foreign din po?

Client: Meron, ilan-ilan. Meron.

Researcher: And may nagt-tour din po ba dito?

Client: Ano ibig mong sabihin nagt-tour?

Researcher: Yung halimbawa, isang school, vivisit kayo sa isang site.

Client: Ah, so book tour ang tawag namin diyan. So meron naman, meron. School, meron. Ang pinaka-recent ay yung kinder, yung daycare. Daycare, yan pumasyal sila. Tapos recent din, may mga teachers. Isang school division silang pumunta rito ng mga teachers. Pumunta sila dito. Tapos ano pa nga ba? May mga grupo na pumapasyal dito. Halimbawa ay grupo ng batchmates ng isang high school or college. Nagbubook sila para sa isang tour. Ang recent dito ay isang batch ng from UP Diliman School of Economic College of Economics, oo, pumasyal sila. Oo. Ayun, UP School of Economics. Yan, pumasyal sila.

Researcher: So ang pagpasok naman po is may logbook naman po?

Client: Meron, may logbook kami. Everyday, may summary kami ng mga visitor. Nakalagay naman. Tapos monthly, may accomplishment report kami. Sinusubmit namin sa office yung monthly summary ng mga bisita. Tapos mga kung ano ba yung mga ginawang programs.

Researcher: Sa NHCP po pinapasok?

Client: Oo, sa Manila. Oo, yun yung aming central office. Kasi kami ay National Historical Commission of the Philippines.

Researcher: Yung mga ano po ba, yung mga nilalogbook niyo po is ine-encode niyo pa po ba sa mga system yan?

Client: Hindi, yung numbers lang.

Researcher: Yung numbers lang po?

Client: Numbers lang. Tapos kung may prominent, yun yung notable or prominent, yun yung nire-record namin. Pero yung not necessary na yung mga pangalan. Pero pag mga book tours, nilalagay din namin.

Researcher: Pag mga book tours po, saan po ang usap? May napunta po ba dito para makipag-usap?

Client: May tumatawag, meron sa Facebook, meron sa email. May walk-in.

Researcher: Ah, may walk-in.

Client: May walk-in, oo.

Researcher: So pag walk-in po na wala pong entrance?

Client: Walang bayad.

Researcher: Pag mga ano po ang book?

Client: Walang bayad.

Researcher: Wala pa rin po?

Client: Wala.

Researcher: Sa schedule-schedule lang din po pala

Client: Schedule lang. Kasi kailangan natin ng booking kasi para alam natin kung kailan natin sila maa-accommodate. Kasi mahirap na kapag may tinanggap tayong book na tour, tapos bigla na lang may sumipot na isang grupo ng ano ng mga visitor. So syempre priority namin kung sino yung nag-book. Yan yung importansya ng book. Malalaman natin kung sino yung priority. Priority lagi yung kung sino yung nag-book. Pero kung nag-walk-in sila tapos wala namang kasabay, okay lang. Pero pag may sumabay at yung isa ay hindi naka-book, priority namin yung naka-book. O kaya pag nagsabay yung booking, sasabihin namin na may kasabay sila.

Researcher: So yung mga history po ng naglologin, dito din po ba hinahandle o NHCP na po?

Client: Yung history nitong underground cemetery, nakukuha po siya through documentary records. Meron din mga interviews, books, ano pa ba, documentary or library research, doon ko rin nakukuha. Meron din sa YouTube kung gusto nating malaman kung ano yung mga field na dito sa underground cemetery ni Shoot. So makikita sa YouTube kung anu-anong mga pelikula ang dito shinoot sa underground cemetery. Buti na lang available sa YouTube yung mga films.

Researcher: So yung diba po nagpapasa po kayo ng report sa NHCP?

Client: Sorry?

Researcher: Nagpapasa po kayo ng report sa NHCP?

Client: Oo.

Researcher: Meron din po ba kayong records dito na kinikita?

Client: Pag sa ano, monthly reports, digital. Naka-digital. Ang meron lang kaming physical copy rito ay yung DTRs. Daily time records..

Client: Oo, tapos yung mga financial reports na pinapasa namin sa office sa Manila, may kopya rin kami. Lahat may physical na kopya.

Researcher: Yung digital po na report na sinasabi ninyo, ano pong format niya?

Client: Ano, Word.

Researcher: Ah, naka-Word po?

Client: Naka-Word siya. Tapos pag kailangan ko ng pirma, co-convert ko sa PDF doon ko pipirmahan ng e-sign form.

Researcher: So punta naman po tayo sa mga operations po. So nagkakaroon po ba kayo ng problema or challenges sa pagkilos niyo dito?

Client: Pagdating sa?

Researcher: Sa mga pag-work po dito. Ay, wait lang po. Ulitin ko lang po ito sa mga services. So booking po siya, nagtu-tour po, and ano pa po?

Client: Okay. So ang mga bagay na sinusulatan ng mga visitor dito ay una yung log sheet. Visitor log sheet. Ang importansya ng log sheet, doon natin makikita kung taga saan sila, anong nationality nila, estudyante ba sila o hindi, anong age range nila, tapos total nila. So sige, ang challenge doon, kukuwentahin mo yung kada isang araw, kukuwentahin mo. Kukuwentahin mo yung araw-araw. Tapos at the end of the month, yung nakuwenta mo everyday, naka-summarize na siya per day. Itototal mo na lang siya buong day. Ang painstaking doon, syempre male-female, ilan yung per age rate, makikita niyo naman yun. Makikita niyo yun mamaya pag pinakita ko sa inyo yung form namin. Pangalawa, na sinasagutan ng mga bisita, optional lang naman ito, evaluation form. So yung evaluation form or feedback form, dito malalaman kung how well we perform as an office. So maraming services yung ino-offer namin: tour guiding, museum tour, usage of resource center. Yung mga books, pinapagamit namin yan as a resource center. Tapos yung mga ginawa nilang photoshoots, video shoots, interviews tulad nito, kailangan ng feedback yan. So kung how well we accommodate

you, yun yon. So pwede kong ipakita sa inyo yung form na ginagamit namin para sa evaluation form. Pwede rin naming ipakita sa inyo yung form kung ano yung service na gusto niyong i-avail. Tulad ng ginawa niyo kanina, pinasulat kayo ng guard no.

Researcher: May mga interview po.

Client: Yun, kung ano bang purpose ng pagpunta niyo rito, nakalagay yun. Isa yun sa sinesend din sa monthly report. Ilan yung nag-interview, ilan yung nag-video shoot, photoshoot. Tapos ano yung grade or rating na binigay sa sayo nung in-accommodate mo sila sa mga services na in-avail nila. Yun yung ilan sa mga sinusulatan. Yung may interaction sa mga bisita. Tapos ano pa nga ba, yun yung mga usually mga documentary things na kailangan may sagot pang feedback or sagot mula sa bisita.

Researcher: So mahalaga naman po yung feedback form. Ang asko lang po is na na-handle po ba siya nang tama, like nasasagutan po ba nang tama nung mga bumisita po dito?

Client: Karamihan nasasagutan nang tama. May ilan na hindi nasasagutan nang maayos sapagkat hindi nila alam na yun palang mga tanong, nasa paligid lang nila. So isa sa mga tanong doon ay "Are you aware of the Citizens Charter?" Ang Citizens Charter, ito ay dito naka-display, pinapakita ano yung mga proseso nitong opisina. Yun yung Citizens Charter. So ang purpose ng Citizens Charter, para malaman ng mga bisita o ng mga kukuha ng available ng services ng office na ito yung proseso, ganito katagal yung duration ng proseso at ito yung mga halaga na babayaran mo. So in our case, less than 5 minutes lang yung process, 5 to 15 minutes lang yung process nung request, libre at naka-display diyan sa labas. Kaya lang ibang bisita, hindi sila aware na yun pala yung Citizens Charter. So kailangan pa naming ituro na doon po yung kaso pag hindi nila tinuro, ang masasagot nila sa feedback form, "Wala kaming nakitang Citizens Charter." Importante yun kasi ang importansya ng Citizens Charter para maiwasan ang mga tinatawag nating mga red tape, mga unnecessary process na wala naman, na hindi naman pala kailangan pero pinapagawa sa iyo. May mga binabayaran ka na hindi naman pala kailangan bayaran. Oo, yun yung importansya ng Citizens kasi it shows the transparency of system and process of the office.

Kasi kapag wala sa Citizens Charter yung isang proseso at naniningil ka, gumagawa ka ng extra paperwork na wala naman sa process, that's already corruption. That's

already red tape. Oo, kaya yun yung laging reklamo ng mga, hindi sa amin ha, sa ibang mga tao na kumukuha ng services from the government. Let's say sa mga lisensya, mahabang pila, tapos may cut-off kahit wala namang dapat cut-off. May babayaran ka kahit wala naman doon sa listahan ng mga dapat bayaran. Pag nagbayad ka, walang resibo. Bawal yun. So kaya yun yung importansya ng Citizens Charter to, if not to eradicate but to lessen the, well, yun ang ari niya, to eradicate corruption and unnecessary bureaucratic processes.

Researcher: So dito po ba sa office na ito, ilan po ang nagwo-work? Kayo lang po ba? Like yung...

Client: Regular personnel, dalawa kami. Dalawa kaming regular personnel, tapos dalawang security guard, dalawang janitorial. So yung security guard and janitorial, they are hired through an agency, a private agency. Oo, so kami lang yung regular directly reporting sa National Historical, sa National Government.

Researcher: Ito po ay funded po ng NHCP, di ba?

Client: Oo.

Researcher: Yes. So ito po ay since free po siya, wala po siyang income?

Client: Yes. Yun yung ang prinsipyo kasi ng National Historical, to keep museums free of charge. Kasi unang-una, kakaunti lang yung nag-aappreciate ng history. So hindi dapat maging hadlang ang entrance fee. So yun. At saka yun yung way we use our taxes. So we preserve historical sites, structures, we promote the values of our heroes, we remember significant historical events. Through these things, mga bayani, mga makasaysayang lugar, we instill patriotism, pagiging makabayan. Kasi sa pagiging makabayan natin, maiiwasan yung mga tao na nagpapadaig sa ibang bansa, pumapanig sa kapakanan ng ibang bansa kumpara sa kapakanan natin. Namimigay ng isla, di ba? Madaling isuko sa ibang banyaga yung sinasabi nating kalayaan. So yun yung importansya ng pagiging makabayan, ang kasaysayan. To instill Filipino nationalism, patriotism, and Tayo bilang isang Pilipino. Kasi kapag malakas yung pagiging makabayan natin, patriotism, yung sense of nationhood, hindi tayo mag-iisip na ipamigay ang mga isla natin. Hindi natin iniisip na ipahamak yung mga nagtatanggol ng bansa natin.

Researcher: So clarify ko lang po ulit, wala pong system na ginagamit po dito. Mga more on Excel lang po and Word po?

Client: Basic Microsoft Office. So ang madalas na software na ginagamit namin, Excel at Word pagdating sa documentary. Reports, financial reports. Doon kami gumagawa ng forms, nandoon yun. So extra na lang ito. Hindi lahat ng museum marunong gumamit ng PowerPoint. So I use PowerPoint to make lectures, talks, presentations. PowerPoint ang gamit ko. Pag naman editing ng mga photos, may Photoshop, ginagamit ko yun to publish pubmats, publication materials online. Meron kami, may ginagamit kaming Photoshop.

Researcher: Sa pagbu-book po, ang gamit is through chat ng Facebook, sa page po na emails and tawag po?

Client: Ano lang, technically verbal pa lang yan. Opo, kasi through chat siya and email. So nagiging official siya kapag naisulat na siya sa booking form. Booking form kami.

Researcher: Meaning po, pag nandito na po talaga sila or...

Client: Not necessarily. Halimbawa, binanggit sa iyo na magbu-book po kami ng tour, halimbawa yan. Isusulat na lang namin sa form, kami na nagsusulat. Kasi ang concern namin doon, kapag nagbigay kami ng Google Form halimbawa or Microsoft Office Form, may possibility na hindi sila sanay gumamit ng interface nun. O kaya para sure, sinusulat ko na lang. Tapos ilalagay namin dito na may booking.

Researcher: So doon po sa pagbu-book, may times po ba na namimiss kayo na nagpapa-schedule po?

Client: Well, minsan may malilimutan akong i-book kasi nga dahil nga due to workload or preoccupancy. Meron akong nakalimutan na hindi ko naisulat na may booking pala. Pero good thing lang naman na hindi naman sila, wala silang kasabay. Tapos lagi ko nang pinapaalalahanan na close kami pag Monday, tapos open lang kami pag Tuesday, Sunday, 9:00 AM, 5:00 PM.

Researcher: 9:00 AM po.

Client: So 8:00 AM, yun yung official start ng office hours namin. Kasi then yung 8:00 AM to 9:00 PM, yun yung isang oras na preparation. So meaning kami naglilinis, naglilinis, check things. To check things para pag nago-open kami ng 9:00, may represent iyong museum.

Researcher: So dito po sa museum, ano po kayong mga future development plan niyo po?

Client: Yung contractor, may kasi dahil bagong museum, bagong redevelop lang yung museum, inopen siya last December 11. Ang plano ay yung mga additional infos ng museum ay ilalagay online through QR. Pero magandang ano rin yung concept yun na hindi siya masyadong technical. QR mo lang, then lahat ng info malalaman mo na doon sa about diyan. Ang medyo high tech ay may mga QR sa ibang museum, hindi sa museum namin. Sa ibang museum na nakita ko na pag nagtapat ka ng QR, yung mga objects sa using screen sa phone mo halimbawa, gagalaw bigla yung mga, may animation na mangyayari sa mga museum objects. Kaya lang museum may ganun. Pero high, ano yun, mataas sa maintenance yun kasi.

Client: Sa ibang museo na nakita ko na pag nagtapat ka ng QR, yung mga objects sa using screen sa phone mo, halimbawa... Gagalaw bigla yung mga may animation na mangyayari sa mga museum objects. Ayala museum, ganoon. Pero mataas sa maintenance yun kasi minsan internet, hindi gumagana, nagka-crash yung kung anumang interface meron sa kanila sa interaction nung dapat gusto nilang ipakita. So yun lang naman nakikita ko, maglalagay yung contractor nito ng QR. Pero kung bukod sa QR, merong mas okay na...

Researcher: Approach po.

Client: Approach. Halimbawa, naisipan niyo na may ibang feature na gusto kayong ilagay sa museum, pwede naman. Tapos dagdag na rin dito yung resource center. Yan, resource center. So and so far ang nakikita ko ngayon ay may nag-iinitiate ng online library. Merong online library meaning pag ini-scan mo yung QR, pa rin QR pa rin. So pag ini-scan mo yung QR, dadalhin kayo sa isang Google Drive na nandoon lahat ng mga PDF ng books. Pero yung books na yon, yung mga free from copyright. Kasi kapag may copyright yung book, hindi mo pwedeng i-reproduce yan without consent. So halimbawa, buhay pa yung author, tapos ginawa mong PDF yung book niya,

nag-available siya online, pwede tayong mga kaso ng copyright. So kung sa resource center, kung balak nating i-adopt yung ganoon, pwedeng gawin ay ano lang kung anong itsura ng book.

Researcher: Picture lang po.

Client: Picture ng book, anong info, anong basic info ng book. Yun lang, pero hindi nila maa-access kung ano yung laman. Unless ang book na yun ay it's okay to reproduce.

Researcher: Yun din po sana itatanong ko kung pwede bang kung sakaling maggawa ng website, gawin pong online yung mga books and oo nga naman po, hindi nga naman po.

Client: Copyright.

Researcher: Akala ko po sa inyo itatanong yan, sa authors nga din po pala.

Client: So kailangan pupuntahan talaga yung mga aklat. Oo kasi syempre kumikita ang mga publishing company through sale of books. So paano sila makakabenta ng book kung meron na silang digital copy? So yun yung concept. So ang magandang gawin, inventory ng lahat ng books na available, book cover lang, info, sinong author, kailan, copyright niya. Then yun lang, wala nang hindi mo na ipakita ang content. So para at least kung ako researcher ako, tapos hindi ko sure kung yung hinahanap ko ay nandito ba sa office na ito, pwede kong makita sa online, sa QR.

Researcher: Pag mag-i-scout po.

Client: Oo, che-check nila kung nandito ba yung book na hinahanap ko. Tapos nakita ko, ito po dahil ang research ko ay about sa Rizal, tamang-tama, nandito yung mga books related kay Rizal, kailangan ko yan.

Researcher: So accessible naman po siya pag pinuntahan po dito face to face na?

Client: Oo.

Researcher: Ask ko lang po, naboborrow po ba?

Client: Hindi, dito lang.

Researcher: Ah, dito lang.

Client: Kasi ibang books diyan, aking personal. Ang sa office lang, yan.

Researcher: Ah, ito po.

Client: Oo, yung books na yan, akin lahat yan.

Researcher: Pero pwede po siyang ma...

Client: Oo, pinagamit ko yan. Pinagamit ko yan. As long as dito lang sa...

Researcher: Opo. Sa office po.

Client: Office lang o gamitin. Personal ko yan. Kaya para hindi humalo yung books ko sa books ng office, magkahiwalay sila ng bookshelf, tapos meron siyang sticker na kung kanino siya.

Researcher: Balik ko lang po doon sa mga artifacts. Hindi pa po kasi ako nakapasok. Ilang artifacts po kaya nandoon?

Client: Hindi ko binibilang kasi kaya hindi ko binibilang, wala talaga kaming artifact na as in na for visuals usually, mga artifacts nandoon. Visuals, mga replica, ganyan. Oo. Pero kung artifact talaga, nandoon sa chapel.

Researcher: Ah, doon na po yan.

Client: Nandoon. So ano ba ang nandoon sa chapel? Nandoon ang piraso ng sample ng construction, building material nitong cemetery, restoration material ng cemetery, dun makikita.

Researcher: Yun po ay hindi na po accessible po ba?

Client: Accessible lang, naka-display doon.

Researcher: I mean, pwede pong pumasok diyan?

Client: Oo.

Researcher: Yung sa ilalim po?

Client: Yes.

Researcher: Ah, pwede rin po?

Client: Pwede. Tapos sumunod, sumunod, maganda ring maging tutal parang ano to eh, brainstorming na rin ito kung anong pwede mong gawin. Inventory ng mga nakalibing.

Researcher: Ah, pwede po yun?

Client: Kasi maraming nagtatanong, ilan po ba yung nakalibing diyan. Pwedeng inventory. So pwede kayong gumawa ng system na nandoon na ka-inventory yung mga nakalibing. So hindi lang basta names and numbers nila, pwedeng mag-input ng details like birthdate nila, kailan sila namatay, photograph nung puntod.

Researcher: Opo.

Client: Tapos kung may merong available photograph sila nung buhay sila, may ilalagay. Tapos put the photograph, kahit brief info tungkol sa kanila. At least hindi lang sila, pangalan lang.

Researcher: Opo.

Client: Nabubuhay sila through stories. So halimbawa, ang may isang pangalan diyan sa crib, pari siya. Pari lang siya. Pari lang siya kasi both sa pangalan niya, kapanganakan niya, kamatayan niya, wala na kaming ibang info about sa kanya. Buti na lang may nag-donate ng frame ng photo niya. May mukha na siya. Meron na siyang itsura. So navi-visualize natin kung sino siya. Sumunod, may info. May nagkwento yung descendant niya. Descendant, hindi dahil may anak ang pari, descendant through his siblings. Oo, na pari din. I mean, naging pari. So nakapag kwento siya na siya ay dating parish priest, originally from Bulacan. Oo, hindi ko alam yun, from Bulacan siya.

Researcher: Oo, layo.

Client: So ibig sabihin, patunay yun na hindi lahat ng nakalibing ay taga-Nagarlan. Di ba? So it gives you story, di ba? Galing Bulacan pero na-assign sa Laguna as parish priest.

Researcher: So kung sakali nga pong ilalagay po namin yung info sa online, ang consent po ay sa inyo po?

Client: Yun lang, data privacy.

Researcher: Data privacy.

Client: Kasi baka may mga... Pwede, dito lang. Dito lang. Oo, dito lang. So dito lang siya para in case na may gustong mag-research about sa mga nakalibing, dito niya lang maa-access. Pwede yun. Ang kagandahan nun, bukod sa may name, info niya, pwede niyo rin ilagay yung location kung saang part ng cemetery siya nakalibing. Pwede yun. So kung gusto niyo, mag-focus lang kayo sa isang... Ilang software bang balak niyong gawin? Marami o isa lang?

Researcher: Isa lang po yun.

Client: So pumili kayo, inventory ng mga nakalibing. Kasi alam niyo yung inventory ng nakalibing, pwede niyo ring pag na-develop niyo ito ng maayos, pwede niyo rin itong ibenta yung software sa mga cemeteries sa ibang lugar. So idea na yun. So yung software na yun ay yan, malalaman kung sino ang nakalibing, picture nung nakalibing, info niya, saang location. Pwedeng maglagay ng coordinates, pang ilang nitso sya. Pwede yun. So pumili kayo, inventory ng cemetery, resource center, data collection ng mga visitor, museum interaction. Mamili kayo kung anong gusto niyo.

Researcher: Mostly po, pwede pong sama-sama na po yan sa isang system.

Client: Ba, pwede ba yan?

Researcher: Opo.

Client: Sige.

Researcher: So parang ma-access niyo po siya dito, pwedeng doon na rin po ang input ng mga...

Client: Anong platform na gagamitin niyo?

Researcher: Yun pa lang po ang aaralin namin. Depende po sa ibinigay niyo pong info sa amin.

Client: I mean, di ba kailangan ano... Ano yun, i-install siya offline o kailangan...

Researcher: Website lang po.

Client: Naka-website.

Researcher: Accessible po siya.

Client: Anong platform online na ginagamit niyo? Anong...

Researcher: Paano bang pwede?

Client: Halimbawa ay siyempre gagawa kayo ng system saan niyo, saang website niyo siya ilalagay.

Researcher: Yun pa rin po ang titingnan ko sa kasi mostly po yung...

Client: Kasi diba may bayad ang website?

Researcher: Opo, yun din po. May bayad po yung iba.

Client: Meron namang mga websites na free na gumawa ka ng sarili mong bagay. Ano nga ba yun? Dot webs or dati ang uso, blogspot. Mga ganito.

Researcher: Blogspot.

Client: Mga blogspot. So kung ano-ano nilalagay nila itong mga interaction. Tapos may dot web, may dot web pa nga. Dot web.com something.

Researcher: Opo.

Client: Na...

Researcher: So yun lang naman po yung sa tanong namin. And so kung sakali nga pong may ma-develop kaming system and na-accept po siya ng panel namin, so okay lang po ba na kayo po ang client? Agree naman po kayo doon?

Client: Oo. Oo kasi syempre i-tetest nyo yan. Kanino pa ba sa amin? Di ba? So i-accept ko yan then ano rin yun, it will benefit you sa inyong...

Researcher: Studies po.

Client: Studies sa requirement ng school niyo. It will also benefit us for information and systems.

Researcher: Opo. So yun nga po, bale pag natapos po namin siyang ma-develop and approve po siya, property po siya ng LSPU. Pero pwede po naming ibigay sa inyo, gagawa lang po ng memorandum letter and yun, free na po siya. Bale ang babayaran niyo nga lang po is yung pag-magho-host ng website. So nagre-research po kami and may nakita naman po kaming free pero hindi pa po namin nate testing. Ang nakalagay po doon, kailangan nagagalaw po yung website na yun.

Client: At least?

Researcher: At least for one week po. That sa isang week, that nagagalaw po siya para maging active. And may nag-suggest po sa amin na gawa na po niyo na na last year po and mga kaibigan pa namin kasi schoolmates, may bayad po siya 6 dollars po per month po. So kung...

Client: So magkano yung 300 all? Nasa 300, 400, 600?

Researcher: So kung sakali pong may bayad, hindi po kasi namin natetesting yung free. Kung sakali pong may bayad, okay lang po ba yan?

Client: Kung may bayad naman, meron kasi ang office namin na budget.

Researcher: Opo.

Client: Ang kailangan lang, may resibo. Paano mag-i-issue ng resibo yung website? Kahit sino mag-i-issue.

Researcher: Mostly po, ang magbabayad nun is ilalagay po yung bank account. Tapos transfer po ng pera, receive po ng receive.

Client: Ah, hindi ba online? Pwede bang online?

Researcher: Online po.

Client: Online banking, pwede.

Researcher: Oo, online banking.

Client: Pwede naman online banking pero dapat magbibigay sila ng bill or resibo. Ayun nga. Para hindi masayang effort niyo, at least na hindi lang siya for compliance. Nakikita niyo siya as isang effective, efficient na website or software na ginawa niyo.

Researcher: Yun din po maganda doon na nakaka proud. Pag ginagamit po, pag...

Client: Oo kasi it serves purpose. Di ba? Pangit naman yung pinaghirapan mong i-coding, tapos hindi pala ginagamit. Di ba? Para kang nagluto ng pagkain tas walang kumain. So same with ano, web softwares, mga ganyan. Kailangan gumagalaw, kailangan ginagamit.

Researcher: So yun lang po naman po lahat. Question po namin.

Client: Ano nga?

Researcher: Yun lang po.

Client: Yun lang?

Researcher: Opo. So maraming salamat po doon sa info niyo, napaka-very. Ang dami kong natutunan. And after po nito, pwede po akong mag-visit?

Client: Oo, gusto niyo umikot para makita niyo yung, ma-visualize niyo kung ano yung possible na gawin.

Researcher: Pwede pong pumasok?

Client: Oo, samahan ko kayo. Anong oras na ba? 11:00. Okay.

Researcher: Thankyou po.

Story Board

Figure A1. Homepage and Accessible Heritage-Story Navigation

Figure A1 presents the visitor's entry into the platform. It shows how the homepage introduces the museum through chaptered visual storytelling and how visitors can use narration, language, subtitle, text-size, and navigation controls to access the platform's principal modules.

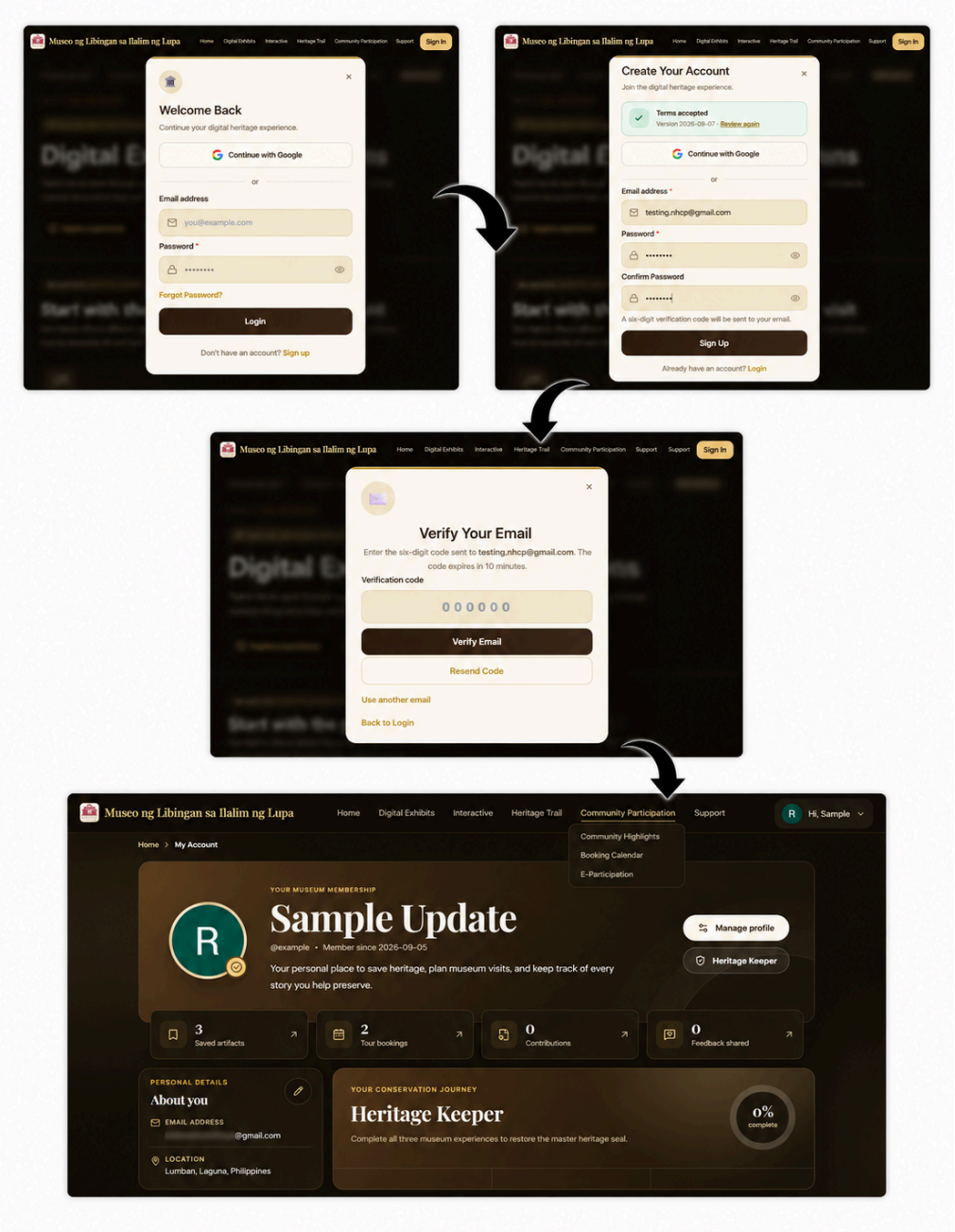


Figure A2. Visitor Registration, Email Verification, and Sign-In

Figure A2 illustrates how a visitor creates and accesses a platform account. The sequence covers account registration, email ownership verification, profile completion, and secure sign-in before the visitor uses account-based functions such as bookmarks, submissions, bookings, and saved progress.

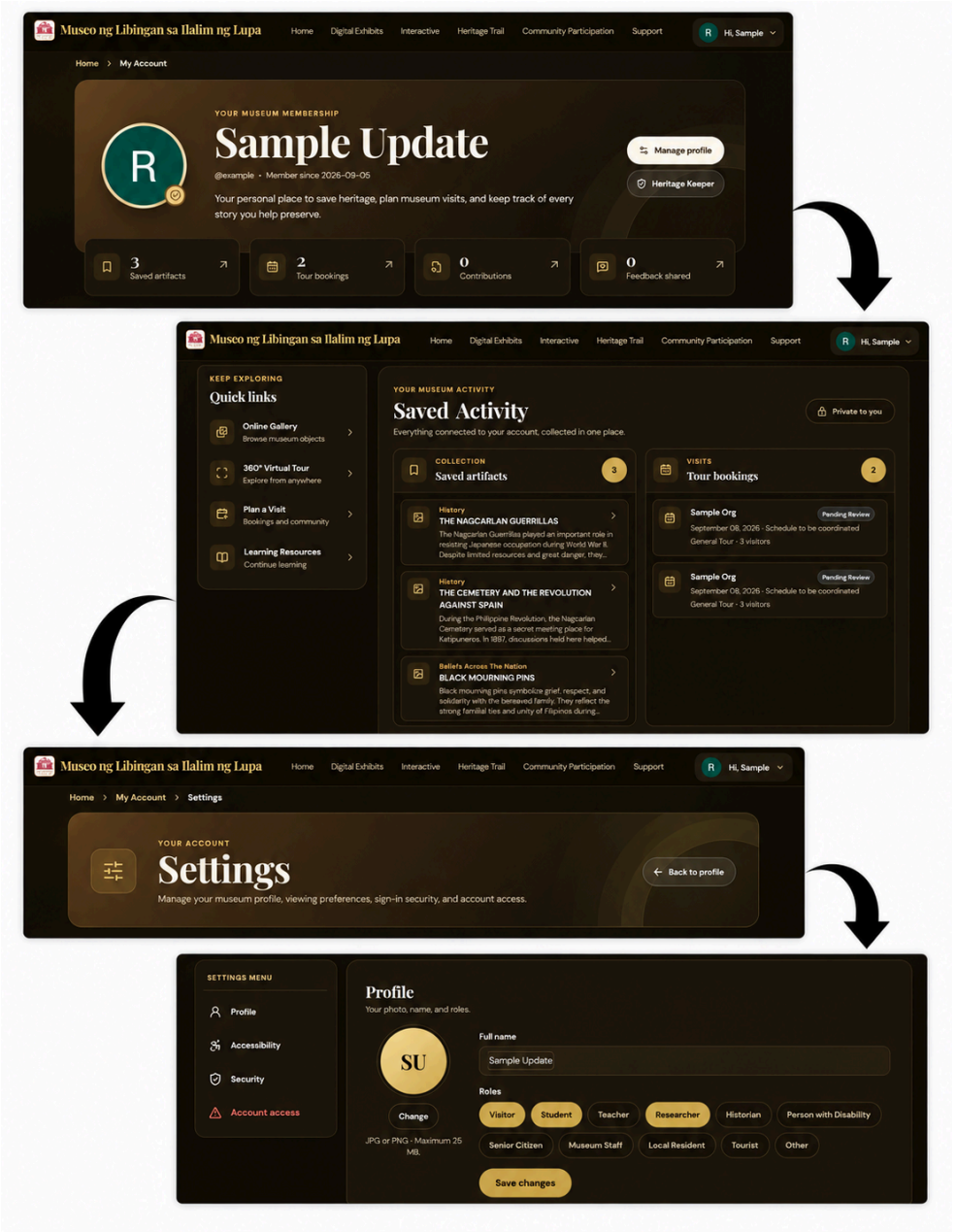


Figure A3. Visitor Account, Saved Content, and Preference Management

Figure A3 shows the registered visitor's personal account workspace. It allows the user to review profile information and account activity, return to saved museum content, monitor available participation records, and manage language, theme, text, profile, and security preferences.

Figure A4. Digital Exhibits Portal, Online Gallery, and Educational Resource Discovery and Access

Figure A4 presents the platform's Digital Exhibits portal and its digital heritage and learning services. Visitors may select the Online Gallery for artifact exploration, use search and filters, open detailed artifact records, inspect available visual or 3D media, and save artifacts to their account. Visitors may also access the Educational Resources library, use search, filtering, sorting, and view controls, open selected resource details, and use available view or download options.

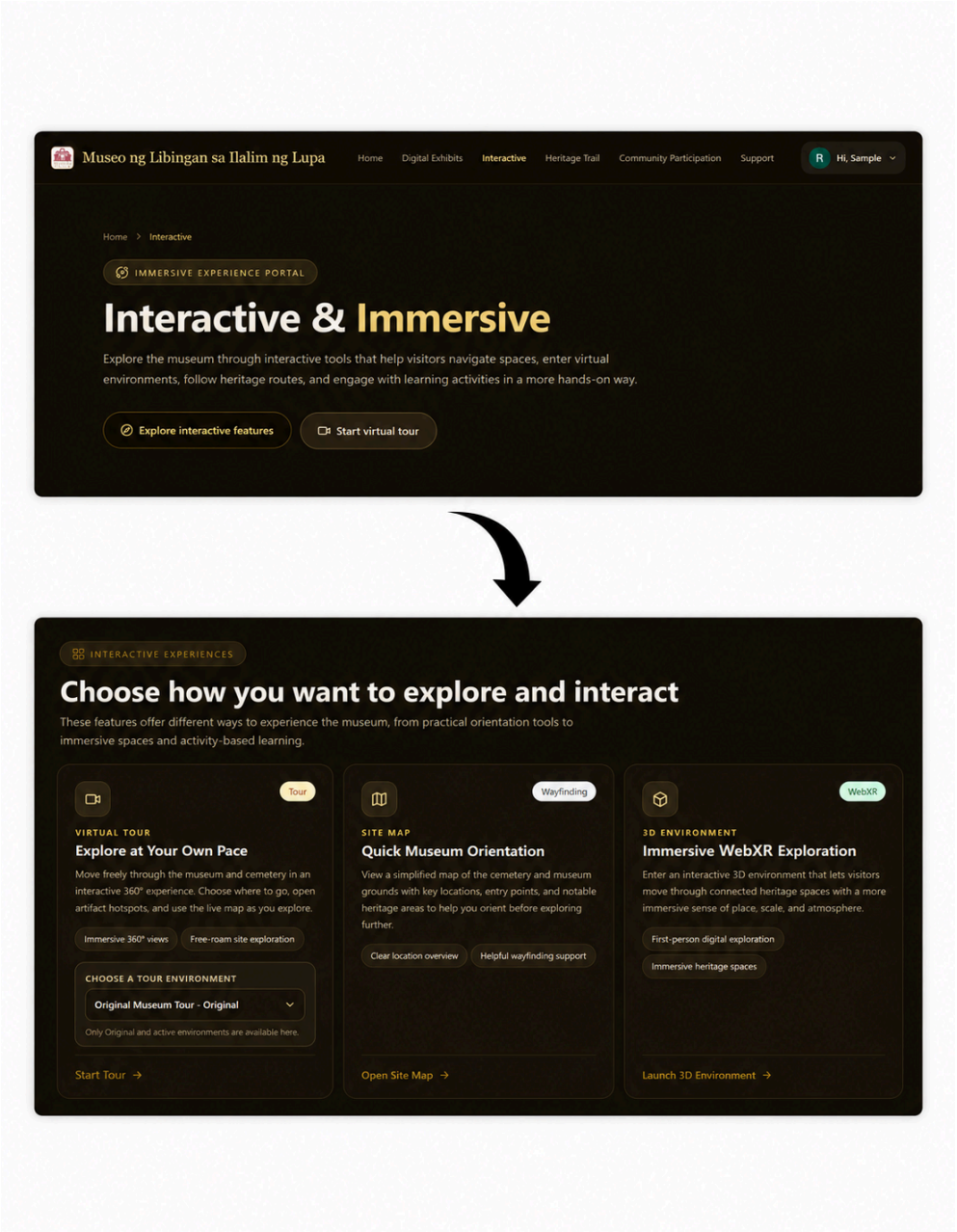


Figure A5. Interactive and Immersive Experience Selection

Figure A5 shows how visitors choose among the platform's major interactive experiences. From the Interactive and Immersive portal, the user may begin a selected 360-degree Virtual Tour, open the Site Map for orientation, or launch the WebXR-enabled 3D Environment.

Figure A6. 360-Degree Virtual Tour Navigation and Artifact Interaction

Figure A6 illustrates the free-roaming Virtual Tour of the cemetery and museum. Visitors can look around panoramic scenes, move between locations through navigation hotspots, use the live minimap for orientation, open artifact or information hotspots, and use available narration, language, subtitles, gyroscope, or VR-mode controls when supported.

Figure A7. Interactive Site Map and Aerial-View Navigation

Figure A7 presents the platform's spatial-orientation tools. Visitors can inspect the museum grounds through the aerial site map, select location markers to view their corresponding information panel, zoom or pan the map, and switch between the illustrated aerial map and the interactive 360-degree aerial panorama.

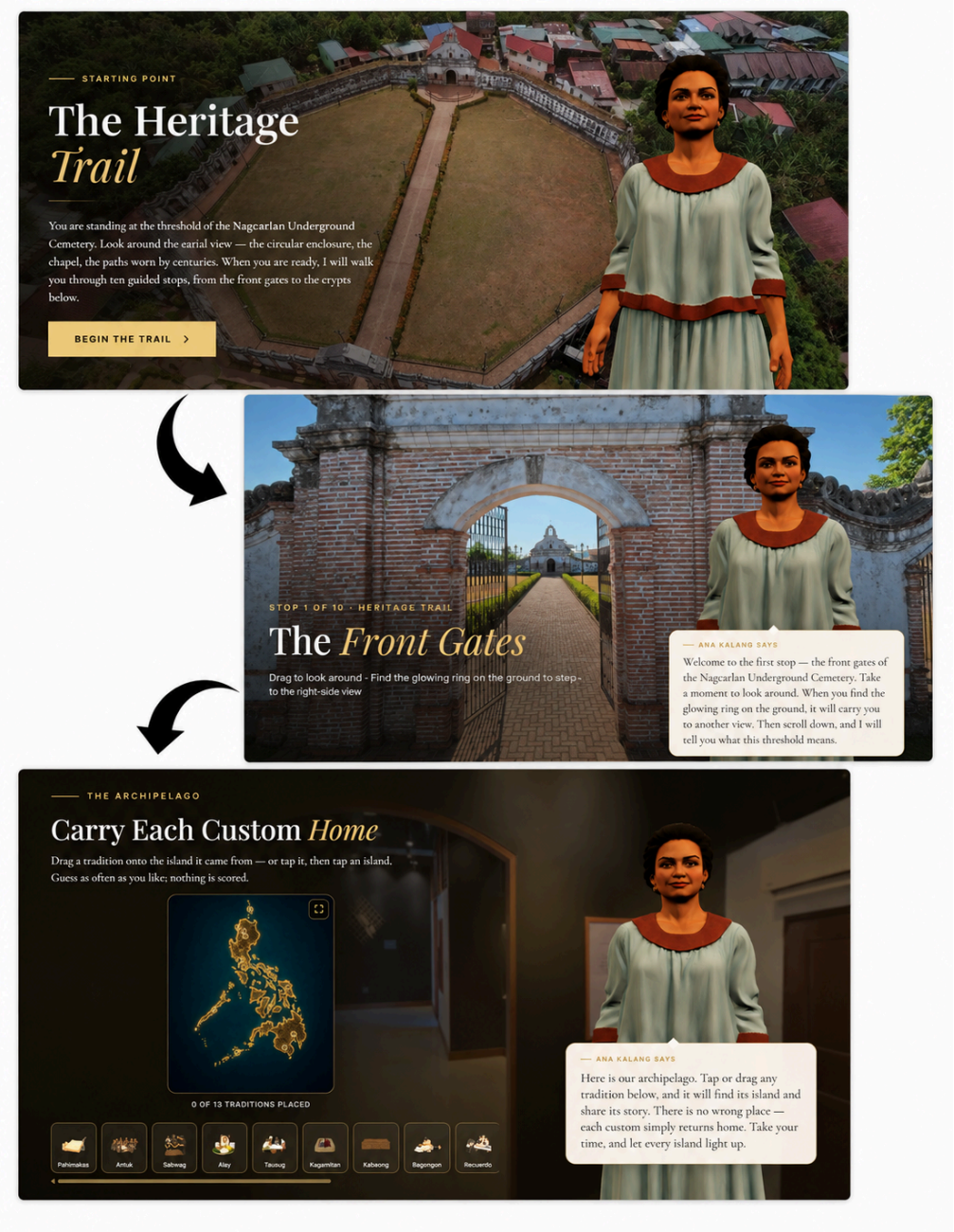


Figure A8. Guided Heritage Trail, Media, and Learning Activities

Figure A8 shows the visitor's progression through the ten-stop Heritage Trail. Each stop combines 360-degree scenes, bilingual narration, historical media, and self-check activities, while completed stops contribute to the visitor's Heritage Keeper progress and earned-fragment state.

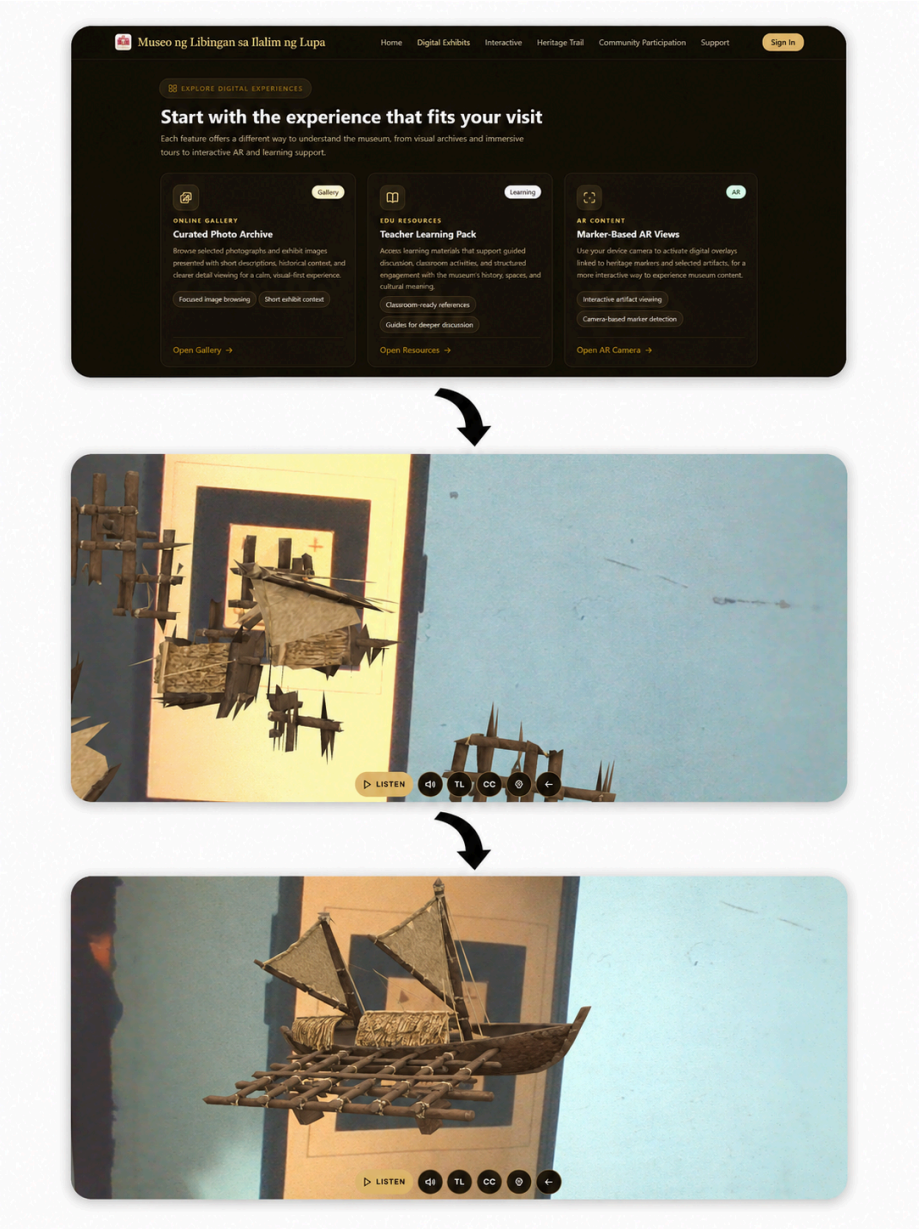


Figure A9. AR Artifact Detection and Interactive Restoration

Figure A9 illustrates the marker-based AR experience. From AR Content Detection, the system requests camera access, detects a registered marker, displays the fragmented 3D artifact with alignment guidance, restores the artifact after the correct viewing angle is maintained, and provides the View in Your Space action or supported device AR viewer.

Figure A10. WebXR 3D Environment Navigation and Interaction

Figure A10 presents the browser-based 3D museum environment. Visitors can select and navigate the Church, Museum, or Underground environment using desktop, touch, or on-screen movement controls, interact with designated objects and hotspots, review the Explorer Record or player-information panel, and enter WebVR or WebAR on compatible devices when evaluated.

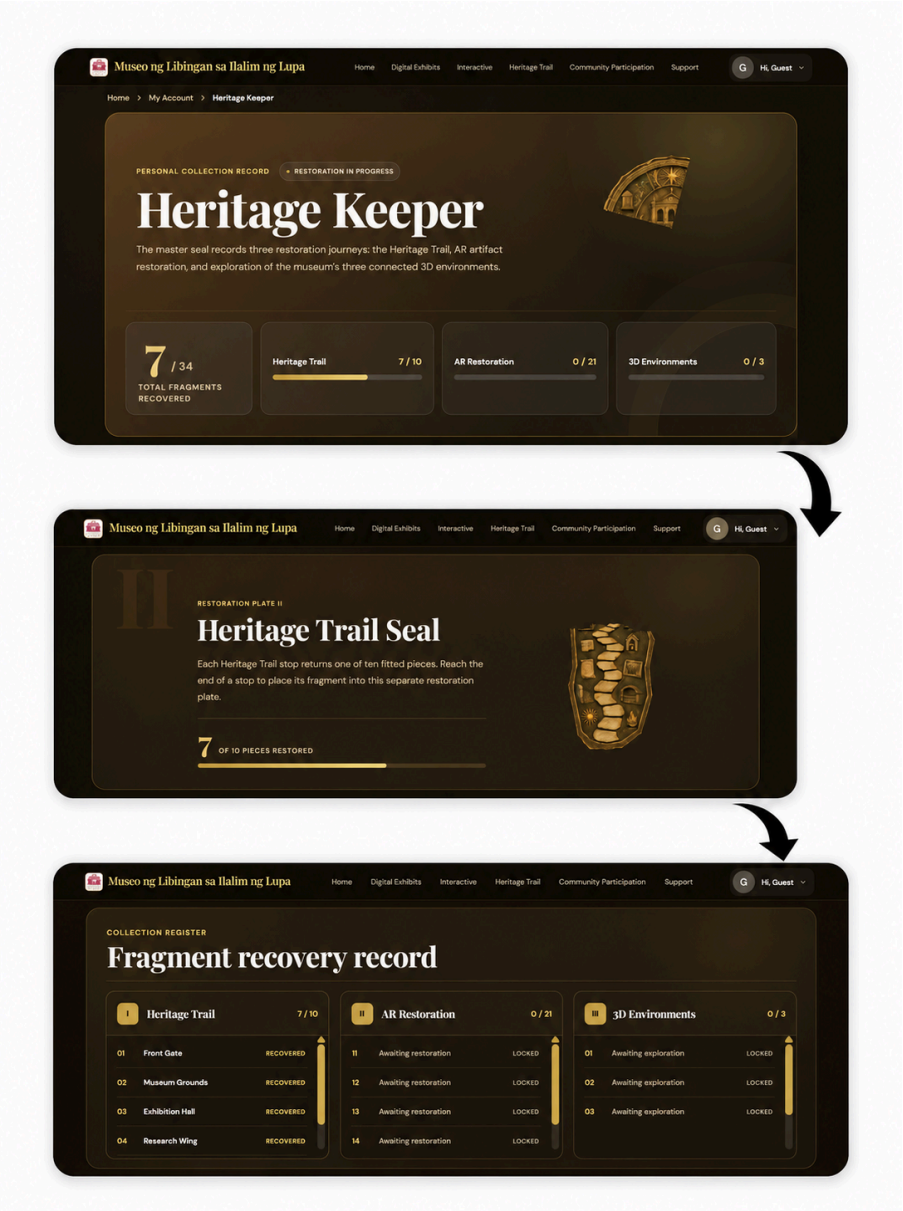


Figure A11. Heritage Keeper Progress and Digital Seal Restoration

Figure A11 shows how a registered visitor's activity is recorded across the Heritage Trail, AR restoration, and WebXR environments. Recovered pieces appear in the AR Restoration Seal, Heritage Trail Seal, 3D Environment Seal, and fragment-recovery ledger, while completion of the required experiences contributes to restoration of the master heritage seal.

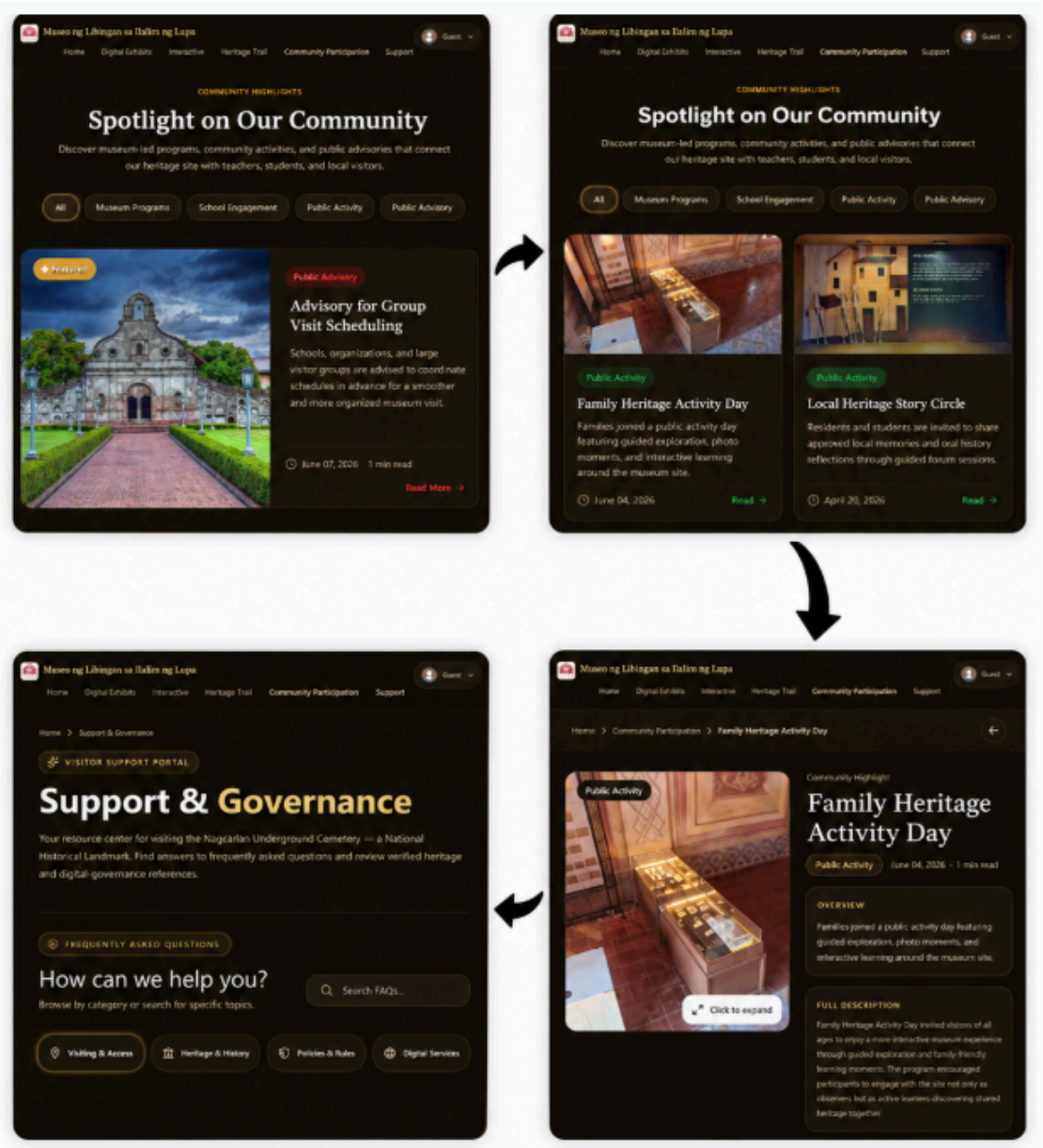


Figure A12. Community Highlights

Figure A12 shows the Community Highlights feature of the developed platform, where users can browse published community-related content and filter entries according to available categories such as museum programs, school engagement, public activity, and public advocacy. These functions were designed to organize community-oriented content, encourage public engagement, and provide users with additional channels for accessing relevant museum information and support services.

Figure A13. E-Participation

Figure A13 presents the E-Participation feature of the developed platform, which provides users with several channels for contributing information and participating in museum-related activities. From the main E-Participation interface, users may access the Visitor Feedback, Teacher Corner, and Stories Submit functions, each intended for a specific type of contribution.

Figure A14. Booking Calendar and Feedback Submission

Figure A14 presents the visitor-facing Community Highlights, Support, guided-visit booking, Visitor Feedback, and E-Participation workflows. Visitors may browse museum programs, school-engagement posts, public activities, or advisories, consult the Support page, submit a Request a Booking, monitor booking status, provide Visitor Feedback, contribute materials through the Teacher Corner, or submit personal and cultural narratives through Share My Story.

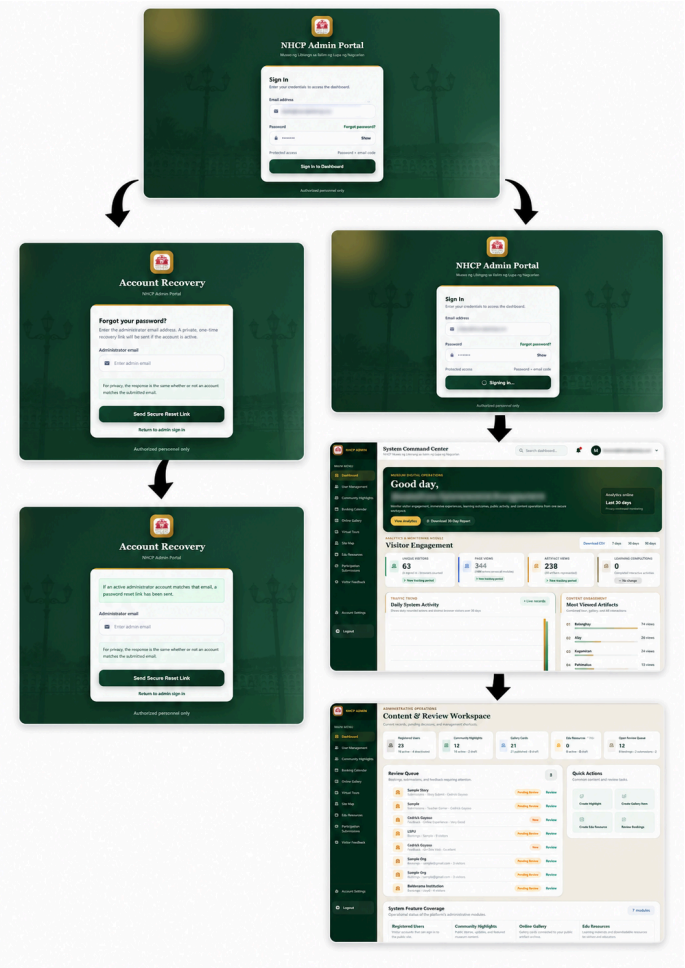


Figure A15. Administrator Authentication, Password Recovery, Dashboard, Analytics, and Review Workspace

Figure A15 shows how authorized museum personnel access the protected NHCP Admin Portal through credential-based authentication and use the Forgot Password, recovery-email, and Reset Password process when necessary. After successful login, the administrator dashboard presents registered-user, content, and review-queue summaries together with Visitor Engagement, Most Viewed Artifacts, Virtual Tour Statistics, Device Usage, Activity Results, Recent User Activity, Engagement by Module, and the Content and Review Workspace with Review Queue and Quick Actions.

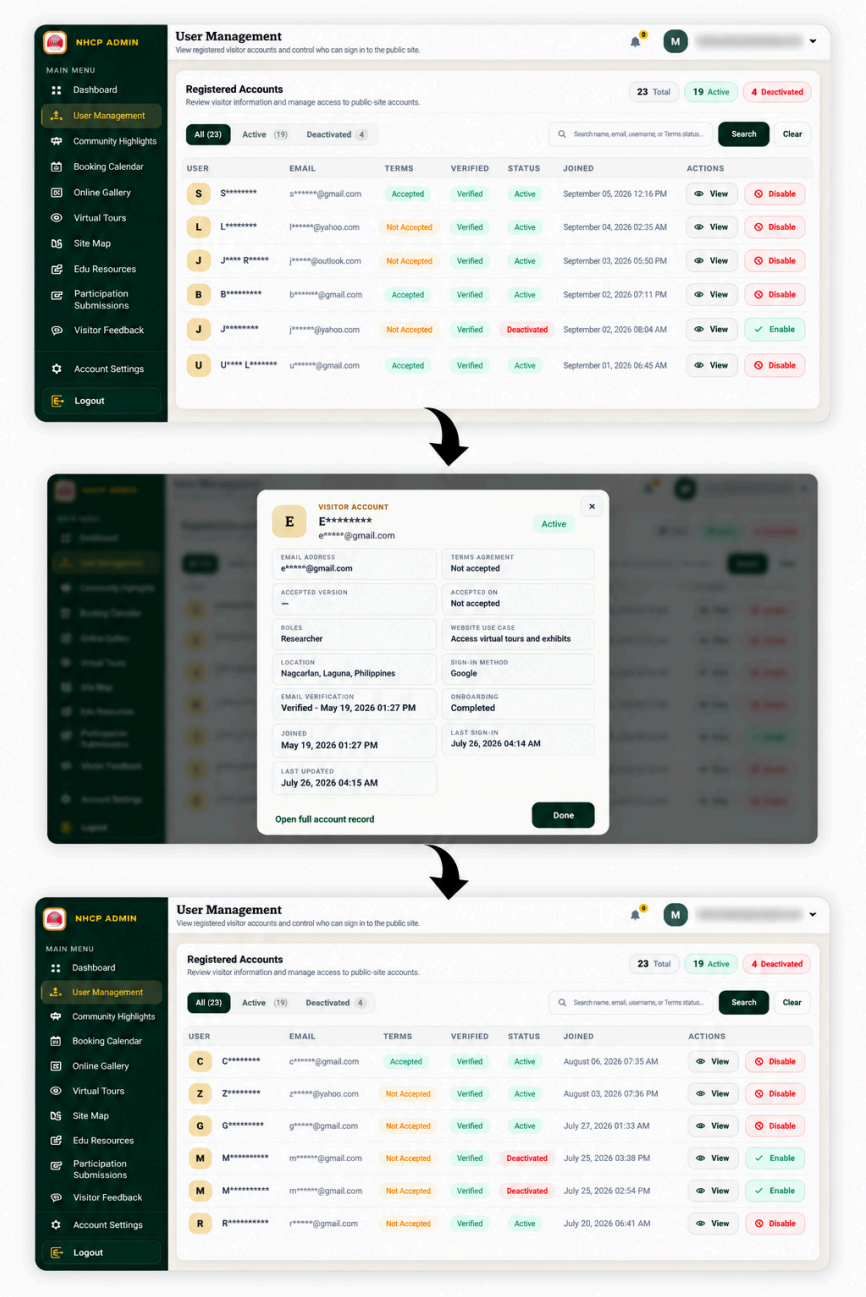


Figure A16. Public-User Account Review and Access Management

Figure A16 shows how an administrator uses User Management to locate and review registered visitor accounts. Authorized personnel can use search controls, inspect an account quick-view or full detail panel and activity summaries, and perform Deactivate or Activate actions to manage public access.

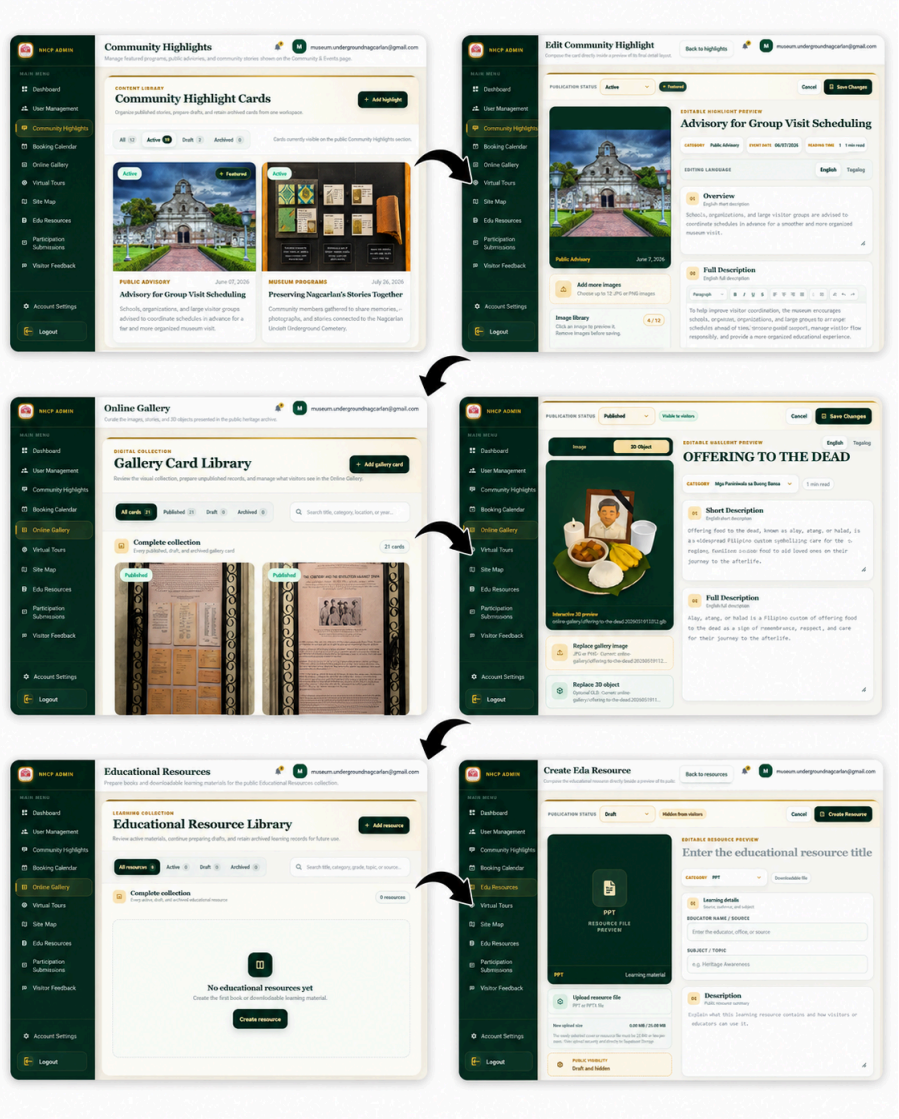


Figure A17. Community Highlight, Online Gallery, and Educational Resource Content Management

Figure A17 presents the content-management workflows for Community Highlight Cards, Gallery Cards, and the Educational Resource Library. Administrators can create or edit content, manage media and descriptive information, review saved records, use Publish or Feature actions, manage Active, Draft, or Archived status, and verify that published content appears in the appropriate public module.

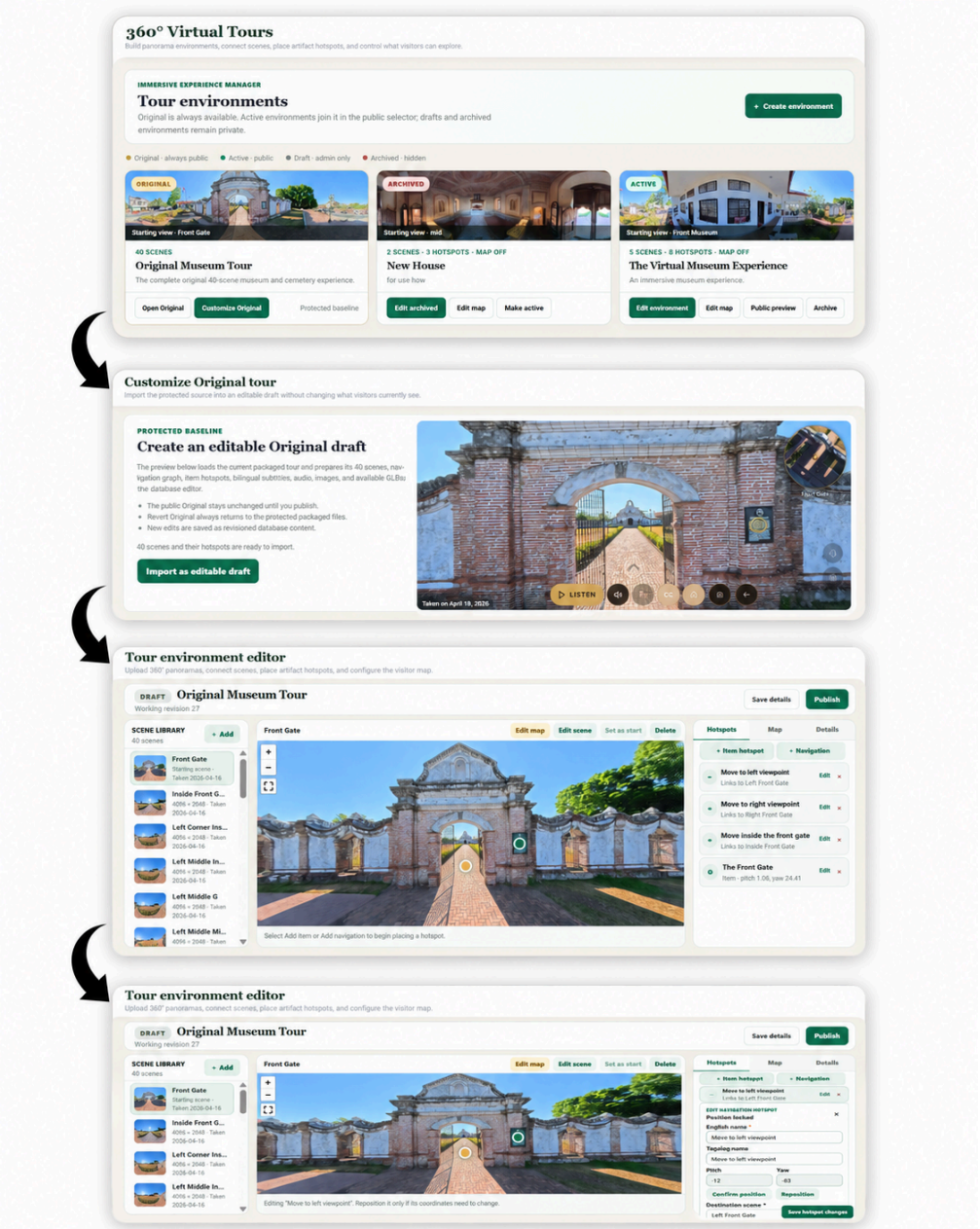


Figure A18. Virtual Tour Environment, Scene, and Hotspot Management

Figure A18 shows how administrators maintain immersive tour content using the Tour Environments library and Virtual Tour editor. Authorized personnel can create a New Environment or use the Original-import workflow, organize scene panoramas and scene settings, create or edit hotspots, preserve active, draft, or archived tours, and make the published environment available in the public tour selector.

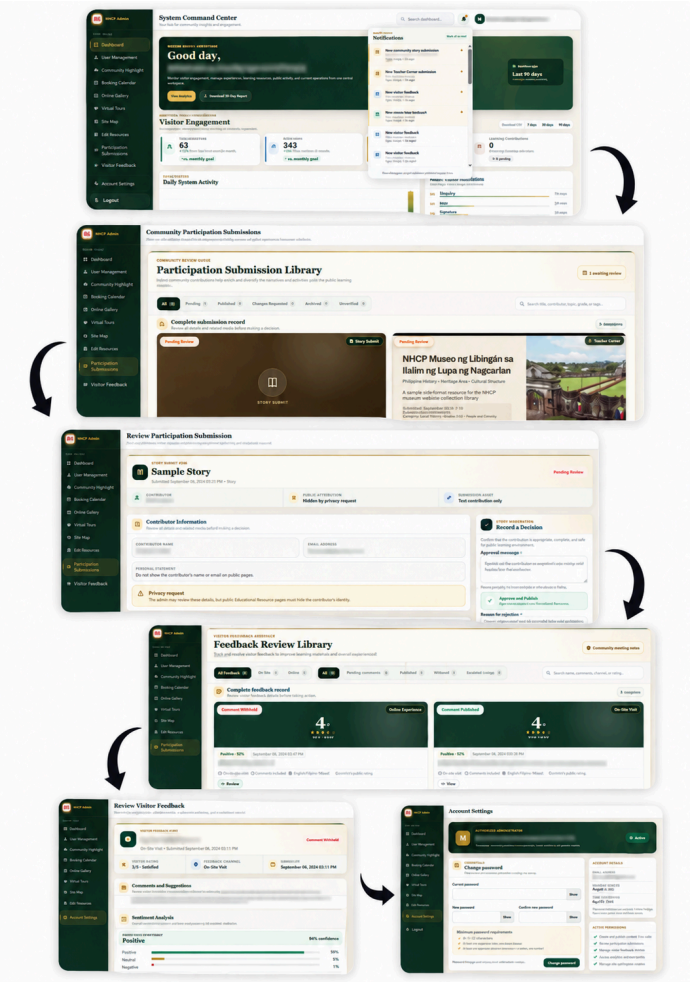


Figure A19. Guided-Visit Booking Review, Participation-Submission Moderation, Visitor-Feedback Review, Notifications, and Account Security

Figure A19 presents the administrator workflows for Guided-Visit Booking Review, Participation-Submission Moderation, Visitor-Feedback Review, Notifications, and Account Security. Administrators can use Booking Operations to review requests and issue Confirm, Reschedule, or Cancel decisions; use the Participation Submission Library to Publish or Archive submissions; use Feedback Analytics and the Feedback Review Library to classify responses; and manage notifications, Administrator Account Settings, password changes, and Sign Out All Sessions.

Flowchart

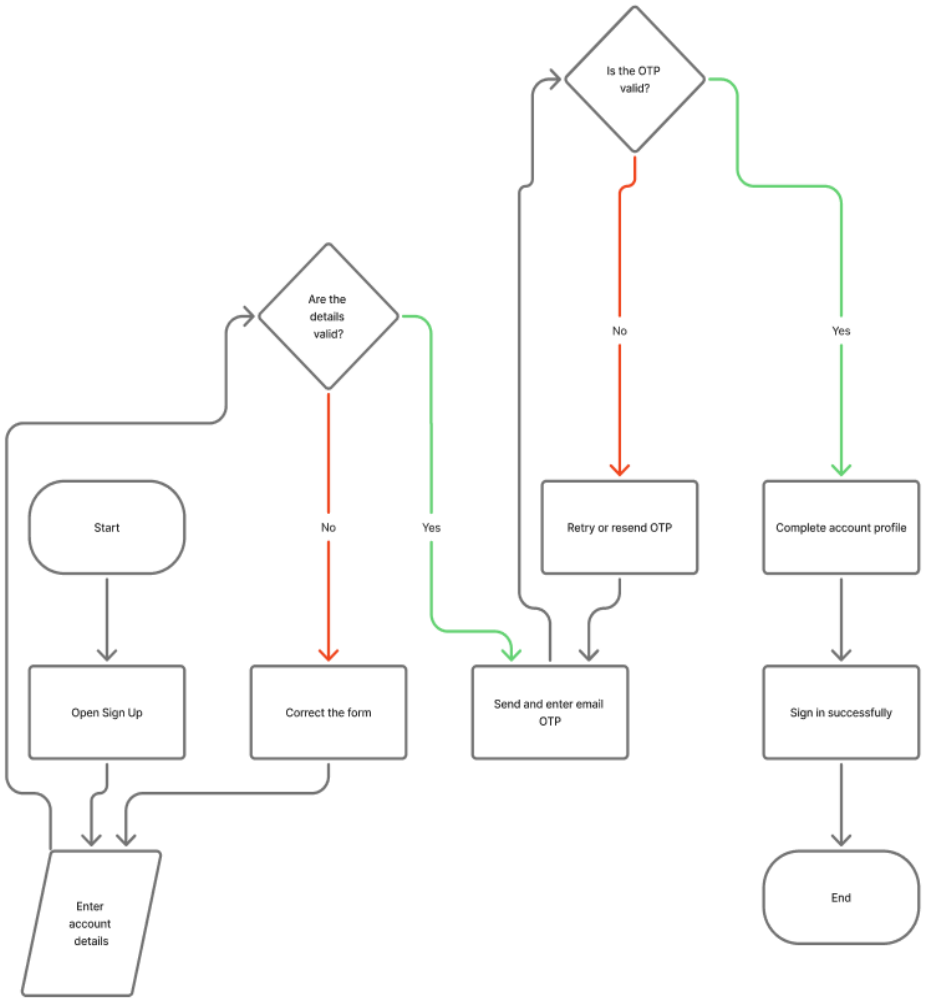


Figure A20. Visitor registration, email verification, and sign-in process.

Figure A20 shows how a visitor creates an account, verifies the registered email address, completes the profile, and signs in to access account-based features.

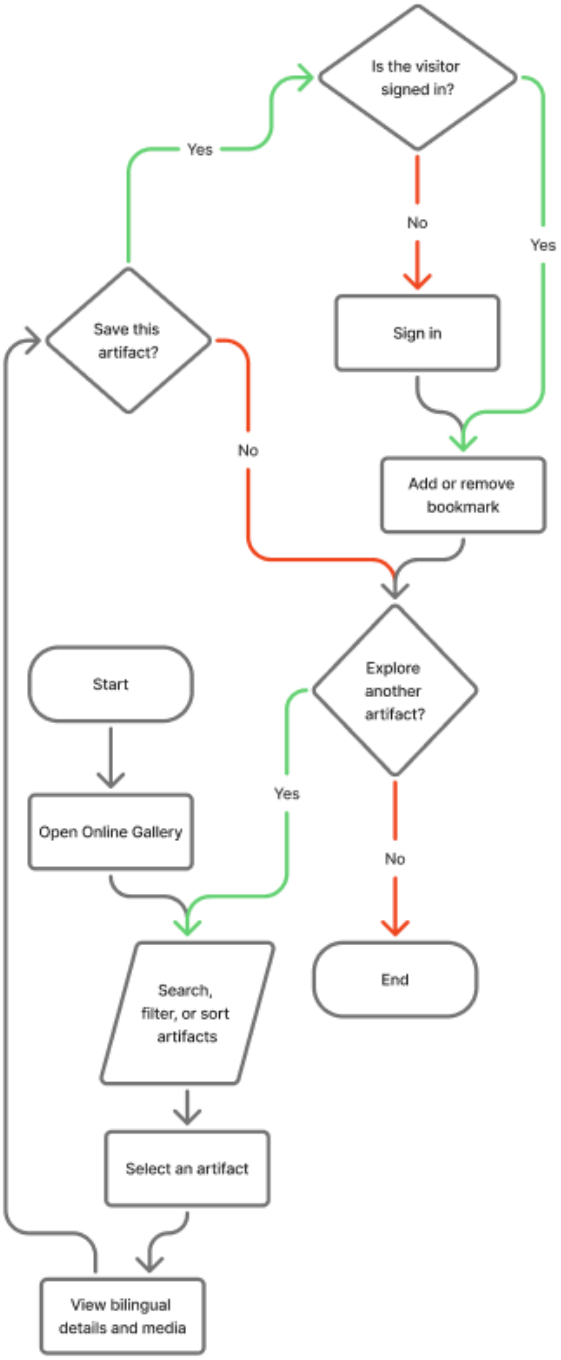


Figure A21. Online Gallery discovery, artifact viewing, and bookmarking process.

Figure A21 shows how a visitor searches the published collection, opens an artifact record, examines available media, and saves the artifact when signed in.

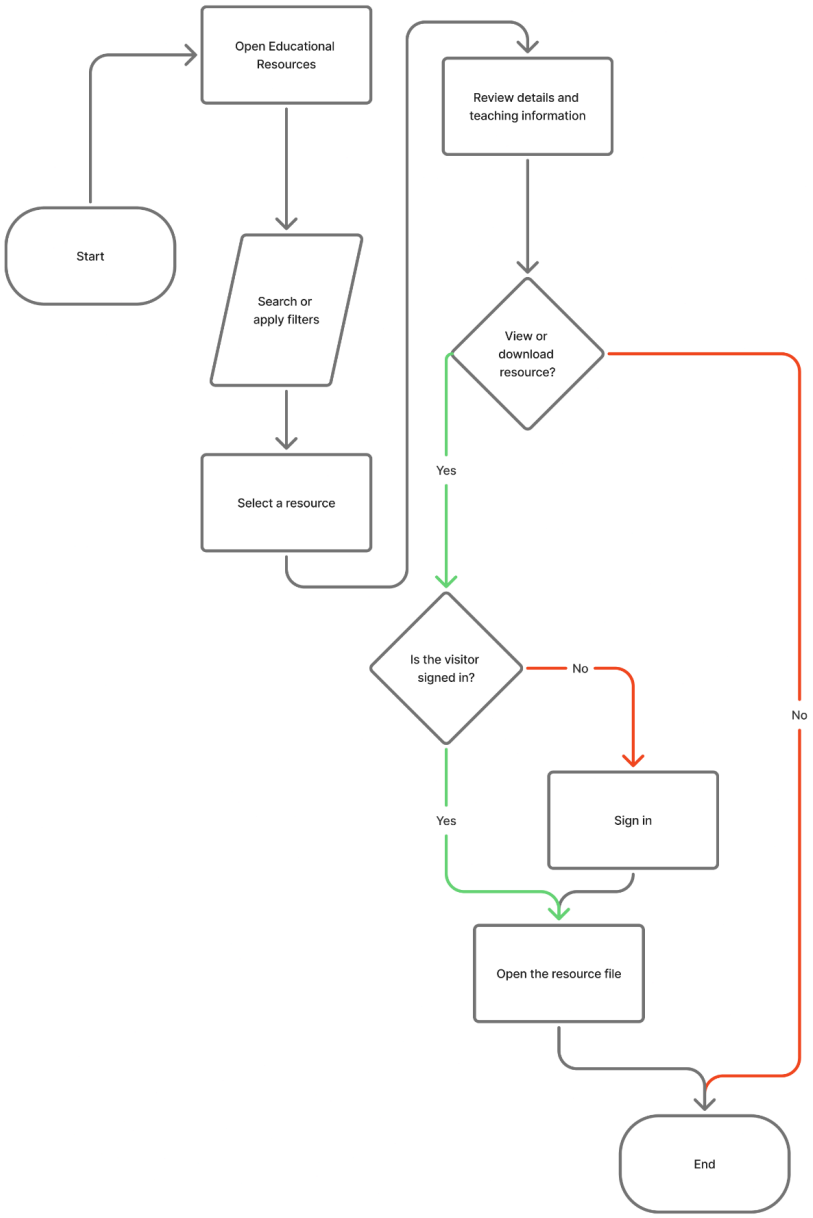


Figure A22. Educational Resource discovery and access process.

Figure A22 shows how a visitor searches the learning library, opens a resource record, signs in when required, and views or downloads the available material.

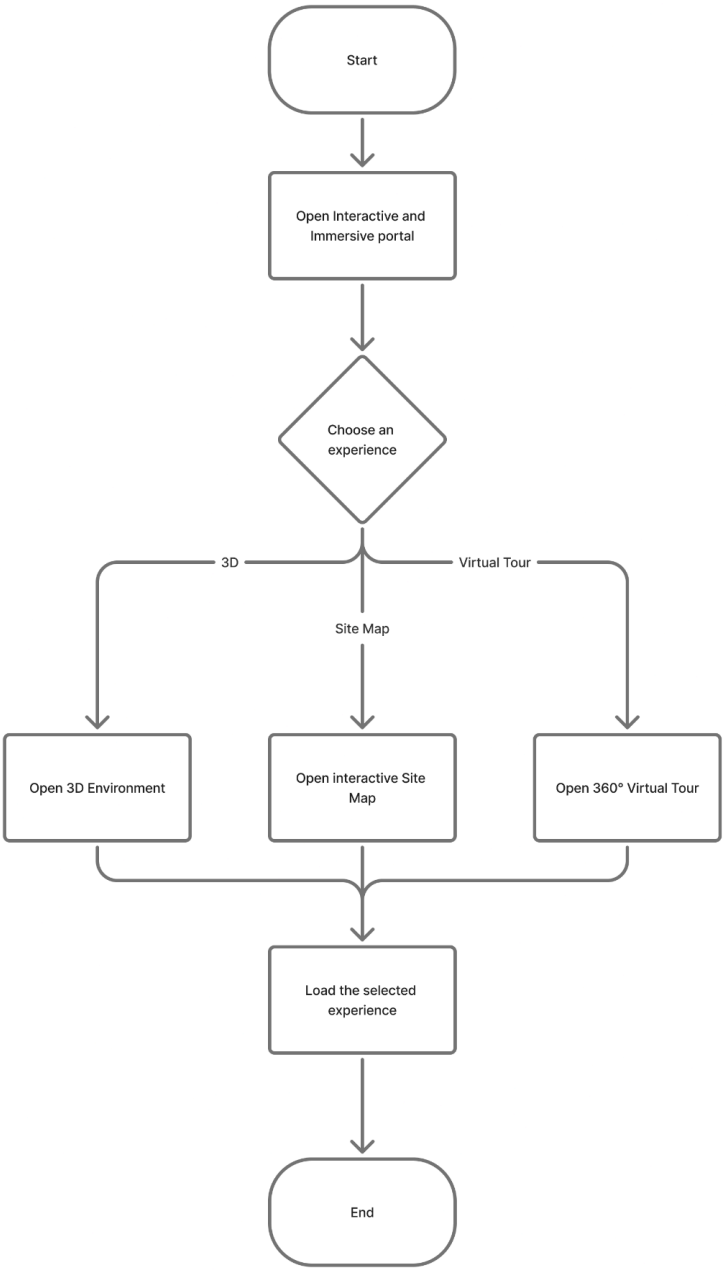


Figure A23. Interactive and immersive experience selection process.

Figure A23 shows how a visitor chooses the 360-degree Virtual Tour, interactive Site Map, or WebXR-enabled 3D Environment from the Interactive portal.

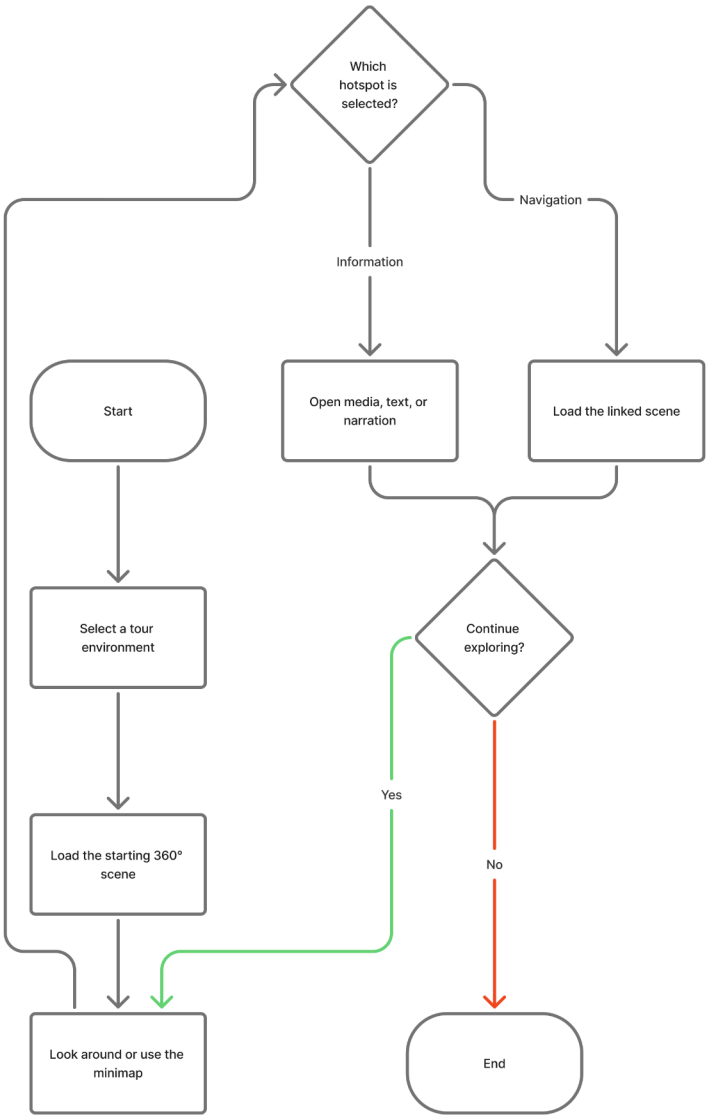


Figure A24. 360-degree Virtual Tour navigation and hotspot process.

Figure A24 shows how a visitor selects a tour, explores panoramic scenes, follows navigation hotspots, opens information hotspots, and uses the minimap and narration controls.

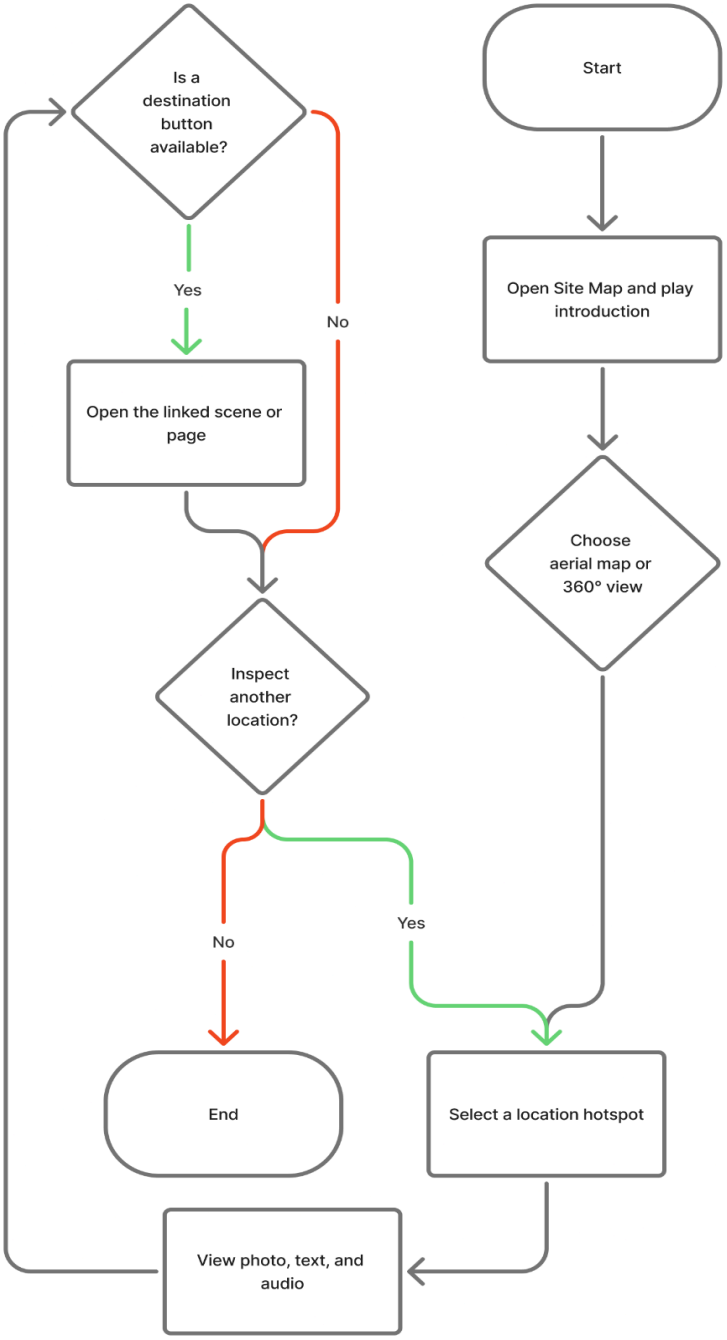


Figure A25. Interactive Site Map and linked-location process.

Figure A25 shows how a visitor switches map views, selects a location hotspot, reads or listens to its information, and opens a linked Virtual Tour scene when available.

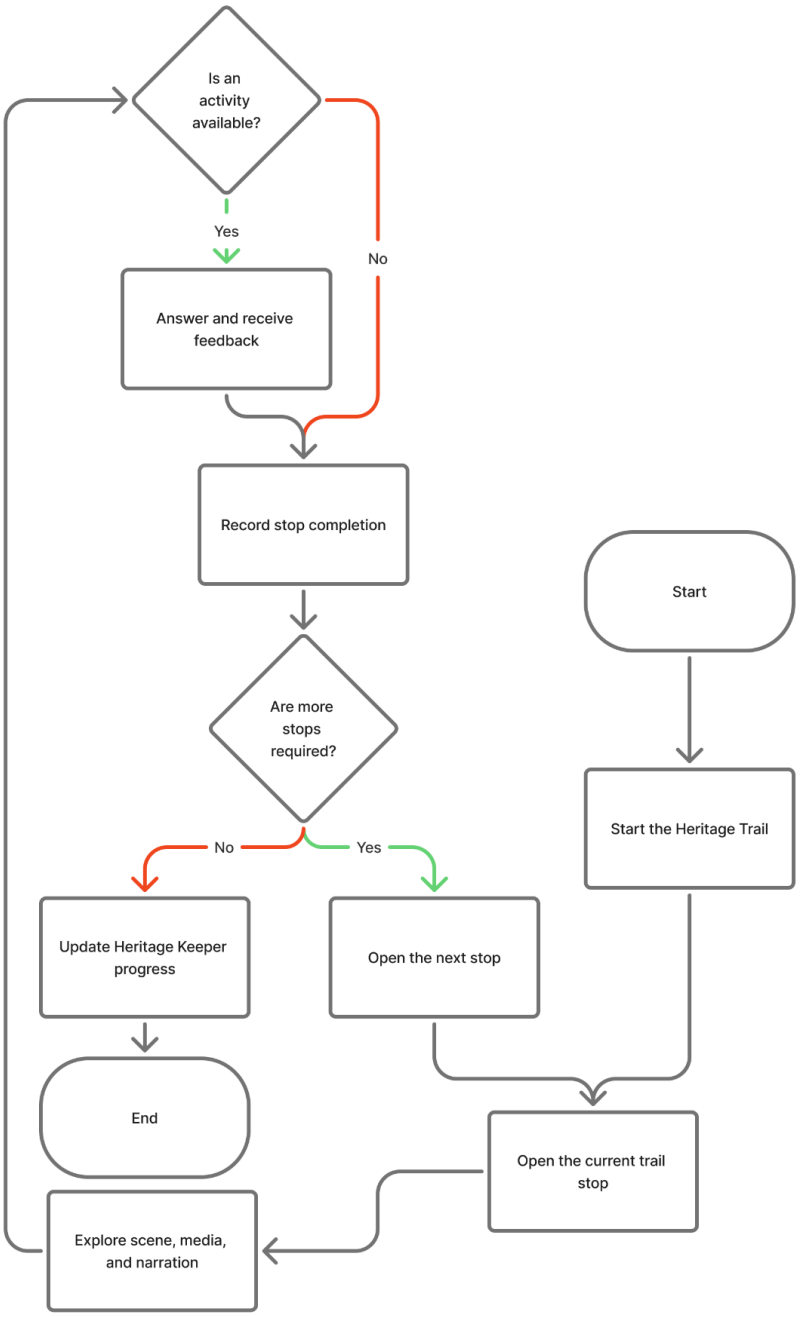


Figure A26. Guided Heritage Trail and learning-progress process.

Figure A26 shows how a visitor enters a trail stop, explores its media and narration, completes any available activity, records progress, and proceeds until the trail is complete.

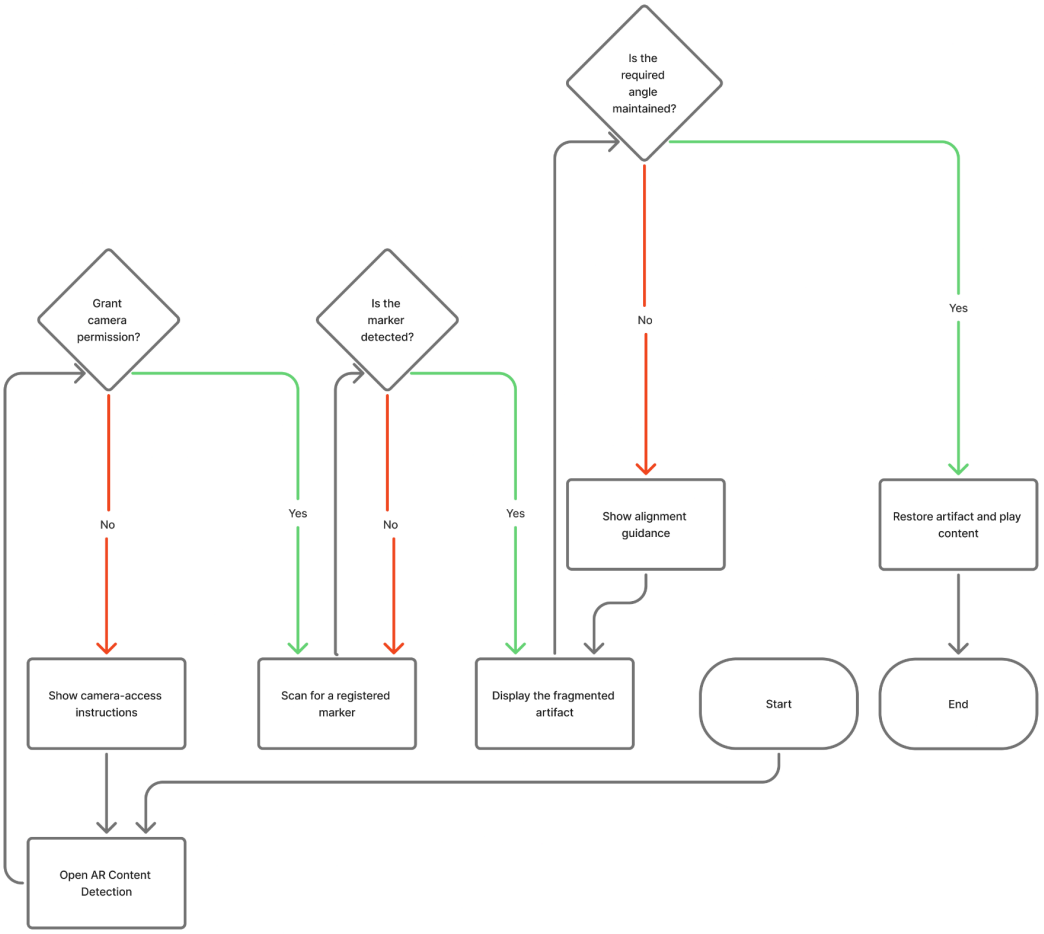


Figure A27. Augmented-reality artifact detection and restoration process.

Figure A27 shows how the visitor grants camera access, scans a registered marker, aligns the fragmented object, completes restoration, and optionally opens the supported real-world AR viewer.

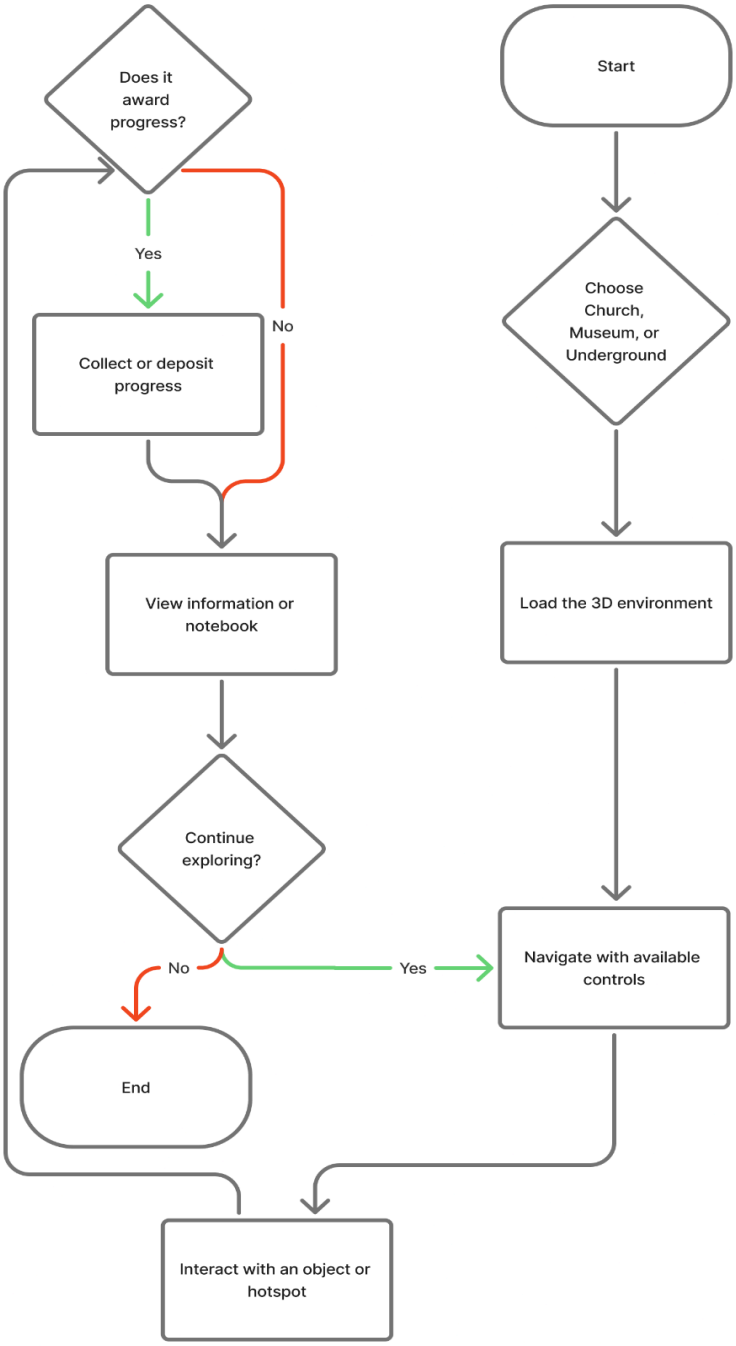


Figure A28. WebXR-enabled 3D environment exploration process.

Figure A28 shows how a visitor loads a heritage environment, navigates with supported controls, interacts with objects, records collectibles or achievements, and optionally enters an immersive XR mode.

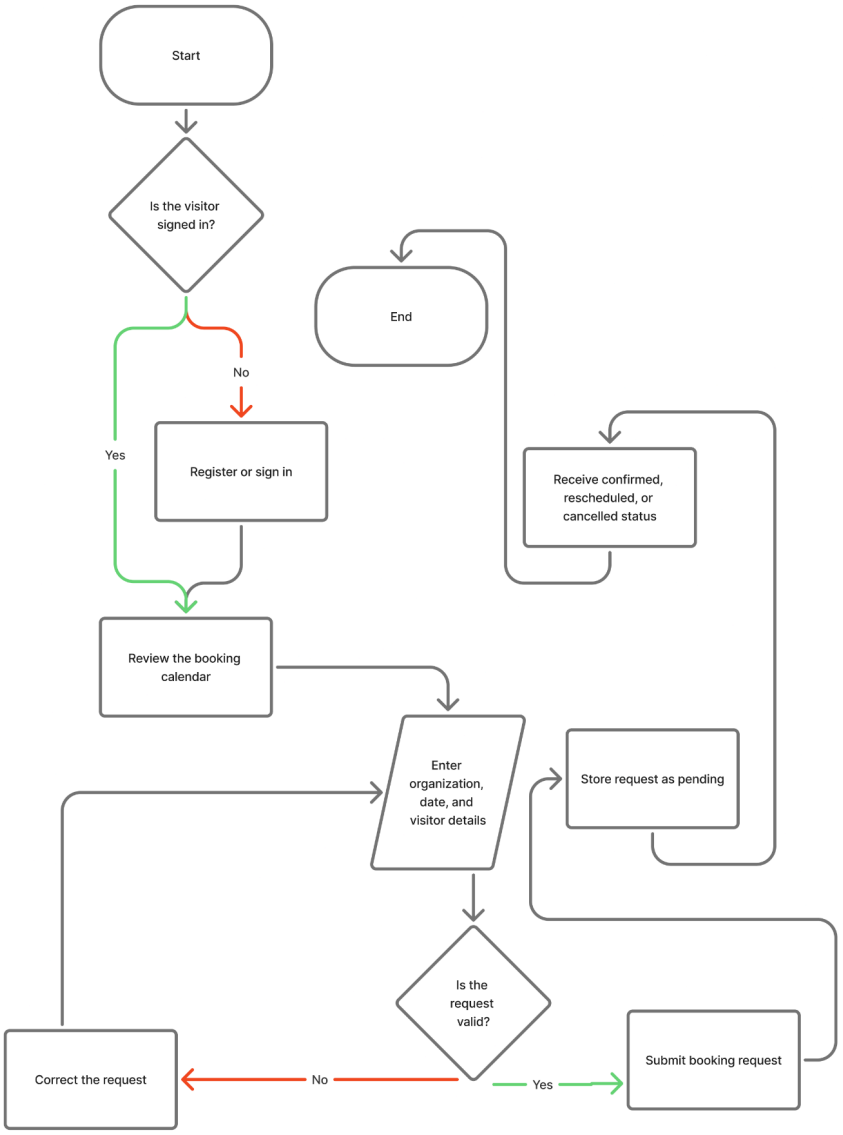


Figure A29. Guided-visit booking request and schedule-status process.

Figure A29 shows how a registered visitor checks the public calendar, submits a valid booking request, receives a pending status, and later views the museum's scheduling decision.

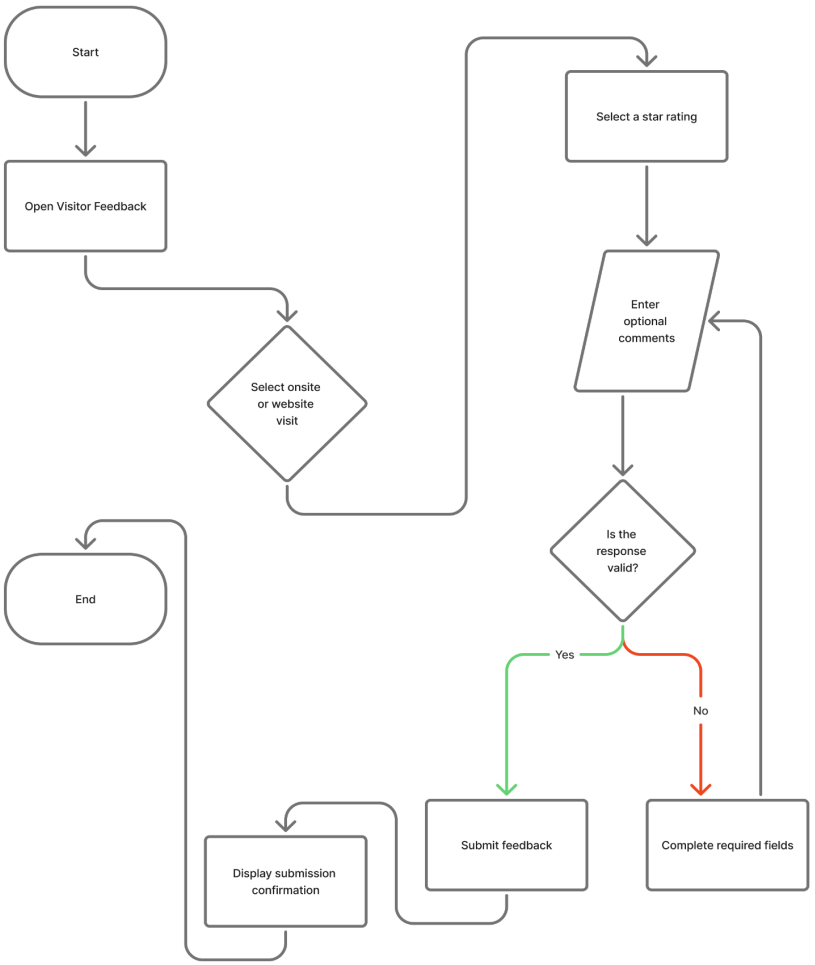


Figure A30. Visitor feedback rating and submission process.

Figure A30 shows how a registered visitor selects the visit channel and rating, supplies optional comments, submits the response, and receives confirmation.

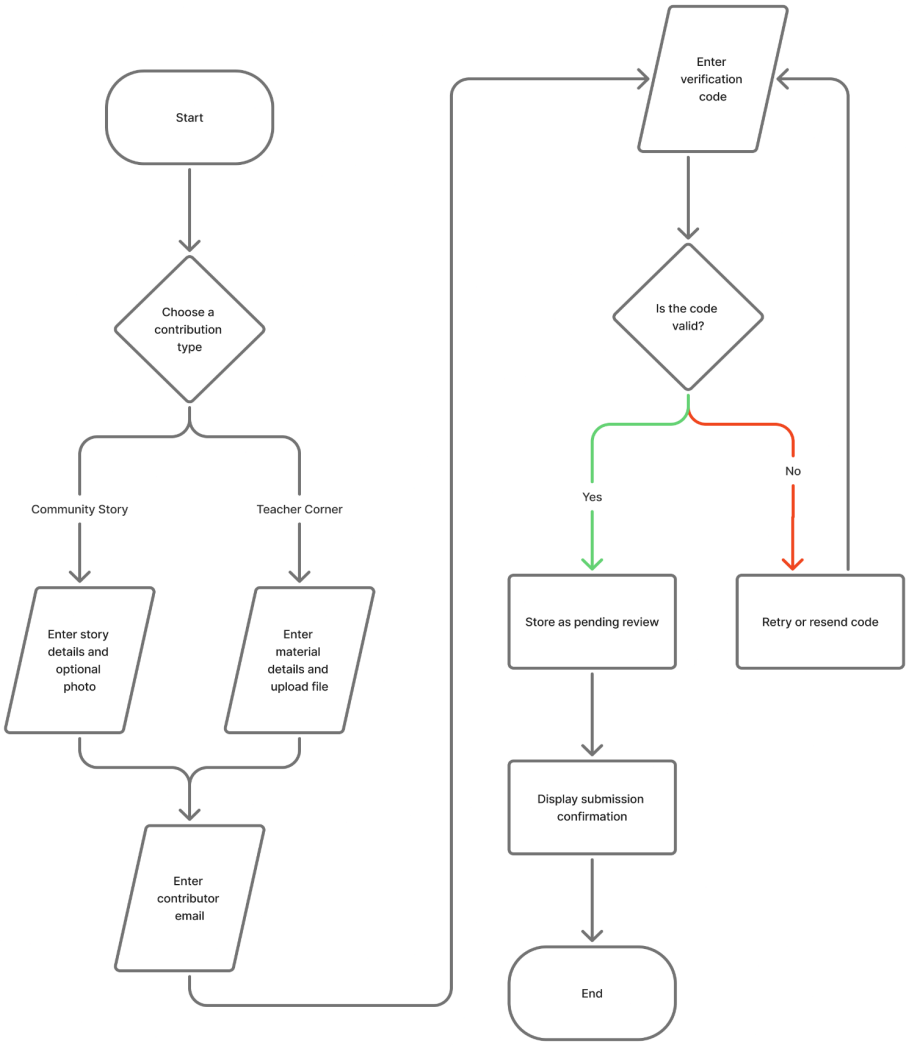


Figure A31. Educational-material and community-story submission process.

Figure A31 shows how a contributor chooses a submission type, provides the required content and file or photo, verifies the email address, and sends the contribution to the administrative review queue.

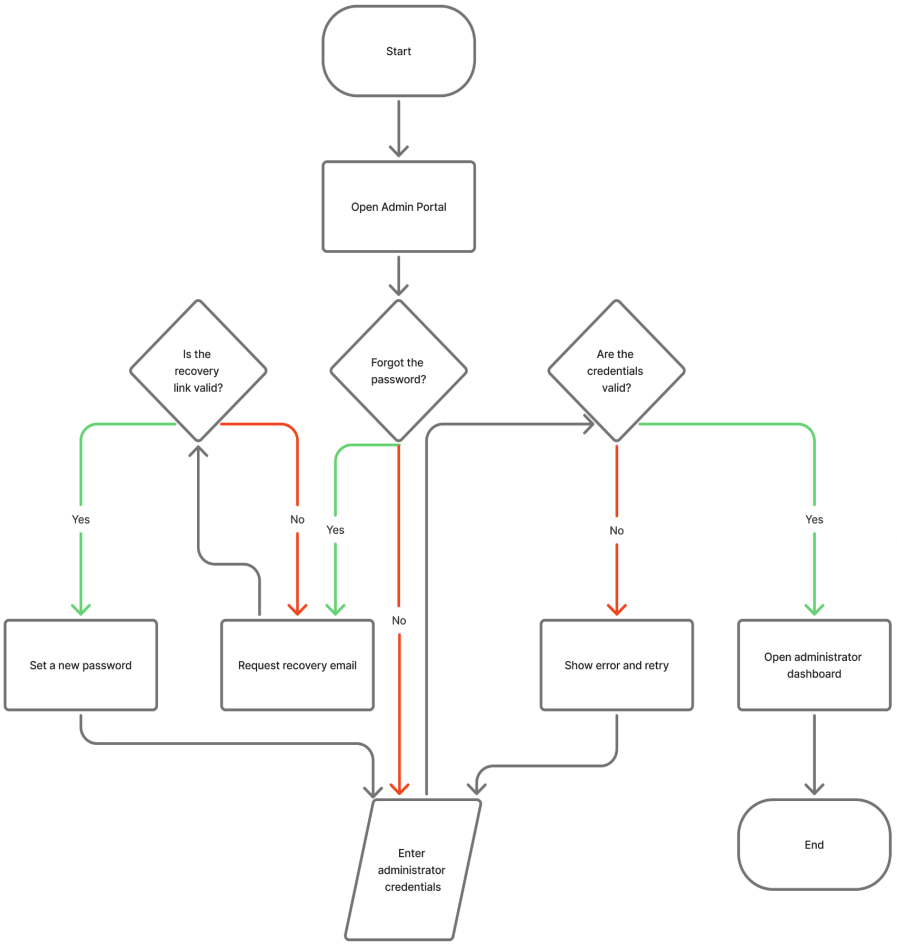


Figure A32. Administrator authentication and password-recovery process.

Figure A32 shows how authorized personnel sign in to the protected portal or request a time-limited recovery link before setting a new password and returning to the login process.

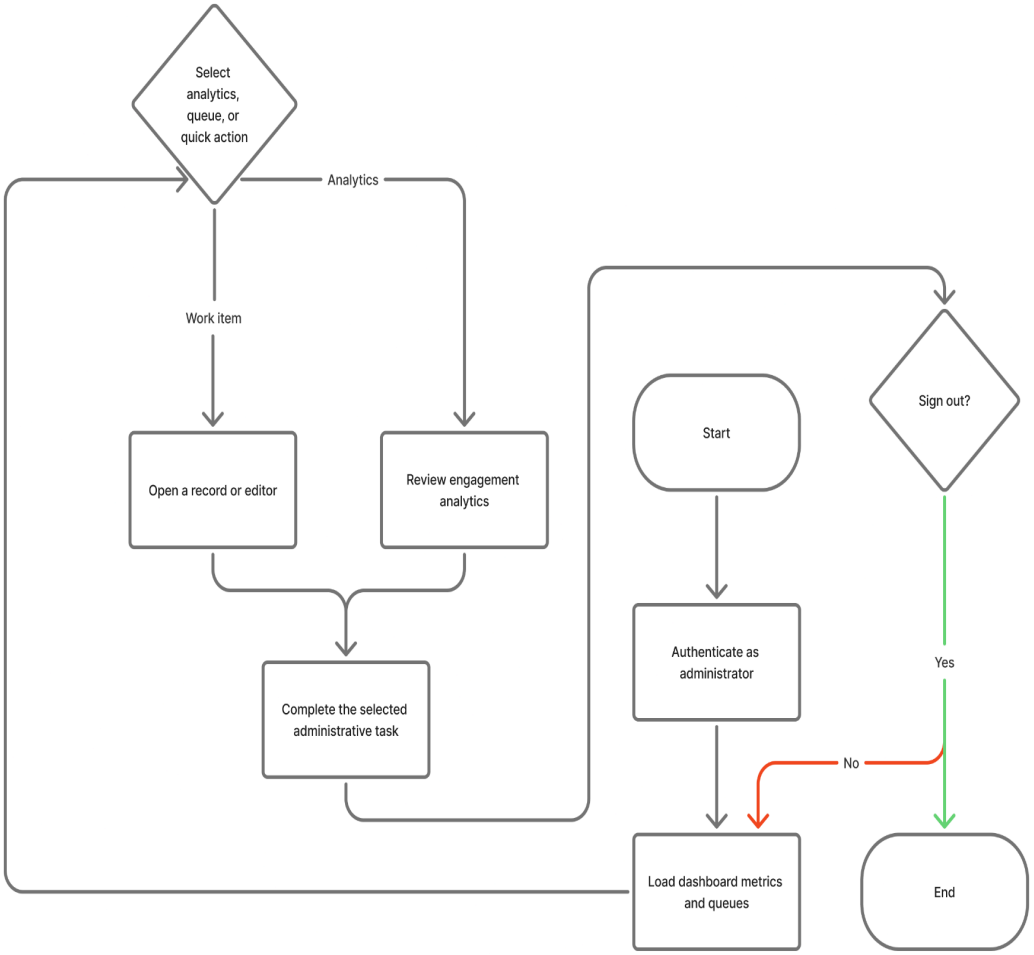


Figure A33. Administrator dashboard and review-workspace process.

Figure A33 shows how an authenticated administrator reviews platform indicators, opens a queue or quick action, completes the selected task, and returns to the dashboard or signs out.

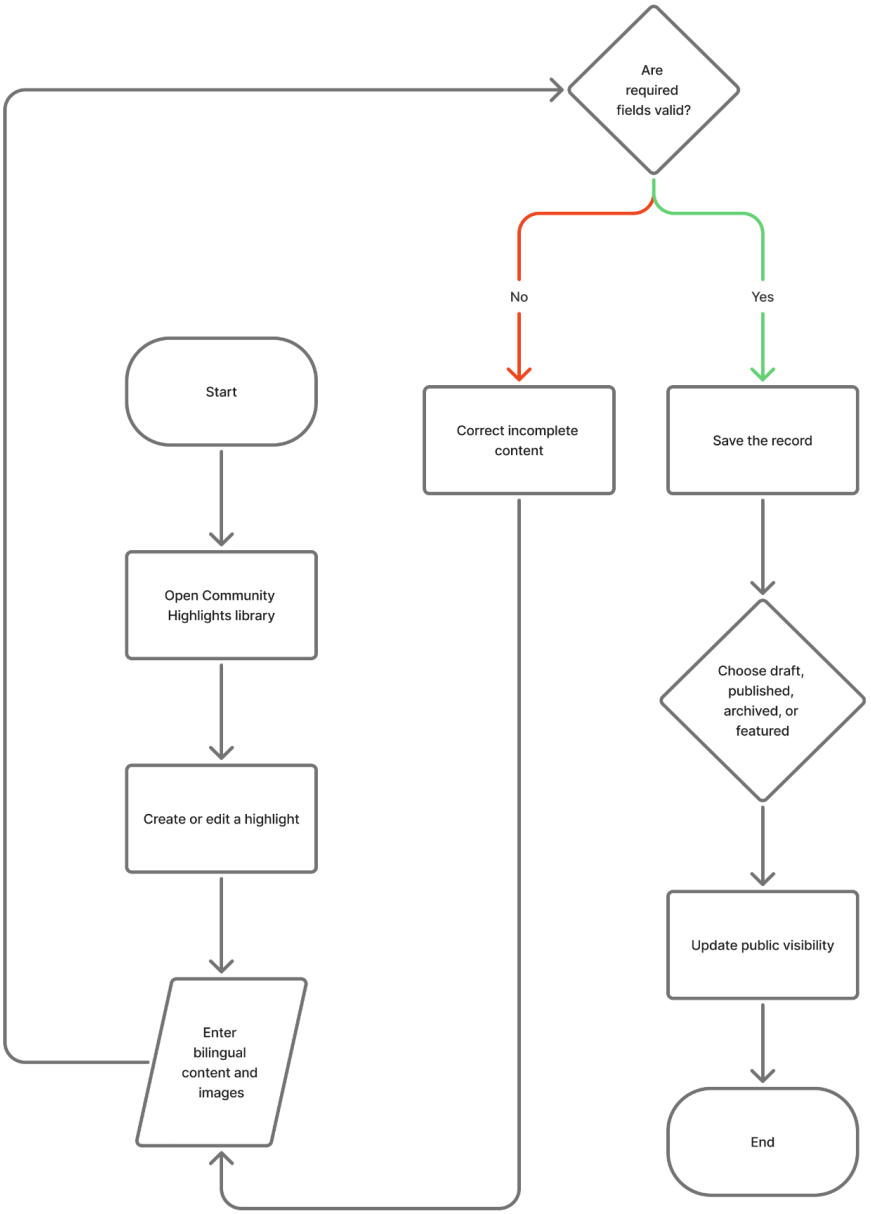


Figure A34. Community Highlight creation, publication, and public-display process.

Figure A34 shows how an administrator creates or edits a bilingual highlight, validates its fields and media, saves the record, controls publication or featured status, and verifies the public result.

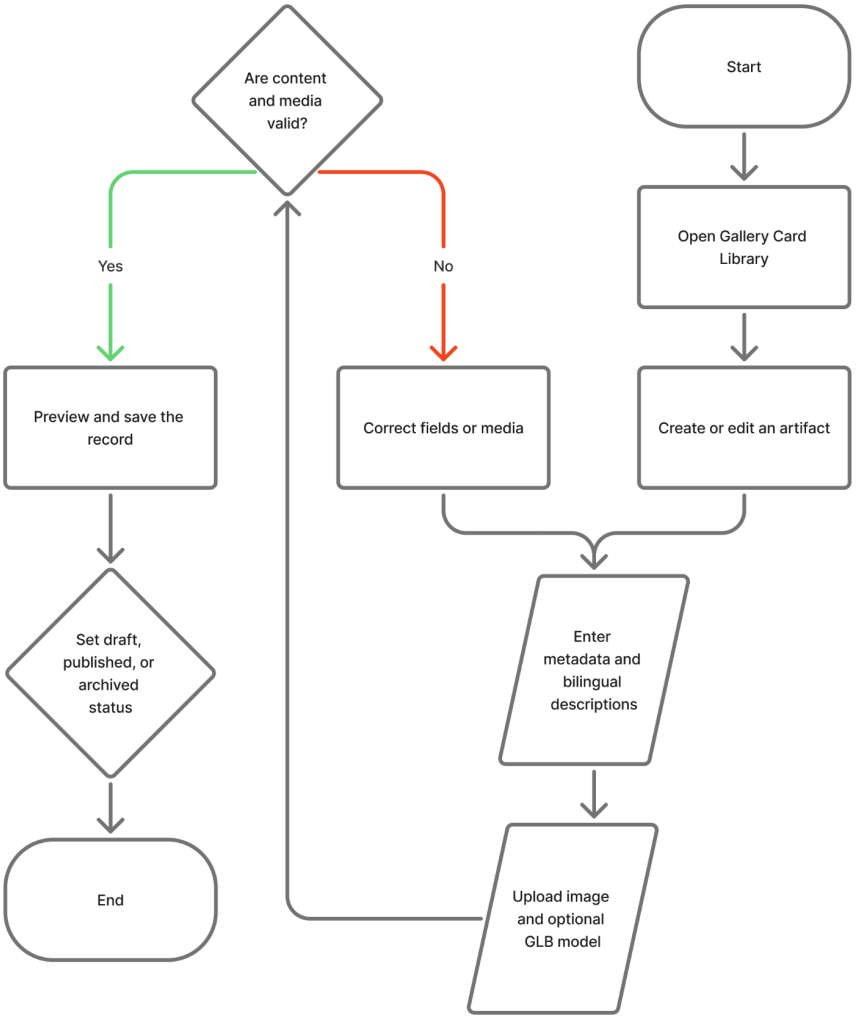


Figure A35. Online Gallery artifact creation and publication process.

Figure A35 shows how an administrator enters bilingual artifact information, uploads and validates media, previews the record, saves its status, and makes a published artifact available to visitors.

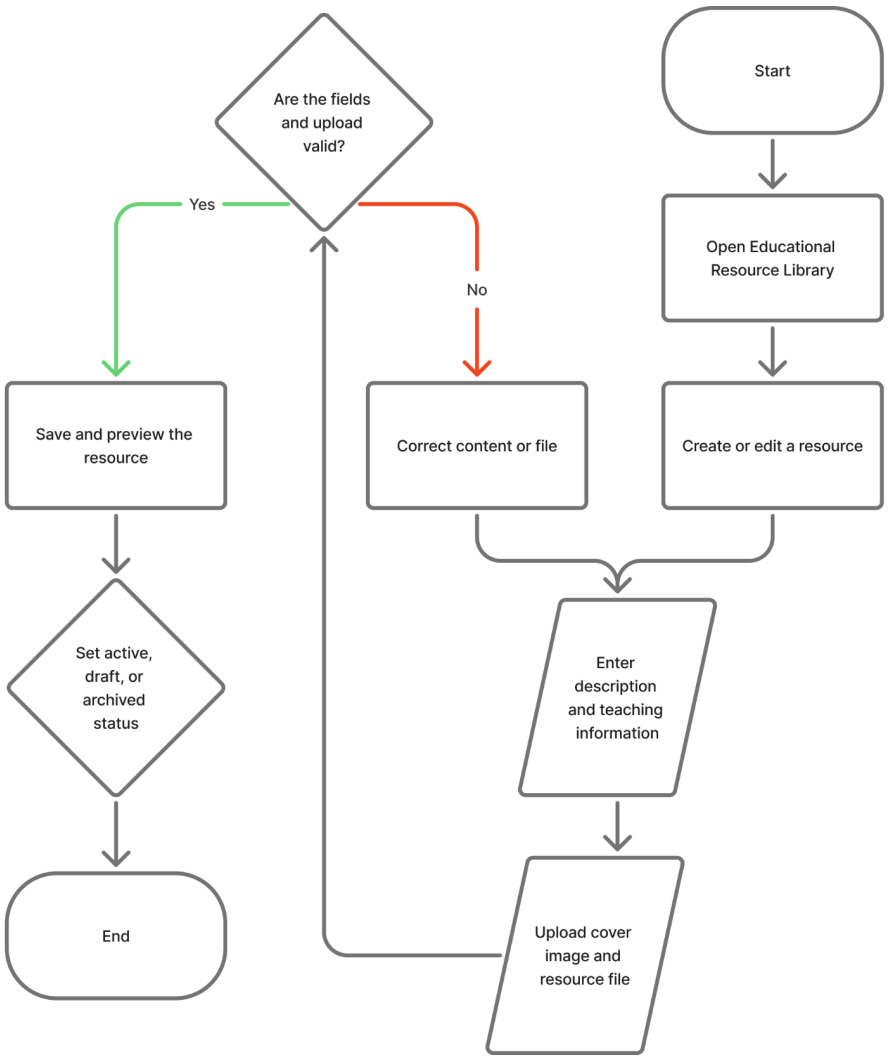


Figure A36. Educational Resource creation and publication process.

Figure A36 shows how an administrator supplies a learning resource and its metadata, validates the upload, saves and previews the record, and controls whether it is active, draft, or archived.

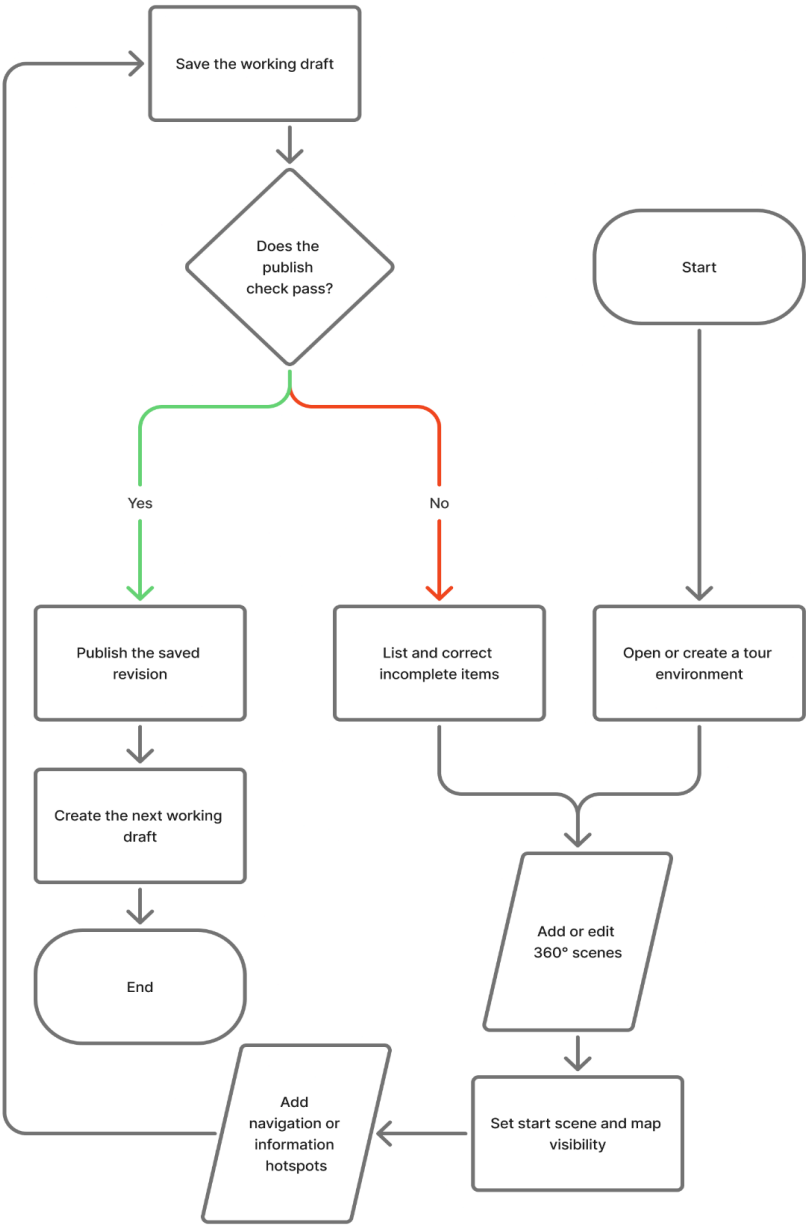


Figure A37. Virtual Tour environment, scene, hotspot, and publication process.

Figure A37 shows how an administrator edits a working revision, adds validated panoramic scenes and linked hotspots, saves all changes, resolves publish-check errors, and publishes an atomic public revision.

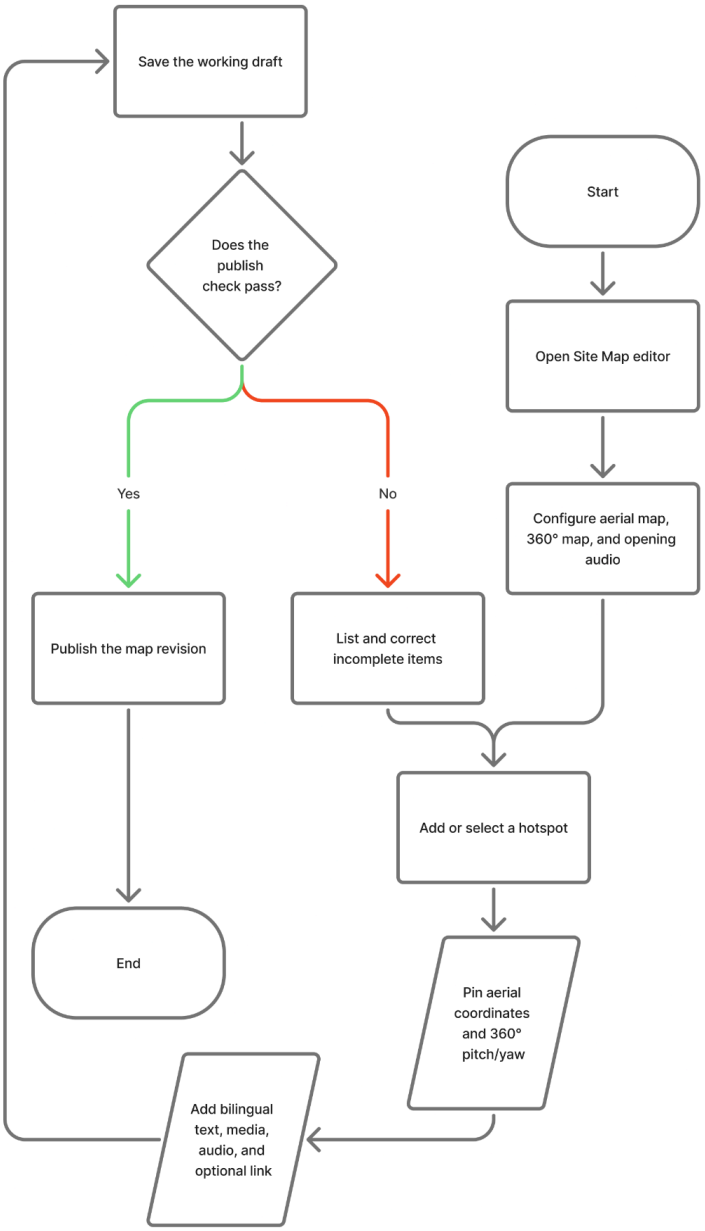


Figure A38. Interactive Site Map configuration and publication process.

Figure A38 shows how an administrator edits the database-backed map draft, configures maps and opening audio, positions related aerial and 360-degree hotspots, saves changes, validates the configuration, and publishes it.

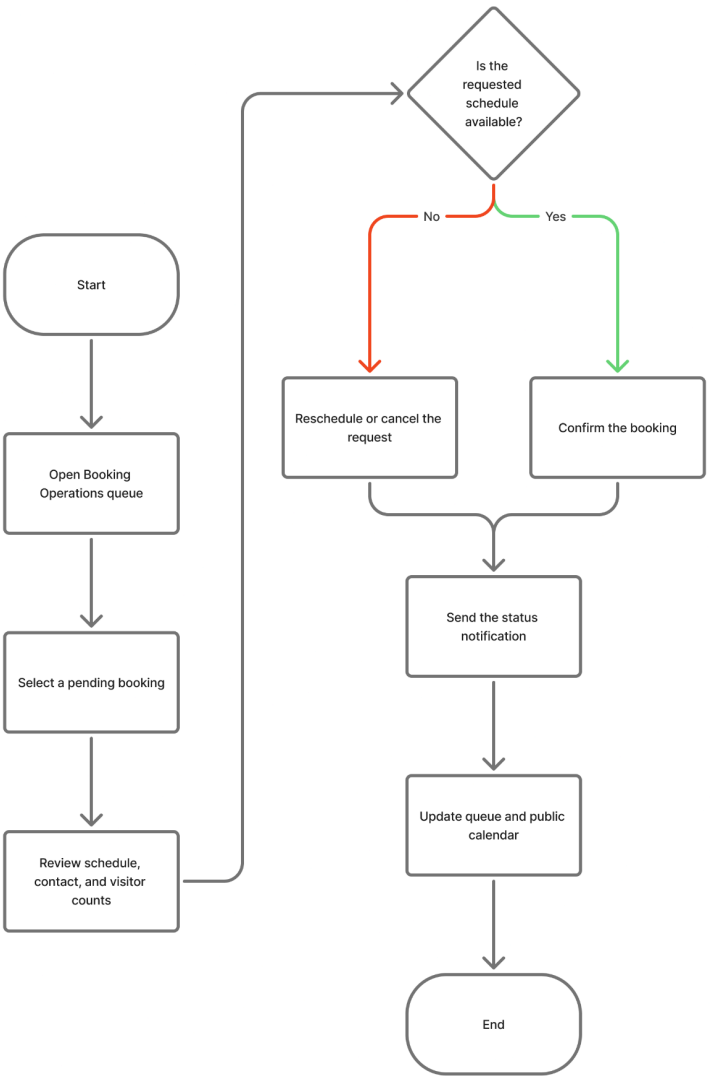


Figure A39. Guided-visit booking review and administrative-decision process.

Figure A39 shows how an administrator opens a pending request, checks its details and schedule, confirms, reschedules, or cancels it, and updates both the requester and booking calendar.

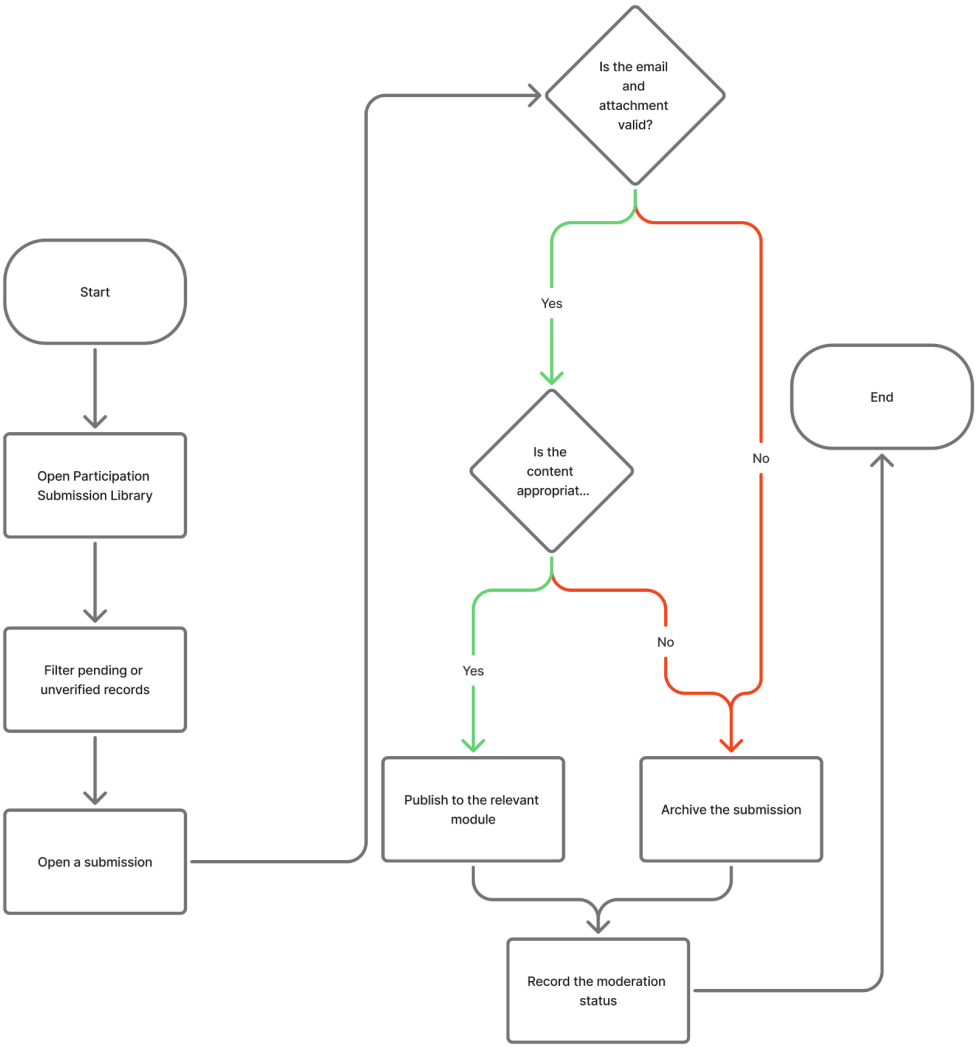


Figure A40. Participation-submission review and moderation process.

Figure A40 shows how an administrator filters the contribution queue, verifies the submission and attachment, decides whether it is appropriate for publication, and records a published or archived status.

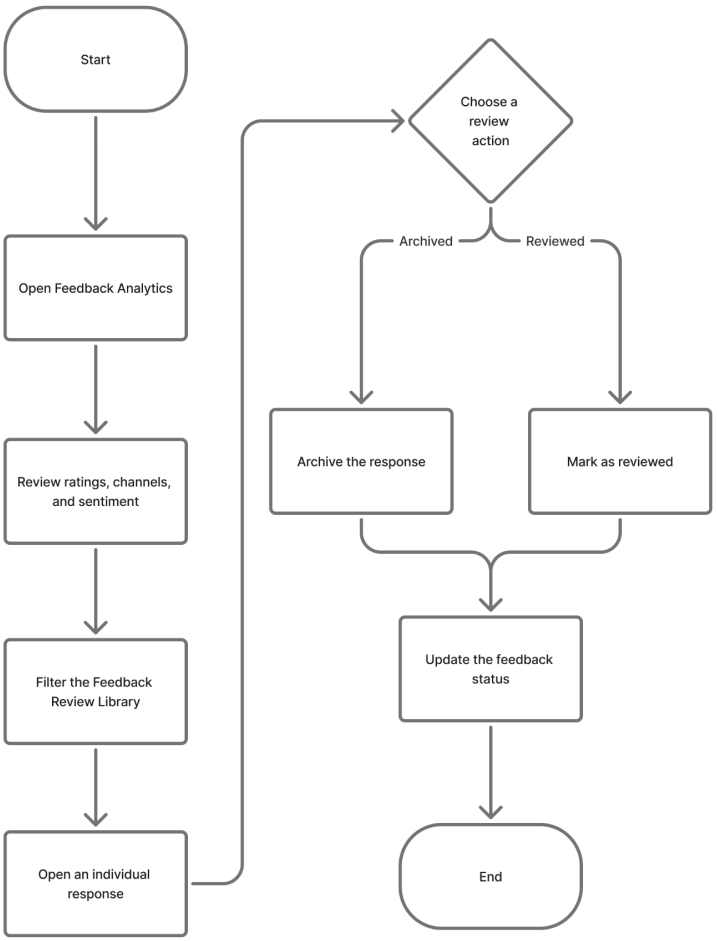


Figure A41. Visitor-feedback analytics and review process.

Figure A41 shows how an administrator examines rating and sentiment summaries, filters feedback records, reviews a response, and marks it reviewed or archived for service-improvement tracking.

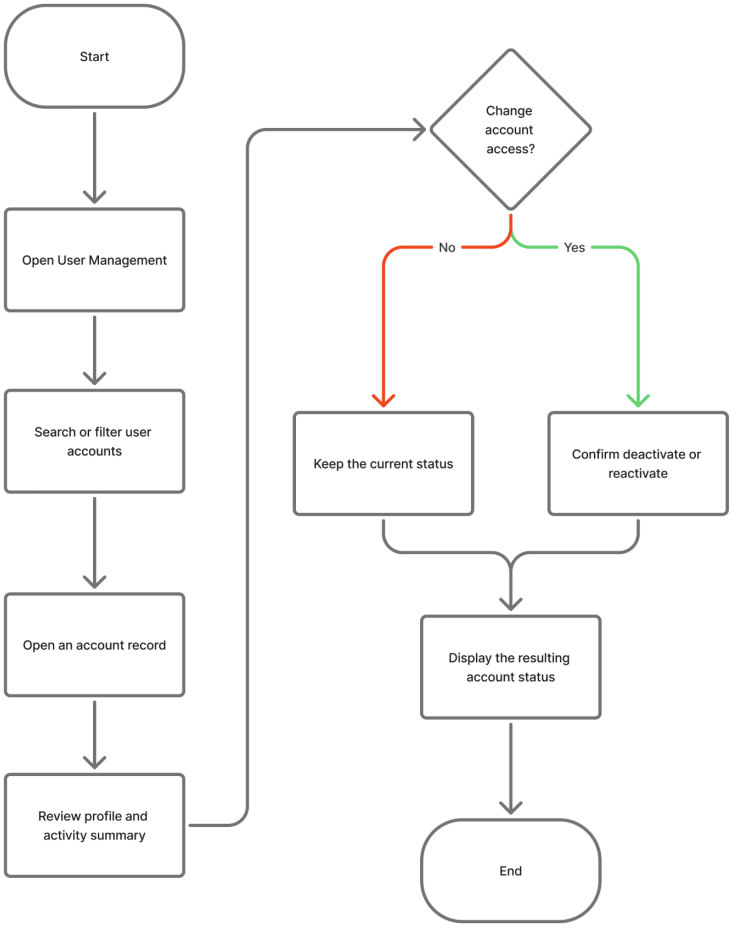


Figure A42. Public-user account review and access-management process.

Figure A42 shows how an administrator locates a registered visitor, reviews account information and activity, and keeps the account active or changes its access status after confirmation.

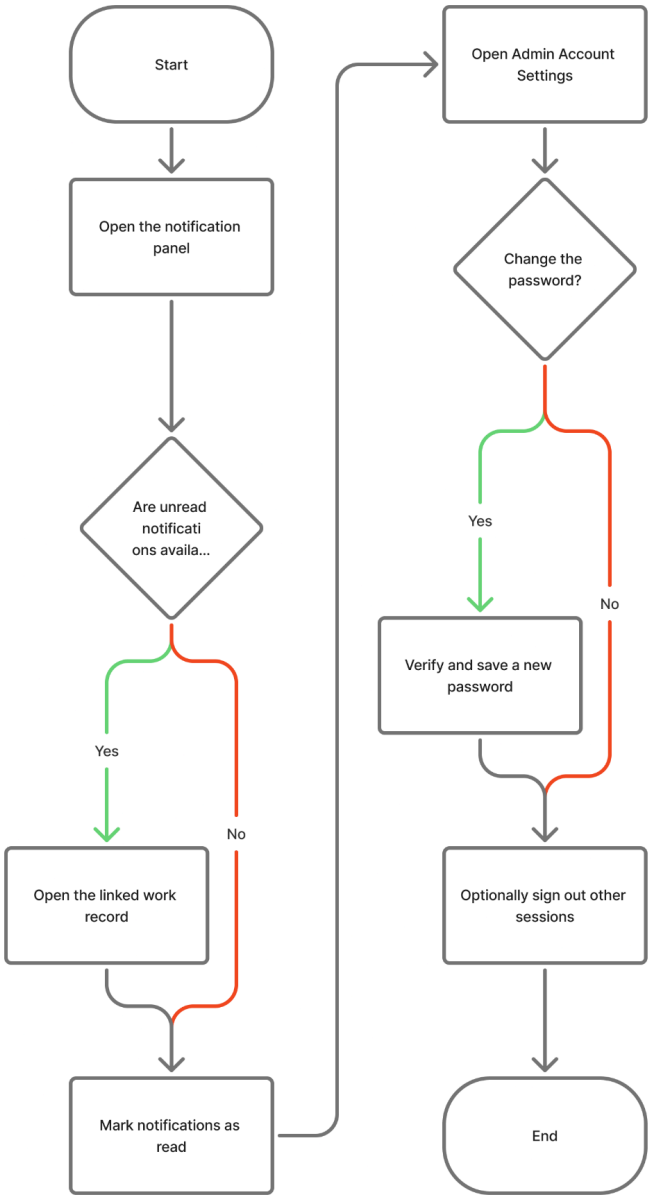


Figure A43. Administrator notifications and account-security process.

Figure A43 shows how an administrator follows a new-work notification, marks notifications as read, changes the administrator password when needed, and invalidates other sessions before signing out.

Use Case Diagram

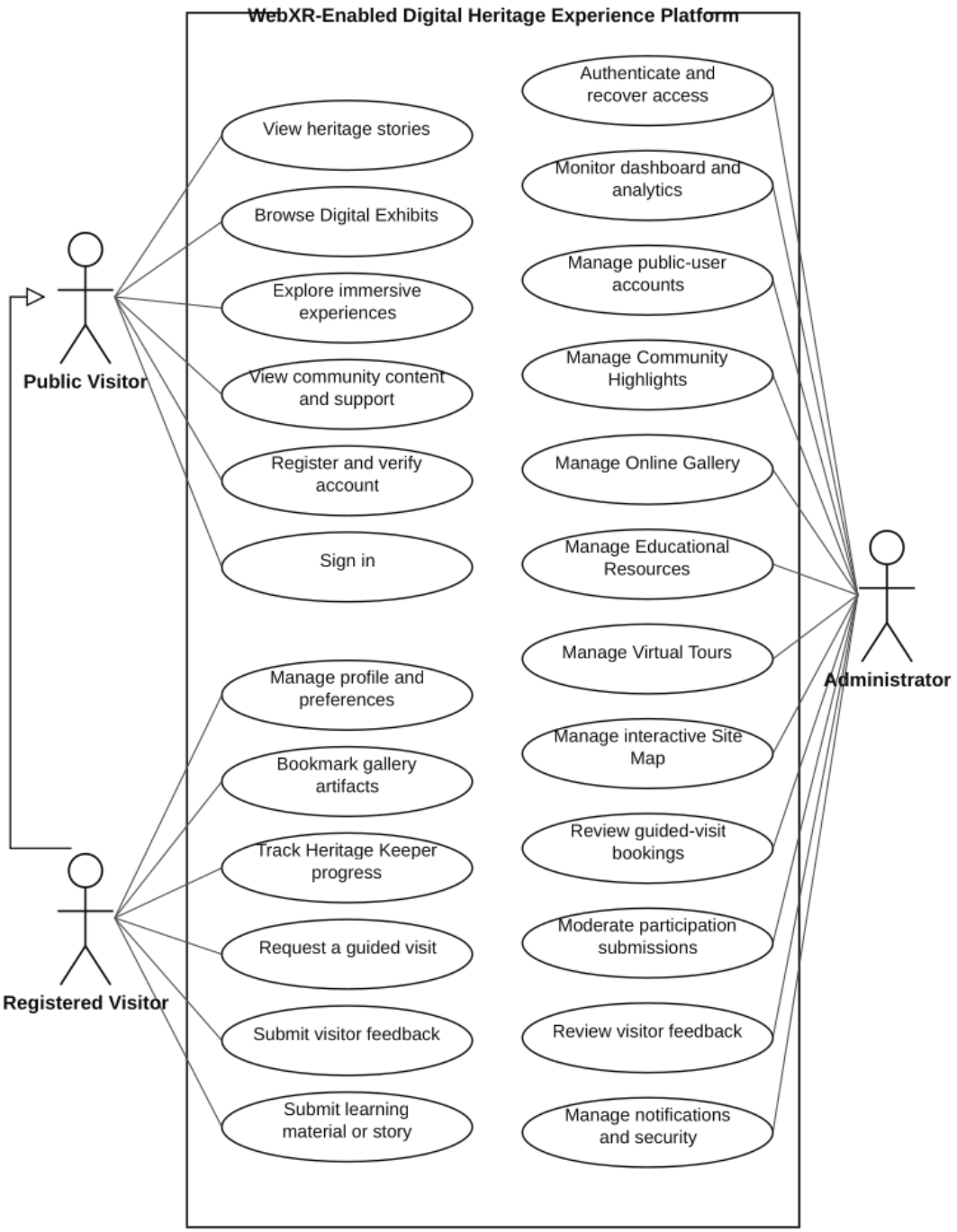


Figure A44. Use Case Diagram of the Platform

Figure A44 shows how Public Visitors access heritage, exhibit, immersive, and community services; Registered Visitors inherit those capabilities and use account-based bookmarking,progress, booking, feedback, and contribution functions;

and Administrators securely manage users, content, immersive modules, analytics, bookings, submissions, feedback, notifications,and account security.

Specification and Design Phase:

This phase presents the technical design of the proposed system before and during development. It includes the system architecture and module specifications that describe the major components and functions of the platform, the system context diagram that shows the interaction between the system and its external users or entities, the data-flow diagram that illustrates how information moves through the system, the data dictionary that defines the data used and stored, and the Hierarchical Input-Process-Output (HIPO) model that shows how system inputs are processed to produce the required outputs.

System Architecture

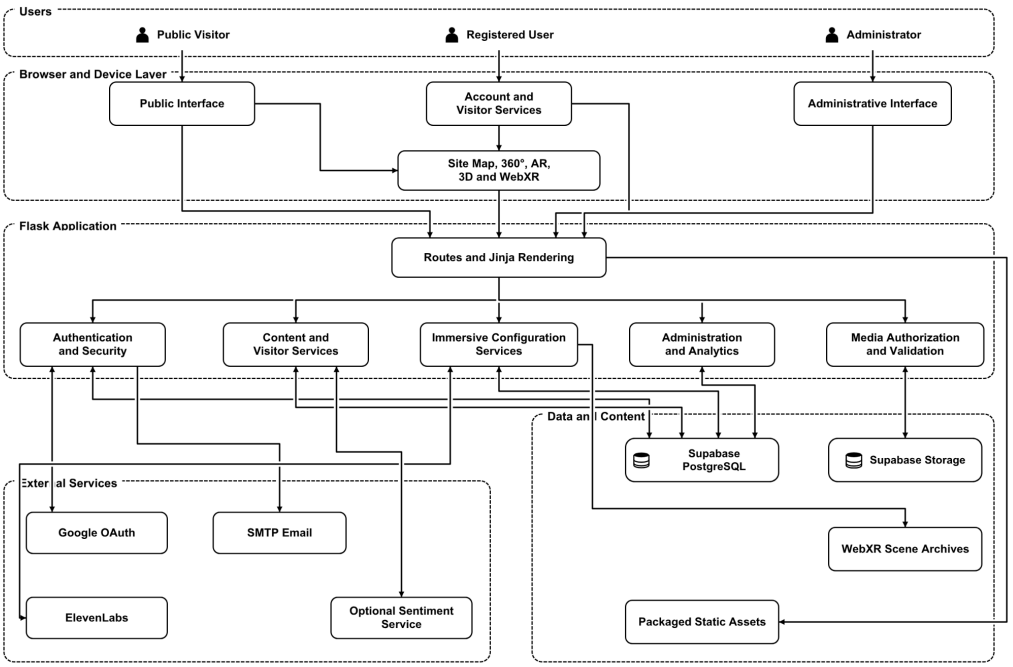


Figure A45. Overall System Architecture

Figure A45 illustrates the overall architecture of the WebXR-Enabled Digital Heritage Platform. Public and registered users access the public website to explore museum information, digital exhibits, educational resources, and immersive experiences, while authorized museum administrators use the secured admin portal to manage users, content, bookings, feedback, and other museum records. Both

interfaces communicate with the Flask application hosted on Vercel, which processes requests, applies system rules and security controls, and connects the platform to Supabase. The Supabase Database stores structured records, while Supabase Storage maintains multimedia files such as images, audio, videos, 360-degree panoramas, and 3D models. The application also communicates with external services for functions such as Google account sign-in, email verification and notifications, and ElevenLabs-generated narration. The arrows represent the exchange of information among users, the platform, and its connected data services.

System Context Diagram

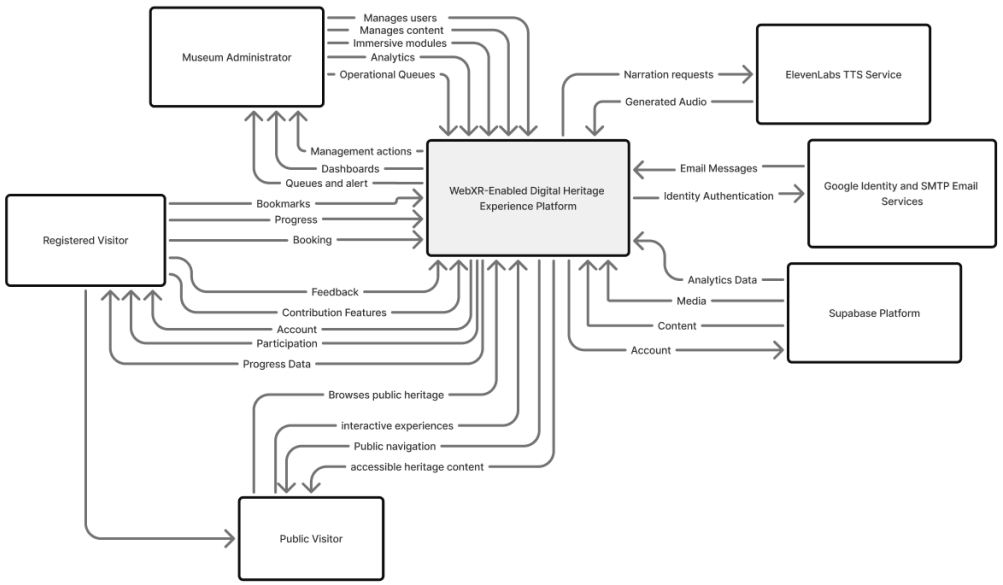


Figure A46. System Context Diagram of the WebXR-Enabled Digital Heritage Platform.

Figure A46 shows the platform's boundary and its relationships with Public Visitors, Registered Visitors, Museum Administrators, Google identity and SMTP email services, the Supabase platform, and the ElevenLabs text-to-speech service. The labeled two-way connections summarize the public content, account, participation, administration, identity, messaging, data, media, analytics, and narration information exchanged across the system boundary.

Data Flow Diagram

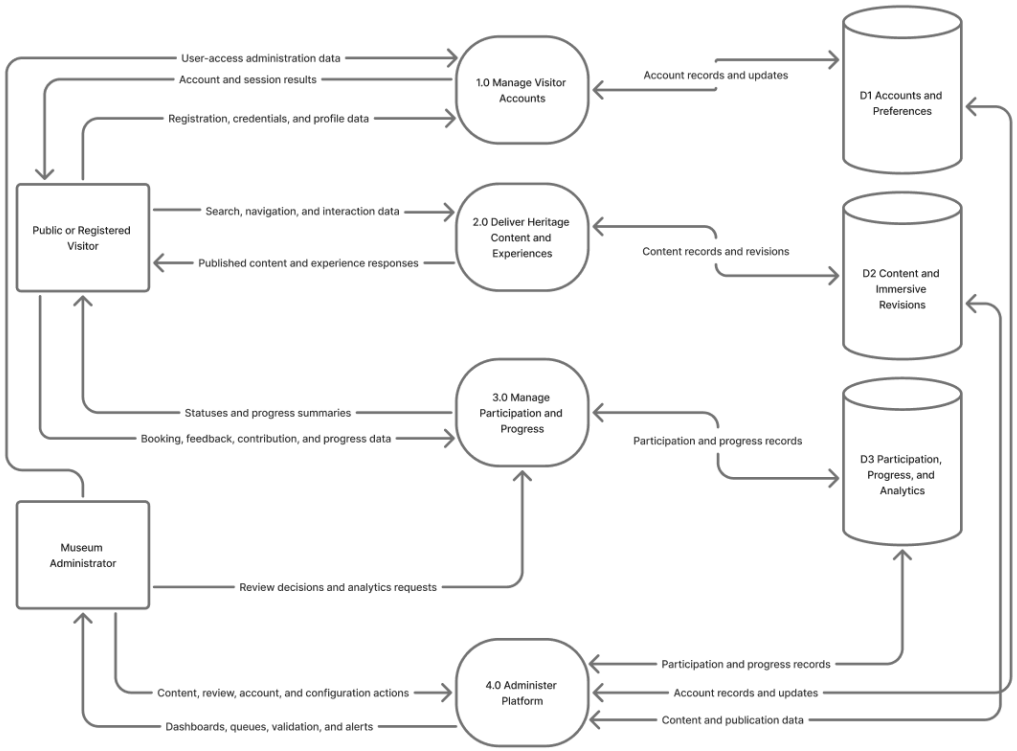


Figure A47. Level 1 data flow diagram of the WebXR-Enabled Digital Heritage Platform.

Figure A47 summarizes how account access, heritage experiences, participation and progress,and platform administration transform visitor and administrator data and read or update the system's account, content, and operational data stores.

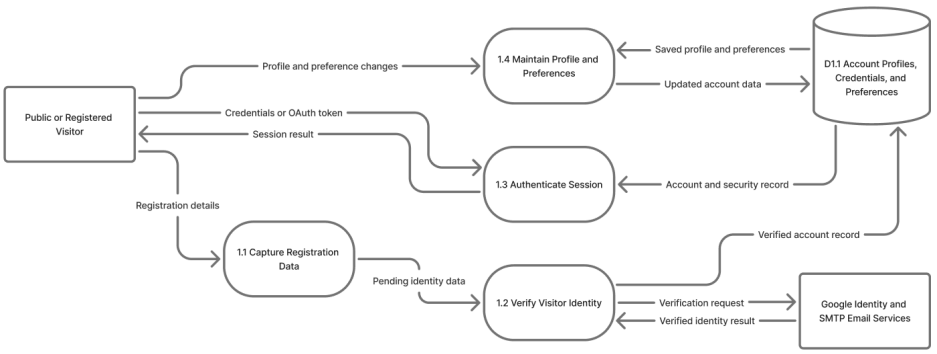


Figure A48. Visitor registration, identity verification, authentication, and profile data flow.

Figure A48 shows how visitor account details are validated through email or Google identity services, stored as account records, used to create an authenticated session, and updated when profile or preference data changes.

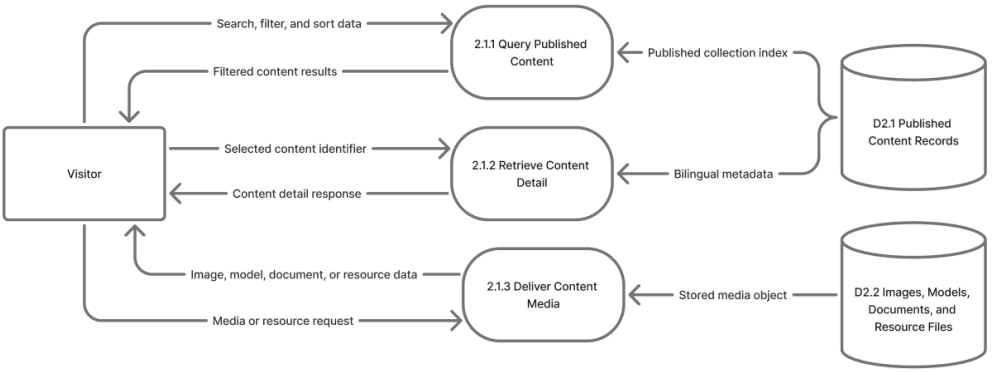


Figure A49. Online Gallery and Educational Resources discovery data flow.

Figure A49 shows how visitor search and selection data are transformed into published collection results, bilingual content details, and the corresponding image, model, document, or learning-resource media.

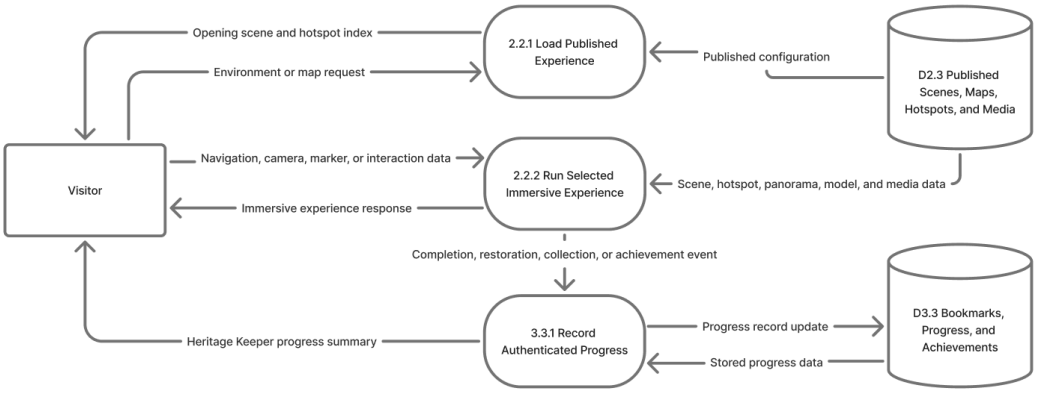


Figure A50. Virtual Tour, Site Map, Heritage Trail, augmented-reality, and WebXR data flow.

Figure 50 shows how visitor environment and interaction data retrieve published scenes, map configuration, hotspots, media, and models, while authenticated completion and achievement events are transferred to the progress store.

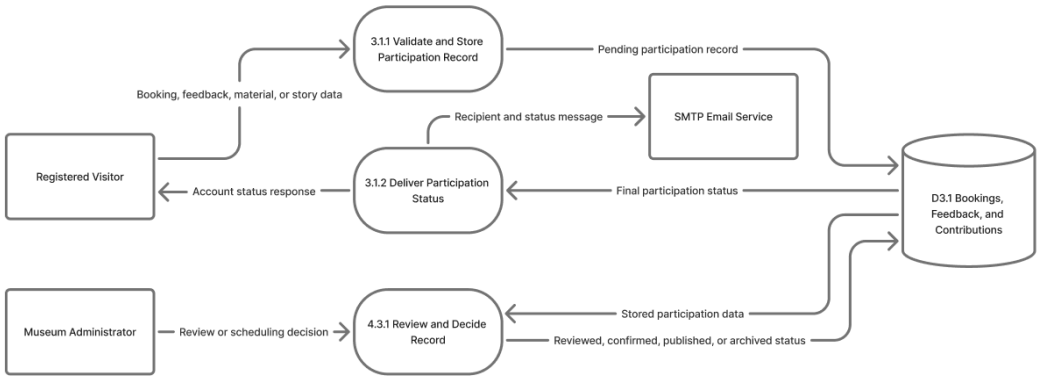


Figure A51. Guided-visit booking, feedback, contribution, review, and notification data flow.

Figure A51 shows how authenticated participation data become stored pending records, how administrators retrieve and decide those records, and how the system returns account or email status information to the registered visitor.

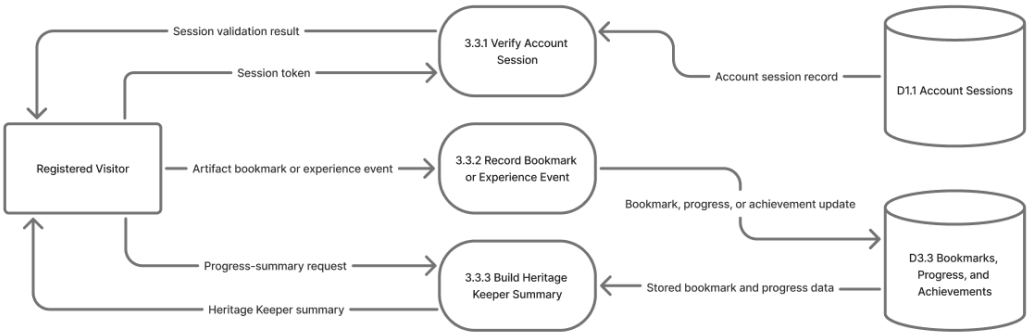


Figure A52. Bookmark, immersive-progress, achievement, and Heritage Keeper data flow.

Figure A52 shows how the registered visitor's session is verified before bookmark or experience data are stored and how the consolidated bookmark, restoration, trail, WebXR, and achievement records are returned as a Heritage Keeper summary.

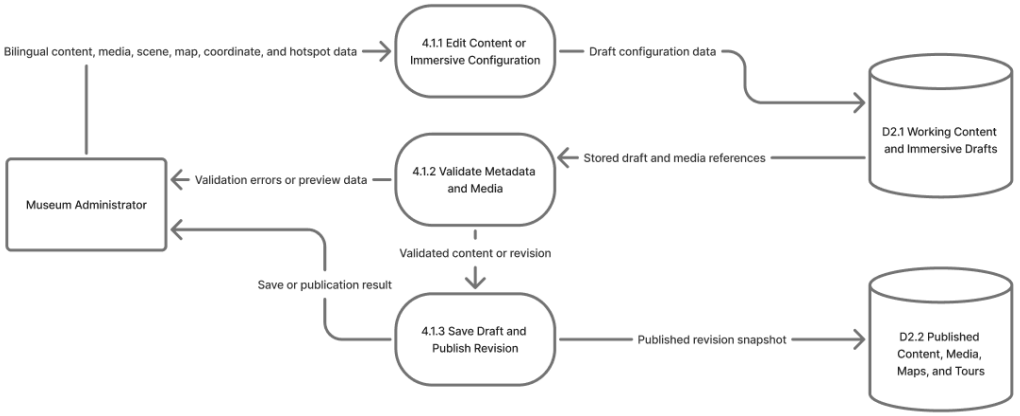


Figure A53. Public and immersive content editing, validation, revision, and publication data flow.

Figure A53 shows how administrator-supplied bilingual metadata, media, scenes, maps,coordinates, and hotspots are validated, stored in a working draft, and promoted to the public content and immersive-revision store after a successful publish check.

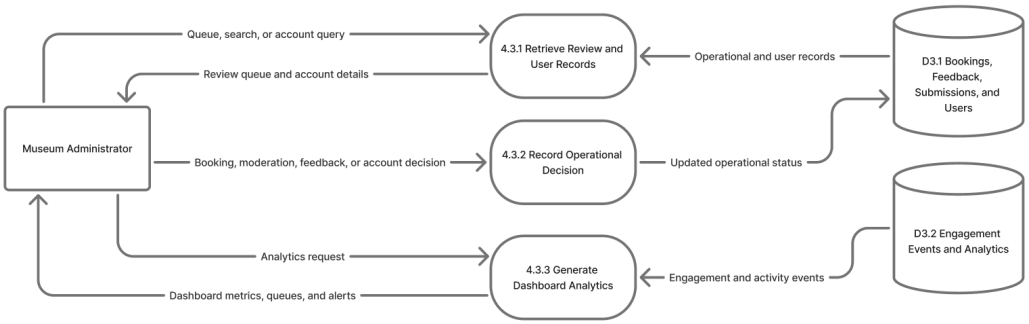


Figure A54. Administrative review, public-user oversight, and analytics dataflow.

Figure A54 shows how the administrator requests operational queues and analytics, receives booking, submission, feedback, account, and engagement data, records review or access decisions, and receives updated dashboard and notification results.

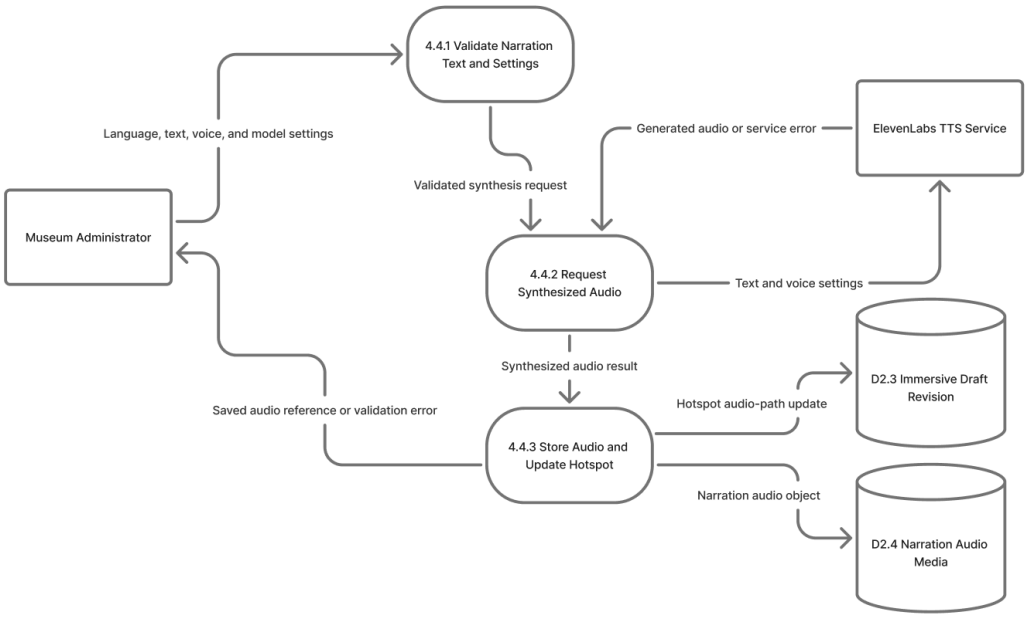


Figure A55. Text-to-speech narration generation and storage data flow.

Figure A55 shows how narration text, language, voice, and model settings are validated and sent to ElevenLabs and how the generated audio object and path are stored in the working immersive revision for later publication.

Data Dictionary

This section presents the data dictionary of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. Each appendix table identifies the final field name, PostgreSQL data type, principal key or nullability constraint, and functional meaning of the field.

Table A1. Data Dictionary for admin_users

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
email	TEXT	Unique; Required	Normalized email address used by the account.
password_hash	TEXT	Required	One-way password hash used to authenticate the account.
is_active	BOOLEAN	Required	Indicates whether the account or category is available for use.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A1 stores administrator credentials and account availability for the protected management portal.

Table A2. Data Dictionary for admin_notifications

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
recipient_email	TEXT	FK → admin_users.email; Composite unique; Required	Email address of the administrator receiving the notification.
event_type	TEXT	Composite unique; Required	Application event that generated the notification or analytics record.
source_id	BIGINT	Composite unique; Required	Identifier of the source record that generated the notification.
title	TEXT	Required	Primary title of the record.
message	TEXT	Required	Short notification message shown to the administrator.
target_url	TEXT	Required	Protected administration page opened from the notification.
read_at	TIMESTAMPTZ	Optional	Date and time when the notification was marked as read.
emailed_at	TIMESTAMPTZ	Optional	Date and time when the notification email was sent.
email_error	TEXT	Required	Last email-delivery error recorded for the notification.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.

Table A2 stores the persistent notification inbox created for each administrator from visitor and community activity.

Table A3. Data Dictionary for admin_password_reset_requests

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
admin_user_id	BIGINT	FK → admin_users.id; Optional	Identifier of the related administrator account.
email_hash	TEXT	Required	HMAC digest of the submitted recovery email address.
request_ip_hash	TEXT	Required	HMAC digest of the client address used for recovery rate limiting.
requested_at	TIMESTAMP TZ	Required	Date and time when the recovery request was submitted.

Records privacy-preserving password-recovery requests for rate limiting and security review.

Table A4. Data Dictionary for admin_password_reset_tokens

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
admin_user_id	BIGINT	FK → admin_users.id; Required	Identifier of the related administrator account.
token_hash	TEXT	Unique; Required	One-way hash of the single-use recovery token.
expires_at	TIMESTAMPTZ	Required	Date and time after which the token is no longer valid.
used_at	TIMESTAMPTZ	Optional	Date and time when the token was consumed; null while unused.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.

Table A4 stores single-use hashed recovery tokens issued to administrator accounts.

Table A5. Data Dictionary for public_users

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
email	TEXT	Unique; Required	Normalized email address used by the account.
username	TEXT	Unique; Required	Unique public username used for account identification.
password_hash	TEXT	Required	One-way password hash used to authenticate the account.
full_name	TEXT	Required	Full display name of the public user.
phone_number	TEXT	Required	Contact telephone number supplied by the user.
organization	TEXT	Required	Organization or institution supplied in the user profile.
is_active	BOOLEAN	Required	Indicates whether the account or category is available for use.
last_login_at	TIMESTAMPTZ	Optional	Date and time of the user's most recent successful sign-in.
email_verified_ at	TIMESTAMPTZ	Optional	Date and time when the user's email address was verified.
email_verificati on_token_h ash	TEXT	Optional	Hash of the current email-verification token.
email_verificati on_expires _at	TIMESTAMPTZ	Optional	Expiration date and time of the email-verification token.
email_verificati on_sent_at	TIMESTAMPTZ	Optional	Date and time when the latest verification message was sent.
email_verificati on_attempt s	INTEGER	Required	Number of failed attempts made against the current verification code.
onboarding_co mpleted_at	TIMESTAMPTZ	Optional	Date and time when the user completed profile onboarding.
website_use_ca se	TEXT	Required	User-selected reason for using the heritage platform.
profile_roles	TEXT	Required	Serialized list of roles or visitor types selected during onboarding.
location_text	TEXT	Required	Free-text location originally supplied by the user.

country_text	TEXT	Required	Country saved in the user's profile.
province_text	TEXT	Required	Province saved in the user's profile.
city_text	TEXT	Required	City or municipality saved in the user's profile.
auth_provider	TEXT	Required	Authentication method associated with the account.
oauth_subject	TEXT	Unique; Optional	Unique external identity subject supplied by the OAuth provider.
local_password_enabled	BOOLEAN	Required	Indicates whether password-based sign-in is enabled for the account.
password_reset_token_hash	TEXT	Optional	Hash of the user's active password- reset token.
password_reset_expires_at	TIMESTAMPTZ	Optional	Expiration date and time of the user's password-reset token.
security_code_hash	TEXT	Optional	Hash of the temporary account- security verification code.
security_code_expires_at	TIMESTAMPTZ	Optional	Expiration date and time of the temporary security code.
security_code_sent_at	TIMESTAMPTZ	Optional	Date and time when the latest security code was sent.
security_code_attempts	INTEGER	Required	Number of failed attempts made against the current security code.
profile_picture	TEXT	Optional	External image URL or storage object path for the user's profile picture.
theme_preference	TEXT	Required	Preferred interface theme: light or dark.
language_preference	TEXT	Required	Preferred interface language: English or Tagalog.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A5 stores public visitor accounts, authentication state, profile information, onboarding data, and interface preferences.

Table A6. Data Dictionary for user_bookmarks

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Required	Identifier of the related public user.
item_type	TEXT	Required	Module type of the bookmarked content.
item_id	BIGINT	Required	Identifier of the bookmarked content record.
created_at	TIMESTAMP TZ	Required	Date and time when the record was created.

Table A6 links a public user to an online-gallery item saved for later viewing.

Table A7. Data Dictionary for tour_bookings

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Required	Identifier of the related public user.
display_title	TEXT	Required	User-facing title assigned to the booking.
scheduled_tour	TEXT	Required	Name of the selected museum tour or visit activity.
booking_type	TEXT	Required	Category of the visiting group or requested tour.
visit_date	DATE	Required	Requested date of the museum visit.
rescheduled_visit_date	DATE	Optional	Replacement visit date when the booking is rescheduled.
start_time	TEXT	Required	Requested starting time stored in display format.
end_time	TEXT	Required	Requested ending time stored in display format.

group_size	INTEGER	Required	Total group size entered when the booking was created.
institution_name	TEXT	Required	School, organization, or institution represented by the group.
contact_person	TEXT	Required	Primary contact person for the booking.
contact_number	TEXT	Required	Telephone number of the booking contact.
tour_agency	TEXT	Required	Tour agency associated with the visit, when applicable.
male_visitors	INTEGER	Required	Number of male visitors included in the booking.
female_visitors	INTEGER	Required	Number of female visitors included in the booking.
pwd_visitors	INTEGER	Required	Number of persons with disabilities included in the booking.
total_visitors	INTEGER	Required	Final total number of visitors included in the booking.
purpose	TEXT	Required	Purpose or objective stated for the museum visit.
notes	TEXT	Required	Additional information supplied with the booking.
location	TEXT	Required	Location associated with the booking or content item.
status	TEXT	Required	Current workflow or publication state of the record.
admin_notes	TEXT	Required	Internal notes recorded by an administrator.
decision_reason	TEXT	Required	Reason given for confirming, cancelling, or rescheduling the request.
reviewed_at	TIMESTAMPTZ	Optional	Date and time when an administrator reviewed the record.

reviewed_by	TEXT	Required	Email or identifier of the administrator who reviewed the record.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A7 stores museum tour requests, visitor counts, schedules, contact details, and administrative decisions.

Table A8. Data Dictionary for heritage_keeper_events

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Composite unique; Required	Identifier of the related public user.
source	TEXT	Composite unique; Required	Origin or provenance recorded for the item or event.
item_key	TEXT	Composite unique; Required	Stable application key for the completed or collected item.
completed_at	TIMESTAMPTZ	Required	Date and time when the milestone was completed.

Table A8 records unique learning or exploration milestones completed by a public user.

Table A9. Data Dictionary for heritage_keeper_awards

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
user_id	BIGINT	PK; FK → public_users.id; Required	Identifier of the related public user.
unlocked_at	TIMESTAMPTZ	Required	Date and time when the award was unlocked.

Table A9 records the date on which a public user unlocked the Heritage Keeper award.

Table A10. Data Dictionary for webxr_player_museum_views

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Composite unique; Required	Identifier of the related public user.
hotspot_key	TEXT	Composite unique; Required	Stable key of the museum hotspot viewed by the user.
viewed_at	TIMESTAMPTZ	Required	Date and time when the hotspot was first viewed.

Table A10 records museum information hotspots viewed by a user in the WebXR experience.

Table A11. Data Dictionary for webxr_player_items

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Composite unique; Required	Identifier of the related public user.
item_key	TEXT	Composite unique; Required	Stable application key for the completed or collected item.
status	TEXT	Required	Current workflow or publication state of the record.
acquired_scene	TEXT	Required	WebXR scene in which the item was collected.
acquired_at	TIMESTAMPTZ	Required	Date and time when the WebXR item was collected.
deposited_at	TIMESTAMPTZ	Optional	Date and time when the WebXR item was deposited.

Table A11 stores collectible WebXR items acquired or deposited by a user.

Table A12. Data Dictionary for visitor_feedback

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Optional	Identifier of the related public user.
visit_type	TEXT	Required	Indicates whether the feedback concerns an on-site or website visit.
rating	INTEGER	Required	Visitor satisfaction rating from one to five.
visitor_name	TEXT	Required	Optional name supplied by the visitor.
comments	TEXT	Required	Written feedback supplied by the visitor.
sentiment_label	TEXT	Optional	Predicted sentiment class for the visitor comment.
sentiment_confidence	NUMERIC(5,4)	Optional	Confidence score assigned to the predicted sentiment.
sentiment_scores	JSONB	Required	JSON object containing the sentiment scores returned by the model.
sentiment_model	TEXT	Required	Name or version of the model used for sentiment analysis.
sentiment_provider	TEXT	Required	Service or library that performed sentiment analysis.
sentiment_analyzed_at	TIMESTAMPTZ	Optional	Date and time when sentiment analysis completed.
status	TEXT	Required	Current workflow or publication state of the record.
reviewed_at	TIMESTAMPTZ	Optional	Date and time when an administrator reviewed the record.
reviewed_by	TEXT	Required	Email or identifier of the administrator who reviewed the record.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A12 stores visitor ratings, comments, review status, and automated sentiment-analysis results.

Table A13. Data Dictionary for analytics_events

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
event_type	TEXT	Required	Application event that generated the notification or analytics record.
module	TEXT	Required	Application module in which the analytics event occurred.
target_id	TEXT	Required	Stable identifier of the page, scene, artifact, or resource involved.
visitor_hash	TEXT	Required	Privacy-preserving visitor identifier used to group anonymous events.
user_id	BIGINT	FK → public_users.id; Optional	Identifier of the related public user.
device_type	TEXT	Required	General device category detected for the visit.
browser_family	TEXT	Required	Browser family detected for the event.
os_family	TEXT	Required	Operating-system family detected for the event.
page_path	TEXT	Required	Application path on which the event occurred.
metadata	JSONB	Required	Bounded JSON object containing event-specific attributes.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.

Table A13 stores privacy-conscious interaction events used for system usage and engagement reports.

Table A14. Data Dictionary for community_highlights

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
title	TEXT	Required	Primary title of the record.
slug	TEXT	Unique; Required	URL-safe unique name used to address the record.
category	TEXT	Required	Classification assigned to the content record.
event_date	DATE	Required	Date associated with the community highlight.
short_description	TEXT	Required	Brief description displayed in highlight listings.
long_description	TEXT	Required	Complete English description of the community highlight.
short_description_tl	TEXT	Required	Brief Tagalog description displayed in highlight listings.
long_description_tl	TEXT	Required	Complete Tagalog description of the community highlight.
read_time_minutes	INTEGER	Optional	Estimated number of minutes required to read the highlight.
image_filename	TEXT	Optional	Stored filename of the primary image.
image_filenames	JSONB	Required	Ordered JSON array of image filenames for the highlight gallery.
status	TEXT	Required	Current workflow or publication state of the record.
is_featured	BOOLEAN	Required	Indicates whether the content is emphasized in featured displays.
is_published	BOOLEAN	Required	Indicates whether the content is publicly visible.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A14 stores bilingual community stories, programs, advisories, dates, images, and publication state.

Table A15. Data Dictionary for online_gallery_categories

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
slug	TEXT	Unique; Required	URL-safe unique name used to address the record.
display_name	TEXT	Unique; Required	Primary display label of the gallery category.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
is_active	BOOLEAN	Required	Indicates whether the account or category is available for use.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.
display_name_en	TEXT	Required	English display label of the gallery category.
display_name_tl	TEXT	Required	Tagalog display label of the gallery category.

Table A15 stores ordered bilingual categories used to organize the online gallery.

Table A16. Data Dictionary for online_gallery_items

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
category_id	BIGINT	FK → online_gallery_categories.id; Required	Identifier of the gallery category containing the item.
title	TEXT	Required	Primary title of the record.
period	TEXT	Required	Historical period associated with the gallery item.
year_label	TEXT	Required	Display year or date range associated with the content.
source	TEXT	Required	Origin or provenance recorded for the item or event.

location	TEXT	Required	Location associated with the booking or content item.
short_caption	TEXT	Required	Primary short caption retained for compatibility.
full_description	TEXT	Required	Primary full description retained for compatibility.
teaching_note	TEXT	Required	Optional instructional note for educators and learners.
tags	JSONB	Required	JSON array of searchable keywords assigned to the gallery item.
image_filename	TEXT	Optional	Stored filename of the primary image.
model_filename	TEXT	Optional	Stored filename of the optional three-dimensional model.
status	TEXT	Required	Current workflow or publication state of the record.
is_published	BOOLEAN	Required	Indicates whether the content is publicly visible.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.
title_tl	TEXT	Required	Tagalog title displayed to visitors.
location_tl	TEXT	Required	Tagalog location label displayed to visitors.
short_caption_tl	TEXT	Required	Tagalog short caption displayed in gallery listings.
full_description_tl	TEXT	Required	Complete Tagalog description of the gallery item.
title_en	TEXT	Required	English title displayed to visitors.
location_en	TEXT	Required	English location label displayed to visitors.
short_caption_en	TEXT	Required	English short caption displayed in gallery listings.
full_description_en	TEXT	Required	Complete English description of the gallery item.

Table A16 stores bilingual artifact records, descriptive metadata, media references, and publication state.

Table A17. Data Dictionary for edu_resources

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
title	TEXT	Required	Primary title of the record.
slug	TEXT	Unique; Required	URL-safe unique name used to address the record.
category	TEXT	Required	Classification assigned to the content record.
educator_name	TEXT	Required	Name of the educator who prepared or contributed the resource.
subject_topic	TEXT	Required	Subject area or learning topic associated with the resource.
author	TEXT	Required	Named author of the educational material.
year_label	TEXT	Required	Display year or date range associated with the content.
description	TEXT	Required	Full description of the content or virtual-tour environment.
file_filename	TEXT	Optional	Stored filename of the downloadable educational resource.
cover_image_filename	TEXT	Optional	Stored filename of the resource cover image.
status	TEXT	Required	Current workflow or publication state of the record.
is_published	BOOLEAN	Required	Indicates whether the content is publicly visible.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A17 stores educational resources, descriptive metadata, downloadable files, and publication state.

Table A18. Data Dictionary for community_participation_submissions

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
user_id	BIGINT	FK → public_users.id; Optional	Identifier of the related public user.
submission_type	TEXT	Required	Identifies the submission as a teacher resource or community story.
title	TEXT	Required	Primary title of the record.
resource_category	TEXT	Required	File or resource category selected by the contributor.
story_category	TEXT	Required	Community-story category selected by the contributor.
submitter_name	TEXT	Required	Name supplied by the person submitting the material.
submitter_email	TEXT	Required	Email address used to verify and contact the contributor.
subject_topic	TEXT	Required	Subject area or learning topic associated with the resource.
description	TEXT	Required	Full description of the content or virtual-tour environment.
story_content	TEXT	Required	Full written content of the submitted community story.
resource_file_filename	TEXT	Optional	Stored filename of the submitted educational resource.
photo_filename	TEXT	Optional	Stored filename of the optional submission photograph.
is_private	BOOLEAN	Required	Indicates whether the contributor requested private handling.
status	TEXT	Required	Current workflow or publication state of the record.
verification_code_hash	TEXT	Optional	Hash of the code used to verify the contributor's email.
verification_sent_at	TIMESTAMPTZ	Optional	Date and time when the verification code was sent.
verification_expires_at	TIMESTAMPTZ	Optional	Expiration date and time of the verification code.
verified_at	TIMESTAMPTZ	Optional	Date and time when the submission email was verified.
reviewed_at	TIMESTAMPTZ	Optional	Date and time when an administrator reviewed the record.
reviewed_by	TEXT	Required	Email or identifier of the administrator who reviewed the record.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A18 stores teacher resources and community stories submitted for verification and administrative review.

Table A19. Data Dictionary for virtual_tours

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
slug	TEXT	Unique; Required	URL-safe unique name used to address the record.
title_en	TEXT	Required	English title displayed to visitors.
title_tl	TEXT	Required	Tagalog title displayed to visitors.
description_en	TEXT	Required	English value stored for description.
description_tl	TEXT	Required	Tagalog value stored for description.
captured_on	DATE	Optional	Capture date retained for compatibility; current scene captures are stored per scene.
status	TEXT	Required	Current workflow or publication state of the record.
is_system_original	BOOLEAN	Required	Indicates whether the record represents the protected packaged original.
published_revision_id	BIGINT	FK → virtual_tour_revisions.id; Optional	Identifier of the revision currently served to visitors.
draft_revision_id	BIGINT	FK → virtual_tour_revisions.id; Optional	Identifier of the working revision currently edited by administrators.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
created_by	TEXT	Required	Email or identifier of the administrator who created the record.
published_at	TIMESTAMPTZ	Optional	Date and time when the revision or content was published.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A19 stores the identity, bilingual metadata, publication pointers, and ordering of each 360-degree tour environment.

Table A20. Data Dictionary for virtual_tour_revisions

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
tour_id	BIGINT	FK → virtual_tours.id; Composite unique; Required	Identifier of the virtual-tour environment owning the revision.
version_number	INTEGER	Composite unique; Required	Sequential revision number within the parent tour or site map.
state	TEXT	Required	Lifecycle state of the revision.
created_by	TEXT	Required	Email or identifier of the administrator who created the record.
published_at	TIMESTAMPTZ	Optional	Date and time when the revision or content was published.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.
map_config	JSONB	Required	JSON object configuring the upper- right minimap and its scene-linked markers.

Table A20 stores immutable publication versions and editable drafts of a virtual tour, including minimap configuration.

Table A21. Data Dictionary for virtual_tour_scenes

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
revision_id	BIGINT	FK → virtual_tour_revisions.id; Composite unique; Required	Identifier of the virtual-tour revision owning the scene or hotspot.
scene_key	TEXT	Composite unique; Required	URL-safe stable key used to address the 360-degree scene.
name_en	TEXT	Required	English name displayed for the scene or hotspot.
name_tl	TEXT	Required	Tagalog name displayed for the scene or hotspot.

captured_on	DATE	Required	Capture date retained for compatibility; current scene captures are stored per scene.
panorama_path	TEXT	Required	Storage path of the equirectangular panorama image.
panorama_width	INTEGER	Required	Pixel width recorded for panorama validation.
panorama_height	INTEGER	Required	Pixel height recorded for panorama validation.
panorama_mime	TEXT	Required	Validated MIME type of the panorama image.
validation_status	TEXT	Required	Result of the panorama format and dimension validation.
initial_pitch	NUMERIC(7, 2)	Required	Initial vertical viewing angle of the scene, in degrees.
initial_yaw	NUMERIC(7, 2)	Required	Initial horizontal viewing angle of the scene, in degrees.
initial_hfov	NUMERIC(6, 2)	Required	Initial horizontal field of view of the scene, in degrees.
is_start_scene	BOOLEAN	Required	Indicates whether visitors enter the tour through this scene.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
created_at	TIMESTAMP TZ	Required	Date and time when the record was created.
updated_at	TIMESTAMP TZ	Required	Date and time when the record was last updated.

Table A21 stores each validated equirectangular panorama and its initial 360-degree viewing orientation.

Table A22. Data Dictionary for virtual_tour_hotspots

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
revision_id	BIGINT	FK → virtual_tour_revisions.id; Required	Identifier of the virtual-tour revision owning the scene or hotspot.
scene_id	BIGINT	FK → virtual_tour_scenes.id; Required	Identifier of the virtual-tour scene containing the hotspot.
hotspot_type	TEXT	Required	Identifies the hotspot as an information item or scene navigation control.
name_en	TEXT	Required	English name displayed for the scene or hotspot.
name_tl	TEXT	Required	Tagalog name displayed for the scene or hotspot.
pitch	NUMERIC(7,2)	Required	Vertical hotspot position in the 360- degree scene, in degrees.
yaw	NUMERIC(7,2)	Required	Horizontal hotspot position in the 360-degree scene, in degrees.
target_scene_id	BIGINT	FK → virtual_tour_scenes.id; Optional	Identifier of the destination scene for a navigation hotspot.
target_pitch	NUMERIC(7,2)	Optional	Vertical viewing angle used after entering the destination scene.
target_yaw	NUMERIC(7,2)	Optional	Horizontal viewing angle used after entering the destination scene.
target_hfov	NUMERIC(6,2)	Optional	Horizontal field of view used after entering the destination scene.
image_path	TEXT	Optional	Storage path of the optional hotspot image.
glb_path	TEXT	Optional	Storage path of the optional GLB three-dimensional model.

subtitle_en	TEXT	Required	English narration or information text for the hotspot.
subtitle_tl	TEXT	Required	Tagalog narration or information text for the hotspot.
audio_en_path	TEXT	Optional	Storage path of the optional English hotspot narration.
audio_tl_path	TEXT	Optional	Storage path of the optional Tagalog hotspot narration.
tts_voice_id	TEXT	Required	Configured text-to-speech voice identifier.
tts_model_id	TEXT	Required	Configured text-to-speech model identifier.
nested_hotspots	JSONB	Required	JSON array of secondary hotspots associated with the item.
is_enabled	BOOLEAN	Required	Indicates whether the hotspot is available to visitors.
sort_order	INTEGER	Required	Numeric position used when displaying records in sequence.
created_at	TIMESTAMP TZ	Required	Date and time when the record was created.
updated_at	TIMESTAMP TZ	Required	Date and time when the record was last updated.

Table A22 stores item and navigation hotspots positioned inside a virtual-tour scene, including bilingual media and target views.

Table A23. Data Dictionary for site_maps

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
is_system_original	BOOLEAN	Required	Indicates whether the record represents the protected packaged original.
published_revision_id	BIGINT	FK → site_map_revisions.id; Optional	Identifier of the revision currently served to visitors.
draft_revision_id	BIGINT	FK → site_map_revisions.id; Optional	Identifier of the working revision currently edited by administrators.
created_by	TEXT	Required	Email or identifier of the administrator who created the record.
published_at	TIMESTAMPTZ	Optional	Date and time when the revision or content was published.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A23 stores the protected site-map record and pointers to its current published and working revisions.

Table A24. Data Dictionary for site_map_revisions

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK; Identity; Required	Unique identifier for the record.
site_map_id	BIGINT	FK → site_maps.id; Composite unique; Required	Identifier of the site-map record owning the revision.
version_number	INTEGER	Composite unique; Required	Sequential revision number within the parent tour or site map.
state	TEXT	Required	Lifecycle state of the revision.
config	JSONB	Required	JSON object containing site-map views, coordinates, hotspot content, media, links, and intro audio.
created_by	TEXT	Required	Email or identifier of the administrator who created the record.
published_at	TIMESTAMPTZ	Optional	Date and time when the revision or content was published.
created_at	TIMESTAMPTZ	Required	Date and time when the record was created.
updated_at	TIMESTAMPTZ	Required	Date and time when the record was last updated.

Table A24 stores revisioned site-map configuration, including aerial/360 media, hotspots, links, and narration settings as JSON.

Hierarchical Input-Process-Output Model

This section presents the Hierarchy plus Input-Process-Output (HIPO) model of theWebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The model organizes the system from its principal function into progressively detailed visitor, account, participation, and administration modules.

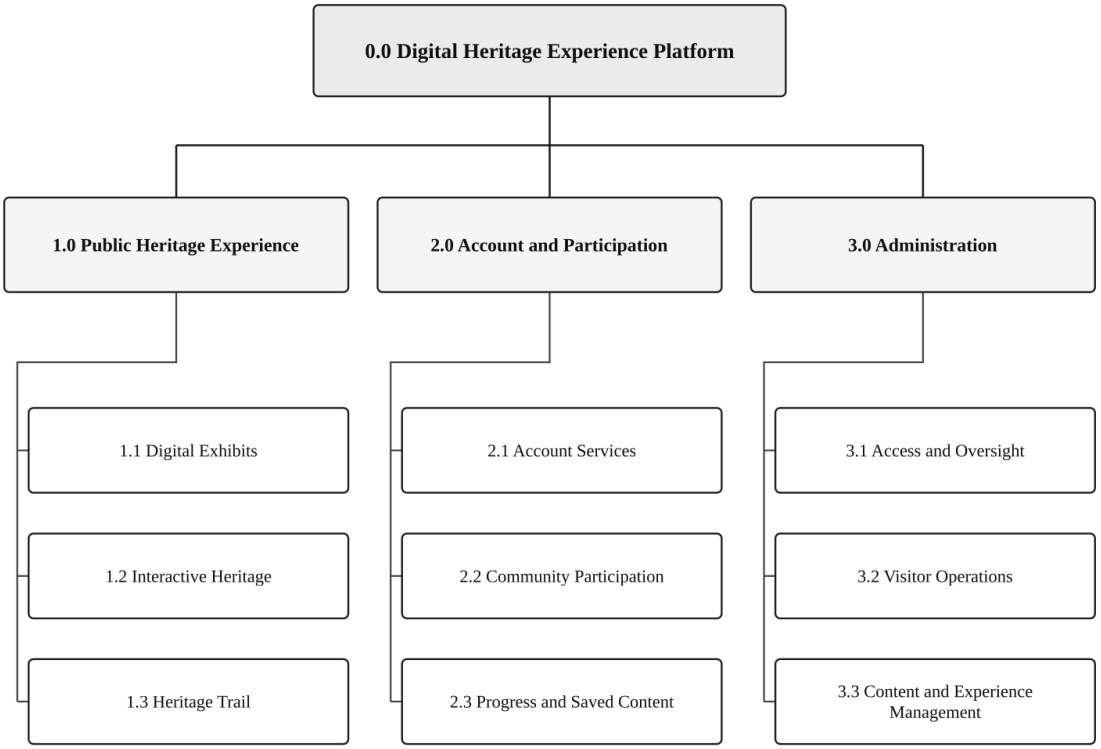


Figure A56. System-Level HIPO Visual Table of Contents.

Figure A56 shows the top-down functional decomposition of the platform into public heritage experiences, account and participation services, and administration functions.

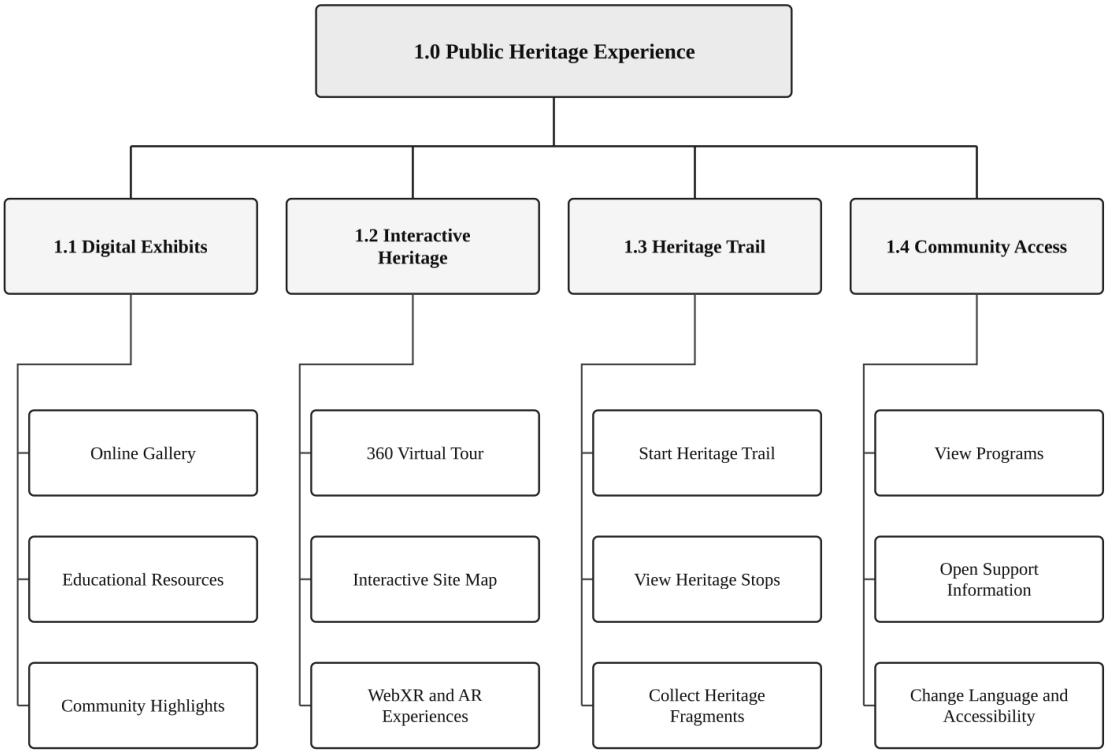


Figure A57. Public Visitor HIPO Hierarchy.

Figure A57 presents the public-facing heritage functions available for viewing collections,opening immersive experiences, following the heritage trail, and accessing visitor information.

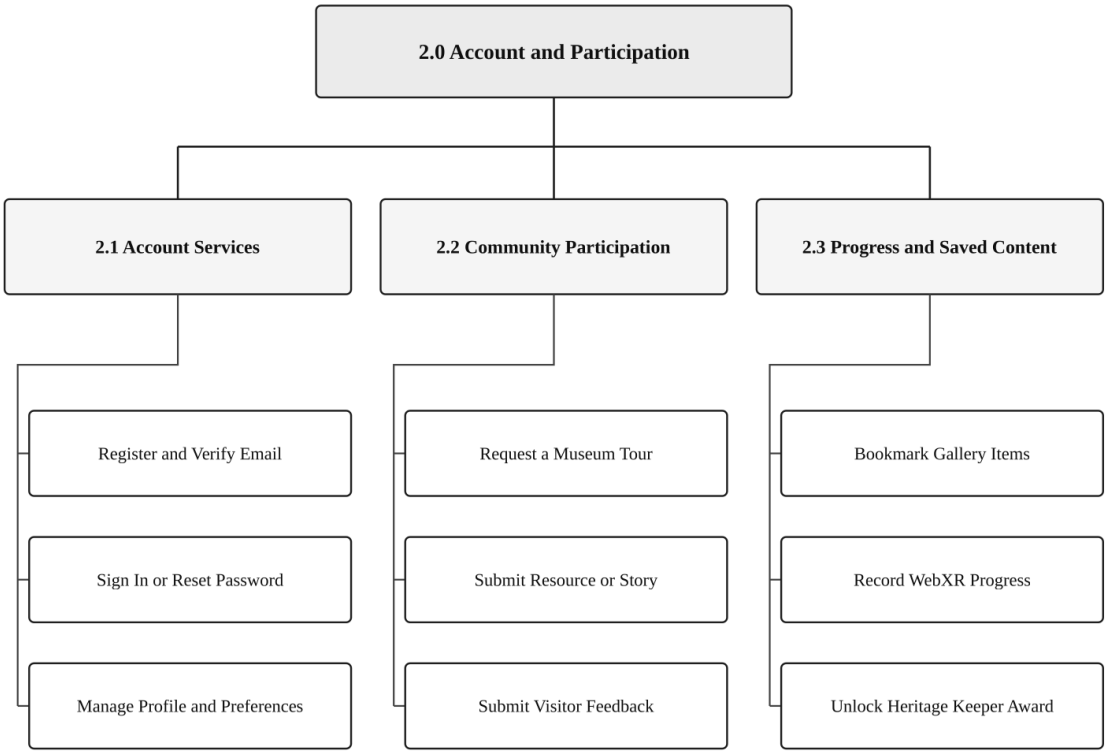


Figure A58. Public Account and Engagement HIPO Hierarchy.

Figure A58 shows the account, community participation, bookmarking, WebXR progress, andHeritage Keeper functions available to public users.

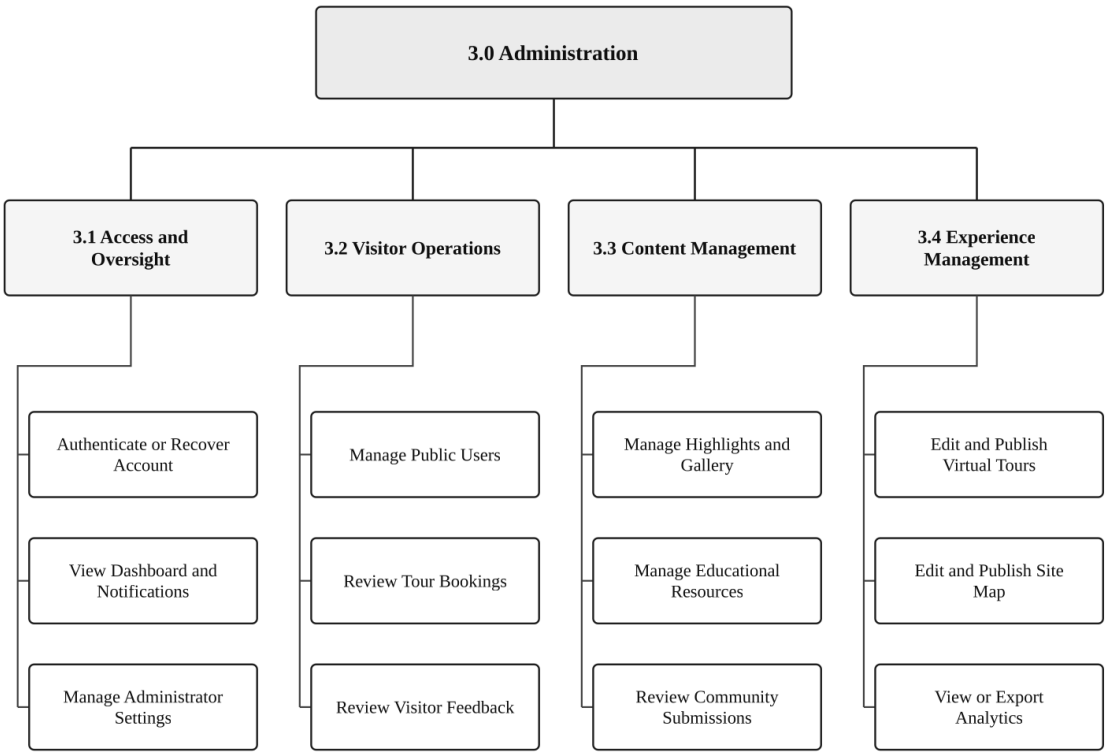


Figure A59. Administrator HIPO Hierarchy.

Figure A59 shows the administrator portal hierarchy for secure access, visitor operations,heritage content management, immersive-experience editing, and analytics reporting.

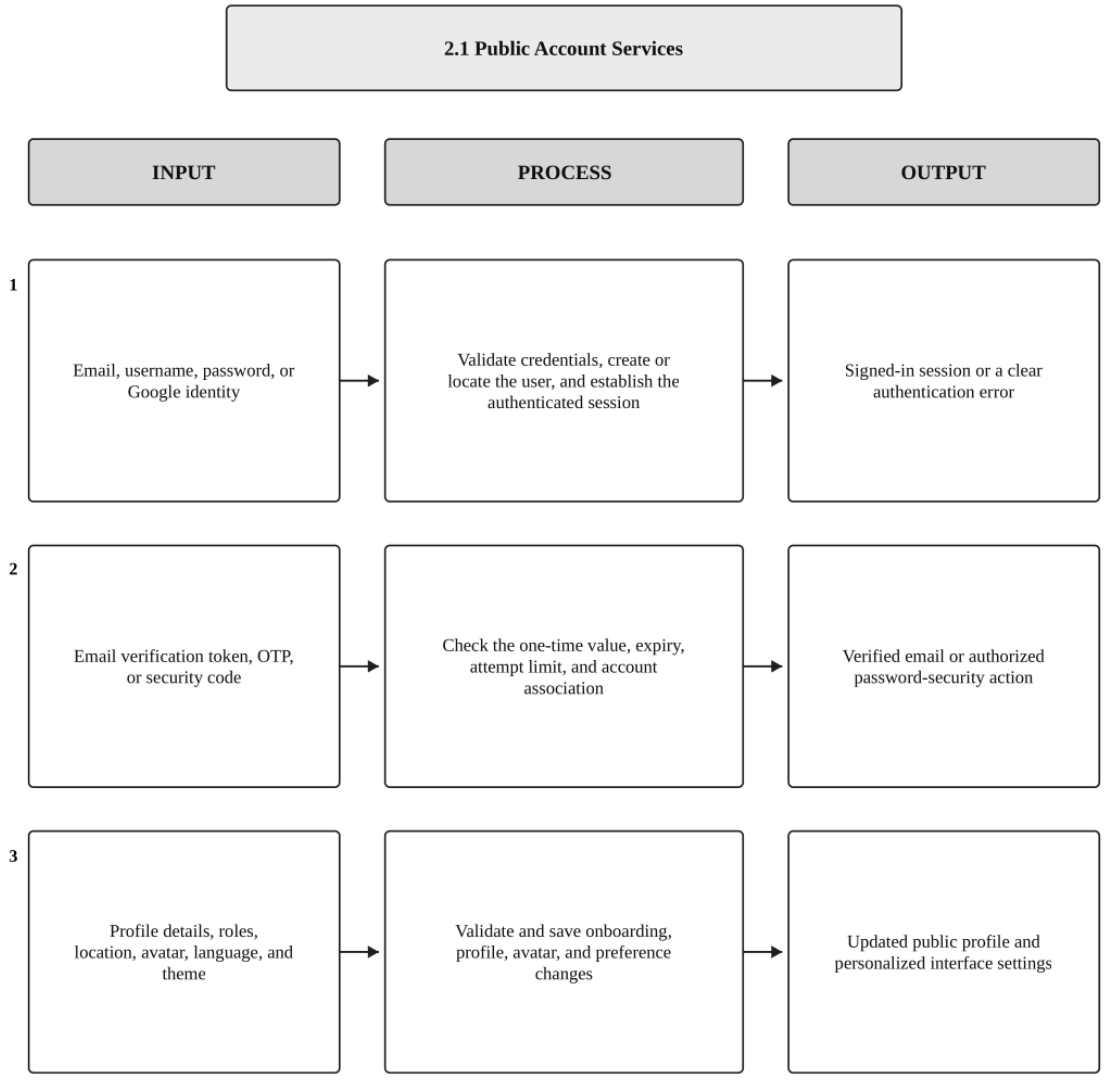


Figure A60. Public Account Services IPO Overview.

Figure A60 identifies the inputs, processing, and outputs used to register, authenticate,verify, secure, and personalize a public account.

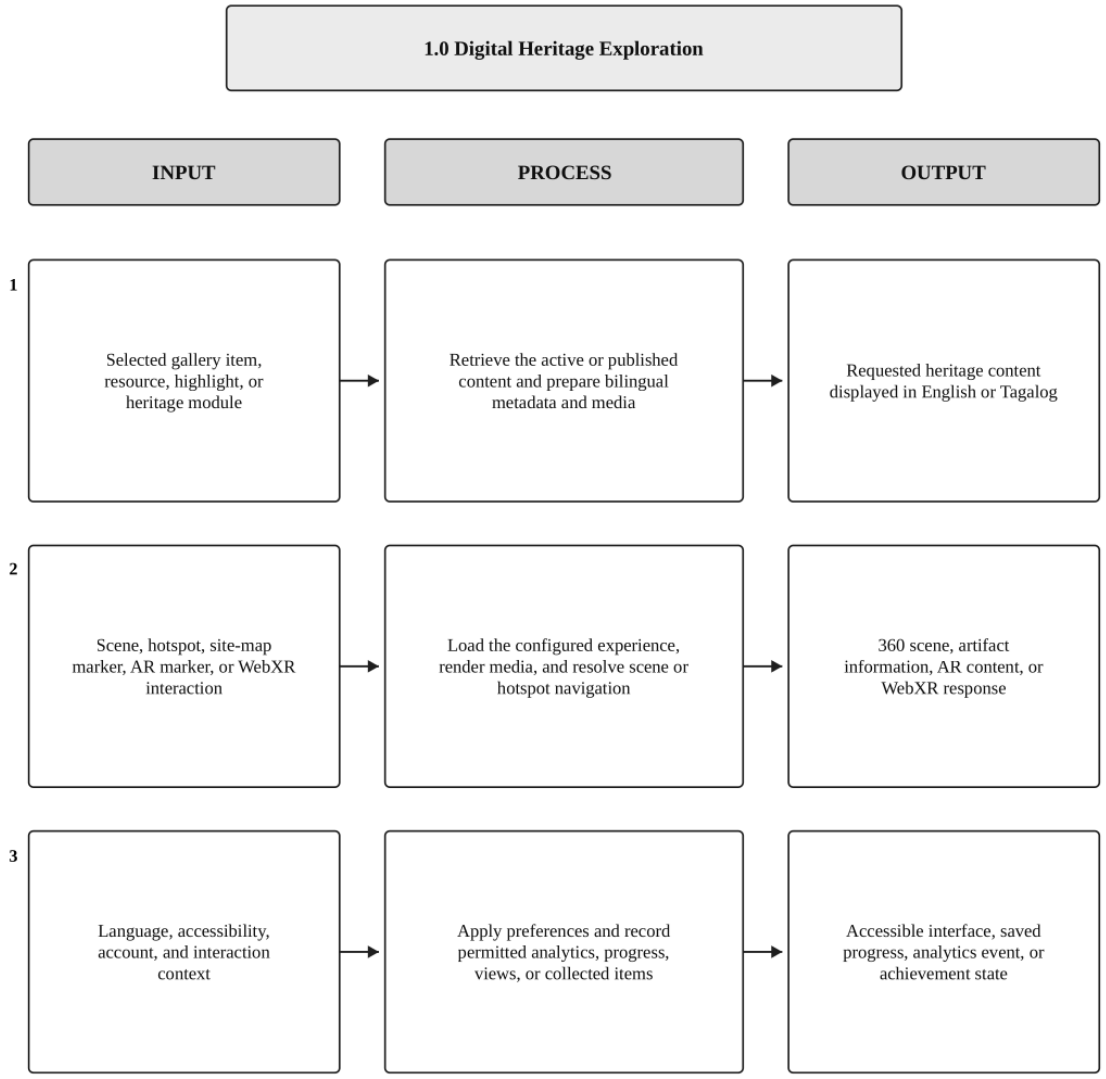


Figure A61. Digital Heritage Exploration IPO Overview.

Figure A61 shows how visitor selections and immersive interactions are processed into bilingual exhibits, 360-degree navigation, augmented-reality content, WebXR responses, and saved progress.

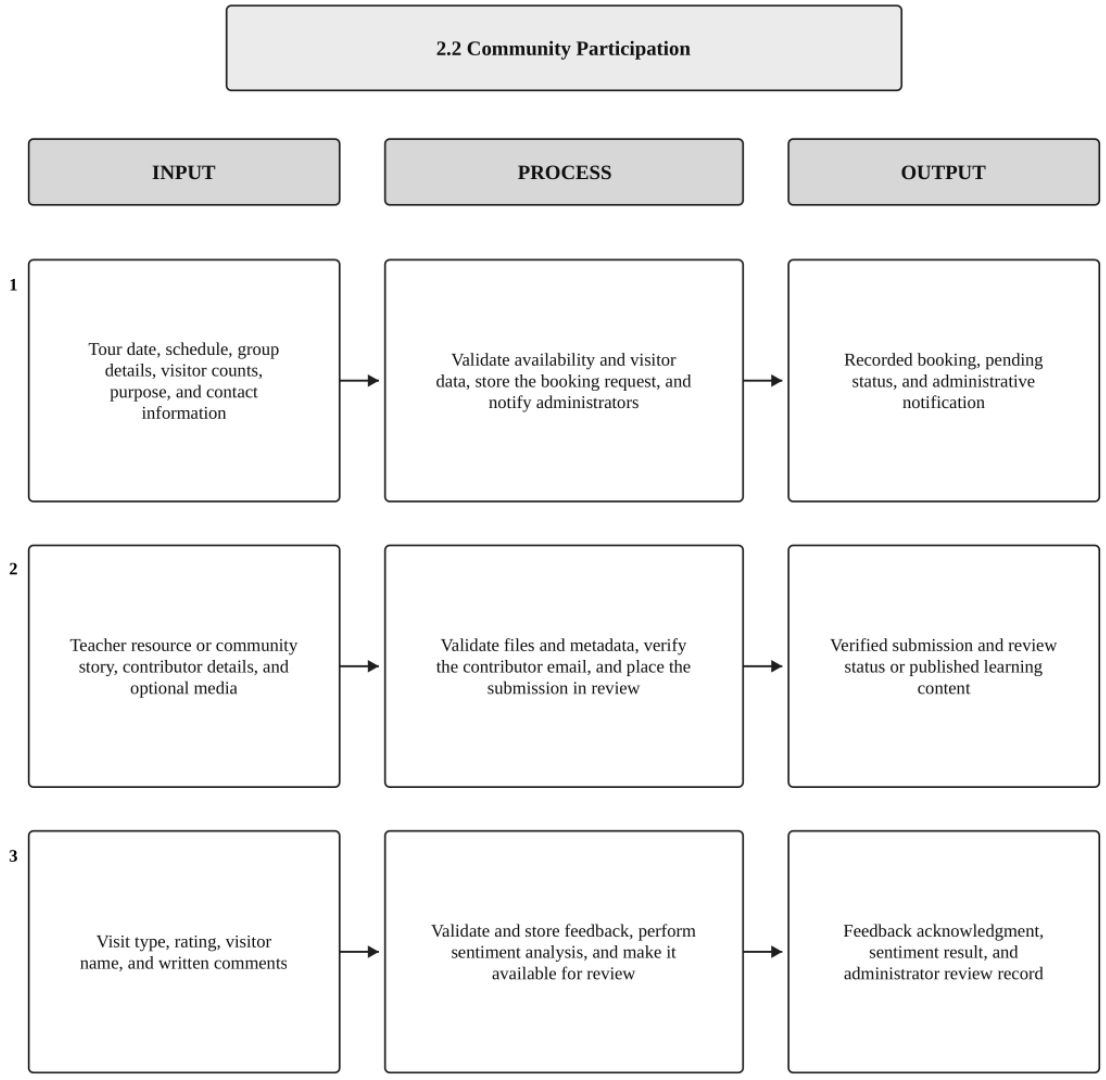


Figure A62. Community Participation IPO Overview.

Figure A62 presents the input, processing, and output relationships for tour requests, community-contributed materials, and visitor feedback.

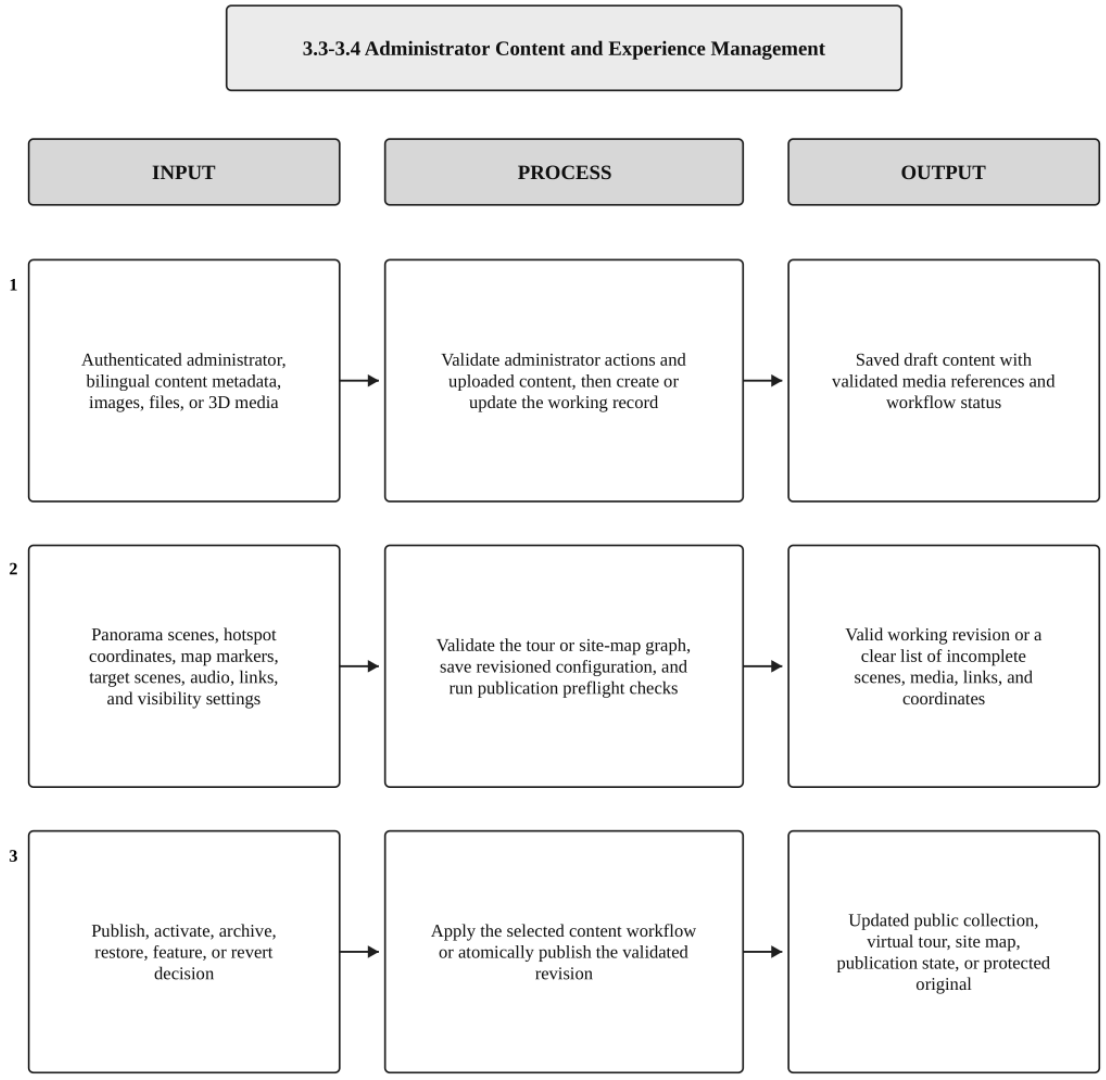


Figure A63. Administrator Content and Experience Management IPO Overview.

Figure A63 shows how administrators transform bilingual metadata, multimedia assets, scenes,hotspots, and publication decisions into validated public content and revisioned immersive experiences.

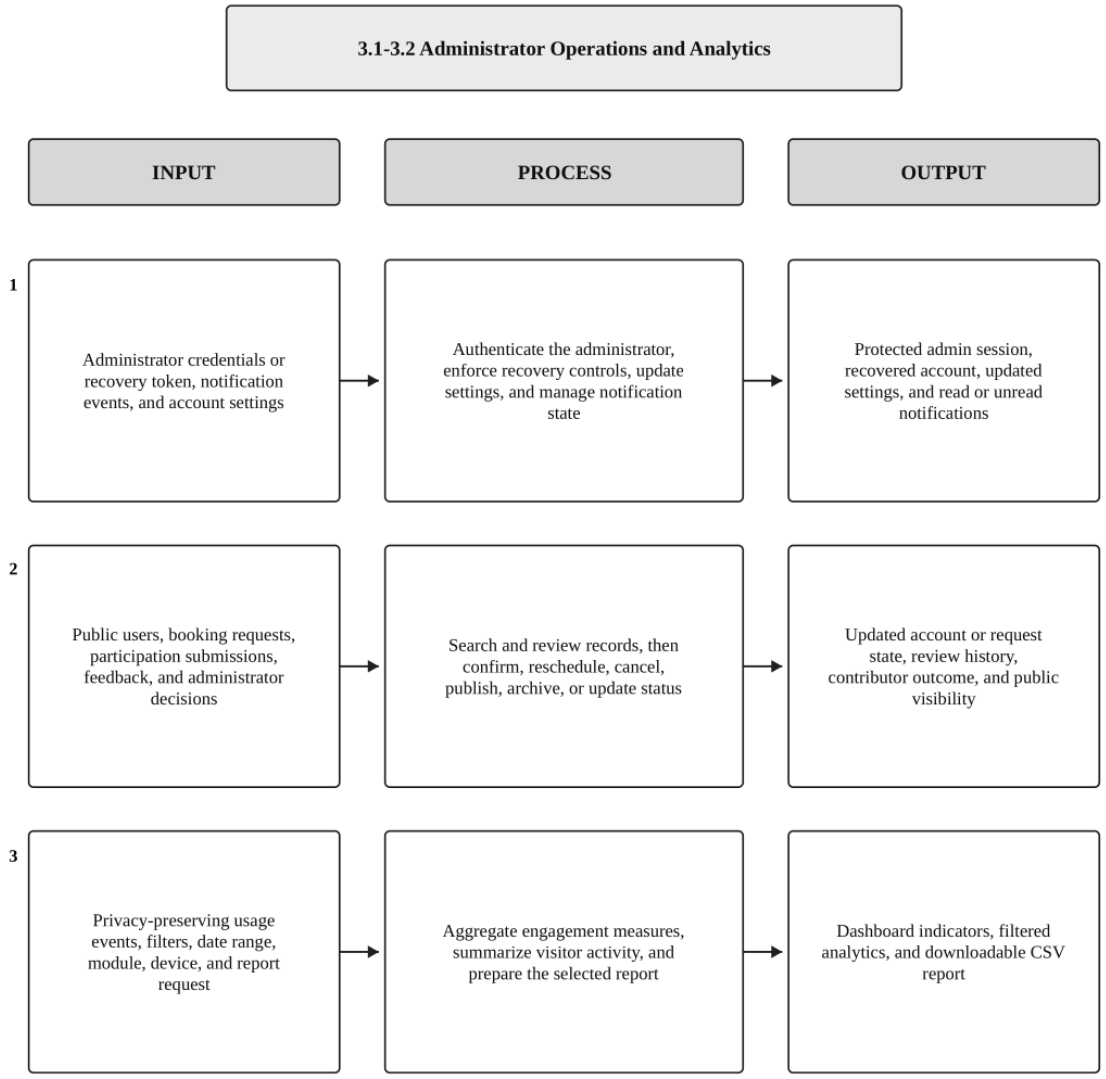


Figure A64. Administrator Operations and Analytics IPO Overview.

Figure A64 presents the inputs, processing, and outputs for administrator access, operational review queues, notifications, booking decisions, and analytics reporting.

Project Schedule

Project Schedule									
2026									
ACTIVITY	**Assigned / Involved Person**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sept**
Phase I. Planning									
Title Formulation	Researchers / Research Coordinator								
Title Proposal Defense									
Phase II. Design									
Brainstorming	Researchers								
System Design Development									
Writing Manuscript Chapter (1-3)									
Phase III. Develop									
Initial System Development	Researchers, Adviser, System Expert, Technical Editor, Statistician								
Research Colloquium									
System Revision									
System Consultation									
Final System, Development									
Phase IV. Test									
Finding System Tester	Researchers, Adviser, System Expert, Technical Editor, IT Experts								
Conduct System Testing									
User Testing									
Phase V. Evaluation									
System Evaluation	Researchers, Adviser, System Expert, Technical Editor, Language Critic, Statistician, Respondents								
Writing Manuscript Chapter (4-5)									
Final Oral Defense									

Hardware and Software Resources

This section presents the hardware and software resources used in the design, development, testing, deployment and operation of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The resources were selected to support the platform's public and administrative interfaces, database-backed content management, multimedia storage, interactive site map, 360-degree virtual tour, marker-based augmented reality, WebXR-enabled three-dimensional environments, bilingual narration, and responsive accessibility features.

Software Resources

A. Core Web Application Technologies

- Python 3
- Flask
- Jinja
- Werkzeug
- HTML5
- CSS3
- JavaScript and ECMAScript Modules
- SQL
- JSON
- YAML

B. Backend Packages

- Supabase Python Client

- Pillow
- PyYAML
- python-dotenv

C. Database, Storage, and Deployment Resources

- Supabase PostgreSQL Database
- Supabase Storage
- Supabase Row Level Security
- Supabase Database Functions
- Vercel

D. Immersive, Mapping, and Multimedia Libraries

- Pannellum 2.5.6
- Three.js
- Spark 2.1.0
- three-mesh-bvh 0.9.2
- model-viewer
- AR.js
- Leaflet 1.9.4
- WebXR Device API
- Web Audio API
- Web Speech API

E. Interface and Presentation Libraries

- Tailwind CSS
- GSAP 3.12.5

- ScrollTrigger
- Lucide Icons
- Google Fonts

F. External Services and Application Programming Interfaces

- Google OAuth 2.0
- SMTP Email Service
- ElevenLabs Text-to-Speech API
- Browser Camera API
- Device Orientation and Motion APIs

Hardware Resources

A. Development and Administrative Workstation

- Desktop or laptop computer
- 64-bit quad-core Intel Core i5, AMD Ryzen 5, or higher
- 8 GB RAM minimum
- 10 GB or more available solid-state storage
- WebGL-capable integrated or dedicated graphics processor
- 1366 × 768 display or higher
- Keyboard and pointing device
- Speakers or headphones
- Broadband internet connection
- Wi-Fi or Ethernet adapter

D. End-User Hardware

- Smartphone, tablet, laptop, or desktop computer

- Current standards-compliant web browser
- Internet connection
- WebGL-capable graphics hardware
- Speakers or headphones
- Camera-enabled mobile device for augmented reality
- Optional WebXR-compatible headset

Testing and Evaluation Instruments

Black Box Testing Sheet

BLACK-BOX TESTING SHEET OF DESIGN AND DEVELOPMENT OF A WEBXR-ENABLED DIGITAL HERITAGE EXPERIENCE PLATFORM FOR THE NHCP MUSEO NG LIBINGAN SA ILALIM NG LUPA NG NAGCARLAN

BLACK-BOX FUNCTIONAL TESTING CHECKLIST

Tester Information

Tester ID	ST-01
Tester Name	
Educational Background / Credentials	
Current Position / Company	
Relevant IT / Testing / Development Experience	
Years of Relevant Experience	
Date of Testing	
System Version / Build	EW01 v1.0 – Testing Build 01

Instructions

- Complete this form while interacting directly with the working system; do not evaluate from a demonstration alone.
- The researchers will provide the approved system URL/build, controlled test accounts, AR markers, and sample test files separately. Do not write passwords in this form.
- Answer each test question by comparing the actual observed result with the stated expected result.
- Record the actual result, then select only one: PASS, FAIL, N/A, or RETEST.
- Mark FAIL—not N/A—when a system problem prevents completion. Explain every FAIL or N/A in Remarks or the execution log.
- Use supplied test data only. Do not use real visitor personal information.

Test Result Codes

- **PASS -** The complete expected result is observed.
- **FAIL -** The observed result differs, or a system problem prevents completion.
- **N/A -** The question is genuinely not applicable to the recorded environment; explain why.
- **RETEST -** A previously failed question is tested again after a reported correction.

Defect Severity

- **Critical** - Core system access or a required end-to-end function is unavailable.
- **Major** - A required feature fails, but unrelated parts remain usable.
- **Minor** - A non-blocking functional, validation, label, or presentation defect.
- **Observation** - A recommendation or note that is not a demonstrated failure.

Test Environment Record

Environment ID	ENV-01
Device / Model	
Operating System / Version	
Browser / Version	
Screen Size / Resolution	
Internet / Network Condition	
Camera / Sensor Available	
WebXR-Compatible Device / Browser	
Test Date / Time	

BLACK-BOX FUNCTIONAL TEST QUESTIONS

Perform each test question on the approved system build. Record the observed result before selecting a status.

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-01	Homepage	Does the system homepage open successfully?	The homepage loads and displays the main Ana Kalang heritage-story interface without broken primary content.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-02	Homepage	Can the tester navigate through the homepage's major sections or chapters?	The expected chapters load and navigation continues without broken links or blocked content.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-03	Homepage	Does the available narration or Listen function operate correctly?	Narration starts, pauses or stops as designed and corresponds to the currently selected chapter.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-04	Homepage	Do the homepage's available Mute, EN/TL, CC, and story text-size controls work?	Each selected control produces its documented change without breaking the page or changing to unrelated content.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-05	Homepage Navigation	Does each major public navigation link open its correct destination?	Each tested link opens the correctly labeled public module without a not-found or blank page.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-06	Responsive Access	Does the homepage remain usable on the assigned mobile, tablet, or desktop environment?	Content and controls remain accessible without critical overlap, blocking horizontal overflow, or unusable navigation.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-07	Registration	Does the registration or Sign Up page open correctly?	The registration form loads and displays the required email, password, and password-confi rmation fields.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-08	Registration	Can a visitor register using valid supplied test information?	The system validates the data, sends the six-digit verification code, and completes the documented verification and account-setup workflow.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-09	Registration Validation	Does the system reject incomplete or invalid registration information?	The system identifies the invalid or required field and prevents creation of an improperly registered account.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-10	Login	Can a verified visitor sign in using valid registered test credentials?	The visitor is authenticated and the appropriate account or intended destination opens.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-11	Login Validation	Does the system reject invalid visitor credentials?	The attempt is rejected with an appropriate message and authenticated access is not granted.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-12	Logout	Does visitor logout end the authenticated session?	The session ends and account-protected functions require authentication when requested again.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-13	Digital Exhibits	Does the Online Gallery section of Digital Exhibits open correctly?	The section loads and published exhibit or gallery content is displayed.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-14	Digital Exhibits	Does opening an gallery item display the correct record?	The selected item's information, images, and available interactive or 3D content are displayed without changing to an unrelated record.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-15	Digital Exhibits	Does the Educational Resources section of Digital Exhibits open and display the correct public resources?	The section loads and displays active or published PPT, Story, DOCS, and PDF resources; draft and archived administrator records are not shown publicly.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-16	Digital Exhibits	Do the Educational Resources search, filters, sorting, grid/list views, details, previews, and downloads work correctly?	The results and count match the search, type, and subject filters; sorting and view changes preserve the selected records, and opening a resource shows its matching details, preview, navigation, related items, and available document download.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-17	Heritage Trail	Does the Heritage Trail open successfully?	The trail loads and makes its introduction and first guided destination available.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-18	Heritage Trail	Does selecting a Heritage Trail stop display the correct content?	The stop opens its corresponding heritage location, 360-degree scene, information, and available narration or activity.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-19	Interactive Site Map	Does the Interactive Site Map open successfully?	The aerial map loads with its available controls and configured location markers.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-20	Interactive Site Map	Does selecting a map marker display the correct location?	The information card, media or links, and selected-marker state correspond to the chosen location.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-21	360-Degree Virtual Tour	Does the 360-degree Virtual Tour open successfully?	The configured initial panorama, controls, and hotspots load without missing primary scene content.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-22	360-Degree Virtual Tour	Does the panorama respond correctly when the tester pans or rotates the view?	The camera follows the supported pointer or touch input in the expected direction without unintended jumps or loss of control.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-23	360-Degree Virtual Tour	Does selecting an available tour hotspot open the correct destination or information?	The matching destination scene, artifact, image, 3D content, or information panel is displayed.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-24	AR Content	Does the AR feature open on the designated supported device?	The AR page prepares its configured models and starts the camera scanner after permission is granted; a denied permission produces an actionable error or retry state.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-25	AR Content Detection	Does the designated AR marker display its corresponding artifact?	The registered marker is detected and the matching configured 3D artifact appears under the required camera and lighting conditions.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-26	AR Recovery	Does AR content respond correctly when the marker is removed and presented again?	Marker-dependent content becomes unavailable after marker loss and the correct artifact is detected again when the marker returns.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-27	3D/WebXR	Does the 3D/WebXR environment open on a compatible browser or device?	The selected 3D scene loads with its environment and controls without an indefinitely blocked loader or missing-scene error.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-28	3D/WebXR Interaction	Do the available viewing, movement, and hotspot controls work?	Supported mouse, keyboard, touch, joystick, collision, and hotspot interactions respond according to the selected scene.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-29	WebXR Compatibility	Does the 3D environment remain usable when immersive VR is unavailable?	The non-immersive 3D experience remains usable and any unsupported immersive-VR state is communicated instead of producing a broken page.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-30	Community Submission	Do the community submission forms open with the correct fields and controls?	The available feedback, Teacher Corner, and community-story forms display their corresponding fields; protected submission requires a signed-in visitor.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-31	Community Submission	Can valid supplied feedback, learning material, or community-story data be submitted?	The system validates and records the assigned submission as received or pending administrator review and displays a confirmation.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-32	Community Validation	Does the system reject incomplete, invalid, unsupported, or oversized community-submission data?	The required correction is identified, the invalid submission is prevented, and no false success is displayed.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-33	Online Booking	Does the online booking schedule open correctly?	The booking calendar and request controls load; the public calendar displays only generic confirmed-booking indicators and no private booking details.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-34	Online Booking	Can a signed-in visitor submit a valid booking request using supplied test data?	A valid future-date request with acceptable visitor counts and required contact information is saved as Pending Review and confirmed to the visitor.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-35	Booking Validation	Does the system reject incomplete or invalid booking information?	The system identifies invalid dates, visitor counts, or required contact fields and prevents an invalid request.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-36	Visitor Account Dashboard	Does the signed-in visitor account dashboard display the correct profile and saved activity?	The dashboard shows the visitor's profile and the corresponding saved gallery bookmarks, tour bookings, participation submissions, visitor feedback, and Heritage Keeper summary without exposing another account's data.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-37	Visitor Account Settings	Can a signed-in visitor update an allowed profile or viewing-preference setting using supplied test data?	The permitted profile, avatar, theme, contrast, language, or text-size change is validated, saved to the correct account, and reflected on the applicable pages.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-38	Heritage Keeper Progress	Does the Heritage Keeper record and display progress from the assigned Trail, AR, or 3D/WebXR activity?	A valid completed activity records its corresponding fragment once, updates the correct progress count and recovery record, and preserves previously earned progress for the signed-in visitor.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-39	Admin Login	Does the administrator login page open correctly?	The protected administrator login interface loads with its required credential fields.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-40	Admin Login	Can an active administrator sign in using valid supplied test credentials?	The administrator is authenticated and the protected dashboard opens.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-41	Admin Login Security	Does the administrator login reject invalid credentials?	Access is denied with an appropriate message and no administrator session is created.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-42	Admin Dashboard	Does the administrator dashboard display the system overview, review queue, recent activity, and analytics correctly?	Dashboard counts and queue links correspond to the available records; analytics load for the selected period, and the requested analytics report downloads as a CSV file.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-43	Admin User Management	Can the administrator open and use the applicable user-management function on a supplied test account?	Authorized user records are displayed and the assigned management action produces the specified account state.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-44	Admin Community Highlights	Can the administrator manage and feature a Community Highlight using a supplied test record?	The administrator can create or edit the record, apply valid draft, active, archived, or restored transitions, set or remove the featured item, and public News displays only the appropriate active content.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-45	Admin Booking Calendar	Can the administrator process a supplied pending booking?	The selected confirm, cancel, or reschedule decision is recorded; required reasons or dates are enforced and a finalized booking cannot be decided again.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-46	Admin Online Gallery	Can the administrator manage an Online Gallery item using a supplied test record?	The administrator can create or edit the item with valid image or 3D media, apply an allowed publish, archive, or restore transition, and the public gallery reflects only the published state.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-47	Admin Virtual Tours	Can the administrator manage an assigned Virtual Tour environment, scene, or hotspot using a supplied draft?	Valid environment metadata, map, scene, hotspot, media, or narration changes are saved; publish validation controls public availability, while archive or restore produces the documented state.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-48	Admin Site Map	Can the administrator update and publish a supplied editable Site Map draft?	Valid map configuration, marker, and uploaded-asset changes are saved; publish validation succeeds only when requirements are met, and the published public Site Map reflects the approved revision.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-49	Admin Edu Resources	Can the administrator manage an Educational Resource using supplied test data and files?	The system validates the category and corresponding PPT/PPTX, DOCX, PDF, or Story cover requirements, saves the resource, and correctly applies draft, active, archived, or restored visibility.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-50	Admin Participation Submissions	Can the administrator review and manage a supplied Teacher Corner or community-story submission?	The selected submission opens with its uploaded content and current status; only the allowed approve, archive, or restore action is accepted, and active approved content appears in the corresponding public section.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-51	Admin Visitor Feedback	Can the administrator review and manage a supplied visitor-feedback record?	The feedback record displays its rating, comments, status, and sentiment information; reanalysis or an allowed reviewed, archive, or restore action updates the correct record and rejects an invalid status transition.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test ID	Module	Test Question	Expected Result	Actual Observed Result	Result
TC-52	Admin Account Settings	Can the administrator securely use the Account Settings function with supplied test credentials?	The page requires the current administrator password, validates a new password or sign-out-all action, rejects invalid input, and ends affected administrator sessions after a successful security change.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-53	Admin Logout and Access Control	Does administrator logout end the session and prevent unauthorized access to protected modules?	Logout clears the administrator session; reopening the dashboard or any management module requires administrator authentication, and a signed-in public visitor cannot view or change administrator records.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST
TC-54	Navigation/Error Handling	Do browser Back and Forward controls preserve normal access to the major modules?	Normal history navigation does not create a critical broken state, expose logged-out protected content, or make the expected public page inaccessible.		[ ] PASS [ ] FAIL [ ] N/A [ ] RETEST

Test Execution and Defect Log

Test ID	Date/Time	Actual Result / Defect	Status	Severity	Evidence / Defect ID	Initials
TC-__						
TC-__						
TC-__						
TC-__						
TC-__						
TC-__						
TC-__						
TC-__						
TC-__						
TC-__						
TC-__						
TC-__						

Retest Record

Test ID	Original Failure / Defect	Correction Reported	Retest Result	Date	Initials
TC-__			[ ] PASS [ ] FAIL		
TC-__			[ ] PASS [ ] FAIL		
TC-__			[ ] PASS [ ] FAIL		
TC-__			[ ] PASS [ ] FAIL		
TC-__			[ ] PASS [ ] FAIL		
TC-__			[ ] PASS [ ] FAIL		
TC-__			[ ] PASS [ ] FAIL		
TC-__			[ ] PASS [ ] FAIL		
TC-__			[ ] PASS [ ] FAIL		

Test Summary

Module / Feature	Total Cases	Passed	Failed	N/A	Pass Rate	Remarks
Homepage / Navigation / Responsive Access	6				___%	
User Registration, Login, and Logout	6				___%	
Digital Exhibits	4				___%	
Heritage Trail	2				___%	
Interactive Site Map	2				___%	
360-Degree Virtual Tour	3				___%	
AR Content	3				___%	
3D/WebXR Environment	3				___%	
Community Submission Forms	3				___%	
Online Booking Schedule	3				___%	
Visitor Account and Heritage Keeper	3				___%	
Administrator Login and Functions	15				___%	
Navigation/Error Handling	1				___%	
OVERALL	**54**				**___%**	

Pass Rate (%) = Number of Passed Test Cases ÷ Total Applicable Test Cases × 100. N/A cases are excluded from Total Applicable Test Cases.

Tester's Overall Recommendation

- [ ] READY FOR USER EVALUATION — no unresolved Critical or Major functional defect remains.
- [ ] READY WITH MINOR REVISIONS — only non-blocking Minor defects or Observations remain.
- [ ] NOT YET READY — one or more Critical or Major defects require correction and retesting.

Tester Comments and Certification

Strengths Observed	
Functional Defects	
WebXR / AR / 3D Observations	
Browser / Device Compatibility	
Other Recommendations	

Certification

I certify that I completed the assigned black-box functional test questions using the working system and recorded the results based on my actual observations in the documented environment.

Tester Signature: _____ Date: _____

Researcher Receiving Results: _____ Date: _____

Compatibility Matrix

Env ID	Device	OS	Browser	WebXR VR	AR Camera	360 Tour	Result / Remarks
ENV-01							
ENV-02							
ENV-03							
ENV-04							
ENV-05							
ENV-06							
ENV-07							
ENV-08							
ENV-09							
ENV-10							

System User Evaluation Questionnaire

SYSTEM USER EVALUATION QUESTIONNAIRE

Design and Development of a WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan

Dear Respondent,

You are invited to evaluate the developed WebXR-Enabled Digital Heritage Platform after using its relevant features. Please answer based on your actual experience during the evaluation session. Participation is voluntary. The information collected will be used solely for academic research and handled confidentially in accordance with the approved research protocol and Republic Act No. 10173 (Data Privacy Act of 2012).

PART I. INFORMED CONSENT

Please select one:
( ) I agree to participate in the study.
( ) I do not agree to participate in the study.

PART II. RESPONDENT INFORMATION

Please provide only the information requested below. Do not write your name. Use the respondent code assigned by the researchers.

Respondent Code	
Respondent Group	( ) Administrative Personnel ( ) Museum Visitor ( ) Person with Disability
Age Group	( ) Below 18 ( ) 18-30 ( ) 31-59 ( ) 60 and above
Device Used	( ) Smartphone ( ) Tablet ( ) Laptop/Desktop ( ) Other: _____
Internet Access During Evaluation	( ) Stable ( ) Intermittent ( ) Limited
Date of Evaluation	

PART III. SYSTEM USER EVALUATION

Directions: After using the relevant system features, place one check mark under the rating that best represents your level of agreement with each statement. Please answer every statement based on your actual experience.

Rating	Response	Filipino Guide
5	Strongly Agree	Lubos na Sumasang-ayon
4	Agree	Sumasang-ayon
3	Neither Agree nor Disagree	Neutral / Walang Pinapanigan
2	Disagree	Hindi Sumasang-ayon
1	Strongly Disagree	Lubos na Hindi Sumasang-ayon

A. SELECTED SOFTWARE QUALITY CHARACTERISTICS

No.	Statement	5	4	3	2	1
1	The platform's major features work as intended. Gumagana ayon sa nilalayong gamit ang mga pangunahing tampok ng plataporma.					
2	The platform's navigation is easy to understand. Madaling maunawaan ang nabigasyon ng plataporma.					
3	The platform responds within an acceptable amount of time. Tumutugon ang plataporma sa loob ng katanggap-tanggap na haba ng oras.					

B. PERCEIVED EASE OF USE

No.	Statement	5	4	3	2	1
4	I find the platform easy to use. Para sa akin, madaling gamitin ang plataporma.					
5	Learning to use the platform is easy for me. Madali para sa akin na matutunan kung paano gamitin ang plataporma.					
6	Using the platform's major features requires little effort from me. Kaunting pagsisikap lamang ang kailangan ko upang magamit ang mga pangunahing tampok ng plataporma.					

C. PERCEIVED USEFULNESS

No.	Statement	5	4	3	2	1
7	The platform is useful for accessing heritage information about the site. Kapaki-pakinabang ang plataporma sa pag-access ng impormasyon tungkol sa pamanang kultural ng lugar.					
8	The platform helps me understand the history and heritage of the site. Nakatutulong ang plataporma upang maunawaan ko ang kasaysayan at pamanang kultural ng lugar.					
9	The platform is a useful resource for learning about the heritage site. Kapaki-pakinabang na sanggunian ang plataporma sa pagkatuto tungkol sa pamanang kultural ng lugar.					

D. ATTITUDE TOWARD USING

No.	Statement	5	4	3	2	1
10	I have a positive attitude toward using the platform. Positibo ang aking pananaw sa paggamit ng plataporma.					
11	I think using the platform is a good idea. Sa palagay ko, magandang ideya ang paggamit ng plataporma.					
12	I like the idea of using the platform as a digital heritage resource. Gusto ko ang ideya ng paggamit sa plataporma bilang digital na sanggunian tungkol sa pamanang kultural.					

E. BEHAVIORAL INTENTION TO USE

No.	Statement	5	4	3	2	1
13	I intend to use the platform again in the future. Nais kong muling gamitin ang plataporma sa hinaharap.					
14	I intend to use the platform when I need information about the heritage site. Nais kong gamitin ang plataporma kapag kailangan ko ng impormasyon tungkol sa pamanang kultural ng lugar.					
15	I plan to use the platform for future museum or heritage-learning activities. Plano kong gamitin ang plataporma sa mga susunod na gawain tungkol sa museo o pagkatuto sa pamanang kultural.					

F. ACTUAL SYSTEM INTERACTION

No.	Statement	5	4	3	2	1
16	I used more than one section of the platform during the evaluation. Gumamit ako ng higit sa isang seksyon ng plataporma habang isinasagawa ang ebalwasyon.					
17	I interacted with the platform's digital heritage content during the evaluation. Nakipag-ugnayan ako sa digital na nilalaman tungkol sa pamanang kultural habang isinasagawa ang ebalwasyon.					
18	I used the platform to access heritage information during the evaluation. Ginamit ko ang plataporma upang ma-access ang impormasyon tungkol sa pamanang kultural habang isinasagawa ang ebalwasyon.					

G. IMMERSIVE DIGITAL EXPERIENCE

No.	Statement	5	4	3	2	1
19	The digital exhibits help me explore the heritage site. Nakatutulong ang mga digital exhibit sa paggalugad ko sa pamanang kultural ng lugar.					
20	The 360° virtual tour helps me visualize the heritage site. Nakatutulong ang 360° virtual tour upang mailarawan ko ang pamanang kultural ng lugar.					
21	The AR/WebXR features are engaging to use. Kawili-wiling gamitin ang mga tampok na AR/WebXR.					
22	The interactive features make the heritage experience engaging. Ginagawang mas kawili-wili ng mga interaktibong tampok ang karanasan sa pamanang kultural.					

H. ACCESSIBILITY

No.	Statement	5	4	3	2	1
23	The platform's text is easy for me to read. Madali para sa akin na basahin ang teksto ng plataporma.					
24	The platform's controls are easy for me to use. Madali para sa akin na gamitin ang mga kontrol ng plataporma.					
25	The accessibility features are useful to me. Kapaki-pakinabang sa akin ang mga tampok para sa accessibility.					
26	The accessibility options help me navigate the platform. Nakatutulong ang mga opsyon sa accessibility sa pag-navigate ko sa plataporma.					

PART IV. COMMENTS AND SUGGESTIONS

1. Which feature of the platform did you find most useful?

2. Which part of the platform was difficult to use or understand?

3. Did you experience any device, browser, internet, accessibility, AR or WebXR difficulty? Please describe.

4. What feature or improvement would you recommend?

5. Other comments or suggestions:

PART V. COMPLETION CHECK

( ) I answered the questionnaire based on my actual experience with the system.
( ) I reviewed my responses for completeness.

Respondent Code: _____ Date: _____

RESEARCH ADVISER VALIDATION AND RECOMMENDATION

As Research Adviser, I have reviewed the questionnaire for alignment with the study objectives, clarity of the statements, appropriateness of the response scale, organization of the evaluation categories, bilingual presentation and suitability for the intended system-user evaluation. Based on this review, the instrument is validated and recommended for actual survey administration, subject to compliance with the study's approved research protocol and any required institutional/ethics approval.

☐ The items are aligned with the intended system evaluation.

☐ The instructions and response options are clear and understandable.

☐ The questionnaire covers the identified evaluation categories.

☐ The instrument is suitable for actual respondent administration.

☐ Comments/suggestions are included to capture qualitative feedback.

☐ The questionnaire may proceed to actual survey administration subject to required institutional approvals.

RECOMMENDED FOR ACTUAL SURVEY ADMINISTRATION

Validated by:

Mia V. Villarica, DIT

Signature: _____ Date: 09 - 07 - 26

Research Adviser

Recommendation:

☐ Approved/Recommended for Actual Survey
☐ For Revision

User’s Manual

This user's manual presents step-by-step procedures for using the public and administrative features of the WebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan.

Manual Coverage

- Public account registration, sign-in, profile, preferences, and saved progress
- Digital exhibits, Online Gallery, educational resources, and support information
- Virtual Tour, Site Map, WebXR 3D Environment, AR detection, and Heritage Trail
- Community participation, tour requests, stories, and visitor feedback
- Administrator dashboards, content libraries, bookings, users, validation, and publishing

Before Using the System

- Use a current desktop or mobile browser and a stable internet connection.
- Allow audio, camera, motion, or full-screen access only for features that require it.
- Use a registered public account to retain bookmarks and interactive progress.
- Use the administrator portal only with an authorized staff account.

Instruction Convention

The word Select covers a mouse click, screen tap, or keyboard activation. Labels in bold identify visible controls. Administrator actions that change status or

remove content must be reviewed in the system's confirmation dialog before they are completed.

Public Homepage

Figure A65. Public Homepage.

Figure A65 shows the public homepage, which provides the main navigation and entry points to the platform's heritage experiences.

Procedure

Step 1. Open the deployed website in a current web browser.

Step 2. Use the navigation bar to open Digital Exhibits, Interactive, Heritage Trail, Community Participation, or Support.

Step 3. Select the language or accessibility controls when an alternative presentation is needed.

Step 4. Select Sign In to use account-only features, or continue as a guest where permitted.

Create a Public Account

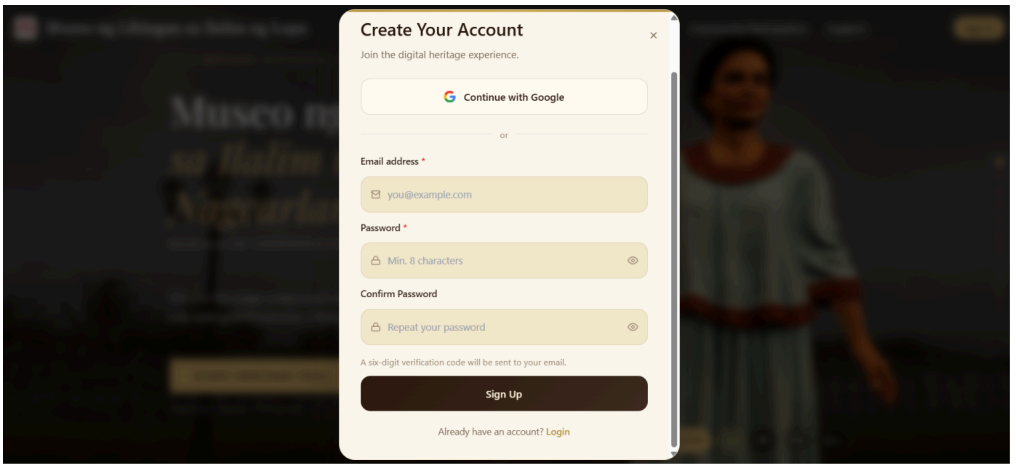


Figure A66. Create a Public Account.

Figure A66 shows the account registration dialog used by a visitor to create a platform account.

Procedure

Step 1. Select Sign Up from the public navigation or sign-in dialog.

Step 2. Enter a valid email address, password, and password confirmation.

Step 3. Select Sign Up and retrieve the one-time verification code sent to the email address.

Step 4. Enter the code and complete the requested profile information.

Step 5. Sign in after the account setup is confirmed.

Sign In and Password Recovery

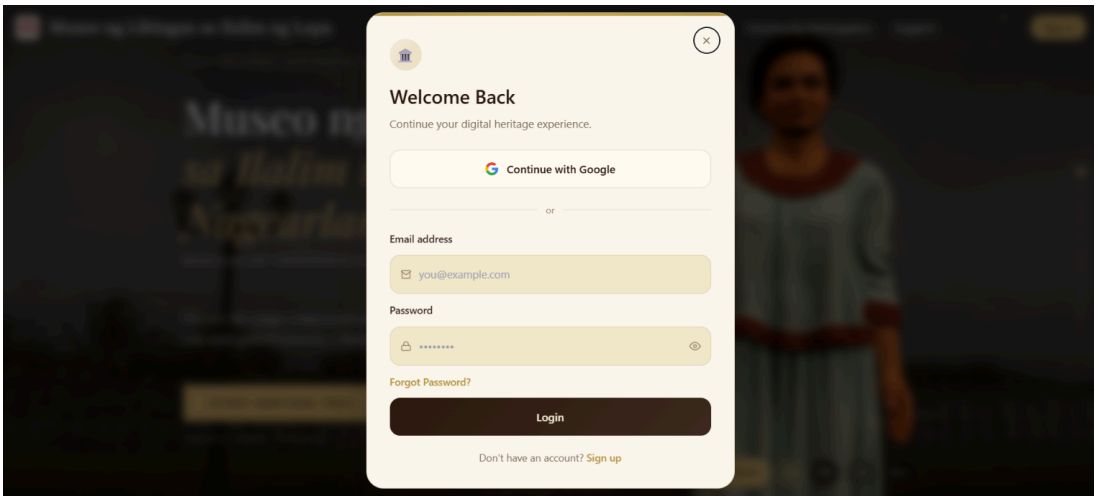


Figure A67. Sign In and Password Recovery.

Figure A67 shows the public sign-in dialog for accessing saved progress and personalized account features.

Procedure

Step 1. Select Sign In and enter the registered email address and password.

Step 2. Select Login, or select Continue with Google when that provider was used for the account.

Step 3. If the password is forgotten, select Forgot Password and enter the registered email address.

Step 4. Open the recovery message, follow the secure link or code, and create a new password.

Digital Exhibit Hub

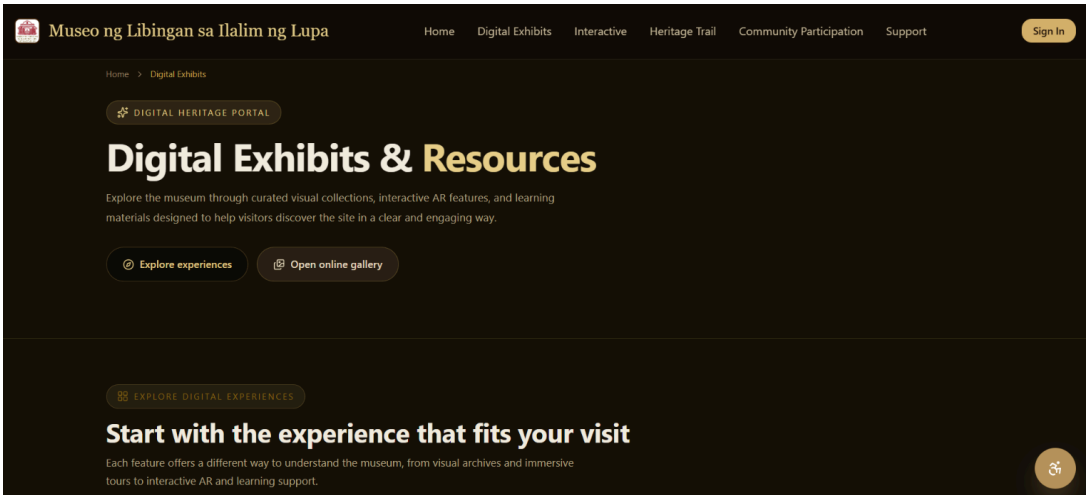


Figure A68. Digital Exhibits Hub.

Figure A68 shows the Digital Exhibits and Resources hub, where visitors choose a curated digital experience.

Procedure

Step 1. Select Digital Exhibits from the main navigation.

Step 2. Review the featured experiences and their short summaries.

Step 3. Select Explore experiences to view the available exhibit categories.

Step 4. Select Open online gallery to browse digitized collection records.

Online Gallery

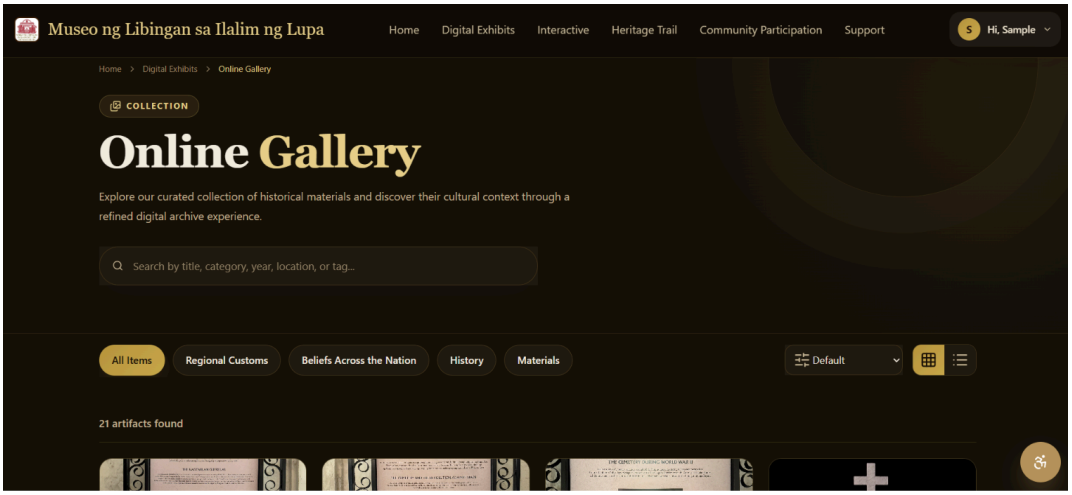


Figure A69. Online Gallery.

Figure A69 shows the searchable Online Gallery used to discover digitized heritage objects and collection records.

Procedure

Step 1. Open the Online Gallery from Digital Exhibits.

Step 2. Enter a title, category, period, person, or tag in the search field.

Step 3. Apply a category or display filter to narrow the results.

Step 4. Select a gallery card to view its bilingual details, images, and available 3D model.

Step 5. Sign in and use the bookmark control when the item should be saved.

Interactive and Immersive Hub

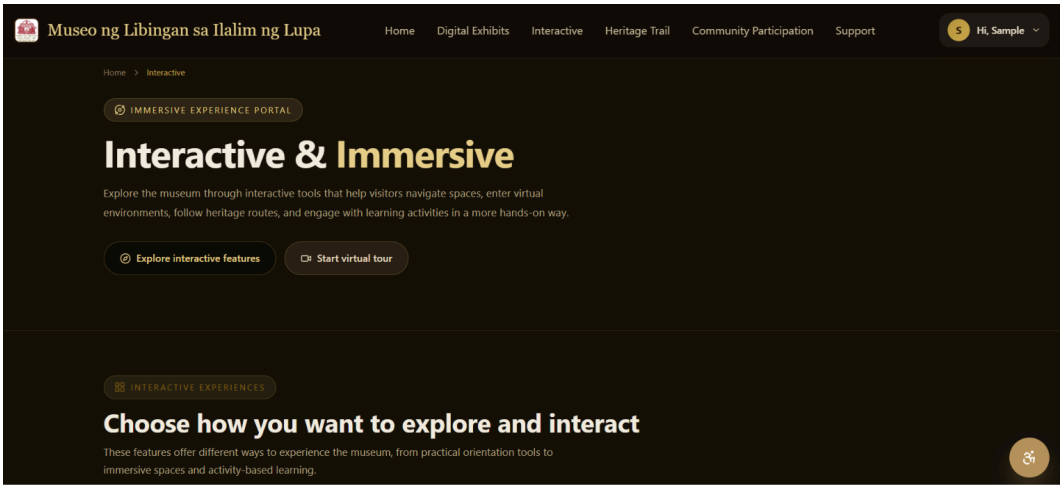


Figure A70. Interactive and Immersive Hub.

Figure A70 shows the Interactive and Immersive hub that links visitors to the site's guided digital experiences.

Procedure

Step 1. Select Interactive from the main navigation.

Step 2. Review the device and interaction notes for each experience.

Step 3. Choose the Virtual Tour, Site Map, 3D Environment, or AR Content Detection experience.

Step 4. Allow full-screen, motion, or camera access only when the selected experience requests it.

360 Degree Virtual Tour

Figure A71. 360-Degree Virtual Tour.

Figure A71 shows the 360-degree Virtual Tour and its scene navigation, hotspot, narration, and minimap controls.

Procedure

Step 1. Open Virtual Tour and wait for the starting panorama to finish loading.

Step 2. Drag the panorama to look around, and use zoom controls when needed.

Step 3. Select a navigation hotspot to move to another scene or an information hotspot to open its content.

Step 4. Use EN or TL, narration, subtitles, and the minimap according to preference.

Step 5. Close the information panel or select Back to leave the experience.

Interactive Site Map

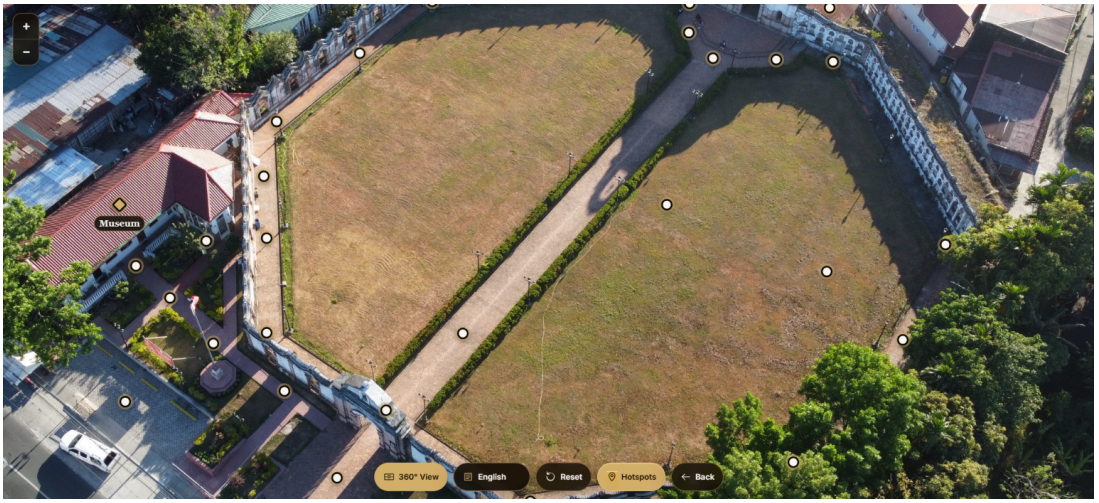


Figure A72. Interactive Site Map.

Figure A72 shows the aerial Site Map, where visible markers identify important areas and linked virtual-tour scenes.

Procedure

Step 1. Open Site Map and select Aerial View or 360-degree View.

Step 2. Turn hotspot visibility on when the markers are hidden.

Step 3. Select a marker to open its bilingual information, photograph, and narration when available.

Step 4. Drag the information panel or the 360-degree view without closing the selected information.

Step 5. Select Open Virtual Tour when the location provides a linked scene.

WebXR 3D Environment

Figure A73. WebXR 3D Environment.

Figure A73 shows the loading entry for the WebXR-enabled 3D Environment before the museum scene and controls appear.

Procedure

Step 1. Open the 3D Environment from the Interactive hub and wait for all assets to load.

Step 2. Use the movement control or keyboard controls to move, and drag or move the device to look around.

Step 3. Approach an interactive object and follow the on-screen action prompt.

Step 4. Collect or deposit eligible objects to update Heritage Keeper progress.

Step 5. Sign in when the system requests an account for saved progress.

AR Content Detection

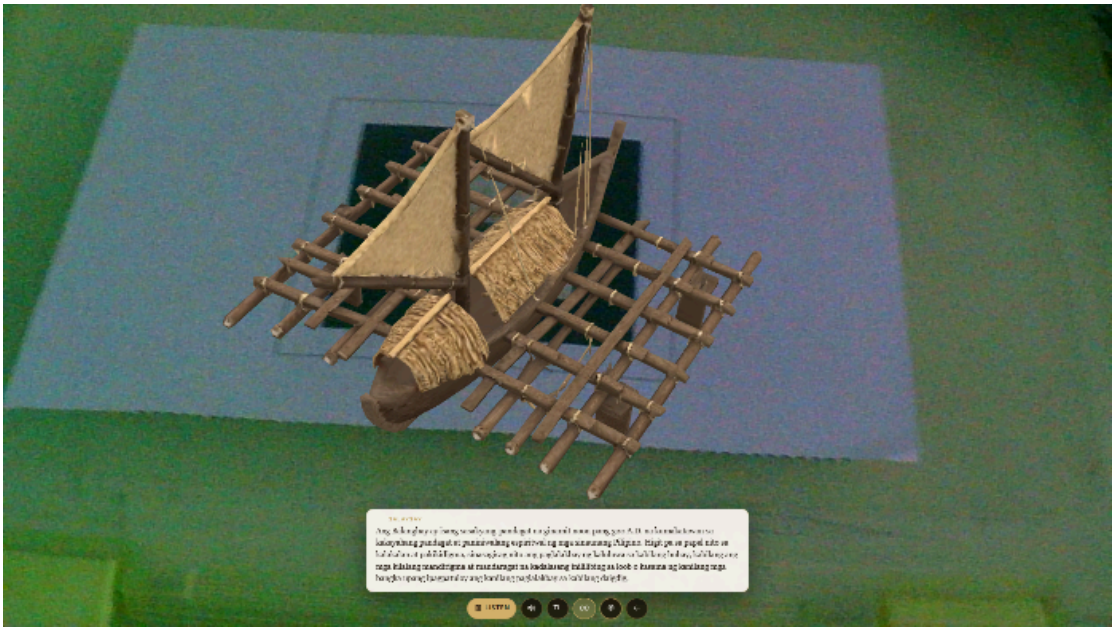


Figure A74. AR Content Detection.

Figure A74 shows the loading entry for marker-based AR Content Detection before camera access and artifact tracking begin.

Procedure

Step 1. Open AR Content Detection from Digital Exhibits or the Interactive hub.

Step 2. Wait for the artifact files to load, then allow camera access.

Step 3. Point the camera at a supported printed marker in a well-lit area.

Step 4. Keep the marker steady until the 3D artifact, information, and narration controls appear.

Step 5. Use the language, audio, subtitle, restoration, or back controls as needed.

Heritage Trail

Figure A75. Heritage Trail.

Figure A75 shows the visual entry to the Heritage Trail, a guided sequence of stories, locations, and collectible seal fragments.

Procedure

Step 1. Select Heritage Trail from the main navigation and start the guided experience.

Step 2. Move through the chapters in sequence and read or listen to each heritage story.

Step 3. Use the map and scene prompts to locate the next stop.

Step 4. Complete the required interactions to collect available seal fragments.

Step 5. Open Heritage Keeper from the account page to review saved progress.

Educational Resources

Figure A76. Educational Resources.

Figure A76 shows the Educational Resources library for finding classroom-ready documents, stories, and reference materials.

Procedure

Step 1. Open Educational Resources from Digital Exhibits or the main navigation.

Step 2. Search by title, author, subject, or topic.

Step 3. Apply the resource-type and subject filters to narrow the library.

Step 4. Select a resource card to view its record and available file.

Step 5. View or download the resource according to the access option shown.

Community Participation

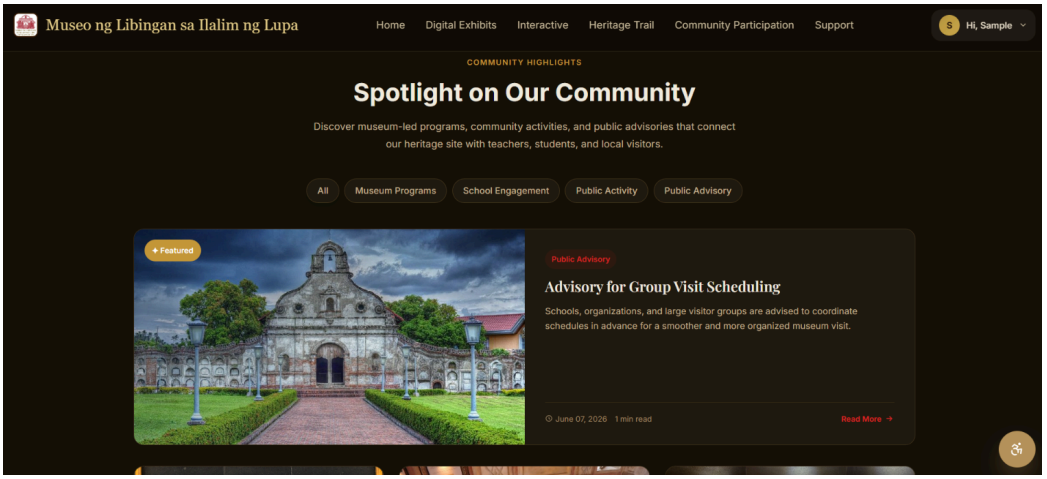


Figure A77. Community Participation**.**

Figure A77 shows the Community Participation page for museum programs, local stories, public activities, and visitor contributions.

Procedure

Step 1. Select Community Participation from the main navigation.

Step 2. Choose Museum Programs, School Engagement, Public Activity, or Public Advisory.

Step 3. Review the available highlights and scheduled activities.

Step 4. Use the booking or contribution action to submit the required information.

Step 5. Complete email verification when the submission workflow requests it.

Support and Governance

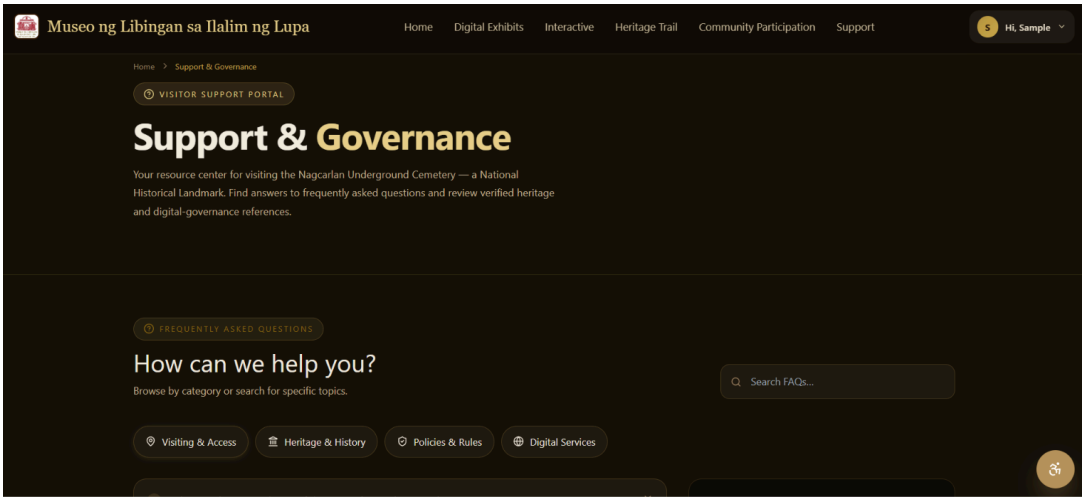


Figure A78. Support and Governance.

Figure A78 shows the Support and Governance page, which groups visitor assistance, heritage guidance, policies, and digital services.

Procedure

Step 1. Select Support from the main navigation.

Step 2. Choose a support category or search for the relevant topic.

Step 3. Read the instructions, policy, accessibility, privacy, or contact information provided.

Step 4. Use the stated contact channel when the concern requires staff assistance.

Public Account Profile

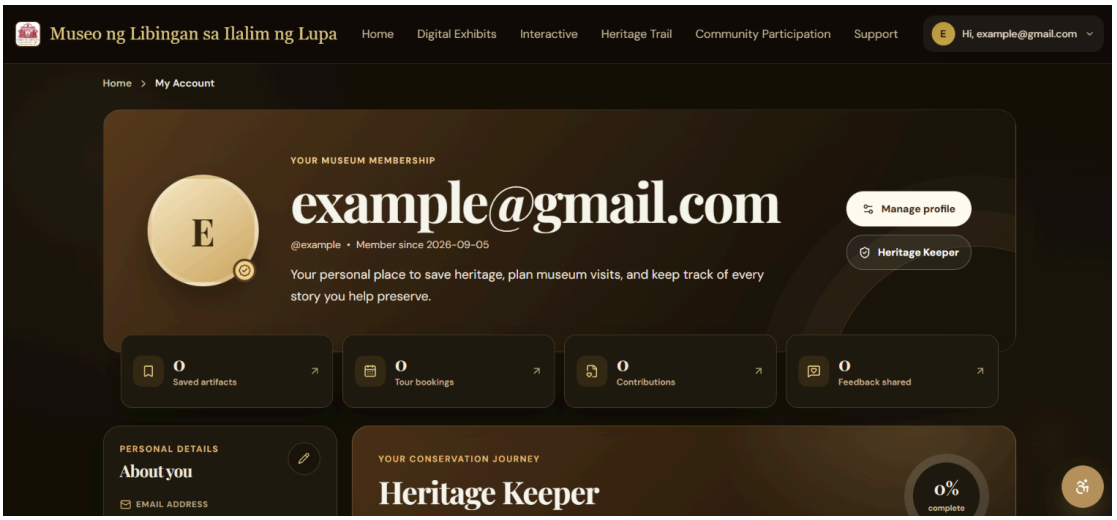


Figure A79. Public Account Profile.

Figure A79 shows the visitor account dashboard with profile details, saved activity, bookings, and feature shortcuts.

Procedure

Step 1. Sign in and select the account name in the public navigation.

Step 2. Open My Account to view the profile dashboard.

Step 3. Review saved bookmarks, booking activity, collection progress, and recent actions.

Step 4. Select Manage Profile, Heritage Keeper, or another shortcut to continue.

Heritage Keeper

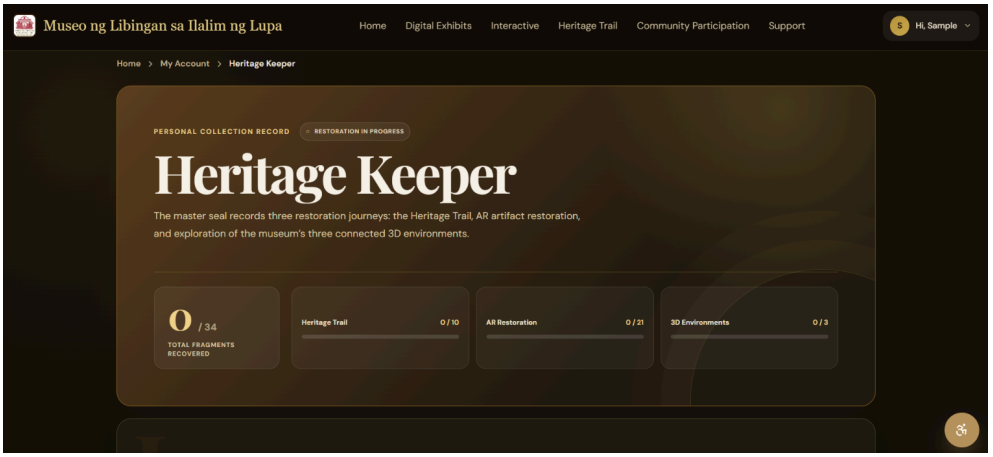


Figure A80. Heritage Keeper.

This A80 figure shows Heritage Keeper, which records seal fragments and progress earned across the platform's interactive experiences.

Procedure

Step 1. Open My Account and select Heritage Keeper.

Step 2. Review the total seal progress and the activity-specific fragment trays.

Step 3. Return to AR Restoration, Heritage Trail, or the 3D Environment to obtain missing fragments.

Step 4. Refresh or revisit Heritage Keeper after completing an eligible activity.

Public Account Settings

Figure A81. Public Account Settings.

Figure A81 shows the public account settings used to update the visitor's profile, preferences, security, and account status.

Procedure

Step 1. Open My Account and select Manage Profile or Settings.

Step 2. Update the profile, avatar, roles, location, language, or theme as needed.

Step 3. Open Security to change the password using the required verification step.

Step 4. Select Save changes and confirm that the success message appears.

Step 5. Use Account access only when signing out sessions or deleting the account is intended.

Administrator Sign In

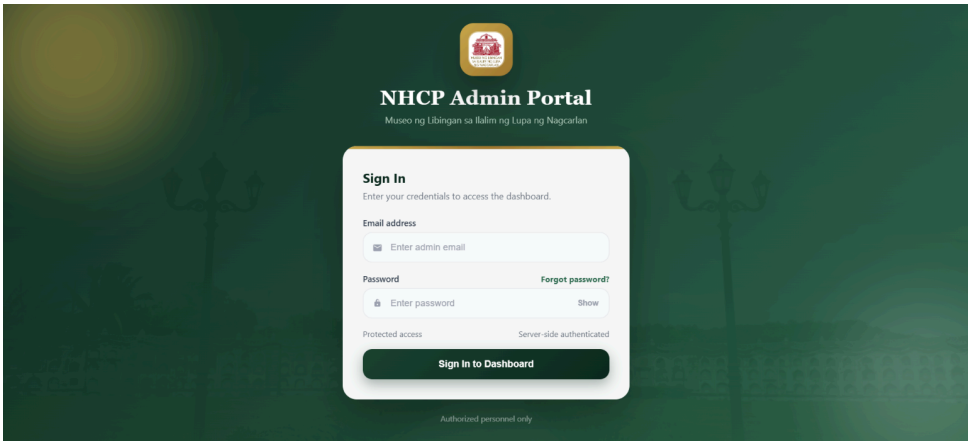


Figure A82. Administrator Sign In.

Figure A82 shows the protected administrator sign-in page for authorized museum personnel.

Procedure

Step 1. Open the administrator portal address provided by the system owner.

Step 2. Enter the authorized administrator email address and password.

Step 3. Select Sign In to Dashboard.

Step 4. Use Forgot password when recovery is required and follow the verified email process.

Step 5. Do not share administrator credentials or leave an authenticated workstation unattended.

Administrator Dashboard

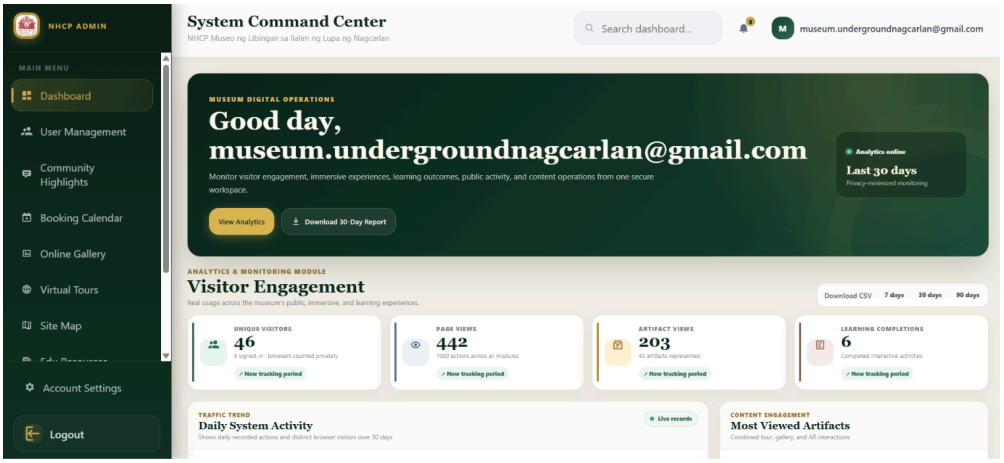


Figure A83. Administrator Dashboard.

Figure A83 shows the System Command Center, which summarizes records, recent activity, queues, and administrator shortcuts.

Procedure

Step 1. Sign in to the administrator portal.

Step 2. Review configuration notices, record totals, visitor engagement, and recent activity.

Step 3. Use the left sidebar to open the required management module.

Step 4. Open notifications to review items that require attention.

Step 5. Use the account menu or Logout when administration is complete.

User Management

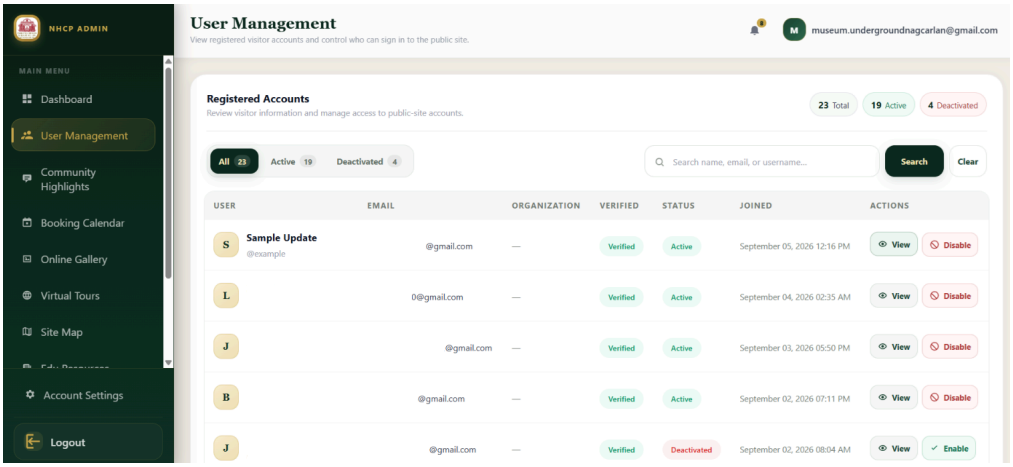


Figure A84. User Management.

Figure A84 shows User Management, where administrators search, inspect, and control registered public accounts.

Procedure

Step 1. Select User Management from the administrator sidebar.

Step 2. Search by name or email, or apply an account-status filter.

Step 3. Open the required user record and verify its details before changing access.

Step 4. Activate or deactivate the account and confirm the action in the confirmation dialog.

Tour Booking Calendar

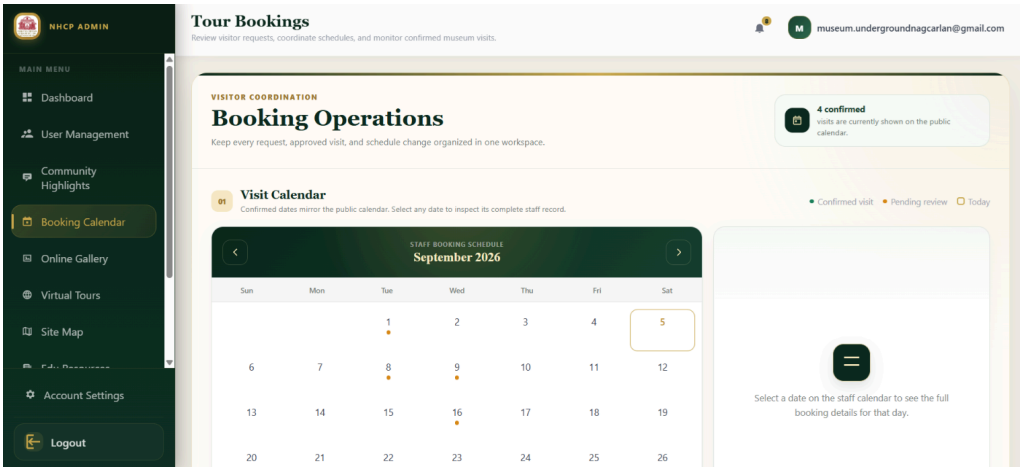


Figure A85. Tour Booking Calendar.

Figure A85 shows Booking Operations, where administrators review the visit calendar and process tour requests.

Procedure

Step 1. Select Booking Calendar from the administrator sidebar.

Step 2. Choose a date or filter the booking list by status.

Step 3. Open a request and review the visitor, schedule, party size, and contact details.

Step 4. Confirm, reschedule, cancel, complete, or mark the request as no-show as appropriate.

Step 5. Enter any required reason and confirm the status change.

Community Highlights Management

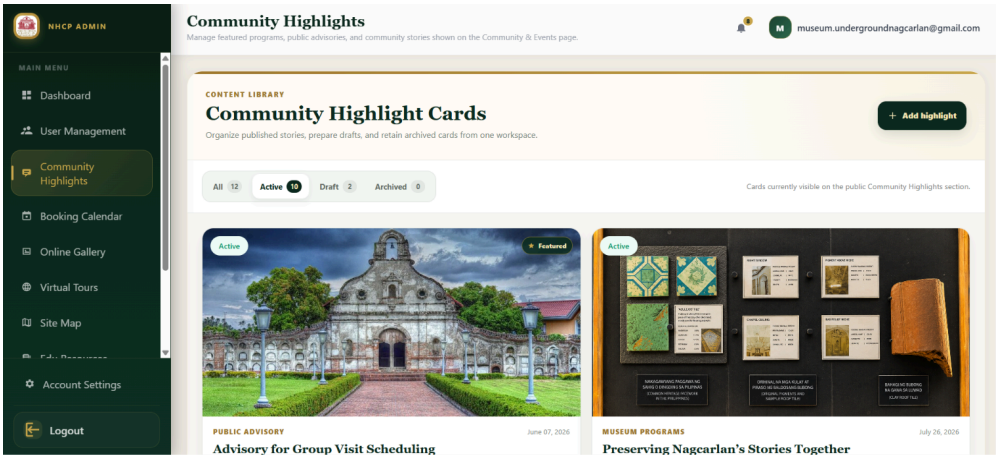


Figure A86. Community Highlights Management.

Figure A86 shows the Community Highlight library used to prepare and publish bilingual museum and community stories.

Procedure

Step 1. Select Community Highlights and choose Add highlight.

Step 2. Enter the English and Tagalog titles, summaries, article text, date, and category.

Step 3. Upload the required image and verify its alternative text and reading time.

Step 4. Save the record, review its preview, and correct incomplete fields.

Step 5. Publish, feature, unfeature, archive, or restore the record after confirmation.

Online Gallery Management

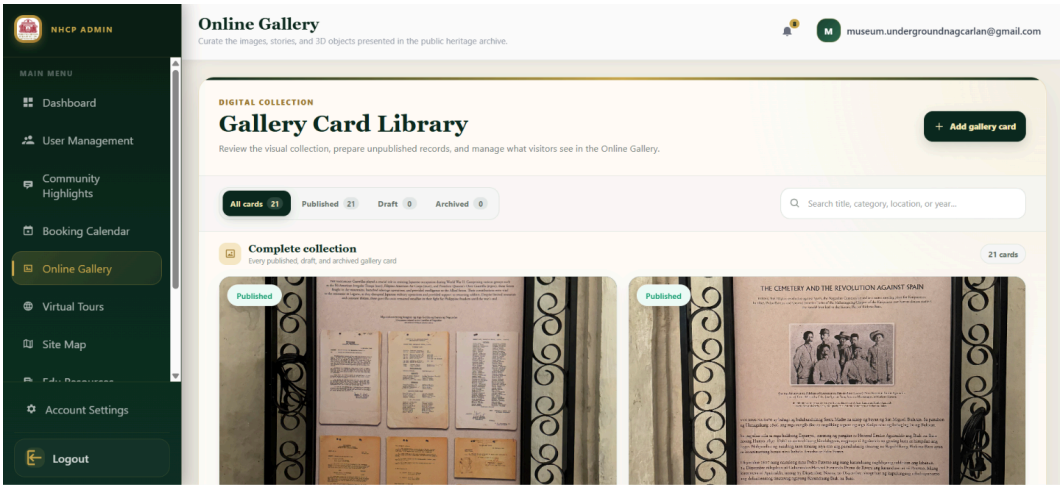


Figure A87. Online Gallery Management.

Figure A87 shows the administrator's Gallery Card Library for preparing public collection records and digital media.

Procedure

Step 1. Select Online Gallery and choose Add gallery item.

Step 2. Enter the bilingual title, description, category, historical metadata, and credits.

Step 3. Upload the supported image and optional 3D model, then verify the preview.

Step 4. Save the draft and correct any validation errors.

Step 5. Activate, publish, archive, restore, or delete the record only after confirmation.

Virtual Tour Management

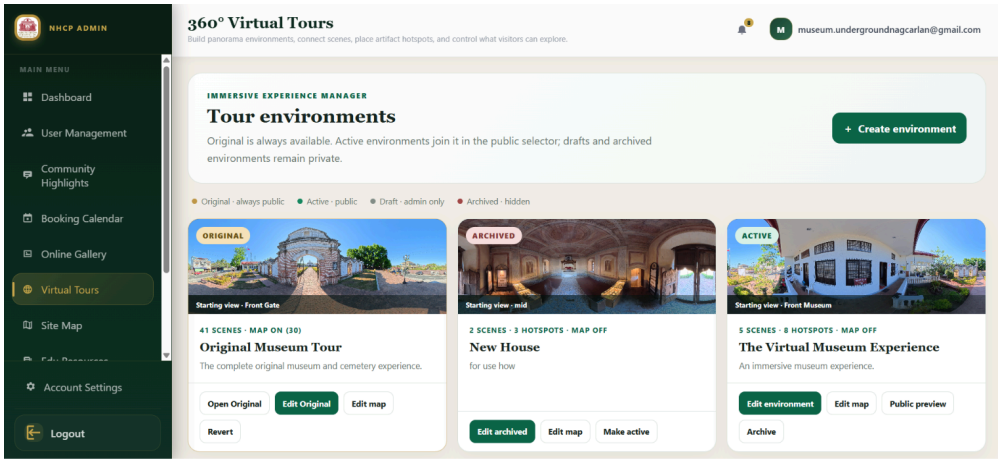


Figure A88. Virtual Tour Management.

Figure A88 shows the 360-degree Virtual Tours library used to create, edit, validate, and publish tour environments.

Procedure

Step 1. Select Virtual Tours and open an environment or choose Create environment.

Step 2. Edit the bilingual metadata and add valid 2:1 panorama scenes within the stated 30 MB limit.

Step 3. Set exactly one starting scene and place item or navigation hotspots using pitch and yaw.

Step 4. Edit the upper-right map, scene links, and per-scene map visibility when required.

Step 5. Save the draft, run preflight validation, and publish only after all reported issues are resolved.

Site Map Management

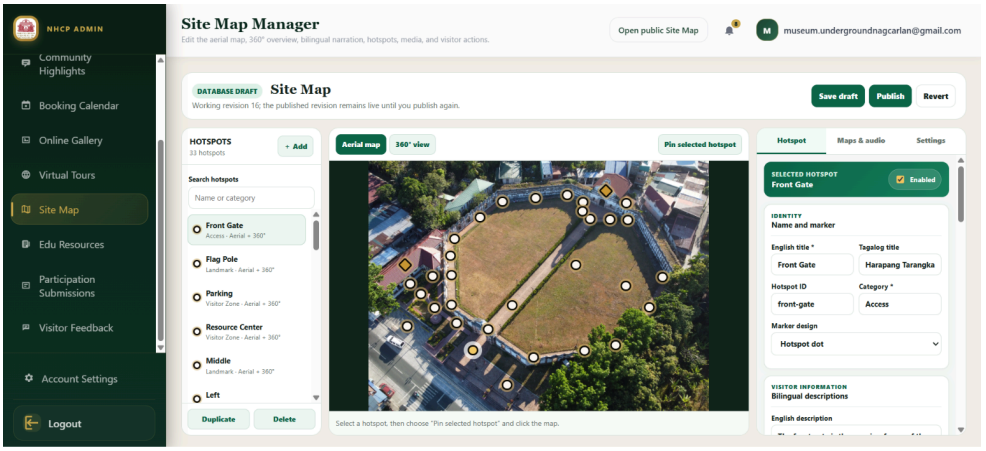


Figure A89. Site Map Management.

Figure A89 shows the Site Map Manager for editing the database revision while preserving the source-file Original.

Procedure

Step 1. Select Site Map and choose Customize or open the working draft.

Step 2. Upload the aerial map, optional 360-degree aerial map, and English or Tagalog starting audio.

Step 3. Add or select a hotspot, then pin its aerial coordinate and 360-degree pitch and yaw.

Step 4. Enter bilingual information, media, narration, related-hotspot status, and an optional action link.

Step 5. Save the draft, run preflight validation, and publish after every issue is corrected.

Educational Resource Management

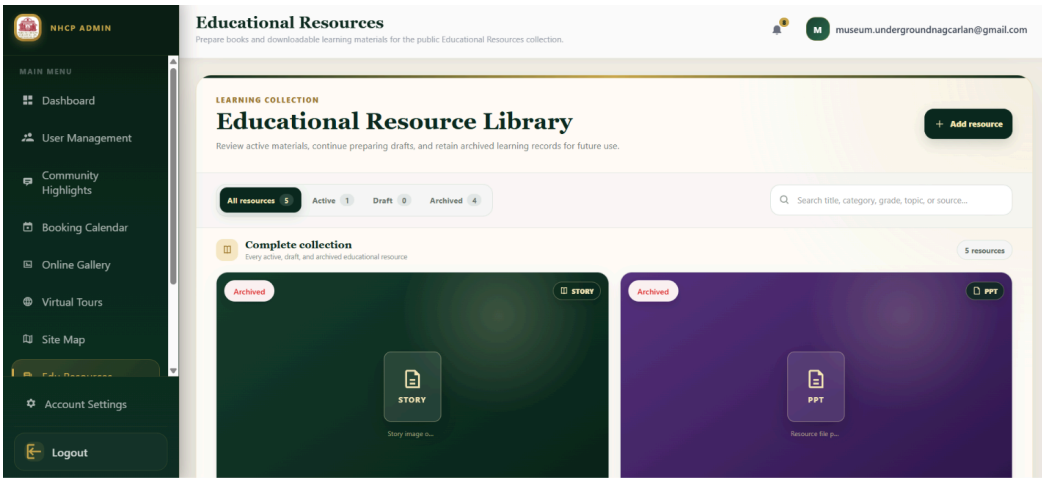


Figure A90. Educational Resource Management.

Figure A90 shows the Educational Resource Library used to organize and publish downloadable learning materials.

Procedure

Step 1. Select Edu Resources and choose Add resource.

Step 2. Enter the title, author, description, subject, topic, resource type, and language.

Step 3. Upload a cover image and the supported resource file or supply the approved resource link.

Step 4. Save the record and review the displayed metadata.

Step 5. Activate, archive, restore, or delete the resource after confirming the action.

Participation Submission Review

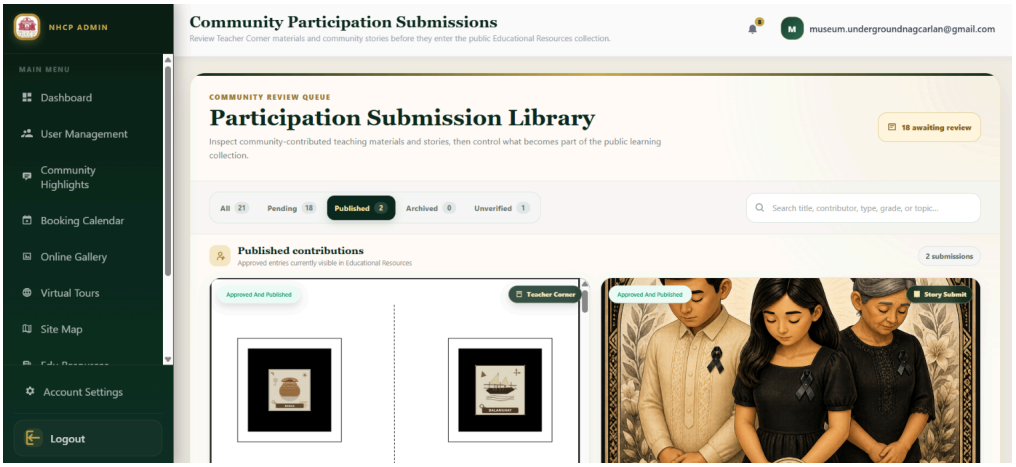


Figure A91. Participation Submission Review.

Figure A91 shows the Participation Submission Library where staff verify visitor-contributed stories and supporting media.

Procedure

Step 1. Select Participation Submissions from the administrator sidebar.

Step 2. Filter the queue by pending, published, archived, or deleted status.

Step 3. Open a submission and review its contributor details, bilingual text, permissions, and media.

Step 4. Approve and publish an acceptable submission, or archive it when further action is required.

Step 5. Confirm each status-changing action before it is applied.

Visitor Feedback Review

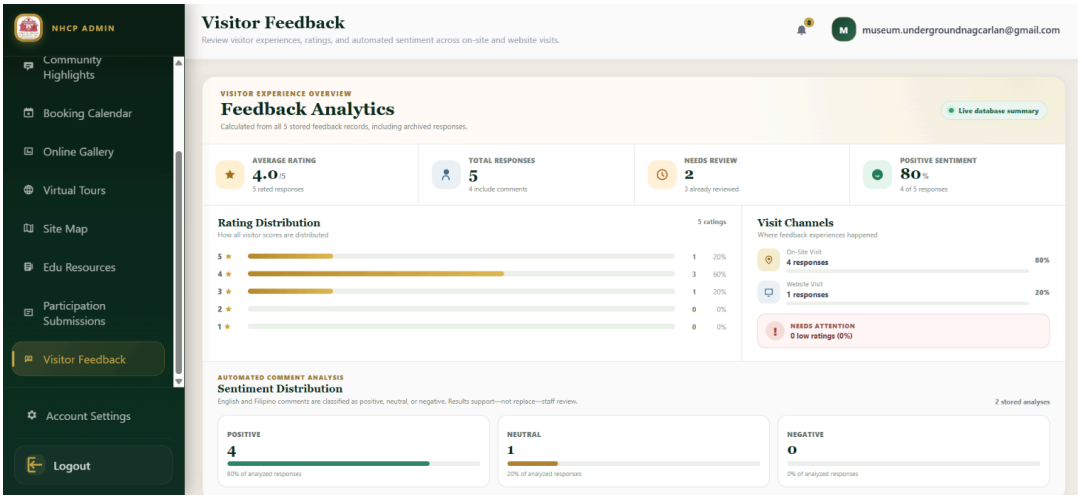


Figure A92. Visitor Feedback Review.

Figure A92 shows the Visitor Feedback dashboard with rating distribution, sentiment summaries, and individual comments.

Procedure

Step 1. Select Visitor Feedback from the administrator sidebar.

Step 2. Review the summary cards, rating distribution, visit channels, and sentiment classification.

Step 3. Filter or search the feedback list and open the required entry.

Step 4. Mark the item as reviewed after staff evaluation.

Step 5. Archive, restore, or delete an item only after confirming the action.

Administrator Account Settings

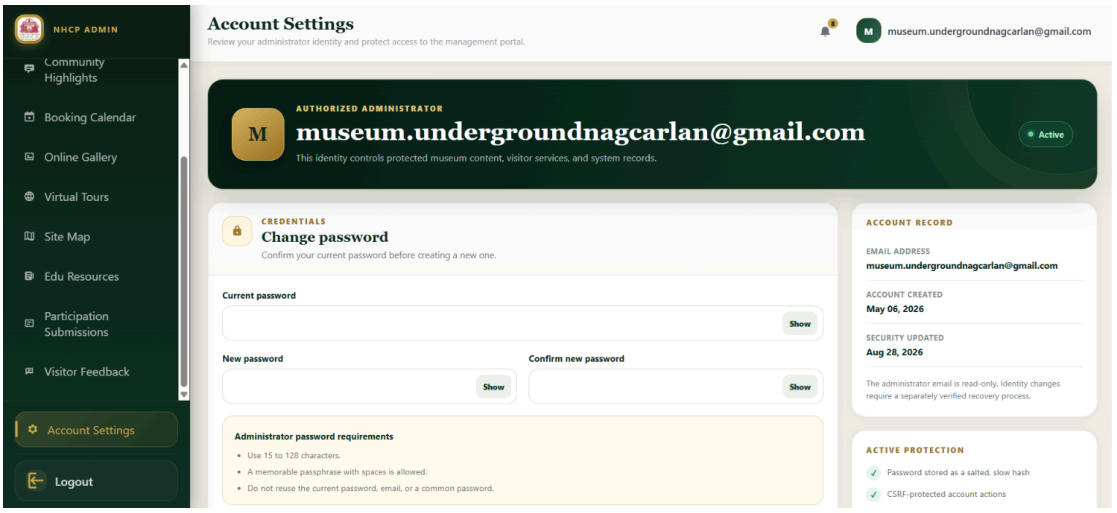


Figure A93. Administrator Account Settings.

Figure A93 shows administrator account settings for reviewing account status and changing the portal password.

Procedure

Step 1. Select Account Settings from the administrator sidebar.

Step 2. Review the signed-in account, security status, and session guidance.

Step 3. Enter the current password, new password, and confirmation.

Step 4. Select Change password and wait for the success confirmation.

Step 5. Reauthenticate when required and sign out after completing administrative work.

APPENDIX B
COMMUNICATION
LETTERS AND FORMS

Communication Letter

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

Date: February 25, 2026

ALBERT VINCENT F. BARRETTO
Officer in Charge
Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan
Brgy. Bambang, Nagcarlan Laguna

SUBJECT: Request for Initial Interview for Capstone Project System Development

Dear Mr. Barretto,

Warm greetings!

We are third-year students of the Bachelor of Science in Information Technology under the College of Computer Studies. As part of the academic requirements for our Capstone Project, we respectfully request permission from your good office to conduct an initial interview and requirements gathering with your organization.

The purpose of this interview is to understand your current system or processes and to identify possible areas where an information system may be designed and developed to help improve operational efficiency. The information gathered will be used strictly for academic purposes only and will be treated with utmost confidentiality.

Should you grant our request, the interview may be conducted at your most convenient date and time, either face-to-face or through an online platform.

We sincerely hope for your kind consideration and support in the completion of our Capstone Project. Thank you very much for your time and cooperation.

Respectfully yours,

Alay, Riz-Anne V.
BSIT3E

Bustria, John Lloyd J.
BSIT3E

Gayoso, Cedrick C.
BSIT3E

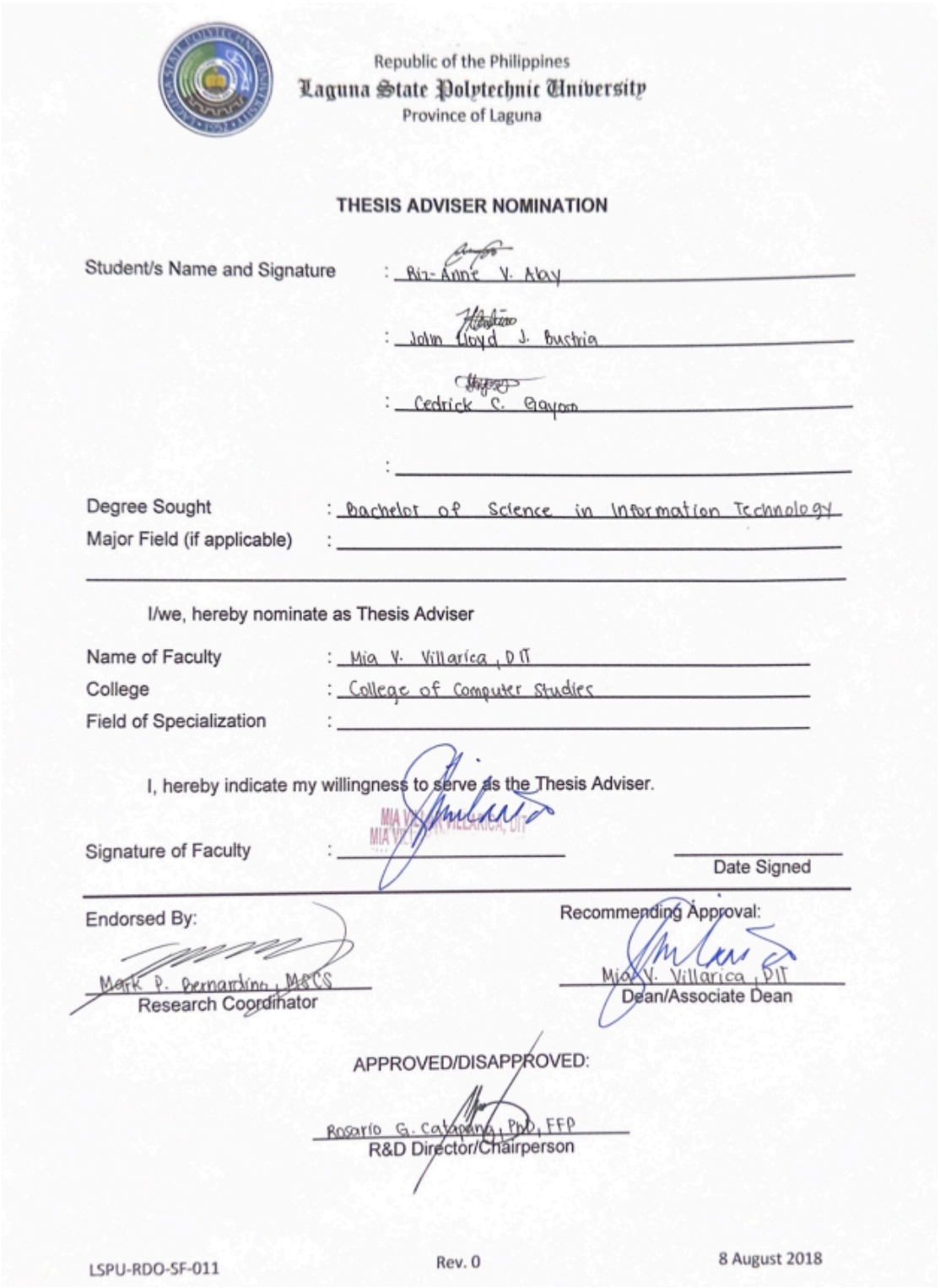
Noted by:

Maria Laureen B. Miranda, LPT, MIT
Capstone Project Subject Instructor
College of Computer Studies

Received by

ALBERT VINCENT F. BARRETTO
National Historical Commission
of the Philippines

Thesis Adviser Nomination Form



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS ADVISER NOMINATION

Student/s Name and Signature : Biz-Anne V. Alay

: John Lloyd J. Bustria

: Cedrick C. Gayon

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) :

I/we, hereby nominate as Thesis Adviser

Name of Faculty : Mia V. Villarica, DIT

College : College of Computer Studies

Field of Specialization :

I, hereby indicate my willingness to serve as the Thesis Adviser.

Signature of Faculty : MIA V. VILLARICA, DIT

Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

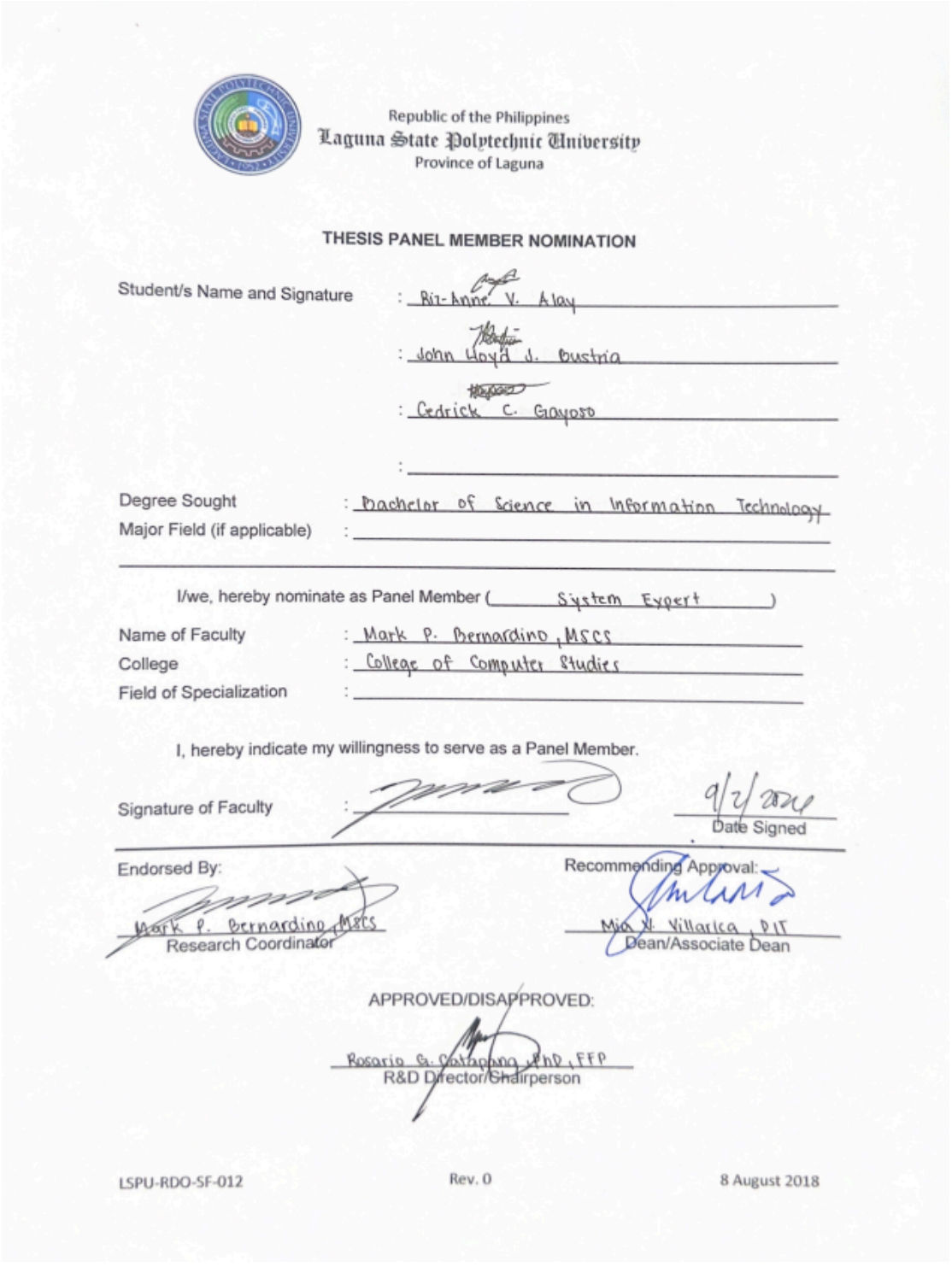
Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catipang, PhD, FFP
R&D Director/Chairperson

LSPU-RDO-SF-011 Rev. 0 8 August 2018

Thesis Panel Nomination Form



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Riz-Anne V. Alay
: John Lloyd J. Bustria
: Cedrick C. Grayoso
:

Degree Sought : Bachelor of Science in Information Technology
Major Field (if applicable) :

I/we, hereby nominate as Panel Member (System Expert)

Name of Faculty : Mark P. Bernardino, MSCS
College : College of Computer Studies
Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : 9/2/2024
Date Signed

Endorsed By:
Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:
Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:
Rosario G. Gothong, PhD, FFP
R&D Director/Chairperson

LSPU-RDO-SF-012 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Riz-Anne V. Alay
: John Lloyd J. Bustria
: Cedrick C. Gayoso
:

Degree Sought : Bachelor of Science in Information Technology
Major Field (if applicable) :

I/we, hereby nominate as Panel Member (Technical Editor)

Name of Faculty : Reynalen C. Justo, LPT, DIT
College : College of Computer Studies
Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia N. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Calapang, PhD, FFP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Ria-Anne V. Alay

: John Lloyd J. Bustria

: Fedrick C. Bayoso

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) :

I/we, hereby nominate as Panel Member (Statistician)

Name of Faculty : Gener F. Mosico, MSIT

College : College of Computer Studies

Field of Specialization : ICT

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty :

Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia N. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Calayagan, PhD, FPP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : Riz-Anne V. Alay

: John Lloyd J. Bustria

: Cedrick C. Gayoso

:

Degree Sought : Bachelor of Science in Information Technology

Major Field (if applicable) :

I/we, hereby nominate as Panel Member (Language Critic)

Name of Faculty : Ma. Cezanne D. Dimaculangan

College : College of Arts and Science

Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : _____ Date Signed

Endorsed By:

Mark D. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD, FFP
R&D Director/Chairperson

Title Defense Rating Sheet

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. EW01	Date: March 25, 2026	Time:
Name of Proponents: Alay, Riz-Anne V. Bustria, John Lloyd J. Gayoso, Cedrick C.		
Research Title: Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	98

VERDICT

[ ] **APPROVED.** Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

MIA Y. VILLANUEVA

Role on the Defense

Signature over Printed Name of Evaluator

Form 3

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. EW01	Date: March 25, 2024	Time:
Name of Proponents: Alay, Riz-Anne V. Bustria, John Lloyd J. Gayoso, Cedrick C.		
Research Title: Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	98

VERDICT

[ ] **APPROVED.** Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. EW01	Date: March 25, 2026	Time:
Name of Proponents: Alay, Riz-Anne V. Bustria, John Lloyd J. Gayoso, Cedrick C.		
Research Title: Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
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PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	89

VERDICT

[✓] **APPROVED.** Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Panel
Role on the Defense

Reynaldo G. Justo
Signature over Printed Name of Evaluator

Form 3

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. EW01	Date: March 25, 2026	Time:
Name of Proponents: Alay, Riz-Anne V. Bustria, John Lloyd J. Gayoso, Cedrick C.		
Research Title: Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	86

VERDICT

[ ] **APPROVED.** Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Role on the Defense

Signature over Printed Name of Evaluator

Form 3

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. EW01	Date: March 25, 2026	Time:
Name of Proponents: Alay, Riz-Anne V. Bustria, John Lloyd J. Gayoso, Cedrick C.		
Research Title: Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	85

VERDICT

[ ] **APPROVED.** Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

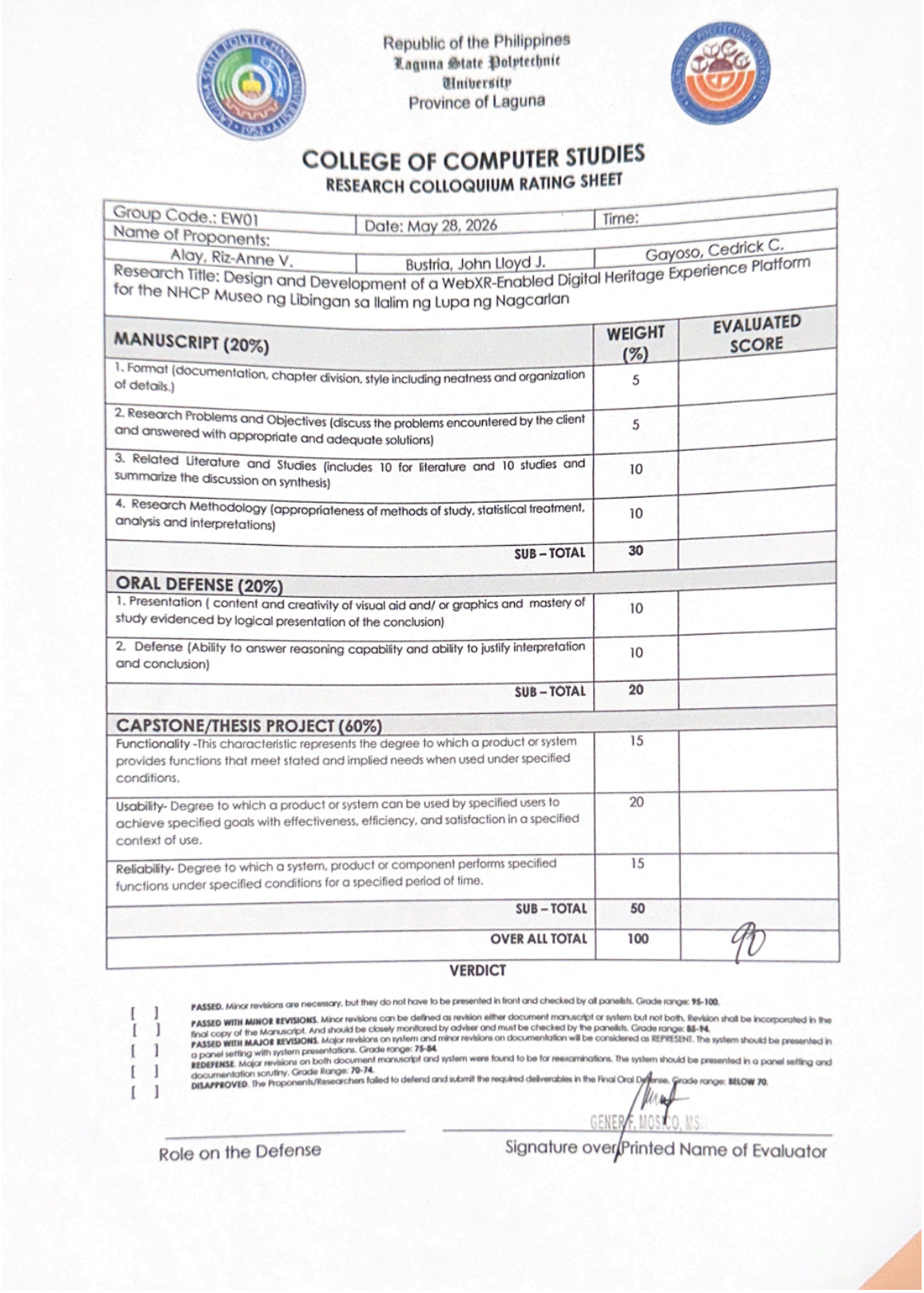
[ ] **DISAPPROVED.** The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Role on the Defense

Signature over Printed Name of Evaluator

Form 3

Research Colloquium Rating Sheet



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: EW01	Date: May 28, 2026	Time:
Name of Proponents:		
Alay, Riz-Anne V.	Bustria, John Lloyd J.	Gayoso, Cedrick C.

Research Title: Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan

MANUSCRIPT (20%)	**WEIGHT (%)**	**EVALUATED SCORE**
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	**20**	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	**50**	
OVER ALL TOTAL	**100**	90

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

GENERE MOSCOSO, MS

Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: EW01	Date: May 20, 2026	Time:
Name of Proponents:		
Alay, Riz-Anne V.	Bustria, John Lloyd J.	Gayoso, Cedrick C.

Research Title: Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan

MANUSCRIPT (20%)	**WEIGHT (%)**	**EVALUATED SCORE**
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	**20**	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	**50**	
OVER ALL TOTAL	**100**	90 1/

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: EW01	Date: May 20, 2026	Time:
Name of Proponents:		
Alay, Riz-Anne V.	Bustria, John Lloyd J.	Gayoso, Cedrick C.
Research Title: Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	**20**	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	**50**	
OVER ALL TOTAL	**100**	

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

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[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: EW01	Date: May 20, 2026	Time:
Name of Proponents:		
Alay, Riz-Anne V.	Bustria, John Lloyd J.	Gayoso, Cedrick C.
Research Title: Design and Development of a WebXR-Enabled Digital Heritage Experience Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	**20**	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	**50**	
OVER ALL TOTAL	**100**	95

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

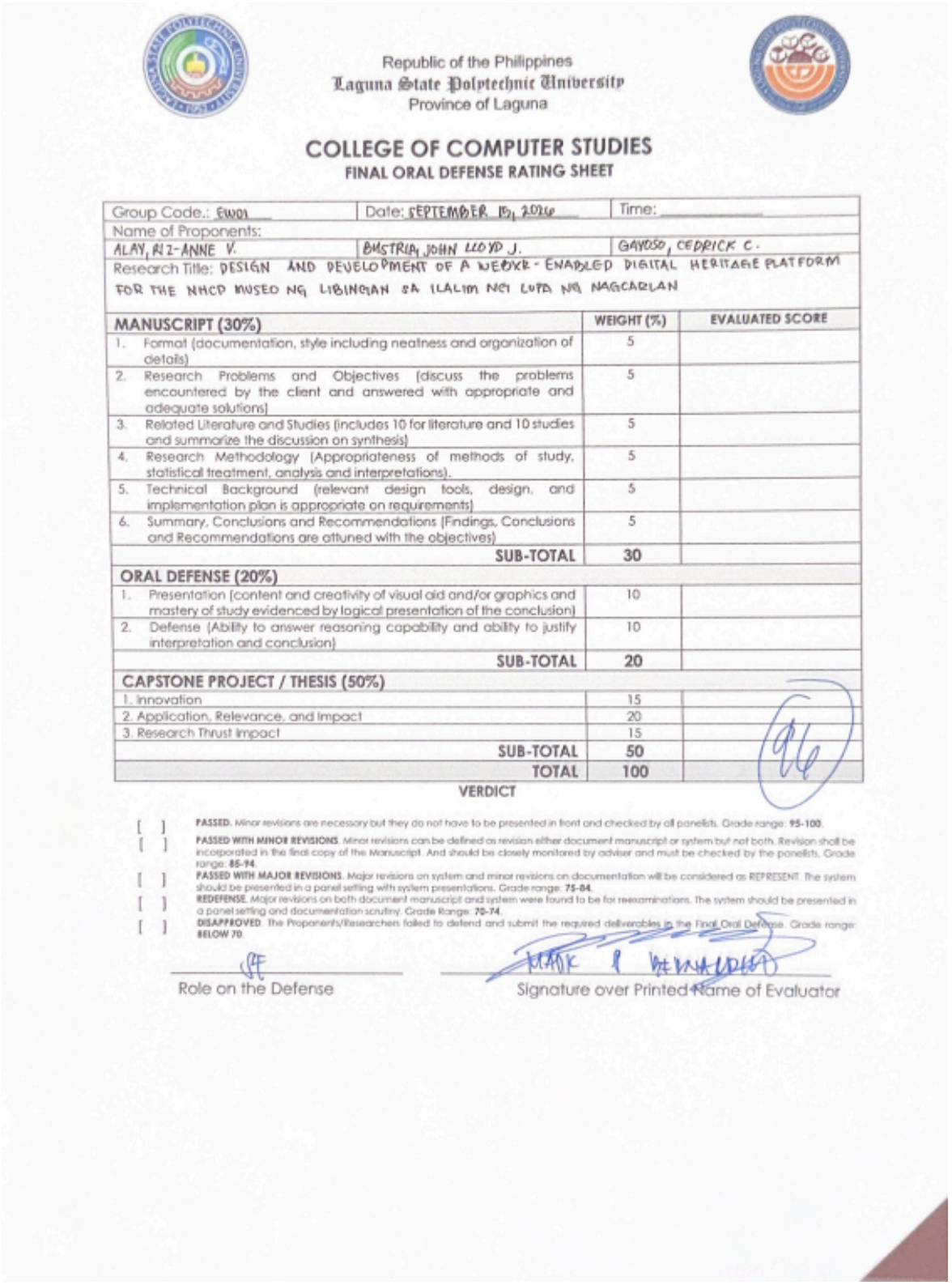
[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Role on the Defense

Signature over Printed Name of Evaluator

Final Oral Defense Rating Sheet



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
FINAL ORAL DEFENSE RATING SHEET

Group Code.: EW01	Date: SEPTEMBER 15, 2016	Time:
Name of Proponents:		
ALAY, RIZ-ANNE V.	BUSTRIA, JOHN LLOYD J.	GAYOSO, CEDRICK C.
Research Title: DESIGN AND DEVELOPMENT OF A WEB/K-ENABLED DIGITAL HERITAGE PLATFORM FOR THE NHCP MUSEO NG LIBINGAN SA ILALIM NG LUPA NG NAGCARLAN		

MANUSCRIPT (30%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	5	
4. Research Methodology (Appropriateness of methods of study, statistical treatment, analysis and interpretations).	5	
5. Technical Background (relevant design tools, design, and implementation plan is appropriate on requirements)	5	
6. Summary, Conclusions and Recommendations (Findings, Conclusions and Recommendations are attuned with the objectives)	5	
SUB-TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	**20**	
CAPSTONE PROJECT / THESIS (50%)		
1. Innovation	15	
2. Application, Relevance, and Impact	20	
3. Research Thrust Impact	15	
SUB-TOTAL	**50**	
TOTAL	**100**	

VERDICT

[ ] **PASSED.** Minor revisions are necessary but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
FINAL ORAL DEFENSE RATING SHEET

Group Code.: EWD1	Date: SEPTEMBER 15, 2026	Time:
Name of Proponents:		
ALAY, RIZ-ANNE V.	BUSTRIA, JOHN LLOYD J.	GLAVOSO, CEDRICK C.
Research Title: DESIGN AND DEVELOPMENT OF A WEBXR-ENABLED DIGITAL HERITAGE PLATFORM FOR THE NHCP MUSEO NG LIBINGLAN SA ILALIM NG LUPA NG NAGCARLAN		

MANUSCRIPT (30%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	5	
4. Research Methodology (Appropriateness of methods of study, statistical treatment, analysis and interpretations).	5	
5. Technical Background (relevant design tools, design, and implementation plan is appropriate on requirements)	5	
6. Summary, Conclusions and Recommendations (Findings, Conclusions and Recommendations are attuned with the objectives)	5	
SUB-TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	**20**	
CAPSTONE PROJECT / THESIS (50%)		
1. Innovation	15	
2. Application, Relevance, and Impact	20	
3. Research Thrust Impact	15	
SUB-TOTAL	**50**	
TOTAL	**100**	95%

VERDICT

[ ] **PASSED.** Minor revisions are necessary but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

CHAIRMAN
Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
FINAL ORAL DEFENSE RATING SHEET

Group Code.: BW01	Date: SEPTEMBER 15, 2026	Time:
Name of Proponents:		
ALAY, RIZ-ANNE V.	BUSTRIA, JOHN LLOYD J.	GAYOSO, CEDRICK C
Research Title: DESIGN AND DEVELOPMENT OF A WEBXR-ENABLED DIGITAL HERITAGE PLATFORM FOR THE NHCP MUSEO NG LIBINGAN SA ILALIM NG LUPA NG MAGCAPLAN		

MANUSCRIPT (30%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	5	
4. Research Methodology (Appropriateness of methods of study, statistical treatment, analysis and interpretations).	5	
5. Technical Background (relevant design tools, design, and implementation plan is appropriate on requirements)	5	
6. Summary, Conclusions and Recommendations (Findings, Conclusions and Recommendations are attuned with the objectives)	5	
SUB-TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	**20**	
CAPSTONE PROJECT / THESIS (50%)		
1. Innovation	15	
2. Application, Relevance, and Impact	20	
3. Research Thrust Impact	15	
SUB-TOTAL	**50**	
TOTAL	**100**	

VERDICT

[ ] **PASSED**. Minor revisions are necessary but they do not have to be presented in front and checked by all panelists. Grade range: **95-100**.

[ ] **PASSED WITH MINOR REVISIONS**. Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94**.

[ ] **PASSED WITH MAJOR REVISIONS**. Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84**.

[ ] **REDEFENSE**. Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74**.

[ ] **DISAPPROVED**. The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70**.

TE
Role on the Defense

Reynaldo C. Justo
Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

FINAL ORAL DEFENSE RATING SHEET

Group Code.: EWD1	Date: SEPTEMBER 15, 2026	Time:
Name of Proponents:		
ALAY, RIZ-ANNE V.	BUSTRIA, JOHN LLOYD J.	GAYOSO, CEDRICK C.
Research Title: DESIGN AND DEVELOPMENT OF A WEBXR-ENABLED DIGITAL HERITAGE PLATFORM FOR THE NMCP MUSEO NG LIBINGAN SA ILALIM NG LUPA NG MAGICARIAN		

MANUSCRIPT (30%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	5	
4. Research Methodology (Appropriateness of methods of study, statistical treatment, analysis and interpretations).	5	
5. Technical Background (relevant design tools, design, and implementation plan is appropriate on requirements)	5	
6. Summary, Conclusions and Recommendations (Findings, Conclusions and Recommendations are attuned with the objectives)	5	
SUB-TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	20	
CAPSTONE PROJECT / THESIS (50%)		
1. Innovation	15	
2. Application, Relevance, and Impact	20	
3. Research Thrust Impact	15	
SUB-TOTAL	50	
TOTAL	100	92

VERDICT

[ ] **PASSED.** Minor revisions are necessary but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

SAT. / D. A.
Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
FINAL ORAL DEFENSE RATING SHEET

Group Code.: EW01	Date: 9/22/24	Time:
Name of Proponents:		
ALAY, RIZ-ANNE V.	BUSTRIA, JOHN LLOYD J.	GAYOSO, CEDRICK C.

Research Title: DESIGN AND DEVELOPMENT OF A WEBXR-ENABLED DIGITAL HERITAGE PLATFORM FOR THE NHCP MUSEO NG LIBINGAN SA ICALIM NG LUPA NG HACIENDA LAN

	WEIGHT (%)	EVALUATED SCORE
MANUSCRIPT (30%)		
1. Format (documentation, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	5	
4. Research Methodology (Appropriateness of methods of study, statistical treatment, analysis and interpretations).	5	
5. Technical Background (relevant design tools, design, and implementation plan is appropriate on requirements)	5	
6. Summary, Conclusions and Recommendations (Findings, Conclusions and Recommendations are attuned with the objectives)	5	
SUB-TOTAL	**30**	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	**20**	
CAPSTONE PROJECT / THESIS (50%)		
1. Innovation	15	
2. Application, Relevance, and Impact	20	
3. Research Thrust Impact	15	
SUB-TOTAL	**50**	
TOTAL	**100**	95%

VERDICT

[✓] **PASSED.** Minor revisions are necessary but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Language Critic
Role on the Defense

Ma. Cezanne Dimaculangan-Villanueva
Signature over Printed Name of Evaluator

APPENDIX C

CURRICULUM VITAE OF SYSTEM TESTERS

Curriculum Vitae of System Testers

Jerem Micah C. Rivera

IT Professional

About Me

I have prior experience managing the Information Technology of the Municipal Government of Nagcarlan, Laguna.

I work diligently and dependably to satisfy my clients and to earn the respect of my coworkers.

I'm innovative and constantly look for the quickest, most realistic solutions for problems without sacrificing the quality of my work.

I excelled at completing IT tasks and projects, becoming into a useful asset and applying my success to yours.

+63 975 305 5690
jerem.rivera@gmail.com
https://m.me/xeride
Nagcarlan, Laguna, PH 4002

EDUCATION

Master of Science in Information Technology [Units Earned]
Laguna State Polytechnic University - Santa Cruz, Laguna
2021 - 2022

Bachelor of Science in Information Technology
Laguna State Polytechnic University - Santa Cruz, Laguna
2013 - 2015

Associate in Information Technology
Laguna State Polytechnic University - Santa Cruz, Laguna
2011 - 2013

Primary and Secondary Education
Cornerstone Christian Academy - Nagcarlan, Laguna (dc.)
2003 - 2011

WORK EXPERIENCE

Information System Analyst II [SG 16] 2024 - present
Municipal Government of Nagcarlan
Rizal Ave. Nagcarlan, Laguna, PH 4002

Computer Operator II [SG 09] 2018 - 2024
Municipal Government of Nagcarlan
Rizal Ave. Nagcarlan, Laguna, PH 4002

Information Technologist II 2016 - 2018
Municipal Government of Nagcarlan
Rizal Ave. Nagcarlan, Laguna, PH 4002

Production Operator 2015 - 2016
SunPower Manufacturing Ltd. (SPML)
FPIP Tanauan, Batangas, PH 4027

On-the-Job Trainee (Mayor's Office) 2014 - 2015
Municipal Government of Nagcarlan
Rizal Ave. Nagcarlan, Laguna, PH 4002

Tech. Support Staff (Student Assistant) 2012 - 2015
Laguna State Polytechnic University - Santa Cruz, Laguna
Bubukal Santa Cruz, Laguna, PH 4009

REFERENCES

ENGR. RANDY B. MOREDO
Municipal Planning and Development Coordinator (MPDC)
Municipal Government of Nagcarlan, Laguna, PH 4002
+63 928 507 4774

MRS. JOAN MILA M. CO
Municipal Accountant
Municipal Government of Nagcarlan, Laguna, PH 4002
+63 918 915 0803

MRS. LERMA V. AUSTRAL
Assistant Municipal Treasurer
Municipal Government of Nagcarlan, Laguna, PH 4002
+63 928 330 1967

ELIGIBILITY

CIVIL SERVICE PROFESSIONAL
March 12, 2017
Laguna State Polytechnic University
Sta. Cruz, Laguna
Rating: 84.6

Microsoft
Office Specialist
https://verify.certiport.com oSbo - uGob
https://verify.certiport.com dC6r - DTe8

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

Jerem Micah C. Rivera

SKILLS / TOOLS

APPLICATIONS

Data Recovery
- R-Studio Recovery
- TestDisk
- EaseUS Apps
- MiniTool Apps

Database Management
- MySQL / MariaDB
- Access ACCDB/MDB
- Microsoft SQL Studio

Cloud Storage Solutions
- FileZilla Pro
- R-Clone
- Google Drive Management

Computer Aided Design (CAD)
- Autodesk AutoCAD
- Sketchup Pro
- V-Ray Renderer
- Enscape Renderer

Cybersecurity
- Awareness Training
- Disaster Preparedness and Mitigation
- IT Incident Management
- IT Security Management

Geographical Information System (GIS)
- GIS Data Collection
- GIS Data Analysis
- Digital Mapping
  - QuantumGIS (QGIS)
  - Google Earth Pro
  - Open Street Maps (OSM)

Microsoft Office Suite
- Word
- Excel
- PowerPoint
- Publisher
- Access
- Visio Pro
- Project Pro

Office and Productivity Automation
- Automation Tools
- Macro
- Visual Basic for Applications (VBA)

Remote Access
- TeamViewer
- Windows RDP
- AnyDesk
- VPN

Virtualization
- VM Management
- VMWare Workstation
- VMWare ThinApp
- Oracle Virtualbox

Web Development
- HTML and CSS
- PHP
- Content Management Systems (CMS)
- WordPress Development

Drone Piloting Skills
Human Resource Management Systems
IT Process Improvement
IT Research
Office Security and Surveillance Systems
Social Media Management
Time Tracking and Project Management

COMPUTER HARDWARE

Hardware Troubleshooting
Hardware Repair and Maintenance
IT Assets Management and Administration
Network Troubleshooting and Repair
Network Management and Administration
PC Building
PC Physical Cleaning
Server Management

COMPUTER SOFTWARE

Digital Assets Management
Digital Data Backup and Archiving
Digital Storage Analysis
File System Analysis
Operating Systems (OS)
- Windows OS
- Windows Server
- Linux Distributions
- Android OS

OS Installation
Performance Benchmarking and Monitoring
Software Installation
System Administration
System Testing and Debugging
System Troubleshooting

MULTIMEDIA

Photo Editing
- Adobe Photoshop CC
- DxO Nik Collection Tools
- Retouch4me AI Tools

Graphic Designs
- Tarpaulin / Large-Format Designs
- T-Shirt Designs
- Logo Designs
- ID Card Layout Design
- Poster Designs

Photography
- Adobe Lightroom
- Adobe Camera Raw

Video Editing
Audio Editing
- Adobe Audition CC
- Audacity

Audio Mixing
- Audio Mixer Operations
- Virtual DJ Pro

Audio/Visual Equipment Management

PRACTICAL SKILLS

Documentation
Email and Calendar Management
Encoding
Reporting
Training and Teaching

PERSONAL SKILLS

Attention to Detail
Communication
Creativity
Critical Thinking
Efficiency
Fast Learning
Flexibility
Leadership
Logical Thinking
Planning and Analysis
Self-Learning
Self-Motivated
Team Oriented
Work Independently

Gerica C. Urriza

Purok 5, Brgy. Taytay,
Nagcarlan, Laguna, 4002

0930-944-8041/0992-592-0106

jericaconcordia@gmail.com

About Me

I have a clear, logical mind with a practical approach to problem solving and drive to see things through to completion. I am aeger to learn , I enjoy overcoming challenges, and i have a genuine interest in making organizations successful.

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

Gerica C. Urriza

Work Experience

2024-2025
MAYOR'S OFFICE ICT UNIT
Municipal Government of Nagcarlan, Laguna

2024
ON-THE-JOB TRAINEE
MAYOR'S OFFICE ICT UNIT
Municipal Government of Nagcarlan, Laguna

2022-2023
COUNSELOR SECRETARY STAFF
Municipal Government of Nagcarlan, Laguna

2021 -2022
BUSINESS PERMIT AND LICENSE OFFICE
Municipal Government of Nagcarlan, Laguna

2019 - 2020
HARLEYS HARDWARE
Maharlika Highway, San Pablo City, Laguna

2015 - 2017
LOTTO OPERATOR
Mabini St. Nagcarlan, Laguna

Education

TERTIARY: 2020 - 2024
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
Pamantasan ng Lungsod ng San Pablo

TERTIARY: 2017 - 2019
COMPUTER PROGRAMMING TECHNOLOGY
Dalubhasaang Lunsod ng San Pablo

SECONDARY: 2010-2014
PLARIDEL NATIONAL HIGH SCHOOL
Nagcarlan, Laguna

INTERMEDIATE: 2004 - 2010
GUERILLA ELEMENTARY SCHOOL
San Pablo City, Laguna

Character References

Ms. Carmela Herrera | 0977-4875-821
Store Supervisor - Allsports - Vista Mall
Sta. Rosa City, Laguna

Mr. Jerem Micah C. Rivera | 0975-305-5690
Computer Operator II
Nagcarlan, Laguna

Ms. Ella Mae Cuello | 0919-7675-901
Store Supervisor - Harleys Home Depot
San Pablo City, Laguna

Noreen Ma. Patrice D. Matienzo

Nagcarlan, Laguna, Philippines
matienzonoreenma.patrice@gmail.com
09051642357

CAREER OBJECTIVE

To secure a challenging role in computer science where I can apply my technical skills and passion for problem-solving to drive innovation. Dedicated to continuously learning and adapting to new technologies. Striving to make a significant impact in the field of computer science through creative thinking and collaborative efforts.

EDUCATION

Laguna State Polytechnic University – San Pablo Campus
San Pablo City, Laguna, 4000
Bachelor of Science in Computer Science
Major in Intelligent System
2020 – 2024

- *Graduated with Latin Honor*

Upland Integrated National High School
Nagcarlan, Laguna, 4002
Technical Vocational Livelihood – Information Communications Technology - Computer Systems Servicing
2018 – 2020

- *Awarded with Honors*
- *Conduct Awardee*
- *Excellence Award in Work Immersion*
- *NCII Certificate – Computer Systems Servicing*
  February 2020

Upland Integrated National High School
Nagcarlan, Laguna, 4002
Junior High School
2014 – 2018

- *Awarded with Honors*

Lazaan – Malinao Elementary School
Nagcarlan, Laguna, 4002
Elementary
2008 –
2014

- *Special Awards*

WORK EXPERIENCE

FRSS Technology & Consultancy Corp. – Sta. Cruz Laguna
2019 – 2020
Sta. Cruz, Laguna, Philippines
Work Immersion

- *Sales inventory*
- *Communication skills*
- *Knowledge in computer hardware/ software*

Special Program for Employment of Students (SPES) Municipality of Nagcarlan
2018 – 2024

- *Encoding data*
- *Organizing documents*
- *Assisting clients*

Clerk - Local Youth Development Office
Municipality of Nagcarlan, Laguna
September 2024-Present

- *Creating Payroll and other documents*
- *Maintaining Records*
- *Critical thinking skills*

PROJECTS

Inventory System | Back-end Developer | *Php, Mysql, HTML*
CCS February 2022

IPCR System | Front-end Developer | Tester | *Javascript, React, Next.js, MongoDB*
April 2023

Unveiling Public Sentiment on The Perspective Of K- 12 Education: A Facebook Sentiment Analysis | Programmer | *Python*
July 2023 – April 2024

Archiving System | Front-end Developer | Tester | *Javascript, React, Node.js, Supabase* January 2024 – April 2024

SEMINARS ATTENDED

ASEAN Data Science Explorers Enablement Session Department of Information and Communications Technology - Region IV-A
April 18, 2024

20 Years of Inspiring Future Technology Leaders!
UP System Information Technology Foundation
November 17, 2022

13th EdukCircle International Convention on Engineering and Computer Technology
UP Diliman April 2023

SKILLS

Languages:
- JavaScript
- Java
- Python
- C#

Frameworks:
- React
- Node.js

Developer Tools:
- Github
- Visual Studio Code
- Anaconda
- Jupyter Notebook
- Google Colab
- Figma

PERSONAL SKILLS & INTERESTS

Computer:
- Microsoft Word
- Microsoft PowerPoint
- Microsoft Excel
- Canva

Languages:
- English (Conversational)
- Filipino (Fluent)

Bernard E. Arciaga

Bachelor of Science in Information Technology

PERSONAL INFO

Address
Sitio Mapalad Barangay,
Pagsawitan Sta Cruz Laguna.

Phone
+639397919937

Email
bernardarciaga44@gmail.com

Date of Birth
May 9, 1996

Birth Place
Santa, Cruz Laguna

OBJECTIVE

To acquire a challenging position in an environment where I can best utilize my skills and education and to obtain a management position, in which I am given the opportunity to play a direct role in the unlimited growth and success of solid organization.

ACHIEVEMENTS & AWARDS

- Proficiency Award in PC Troubleshooting and Networking
- Leadership Award
- Department Service Award

SEMINAR & TRAININGS

- Leadership Training & Team building seminar. (May 13, 2019)
- Youth for Information Technology UP Diliman. (September 26, 2019)

ACADEMIC

Philippine Women's University
Bachelor of Science in Information Technology

- On-The-Job-Training

Certification 486 hours On-The-Job-Training
on October 2020 To December 2020.
Supervised by Technical Support, Department at Rizalnet Enterprises M.H del pilar St. Sta Cruz laguna.

MR. MARK BRYAN P. ROAN
Technical Support

MR. ROMEL S. CALACALSADA
Officer-in-Charge

SKILLS

Proficient In

- Cloud-based System (cloud computing)
- Network Administration and Security
- CCTV NVR & IP Camera Installation include "Software and Hardware" Troubleshoot
- Optimize the IT infrastructure
- Diagnose and resolve hardware, software, and network issues
- Hardware and Software Installation and Configuration
- Operating Systems (Windows, Mac, Linux)
- Firewall Security System

Other Skills

- Configuration, administration and maintenance of local area network, hardware and IT services for the company
- Knowledge of Microsoft Office Applications and Point of Sale (POS) System
- Ability to troubleshoot program and system malfunctions
- Skills in addressing system administration issues and network concerns

WORKING EXPERIENCE

- *Management Information System (MIS) (2025-present)* in (First United Farmers Bank, Inc.)
- *Information Technology Specialist (2022-2025)* in (Noli Real Estate Properties Corporation) 3 years of professional working experience in Information Technology Specialist

JANSSEN REYES ESPINO

Committed graduate of Bachelor of Science in Information Technology in Laguna State Polytechnic University who has experience in the field of telecommunication activation and data encoding skills that I have learned in the program and internship to elevate self-growth and to contribute to future organizations. Passionate about precision and quality of work.

ACHIEVEMENTS AND PROFESSIONAL AFFILIATIONS

- Civil Service Exam Professional Passer, March 2025
- Cisco Networking Academy Operating Systems Basics - Completer, December 2024
- eLGU Admin Training and System Setup
- System Administrator in phpMyAdmin completer
- Graduated First Honorable Mention, 2015

EXPERIENCES

- **MUNICIPALITY OF SANTA CRUZ IT STAFF**
  JULY 2025-PRESENT
- **NESTLE BRAND AMBASSADOR**
  JULY 2022 - OCTOBER 2022
- **NATIONAL ELECTION SUPPORT STAFF**
  MAY 2022
- **BARANGAY ELECTION THIRD MEMBER**
  OCTOBER 2023
- **PLDT DATA ENCODER AND SERVICE ACTIVATION**
  JANUARY 2025-JUNE 2025
- **SENATORIAL ELECTION SUPPORT STAFF**
  MAY 2025

INTERNSHIP

CENTEGRA TECHNOLOGY INC. PLDT HOME DISTRIBUTOR
Banca-Banca, Victoria, Laguna

- Took internship in the company for 600 hours as data encoder and service activation.
- Validated customer's application before passing their service order to main office.
- Calling out customers to confirm the status of their internet and voice data.
- Data activation using third party apps.
- Manual service activation.

SKILLS

- **FOUNDATIONAL KNOWLEDGE IN MICROSOFT EXCEL**
- **DATA ENTRY AND ENCODING**
- **PROFICIENT IN COMPUTER APPLICATIONS**
- **ADOPTABLE TO NEW TECHNOLOGIES**
- **BASIC UNDERSTANDING OF COMPUTER MAINTENANCE AND REPAIR**
- **DETAIL ORIENTED**
- **COMMUNICATION SKILLS**
- **TIME MANAGEMENT**
- **ENTRY-LEVEL EXPERIENCE IN COMPUTER HARDWARE SERVICES**
- **COMPUTER TROUBLESHOOTING**

EDUCATION

Laguna State Polytechnic University, 2021-2025
Bachelor of Science in Information Technology

Laguna Senior High School, 2019-2021
Senior High School
Science, Technology, Engineering, and Mathematics

Pedro Guevara Memorial National High School 2015-2019
Junior High School

Bagumbayan Elementary School, 2009-2015
Grade 1-6

CONTACT INFORMATION

0975-353-7625
ssen.jre@gmail. com
Sitio 1 Bagumbayan Santa Cruz, Laguna

CHARACTER REFERENCE

Asher H. Pasco
0925-301-6826
Education Program Supervisor
San Pedro City Laguna

JONAS T. SANTOS

Barangay. Jasaan, Santa Cruz, Laguna, Philippines, 4009
Jonassantos122113@gmail.com | 09675977572

- IT professional with three years of experience in technical support, system administration, and IT operations in the Municipal Government of Santa Cruz, Laguna. Skilled in troubleshooting hardware, software, printers, and network issues, providing IT support across multiple municipal offices, and maintaining systems to ensure efficient operations. Known for strong problem-solving abilities, adaptability, and delivering reliable technical solutions that improve productivity and service delivery.

Work Experience

- **MUNICIPAL GOVERNMENT OF SANTA CRUZ**

Management Information Systems office | MIS IT Staff **2025 – Present**

- Provide technical support for hardware, software, printers, and network issues across municipal offices.
- Install, configure, and maintain computer systems, peripherals, and user accounts.
- Perform system administration, preventive maintenance, and software updates.
- Troubleshoot network connectivity issues and support IT projects and system implementations.
- Maintain IT asset inventory, service records, and technical documentation.

Municipal Administrator's Office / Municipal ICT Office | Administrative Aide **2024 – 2025**

- Performed administrative and clerical functions to support daily office operations.
- Prepared and maintained records, reports, and official correspondence.
- Assisted with document processing, office coordination, and administrative support.

Municipal Government of Santa Cruz | DOLE - Government Intern **2023 – 2024**

- Municipal Administrator's Office / Municipal ICT Office

- **PROVINCIAL GOVERNMENT OF LAGUNA | ON-THE-JOB TRAINING** **2022**
  - Laguna Parole and Probation Office - R4A

Key responsibilities

- Provide technical and systems support for multiple LGU information systems, including the ***General Services Office Management System, Motorized Tricycle Operators Permit System***, and ***EBOSS Queueing System.***
- Troubleshoot and resolve system issues, perform database record updates and corrections, manage user accounts, and generate operational reports.
- Perform source code modifications, including layout adjustments (margins and page breaks), to improve system-generated reports and document formatting.
- Administer and maintain EBOSS Queueing System kiosks, ensuring reliable operation and resolving system-related concerns.
- Perform daily backups of LGU system databases to ensure data integrity, security, and disaster recovery readiness.
- Provide technical support for municipal offices by installing, maintaining, and troubleshooting computers, printers, network connectivity, and other IT equipment.
- Prepare and process Purchase Requests (PRs) for MIS office supplies, IT equipment, maintenance requirements, and other operational needs.
- Manage inventory and monitoring of office supplies, computer equipment, and IT tools to ensure availability and accountability.

Technical Skills & Abilities

- IT Technical Support
- Hardware Installation, Configuration & Troubleshooting
- Backup and recovery deleted files
- Software Installation & Maintenance
- Printer Installation & Troubleshooting
- Network Troubleshooting
- System Administration
- System User Account Creation and Management
- Operating System Installation & Configuration (Windows)
- Technical Documentation
- Computer System Maintenance
- Problem Solving & Troubleshooting
- Analytical Thinking
- Customer Service & End-User Support
- Communication Skills & Team Collaboration
- Time Management
- Attention to Detail
- Adaptability & Continuous Learning
- Organizational Skills
- Ability to Work Under Pressure

Education

- **TERTIARY LEVEL:** **2018 – 2022**
  Bachelor of Science in Information Technology
  Major in Service Management Program
  Laguna State Polytechnic University – Santa Cruz Campus
- **SENIOR HIGH SCHOOL:** **2016 – 2018**
  Science, Technology, Engineering and Mathematics – STEM
  AMA Computer Collage – Santa Cruz Campus

Civil Service Eligibility

- **Career Service Subprofessional Eligibility**
  Passed
  March 2026

Personal Information

- Age: 26 years old
- Birthdate: May 11, 2000
- Birthplace: Santa Cruz, Laguna
- Sex: Male
- Height: 4'11"
- Weight: 40kg
- Citizenship: Filipino
- Civil status: Single
- Religion: Roman Catholic
- Languages: English/Tagalog

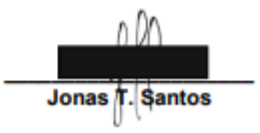
Seminars & Training

- **Introduction to Networking**
  Provincial Government of Laguna - Miso Training Room
  April 30, 2026
- **E-LGU User's Training**
  DICT- CALABARZON
  April 23, 2026
- **Web Development (Laravel & MySQL) & Server Administration**
  Provincial Government of Laguna - Miso Training Room
  January 5 – 16, 2026
- **Data Analytics Essentials**
  December 5 – 11, 2025
  DICT-ITU DTC (Net. Acad. Program)
- **Gender Sensitivity and Safe Space Act of the Philippines**
  LGU Santa Cruz
  December 2, 2025
- **E-LGU Admin Training and System Set-Up**
  DICT- CALABARZON
  November 25 – 28, 2025
- **Google Workspace for Educators/LGU Employees**
  DICT – CAR
  October 27, 2025
- **Cybersecurity and PNPKI Awareness at LGU - Santa Cruz**
  DICT - Calabarzon
  August 11, 2025
- **Basic & Intermediate Geographic Information System (GIS) Training**
  Uplink Integrated Solution Inc.
  January 30 – February 1, 2025
- **City/Municipal-Level Training Coordination, and**
  Field Visit of LGU on the Adoption of the Philippines Standard Industrial Classification
  PHILLIPPINES STATISTICS AUTHORITY
  July 6, 2024
- **Computer System Servicing Training**
  Provincial Government of Laguna - Miso Training Room
  October 16 – 20, 2023
- **Installing and Configuring Computer System & Introduction to Computer System Servicing**
  TESDA Online Program
  July 2022

Character Reference

Reference available upon request.

I hereby certify that the statement above are true and correct.



Jonas T. Santos

CHRISTIAN DALE JUMARANG PADILLA

LYNVILLE RESIDENCES, SAN JUAN
SANTA CRUZ LAGUNA

CDJPADILLA29@GMAIL.COM

09308806944

Objectives: Seeking an IT role to utilize my skills along with the growth of the organization as well as to enhance my knowledge about new and evolving trends in the IT sector.

TECHNICAL SKILLS

Computer Hardware & Software Troubleshooting

Network Troubleshooting

Database Design and Implementation

Database Backups and Restoration

PERSONAL SKILLS

Good oral and written communication skills (English and Filipino)

Strong communication and presentation skills

Ability to work independently or as a part of a team

Programming: Proficient in Visual Basic, PHP C++, JAVA, HTML, CSS, SQL, GIT, CLI, Google Firebase

Operating System: Windows

Networks: LAN/WAN, Wi-Fi, VPN, DNS, MAC, IP Addresses

Applications: Microsoft 365, Microsoft SQL Server, Dreamweaver, Anydesk, TeamViewer

SOCIAL

Viber: Christian Dale Padilla

CHARACTER REFERENCE

Available upon request.

WORK EXPERIENCE/CERTIFICATIONS

Head IT

First United Farmers Rural Bank | April 2022 - Present

- Installation, configuration and software updates of the company's systems application to maintain healthy transactions on a daily basis. Handles contacts with third party vendor relationships or service providers for technology solutions improving services and bank's technical matter.
- Database restoration and daily backups of Microsoft SQL Server including Windows Server management. Writes and performs basic to complex SQL queries to check system's error, analyzing large datasets and optimization of data retrieval to support the company's business intelligence and decision making.
- Restoring IT infrastructure, data backups including cloud-based solutions along with the Business Continuity plans (BCP) and Disaster Recovery Plans (DRP). Ensuring operational stabilization, minimize downtimes and protects servers during disruptions or emergencies.
- Handles support metrics (ticket volumes and bank's emerging technical issues) in a timely manner.
- Level 1 and Level 2 IT support operations for ensuring high availability of critical finance applications such as core banking platforms, network connectivity, workstation issues and ensuring uninterrupted operations.
- Strengthened cybersecurity practices and access control measures and enforcing security policies, authentication protocols for compliance of the company's standard regulations. Monitors Windows firewalls, VMware virtual environments (for QA and Testing virtualization technologies) from unauthorized network access, securing system operations.
- Developed and implemented IT support policies and standard operating procedures.

IT Operations – Banking

- Creating monthly Financial Reports, Comprehensive Credit and Equity Exposure Reports, to be submitted for mandatory submissions of Bangko Sentral ng Pilipinas.
- Knowledgeable in Bank's XML programming language/Application Programming Interface.
- Gives assistance to the Bank's Compliance Officer regarding revisions for regulatory submissions of bank's Certificate of memorandum and other documents.

Meetings/Seminars:

- **Rural Bank Strengthening Forum Program Technology and Innovation Forum Theme: Through Innovations in Satellite and Cloud Computing** - Bangko Sentral ng Pilipinas Assembly Hall Malate, Manila – May 2024 (In Person). Participants: IT Head, Governance, Bank Executives, Chief Compliance Officer, International and Local Service Providers of Banking Systems
  - Gained insights into satellite and cloud computing solutions that enhance digital transformation and operational efficiency in financial services.
- **2025 BSP Cybersecurity & IT Risk Fundamentals Roadshow** September 2025 – BSP Lucena (In Person) Participants: IT Head, Governance, Bank Executives, Chief Compliance Officer
  - Hands on threat detection, network security and risk management practices.
- **Basic Concepts of Credit Risk and Customer Relationship Management Framework** – April 2026
  -Completed specialized training in developing strong analytical skills in risk assessment for compliance and financial business.

EDUCATION

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

University of Perpetual Help System Laguna | 2017 – 2021

CLASS OF 2021

GLADY MAY G. AMERICA

OBJECTIVES

Seeking a challenging opportunity where i will
able to utilize my strong organizational skills,
educational background and ability to work well
with people, which allow me to grow personally and professionally.

EDUCATIONAL BACKGROUND

DALUBHASAAN NG LUNSOD NG SAN PABLO
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
2012-2016

TIGBINAN NATIONAL HIGH SCHOOL
HIGH SCHOOL DIPLOMA
2012-2016

FLORENCE ELEMENTARY SCHOOL
2002-2008

WORK EXPERIENCE

I.T. Personnel
First United Farmers Rural Bank, Inc.
San Pablo City, Laguna
May 2023 - Up to Present

SUPERVISOR
Cellstuff Premier Corporation
Lipa City, Batangas
May 2021 - May 2022

Data Encoder / Office Staff
Ultimart Supermarket
San Pablo City, Laguna
March 2019 - August 2020

Data Encoder / Office Staff
Metro Savers Department Store
San Pablo City, Laguna
August 2016 - February 2019

ABOUT ME

Brgy. III-F, San Pablo City, Laguna
g.america@myyahoo.com
0991-499-2038
May 20, 1996
30 Years Old
Single

SKILLS

- Typing
- Data Entry
- Time Management
- Microsoft (Word, Excel, Powerpoint)

CRIS D. ORENCIA

APPIAN DEVELOPER | LOW-CODE APPLICATION DEVELOPMENT

PROFESSIONAL SUMMARY

Appian Developer with hands-on experience in designing custom workflows, automating processes, and integrating systems using Appian's low-code platform. Focused on delivering solutions that improve efficiency, meet business needs, and enhance user engagement. Comfortable working both independently and as part of cross-functional teams to achieve project goals.

PROGRAMMING & TOOLS

- SQL for querying Appian databases and external systems.
- HTML/CSS for custom UI components.
- JSON for data transformation in integrations.
- JIRA for task management and documentation.

EDUCATION

Bachelor of Science in Information Technology
Laguna University
2017 - 2021

TRAINING

- 77 Career Development Program - Appian Developer Training (2021)
- 77 Career Development Program - Sysprog Training (2024)
- 77 Career Development Program - Python Training (2025)

AGILE & COLLABORATION

- Familiar with Scrum and Kanban methodologies.
- Cross-functional collaboration with business analysts, project managers, and stakeholders.

DEPLOYMENT

- Application deployment across Dev, Test, and Production environments.
- Environment configuration, troubleshooting, and versioning.

PROFESSIONAL EXPERIENCE

SEVEN SEVEN GLOBAL SERVICES INC.
Developer 1 - Application Developer (Appian) |
April 2022 - 2024

- Designed and developed an Appian-based Disbursement System to streamline and automate payment processing for disbursements in a financial services environment.
- Implemented Appian Process Models to automate the flow of funds.
- Integrated third-party payment gateways using RESTful Web Services and Appian Integration Objects to trigger external transactions and synchronize status updates.
- Developed and maintained automated notifications and alerts within the application to ensure timely approvals, status updates, and compliance with regulations.
- Implemented business rules and conditions for automatic payment approvals based on predefined criteria such as payment amounts, account verifications, and fund availability.
- Collaborated with business stakeholders to gather requirements and ensure that the system met all regulatory and compliance requirements for disbursements.
- Conducted unit testing, system integration testing (SIT), and user acceptance testing (UAT) to ensure a seamless and secure disbursement process.

APPIAN CORE SKILLS

Appian Platform: Appian Designer and Appian Cloud; applications, process models, and reports.

Process Automation: Creating, managing, and optimizing process models and workflow automation.

Rules Engine: Expression Rules, Decision Rules, and Business Rules for business logic and automated decisions.

Integrations & APIs: SOAP/REST Web Services, Integration Objects, Web API, RESTful APIs, and third-party integrations.

Data Management: Data Types, Records, data stores, queries, and reports for real-time data.

Security: Roles, permissions, access controls, authentication, authorization, and security policies.

APPENDIX D

CURRICULUM VITAE OF RESEARCHERS

Curriculum Vitae of Researchers

ALAY, RIZ-ANNE V.

Poblacion 2, Nagcarlan, Laguna
rizannevalay@gmail.com +639685210670

PERSONAL INFORMATION

Full Name: Riz-Anne Vergara Alay
Age: 21 years old
Gender: Female
Date of Birth: November 22, 2004
Place of Birth: Nagcarlan, Laguna
Citizenship: Filipino
Religion: Roman Catholic
Civil Status: Single

SKILLS

Software Proficiency: Microsoft Office, Canva, Google Workspace
UI/UX & Layout Design: Interface wireframing, Digital layout optimization, Graphic Asset Creation Software Proficiency
Programming Languages: HTML, CSS, Java, Python
Core IT Operations: System Configuration, Database Management

EDUCATION

COLLEGE
Bachelor of Science in Information Technology
Major in Web and Mobile Application Development
Laguna State Polytechnic University
Bubukal, Sta. Cruz, Laguna
2023 -

SENIOR HIGH SCHOOL
Technical Vocational Livelihood Track (Information and Communication Technology)
Laguna State Polytechnic University
Bubukal, Sta. Cruz, Laguna
2021 - 2023

JUNIOR HIGH SCHOOL
Rizal Standard Academy
Rizal Avenue, Nagcarlan, Laguna
2017 - 2021

AFFILIATIONS

STUDENT ORGANIZATION OF THE COLLEGE OF COMPUTER STUDIES - SANTA CRUZ CAMPUS
Business Manager
Laguna State Polytechnic University
2024 - 2025

REFERENCES

MIA VILLARICA, DIT
Associate Dean
College of Computer Studies
Laguna State Polytechnic University
Sta. Cruz, Laguna

REYNALEN JUSTO, LPT, DIT
Professor
College of Computer Studies
Laguna State Polytechnic University
Sta. Cruz, Laguna

BUSTRIA, JOHN LLOYD J.

Brgy. Calumpang, Nagcarlan, Laguna
lloydbustria9@gmail.com +639692991918

PERSONAL INFORMATION

Full Name: John Lloyd Jovellano Bustria
Age: 21 years old
Gender: Male
Date of Birth: June 03, 2005
Place of Birth: Nagcarlan, Laguna
Citizenship: Filipino
Religion: Roman Catholic
Civil Status: Single

EDUCATION

COLLEGE
Bachelor of Science in Information Technology
Major in Web and Mobile Application Development
Laguna State Polytechnic University
Bubukal, Sta. Cruz, Laguna
2023 -

SENIOR HIGH SCHOOL
Technical Vocational Livelihood Track (Information and Communication Technology)
Laguna State Polytechnic University
Bubukal, Sta. Cruz, Laguna
2021 - 2023

JUNIOR HIGH SCHOOL
Calumpang National Highschool
Calumpang, Nagcarlan, Laguna
2017 - 2021

SKILLS

Web Development
- HTML, CSS, JavaScript, PHP and Python
- Responsive and Mobile Web Development
- Database Integration and Management

UI/UX Design
- User Interface and Layout Design
- Wireframing and Prototyping
- User-Centered Design

IT & Professional Skills
- Technical Troubleshooting
- Problem-Solving and Critical Thinking
- Communication, and Time Management

AFFILIATIONS

STUDENT ORGANIZATION OF THE COLLEGE OF COMPUTER STUDIES - SANTA CRUZ CAMPUS
Treasurer
Laguna State Polytechnic University
2024 - 2025

REFERENCES

MIA VILLARICA, DIT
Associate Dean
College of Computer Studies
Laguna State Polytechnic University
Sta. Cruz, Laguna

REYNALEN JUSTO, LPT, DIT
Professor
College of Computer Studies
Laguna State Polytechnic University
Sta. Cruz, Laguna

GAYOSO, CEDRICK C.

Luisiana, Laguna, Philippines
cedrick.gayoso@gmail.com +639158745801

PERSONAL INFORMATION

Full Name: Cedrick Capili Gayoso
Age: 21 years old
Gender: Male
Date of Birth: May 21, 2005
Place of Birth: Sta. Cruz, Laguna
Citizenship: Filipino
Religion: Iglesia Filipina Independiente (IFI)
Civil Status: Single

EDUCATION

COLLEGE
Bachelor of Science in Information Technology
Major in Web and Mobile Application Development
Laguna State Polytechnic University
Bubukal, Sta. Cruz, Laguna
2023 -

SENIOR HIGH SCHOOL
Science, Technology, Engineering, and Mathematics (STEM)
Luis Bernardo Memorial High School Inc.
Luisiana, Laguna
2021 - 2023

JUNIOR HIGH SCHOOL
Luis Bernardo Memorial High School Inc.
Luisiana, Laguna
2017 - 2021

SKILLS

Technical Skills:

- HTML
- CSS
- JavaScript
- PHP
- MySQL
- Python
- Java
- C#
- Flutter (Basic)
- Android Studio
- WordPress
- GitHub
- Database Management
- System Integration
- AutoCAD 2D
- Basic Networking

Other Skills:

- Troubleshooting
- Critical Thinking
- Adaptability
- Time Management
- Attention to Detail

SEMINARS AND TRAININGS

AutoCAD 2D Training

REFERENCES

MIA VILLARICA, DIT
Associate Dean
College of Computer Studies
Laguna State Polytechnic University
Sta. Cruz, Laguna

REYNALEN JUSTO, LPT, DIT
Professor
College of Computer Studies
Laguna State Polytechnic University
Sta. Cruz, Laguna